FRACTIONAL DE GIORGI RIGIDITY IN DIMENSION EIGHT NEAR ONE

Abstract. We present an argument for one-dimensional symmetry of bounded entire solutions of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^8$ that are strictly monotone in one direction, for s sufficiently close to one. The proof uses the classical stable Allen–Cahn classification and the stable Bernstein theorem in $\mathbb{R}^7$. Its analytic core consists of intrinsic sheet separation, two-component interaction estimates, quantitative profile preparation, signed row capacities, and curvature retention on a complete intrinsic limit. A residual calculation and a global physical test show that this limit is minimal and stable, giving stable classification in $\mathbb{R}^7$ near the local endpoint. The directional limits of a monotone solution then yield global minimality in $\mathbb{R}^8$; a monotone perimeter blow-down and asymptotic flatness give the claimed symmetry. No energy-growth bound, prescribed end states, or restriction on the number of transition components is imposed.

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Contents

1. Introduction 1
2. Compactness and the geometry of the zero set 3
3. Interaction forces and quantitative separation 8
4. Quantitative preparation of the transition region 32
5. Capacities and the normal equation 55
6. The finite-core operator and curvature retention 74
7. Complete stable minimal limits 87
8. Stable rigidity and the dimension-eight conclusion 100
References 105

1. Introduction

The De Giorgi symmetry problem asks when monotonicity of an entire phase transition forces its level sets to be hyperplanes. We study the fractional Allen–Cahn equation in dimension eight for orders sufficiently close to the local endpoint. The argument first establishes stable classification in dimension seven and then identifies the directional limits of a monotone solution in dimension eight.

The equation is written with the Fourier normalization

$$L_s u := (-\Delta)^s u = u - u^3, \qquad \widehat{L_s u}(\xi) = |\xi|^{2s}\widehat{u}(\xi). \tag{1.1}$$

The local limit is $L_1 = -\Delta$. A bounded entire solution is *stable* if

$$Q_{s,u}(\phi) := \int_{\mathbb{R}^d} |\xi|^{2s}|\widehat{\phi}(\xi)|^2 \,\mathrm{d}\xi - \int_{\mathbb{R}^d} (1-3u^2)\phi^2 \,\mathrm{d}x \geq 0 \quad (\phi \in C_c^\infty(\mathbb{R}^d)). \tag{1.2}$$

Here d denotes an arbitrary ambient dimension in preliminary analytic statements. In the principal geometric argument we always use

$$N = 8, \qquad n = N - 1 = 7, \qquad m = n - 1 = 6.$$

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Thus the monotone solution lives in $\mathbb{R}^8$, its stable end profiles in $\mathbb{R}^7$, and the curvature-rescaled carrier is six-dimensional. Keeping these dimensions separate is essential.

Let $q_s : \mathbb{R} \to (-1, 1)$ be the increasing centered layer,

$$L_s q_s = q_s - q_s^3, \quad q_s(0) = 0, \quad q_s(-\infty) = -1, \quad q_s(+\infty) = 1.$$

Existence and uniqueness of the layer are given by [3, Theorem 2.4]; at the local endpoint $q_1(t) = \tanh(t/\sqrt{2})$. A *classified stable solution* means a constant well ± 1 or a rotated translate $q_s(e \cdot x - a)$, $|e| = 1$. The zero constant is a solution but is unstable.

1.1. **Main result.**

Theorem 1.1 (Dimension-eight rigidity)**.** *Under the conclusions of Theorems 1.2 and 1.3, there exists $s_0 \in (1/2, 1)$ such that for every $s \in (s_0, 1)$, every bounded entire solution of*

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^8, \qquad \partial_8 u > 0,$$

has the form

$$u(x) = q_s(e \cdot x - a), \qquad e \in \mathbb{S}^7, \quad e_8 > 0.$$

There is no hypothesis on its energy growth, zero-set geometry, number of components, or uniform convergence to its end profiles.

The threshold is existential; no numerical value is asserted. The absence of a prescribed directional limit is important: these limits will be classified as stable solutions in one lower dimension and will then yield the minimizing property of the original solution.

1.2. **The classical rigidity theorems.**

We use the following two recent classical rigidity results. We refer to them as CL7 and SB7, respectively, throughout the proof.

Theorem 1.2 (Classical stable classification, [1, Theorem 1.2])**.** *Every bounded stable entire solution of $-\Delta v = v - v^3$ in $\mathbb{R}^7$ is a well or a one-dimensional layer.*

The cited theorem is stated for C^2 solutions taking values in $[-1, 1]$. Proposition 2.2, applied at $s = 1$, supplies both properties for every bounded entire solution, so its hypotheses give the formulation above. The unrestricted stable statement is essential. A classification under an energy-density bound would not suffice, since the translated limits used below satisfy no such additional hypothesis.

Theorem 1.3 (Stable Bernstein theorem, [2, Theorem 1.1])**.** *Every smooth complete connected boundaryless two-sided stable minimal immersion $F : M^6 \to \mathbb{R}^7$ is flat. Stability means*

$$\int_M |A_F|^2 \phi^2 \, \mathrm{d}\mu_F \le \int_M |\nabla \phi|^2 \, \mathrm{d}\mu_F \quad (\phi \in C_c^\infty(M)). \tag{1.3}$$

No properness, embeddedness, or Euclidean area-growth assumption is added.

The immersed formulation is likewise essential: the geometric limit is constructed as a complete intrinsic carrier, rather than as an embedded ambient limit. The precise hypotheses of the other cited results are specified where they are applied.

1.3. **Outline of the proof.**

We first prove uniform solution compactness and the preservation of stability. CL7 then supplies, at every moving zero center of a hypothetical counterexample sequence $s_j \to 1$, a centered local layer. This gives global regular zero geometry and a curvature supremum $\kappa_j \to 0$, but gives no comparison between κ_j and $1 - s_j$. The flat-zero case is separated before any curvature division.

For positive curvature, intrinsic separation is followed by the first force estimate

$$\mathcal{F}_j := e^{-\sqrt{2}\delta_j} + (1 - s_j)\delta_j^{-2s_j} = o(\kappa_j). \tag{1.4}$$

The proof of this estimate must precede quantitative chamber preparation. Nested natural windows then give an $O(\kappa_j)$ comparison on complete buffers. We define the actual phase resolvent, its signed interaction rows, their capacities, and a single fixed-profile gauge for the vector normal equation. Capacity mapping and the complementary Schur calculus lead to C^1 compactness of

curvature-scaled normals. The marked curvature is retained on a complete intrinsic immersion. Resonant residual testing proves minimality, and one unsigned physical amplitude attached to an entire intrinsic test proves stability. SB7 then rules out the nonzero curvature mark.

This gives unrestricted stable classification in $\mathbb{R}^7$ near $s = 1$. Finally we pass to the two end profiles of the original monotone solution, use their minimizing character to obtain a minimizer in $\mathbb{R}^8$, and use its monotone classical-perimeter blowdown together with asymptotic flatness to finish the dimension-eight lift. The arbitrary-minimizer classification valid only through dimension seven is not used in dimension eight.

1.4. **Conventions and asymptotic quantifiers.** After discarding finitely many terms, write

$$s_j \geq s_b > \tfrac{3}{4}, \quad p_j = 2s_j, \quad \lambda_j = 1 - s_j, \quad \kappa_j > 0, \quad \kappa_j \longrightarrow 0.$$

A uniform $o(1)$ is uniform over moving marked zeros and over all charts needed for each fixed compact intrinsic region after curvature rescaling. Constants may depend on a fixed depth, atlas, derivative order, or window enlargement chosen before $j \to \infty$; these parameters are exhausted in the explicit order stated at the relevant step. No rate linking λ_j to κ_j is assumed. We use ν for the phase-oriented normal $\nabla u/|\nabla u|$, and N_i for a consistently upward normal on a locally ordered sheet; thus $\nu_i = \sigma_i N_i$, with alternating $\sigma_i \in \{-1, 1\}$. A physical tangential variable is $x \in \mathbb{R}^6$; its curvature-scaled version is $y = \kappa_j x$.

2. Compactness and the geometry of the zero set

2.1. **Uniform regularity and one-dimensional layers.** For $0 < s < 1$ the Fourier normalization is

$$L_s u(x) = c_{d,s}\,\mathrm{PV}\int_{\mathbb{R}^d} \frac{u(x) - u(x+z)}{|z|^{d+2s}}\,\mathrm{d}z, \qquad c_{d,s} = \frac{4^s\Gamma(d/2+s)}{\pi^{d/2}|\Gamma(-s)|}. \tag{2.1}$$

In particular $c_{d,s} \asymp 1 - s$ near one. Let $P_t^s = e^{-tL_s}$ and $R_s = (1 + L_s)^{-1} = \int_0^\infty e^{-t} P_t^s\,\mathrm{d}t$.

Lemma 2.1 (Uniform near-local resolvent estimate)**.** *For $s \in [s_b, 1]$ with $s_b > 1/2$,*

$$\|R_s g\|_\infty + \|\nabla R_s g\|_\infty \leq C_{d,s_b}\|g\|_\infty.$$

The resolvent commutes with bounded weak derivatives.

Proof. For $s < 1$, the completely monotone function $\zeta \mapsto e^{-t\zeta^s}$ has a probability-measure representation $\int_0^\infty e^{-r\zeta}\nu_{t,s}(\,\mathrm{d}r)$. Thus the semigroup kernel is the Gaussian mixture $p_{t,s} = \int G_r \nu_{t,s}(\,\mathrm{d}r)$, where $G_r(x) = (4\pi r)^{-d/2} e^{-|x|^2/(4r)}$. The gamma integral and Tonelli give

$$\int_0^\infty r^{-b}\nu_{t,s}(\,\mathrm{d}r) = \frac{1}{\Gamma(b)}\int_0^\infty \zeta^{b-1}e^{-t\zeta^s}\,\mathrm{d}\zeta = \frac{\Gamma(b/s)}{s\Gamma(b)}t^{-b/s}.$$

For $b = 1/2$ and Gaussian scaling,

$$\|\nabla p_{t,s}\|_1 \leq C_d \frac{\Gamma(1/(2s))}{s\Gamma(1/2)}t^{-1/(2s)}.$$

This is also valid at $s = 1$ using $\nu_{t,1} = \delta_t$. Its coefficient is bounded on $[s_b, 1]$, and $\int_0^\infty e^{-t}t^{-1/(2s)}\,\mathrm{d}t$ is uniformly finite because $s_b > 1/2$. Positivity and unit mass give the zeroth-order bound. Differentiation under the integrals gives the first order bound and commutation in distributions.

For completeness, let u, F be bounded distributional solutions of $(1 + L_s)u = F$, and mollify them in space to obtain u_ϵ, F_ϵ. Semigroup differentiation and integration over $[h, T]$ give

$$e^{-h}P_h^s u_\epsilon - e^{-T}P_T^s u_\epsilon = \int_h^T e^{-t}P_t^s F_\epsilon\,\mathrm{d}t.$$

The mollified functions are uniformly continuous, so the first term tends to u_ϵ as $h \downarrow 0$. The L^∞ contraction makes the terminal term vanish as $T \to \infty$. Hence $u_\epsilon = R_s F_\epsilon$. Removing the spatial mollifier in distributions, using the integrable mass-one resolvent kernel, gives $u = R_s F$. □

Proposition 2.2 (Uniform smoothness and the universal bound)**.** *Every bounded entire solution of* (1.1) *with* $s \in [s_b, 1]$ *satisfies* $|u| \le 1$*. For every fixed integer* k*,*

$$\|D^k u\|_\infty \le C_{d,s_b,k}.$$

Proof. Write $u = R_s(2u - u^3)$. Initially fix an individual finite bound $M \ge \|u\|_\infty$. Lemma 2.1 gives a uniform first weak derivative bound depending on M. Inductively, the product rule gives bounded derivatives of $2u - u^3$ through order k whenever $u \in W^{k,\infty}$; applying the resolvent to those derivatives yields $u \in W^{k+1,\infty}$. One extra derivative supplies a continuous representative of any desired order.

If $a := \sup u > 1$, translate almost-maximizing points. The preceding bounds for this fixed solution yield a locally smooth limit U with $U(0) = a = \sup U$. The equation passes using the bounded nonlocal tail at fixed s, or directly at $s = 1$. At a global maximum $L_s U(0) \ge 0$, whereas $a - a^3 < 0$. The infimum argument gives $u \ge -1$. Repeating the bootstrap with $M = 1$ gives the asserted universal constants. □

Lemma 2.3 (Operator limit with its coefficient)**.** *For* $0 < s < 1$ *and a uniformly bounded family in* $C^{2,\beta}(\mathbb{R}^d)$*,* $0 < \beta < 1$*,*

$$\|L_s u + a_{d,s}\Delta u\|_\infty \le C c_{d,s}\left(\beta^{-1}[D^2 u]_{C^{0,\beta}} + s^{-1}\|u\|_\infty\right), \qquad a_{d,s} = \frac{c_{d,s}|\mathbb{S}^{d-1}|}{4d(1-s)} \to 1.$$

We set $a_{d,1} = 1$ *by continuous extension; at* $s = 1$ *the operator identity is exactly* $L_1 = -\Delta$*.*

Proof. Symmetrize (2.1) and expand $2u(x) - u(x+z) - u(x-z) = -D^2u(x)[z,z] + O(|z|^{2+\beta})$ on the unit ball. The angular integral of $z_a z_b/|z|^2$ is $\delta_{ab}|\mathbb{S}^{d-1}|/d$, while $\int_0^1 r^{1-2s}\,\mathrm{d}r = [2(1-s)]^{-1}$. Writing $H = [D^2u]_{C^{0,\beta}}$, the remainder is bounded by

$$C c_{d,s} H \int_0^1 r^{1+\beta-2s}\,\mathrm{d}r = \frac{C c_{d,s} H}{2+\beta-2s} \le \frac{C c_{d,s} H}{\beta}.$$

The outer integral is bounded by $C c_{d,s}\|u\|_\infty/s$. Finally $|\Gamma(-s)| = \Gamma(1-s)/s$ and $(1-s)\Gamma(1-s) \to 1$ give $a_{d,s} \to 1$. □

Corollary 2.4 (Moving-center compactness)**.** *If* $s_j \to 1$ *and* u_j *are bounded entire solutions, arbitrary translations and rotations have a subsequence converging locally in every fixed* C^k *topology to a bounded solution of* $-\Delta U = U - U^3$*. Stable solutions have stable limits. The analogous compactness statement holds at each fixed* $s \in [s_b, 1)$*.*

Proof. Translations and rotations preserve the equation, stability, and all uniform derivative bounds. A diagonal Arzelà–Ascoli argument therefore gives a subsequence converging in every fixed local C^k topology. The bounds for D^2u_j and D^3u_j also give a uniform global $C^{0,\beta}$ seminorm for D^2u_j: use the third derivative for distances at most one and twice the second-derivative bound for larger distances. Lemma 2.3 now passes the equation. For a fixed compactly supported smooth test ϕ,

$$|\xi|^{2s_j}|\widehat{\phi}(\xi)|^2 \le (1+|\xi|^2)|\widehat{\phi}(\xi)|^2 \in L^1(\mathbb{R}^d).$$

Dominated convergence passes the kinetic term in (1.2), and local uniform convergence passes its potential term. At a fixed fractional order, the symmetrized singular integral has an integrable quadratic bound near the origin; the part beyond radius T is bounded uniformly by $C_{d,s}T^{-2s}$. Local smooth convergence followed by $T \to \infty$ passes the equation, and the same test argument passes stability. □

Lemma 2.5 (Positive Jacobi fields and zero-free stable solutions)**.** *A positive smooth solution* w *of* $L_s w = (1 - 3u^2)w$*, with* $w \in L^1((1+|x|)^{-d-2s}\,\mathrm{d}x)$*, implies stability of* u*. A bounded nonnegative stable entire solution is identically* 1*; a nonpositive one is identically* -1*.*

Proof. First let $0 < s < 1$ and write a compact smooth test as $\phi = w\eta$, where $\eta = \phi/w$ is smooth and compactly supported. Testing the Jacobi equation against $w\eta^2$ and using

$$(w_x\eta_x - w_y\eta_y)^2 - (w_x - w_y)(w_x\eta_x^2 - w_y\eta_y^2) = w_x w_y(\eta_x - \eta_y)^2$$

gives the ground-state identity

$$Q_{s,u}(w\eta) = \frac{c_{d,s}}{2} \iint \frac{w(x)w(y)(\eta(x)-\eta(y))^2}{|x-y|^{d+2s}}\,\mathrm{d}x\,\mathrm{d}y \geq 0. \tag{2.2}$$

Here $w_x = w(x)$ and $\eta_x = \eta(x)$. Local smoothness controls the diagonal singularity. In the tail, at least one variable lies in the fixed support of η, and the assumed weighted integrability of w controls the other variable. Thus truncation and passage to the limit justify the identity. Only a positive lower bound for w on the support of ϕ is needed. At $s = 1$, integration by parts gives instead $Q_{1,u}(w\eta) = \int w^2|\nabla\eta|^2$.

For the sign assertion set $b = \inf u \geq 0$. If $b = 0$, translating almost-minima gives a nonnegative stable limit attaining zero. The strong maximum principle implies it is zero everywhere. But for a nonzero compact test $\phi_R(x) = \phi(x/R)$,

$$Q_{s,0}(\phi_R) = R^{d-2s}\int |\xi|^{2s}|\widehat{\phi}|^2 - R^d \int \phi^2 < 0$$

for large R. Hence $b > 0$. If $b < 1$, a translated minimum b satisfies $L_sU(0) \leq 0 < b - b^3$, another contradiction. Thus $b = 1$. Replace u by $-u$ for the other sign. □

Corollary 2.6 (Uniform well clearing)**.** *Let v_j be bounded stable entire solutions in $\mathbb{R}^d$, with either $s_j \to 1$ or $s_j = s \in [s_b, 1)$ fixed. If*

$$\operatorname{dist}(z_j, \{v_j = 0\}) \longrightarrow \infty,$$

where distance from the empty set is $+\infty$, then, for every fixed R, k,

$$\min_{\sigma\in\{-1,1\}} \|v_j(z_j + \cdot) - \sigma\|_{C^k(B_R)} \longrightarrow 0.$$

For large j, the appropriate well has the sign of $v_j(z_j)$.

Proof. If the conclusion failed, pass to a subsequence with a fixed failed tolerance and with the sign of $v_j(z_j)$ fixed. Every fixed ball is eventually zero-free and has this sign. Corollary 2.4 gives a stable smooth limit of the corresponding sign on all of $\mathbb{R}^d$. Lemma 2.5 identifies it with that well, contradicting the failed tolerance. The same argument covers an empty zero set. □

Lemma 2.7 (Continuity of the centered layer near one)**.** *As $s \to 1$, $q_s \to q_1$ in every fixed local C^k topology, and $q_s'(0) \to 1/\sqrt{2}$.*

Proof. Use the existence and uniqueness of the centered layer recalled in Section 1. Its positive derivative is bounded by Proposition 2.2 and solves the Jacobi equation, so Lemma 2.5 gives stability. Compactness yields a nondecreasing classical stable limit q with $q(0) = 0$; Lemma 2.5 excludes $q \equiv 0$. The finite end limits satisfy $\ell_- \leq 0 \leq \ell_+$. For any sequence $t_i \to \pm\infty$, monotonicity gives local uniform convergence of $q(t_i + \cdot)$ to $\ell_\pm$; compactness improves it to local smooth convergence and preserves stability. Lemma 2.5 therefore gives $\ell_- = -1$, $\ell_+ = 1$, and the smooth convergence also gives $q'(t) \to 0$ at both ends. Multiplication of the classical equation by q' yields

$$\frac{(q')^2}{2} = \frac{(1-q^2)^2}{4} + C.$$

Evaluation at either end gives $C = 0$. Monotonicity and the centered initial condition identify $q(t) = \tanh(t/\sqrt{2})$ by ODE uniqueness. Every subsequential limit is the same, proving the asserted convergence and the derivative limit. □

2.2. **Geometry from the classical classification.**

Proposition 2.8 (Uniform geometry from arbitrary moving zeros)**.** *Using Theorem 1.2, let $s_j \to 1$ and let u_j be bounded stable entire solutions on $\mathbb{R}^7$ that are not classified. For all sufficiently large j, their zero sets Σ_j are nonempty smooth, ambiently closed, embedded, boundaryless, two-sided hypersurfaces, each component intrinsically complete, and*

$$\inf_{\Sigma_j} |\nabla u_j| \geq \frac{1}{2\sqrt{2}}, \qquad \sup_{\Sigma_j} |A_j| \longrightarrow 0. \tag{2.3}$$

For every fixed R, k,

$$\sup_{z\in\Sigma_j}\inf_{e\in\mathbb{S}^6}\|u_j(z+\cdot)-q_{s_j}(e\,\cdot)\|_{C^k(B_R)}\longrightarrow 0. \tag{2.4}$$

For every fixed R, k there is also uniform qualitative clearing:

$$\lim_{D\to\infty}\limsup_{j\to\infty}\sup_{\operatorname{dist}(x,\Sigma_j)\geq D}\min_{\sigma\in\{-1,1\}}\|u_j(x+\cdot)-\sigma\|_{C^k(B_R)}=0,$$

with the supremum over an empty set taken to be zero.

Proof. Lemma 2.5 excludes an empty zero set. Given any sequence $z_j\in\Sigma_j$, compactness and CL7 identify every local limit of $u_j(z_j+\cdot)$ as a centered classical layer. The zero at the origin excludes wells and fixes its translation. Its gradient has magnitude $1/\sqrt{2}$. If the first inequality of (2.3) failed at any zeros, those zeros could be selected as the centers, contradicting this limit. The regular-value theorem gives the asserted smooth geometry, with global normal $\nu_j=\nabla u_j/|\nabla u_j|$.

Closedness follows from continuity. To prove completeness, let (x_i) be an intrinsic Cauchy sequence in one connected component. It is ambient Cauchy and converges to a point x of the closed level set. Choose a regular graph chart at x and a smaller graph neighborhood inside it. The tail lies in the smaller neighborhood; there, lifting straight segments in the graph base shows $d_{\Sigma_j}(x_i,x)\leq C|x_i-x|\to 0$. These graph paths also put x in the same connected component. Thus every component is intrinsically complete.

For unit tangent vectors v, w the second fundamental form, up to the choice of sign, is

$$A_j(v,w)=\frac{D^2u_j(v,w)}{|\nabla u_j|}.$$

Its supremum is finite by the uniform derivative and gradient bounds. If it failed to tend to zero, choose zeros with curvature bounded below by a positive constant. Their translated C^2 limit is a centered layer, whose tangential second derivatives vanish, a contradiction. No maximum of $|A_j|$ needs to be attained. Failure of (2.4) similarly selects a bad moving center; Lemma 2.7 and classical layer compactness contradict the positive error.

Finally, failure of the quantified clearing statement would select a subsequence of centers whose distances from Σ_j tend to infinity and whose errors exceed a fixed positive tolerance. Corollary 2.6 rules out this sequence. □

Remark 2.9 (The scale of the compactness estimates). The estimates are global because their negations can be centered at arbitrary bad points. Nevertheless they only control fixed physical balls. They do not imply an $O(\kappa_j)$ approximation, nor a vanishing error after division by κ_j^2, nor survival of a nonzero curvature mark after rescaling. The remaining analytic sections address those precise scale-sensitive questions.

2.3. **Intrinsic separation and the zero-curvature alternative.** Hereafter the stable equation is in $\mathbb{R}^7$, the tangential dimension is $m=6$, and

$$p_j=2s_j\longrightarrow 2,\qquad \lambda_j=1-s_j\longrightarrow 0,\qquad k=\sqrt{2}.$$

Thus λ_j denotes the small fractional parameter, never the exponential decay rate. We use the preparation already obtained at every moving zero: uniform C^2 convergence on each fixed ball to a centered one-dimensional layer, uniform physical derivative bounds, regularity of zero, completeness of the embedded zero components, and qualitative well clearing. The next argument uses this preparation rather than a second classification theorem.

Lemma 2.10 (Separation including returning branches)**.** *Suppose $0<\kappa_j=\sup_{\Sigma_j}|A_j|\to 0$. Fix a sufficiently small $a>0$, assign intrinsic distance $+\infty$ between components, and put*

$$\delta_j=\inf\{|X-Y|: X,Y\in\Sigma_j,\ d_{\Sigma_j}(X,Y)\geq a/\kappa_j\}. \tag{2.5}$$

The empty infimum is $+\infty$. Then $\delta_j\to\infty$. Moreover,

$$\operatorname{reach}(\Sigma_j)\geq c\min\{\kappa_j^{-1},\delta_j\}. \tag{2.6}$$

Proof. Fix $M < \infty$. Center at any $X \in \Sigma_j$ and rotate its approximate layer direction to the last coordinate, writing points as (x, t). Uniform preparation on $B_{10(M+1)}$ and Lemma 2.7 give, on $B^6_{2(M+1)} \times [-1, 1]$, opposite signs at $t = -1$ and $t = 1$ and $\partial_t u_j \geq c > 0$ for large j. Hence every vertical fiber has exactly one zero $t = h_j(x)$. The implicit function theorem and C^1 preparation give $\|h_j\|_{C^1} \to 0$, uniformly in X. Preparation also excludes zeros in B_M with $|t| \geq 1$. Thus every $Y \in \Sigma_j$ with $|X - Y| \leq M$ belongs to this graph. Lifting the straight segment in its base joins X to Y with length at most $\sqrt{1 + \|Dh_j\|^2_\infty}\, M$. This is bounded independently of j, whereas $a/\kappa_j \to \infty$. Such a pair cannot satisfy (2.5). Since M was arbitrary, $\delta_j \to \infty$, including pairs on returning branches of one global component.

Choose, for example, $a < 1/8$ and $c < 1/8$. Closedness supplies a nearest point to every $Z \in \mathbb{R}^7$. Suppose that Z has two distinct nearest points X, Y at distance $r < c \min(\kappa_j^{-1}, \delta_j)$. Then $|X - Y| \leq 2r < \delta_j$, so the definition of δ_j gives $d_{\Sigma_j}(X, Y) < a/\kappa_j$. In particular they are on the same component. Completeness and Hopf–Rinow supply a minimizing unit-speed geodesic $\gamma : [0, \ell] \to \Sigma_j$ between them. Along it, $|\gamma(t) - Z| \leq r + \ell$ and $|\gamma''(t)| \leq \kappa_j$, so

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} \frac{|\gamma(t) - Z|^2}{2} = 1 + (\gamma(t) - Z) \cdot \gamma''(t) \geq 1 - \kappa_j r - a > \tfrac{1}{2}.$$

Both endpoint derivatives vanish because $Z - X$ and $Z - Y$ are normal. Strict convexity makes this impossible, proving (2.6).

The same calculation also identifies the foot of each short normal segment. Let $Z = X + t\nu_j(X)$ with $|t| < c \min(\kappa_j^{-1}, \delta_j)$, and let Y be a nearest point. Then $|X - Y| \leq 2|t| < \delta_j$. If $Y \neq X$, the minimizing geodesic between them has length less than a/κ_j, and the preceding strict-convexity bound holds with r replaced by $|t|$. The endpoint derivatives again vanish, a contradiction. Therefore $Y = X$. Consequently these normal segments cannot intersect. Since $|t|\kappa_j < 1$, the differential of the normal map is also invertible, giving a smooth tubular parametrization of the asserted neighborhood. Indeed every point in that neighborhood has a nearest point, and its displacement from that point is normal, so it lies in the image of the normal map. □

Proposition 2.11 (A flat zero set already gives an exact layer)**.** *Fix $s \in [s_b, 1)$ in the working near-one interval. A bounded stable entire solution in $\mathbb{R}^7$ whose nonempty zero set is an ambiently closed embedded boundaryless C^2 hypersurface with $A = 0$ is a one-dimensional layer, up to translation, rotation, and sign.*

Proof. Each connected component is closed in $\mathbb{R}^7$, because components are closed in the ambiently closed zero set. Vanishing second fundamental form makes its tangent plane constant, so it lies in one affine hyperplane. The local chart theorem makes it open in that hyperplane; it is also closed there. Connectedness of the hyperplane therefore makes the component the entire hyperplane. Distinct components are disjoint, hence their planes are parallel. We first exclude two zero planes $t = a < b$ without any separation hypothesis.

Write $w = \partial_t u$, $H = (-\Delta)^s - f'(u)$, and $\mathcal{Q}(\phi) = \langle \phi, H\phi \rangle$. Define the tangential translation hull to be the closure of $\{u(\cdot + (z, 0)) : z \in \mathbb{R}^6\}$ in $C^2_{\mathrm{loc}}(\mathbb{R}^7)$. Fixed-order compactness in Corollary 2.4 makes it compact. Every member U solves the same equation, is stable, and retains both zero planes. Lemma 2.5 excludes $U \equiv 0$. If $\partial_t U$ were one-signed, its integral from a to b on every fiber would vanish. The strong maximum principle for a nonnegative fractional Jacobi field then gives $\partial_t U = 0$; the zero plane would give $U = 0$. Consequently every hull member has a transverse derivative taking both signs.

This sign change gives a quantitative defect uniformly over the compact hull. For each member choose two balls with disjoint closures on which its transverse derivative has opposite strict signs. The defect integral over this pair of balls is positive and continuous in the local C^1 topology, since its kernel is separated from the diagonal. It therefore has a positive lower bound in a neighborhood of that member. A finite subcover gives common constants $L, c_0 > 0$: take B_L to contain every selected pair and take the minimum of their positive lower bounds. Thus, for every tangential translate,

$$c_{7,s} \iint_{B_L \times B_L} \big(|w(X)w(Y)| - w(X)w(Y)\big)|X - Y|^{-7-2s}\, \mathrm{d}X\, \mathrm{d}Y \geq c_0.$$

Here s is fixed; no lower bound uniform as $s \uparrow 1$ is asserted. Choose $\eta \in C_c^\infty(B_2)$ with $0 \le \eta \le 1$ and $\eta = 1$ on B_1, and set $\eta_R(X) = \eta(X/R)$. Testing the Jacobi equation against $w\eta_R^2$, using the algebra in Lemma 2.5 (which does not require a sign for this identity), gives

$$\mathcal{Q}(w\eta_R) = \frac{c_{7,s}}{2} \iint w(X)w(Y)(\eta_R(X) - \eta_R(Y))^2 |X - Y|^{-7-2s} \, dX \, dY.$$

The derivative w is bounded by Proposition 2.2. Scaling the cutoff energy therefore bounds the absolute value of the right side by $C_s \|w\|_\infty^2 R^{7-2s}$. Replacing $w\eta_R$ by $|w|\eta_R$ subtracts exactly

$$c_{7,s} \iint \big(|w(X)w(Y)| - w(X)w(Y)\big)\eta_R(X)\eta_R(Y)|X - Y|^{-7-2s} \, dX \, dY.$$

The defect is integrable at the diagonal: if $w(X)$ and $w(Y)$ have opposite signs, each has magnitude at most $\|\nabla w\|_\infty |X - Y|$, so the numerator is at most $2\|\nabla w\|_\infty^2 |X - Y|^2$. For $R \ge 4L$, a grid of spacing $3L$ in the tangential ball of radius $R/2$ gives $N_R \ge c_L R^6$ disjoint balls $B_L((z_i, 0)) \subset B_R$. On each ball $\eta_R = 1$, and the translated defect bound contributes at least c_0. The defect integrand is everywhere nonnegative, so retaining only these disjoint contributions and applying stability gives

$$0 \le \mathcal{Q}(|w|\eta_R) \le C_s R^{7-2s} - c_0 N_R \le C_s R^{7-2s} - c_s R^6.$$

Here $|w|\eta_R$ is compactly supported and Lipschitz, hence belongs to H^s; density of compact smooth tests and boundedness of the potential extend stability to it. Since $2s > 1$, the last upper bound is negative for large R. There is therefore exactly one zero plane.

Translate the unique zero plane to $t = 0$. Each open halfspace is connected and has one sign. Equal signs would make u globally one-signed with a zero, contradicting Lemma 2.5. Orient the normal so that the upper sign is positive. The fixed-order case of Corollary 2.6, applied to arbitrary centers with $t_j \to \pm\infty$, gives

$$\sup_{x \in \mathbb{R}^6} |u(x,t) - 1| \to 0 \quad (t \to +\infty), \qquad \sup_{x \in \mathbb{R}^6} |u(x,t) + 1| \to 0 \quad (t \to -\infty).$$

The fractional Gibbons theorem [10, Theorem 2] applies with $L = -(-\Delta)^s$ and $f_{\mathrm{FV}}(r) = r^3 - r$. Indeed $\inf_{|r| \ge 1} f'_{\mathrm{FV}}(r) = \inf_{|r| \ge 1}(3r^2 - 1) = 2 > 0$, and Proposition 2.2 supplies $u \in W^{3,\infty}$. Thus u depends only on t. The strict monotonicity proved in the same uniform-limit argument [10, Section 3, equation (25)], together with the centered layer uniqueness recalled in Section 1, identifies $u(t) = q_s(t)$. □

This proposition is used before any division by the actual curvature supremum. Artificial positive curvature bounds are not needed for this case, and a nonzero curvature mark is not claimed when $A = 0$.

3. Interaction forces and quantitative separation

3.1. **Selection of a shortest off-diagonal gap.** Throughout this section $f(r) = r - r^3$. Keep the intrinsic cutoff a from Section 2.3 sufficiently small for the tangent-chart continuation below. Set

$$F_j = F_{e,j} + F_{a,j}, \qquad F_{e,j} = e^{-k\delta_j}, \qquad F_{a,j} = \lambda_j \delta_j^{-p_j}. \tag{3.1}$$

All powers and exponentials are zero at $\delta_j = +\infty$. If $\kappa_j \delta_j \ge c > 0$, then

$$\frac{F_j}{\kappa_j} \le C\{\delta_j e^{-k\delta_j} + \lambda_j \delta_j^{1-p_j}\} \to 0.$$

It remains to rule out a subsequence satisfying

$$F_j \ge c\kappa_j > 0. \tag{3.2}$$

This implies

$$\kappa_j \delta_j \le C\delta_j F_j \to 0. \tag{3.3}$$

No estimate of order $O(\kappa_j)$ for the actual chamber error is used in this contradiction argument.

Lemma 3.1 (Almost-minimum pins and complete thin windows)**.** *Under (3.3), choose $\eta_j \downarrow 0$ with $\eta_j\delta_j \to 0$. After rigid motions there are consecutive actual zeros $0, d_j e_7$ such that*

$$\delta_j \le d_j \le (1+\eta_j)\delta_j, \qquad h_{1,j}(0) = 0, \quad h_{2,j}(0) = d_j, \quad Dh_{1,j}(0) = 0, \tag{3.4}$$

$$|Dh_{2,j}(0)| \le C\sqrt{\kappa_j\delta_j}(\sqrt{\eta_j} + \eta_j). \tag{3.5}$$

For each fixed working radius and each sufficiently large fixed normal buffer, all zeros in the associated thin window are a finite ordered alternating family of graphs over the same base. With $\ell_j = (\delta_j/\kappa_j)^{1/2}$, these windows contain every prescribed fixed multiple of ℓ_j, after enlarging the fixed normal buffer. Their estimates are

$$|D^2 h_{i,j}| \le C\kappa_j, \qquad |Dh_{i,j}| \le C_B\sqrt{\kappa_j\delta_j} =: \nu_{B,j} \to 0, \qquad h_{i+1,j} - h_{i,j} \ge \delta_j. \tag{3.6}$$

A window of height $O(B\delta_j)$ has $O(B)$ central-height labels; no bound independent of B is required.

Proof. An admissible pair at distance at most $(1+\eta_j)\delta_j$ exists by the definition of the infimum. We first record the local graph radius used to move its endpoint. There is a dimensional $c_g > 0$ such that the actual branch through a point with $|A| \le \kappa$ is a graph over its tangent disk of radius c_g/κ. While its slope is at most one, the curvature formula gives $|D^2 h| \le C\kappa$ and hence $|Dh(x)| \le C\kappa|x|$. Choosing c_g small improves the slope bound to $1/2$. Lifted radial paths then have length at most $\sqrt{2}c_g/\kappa$. Completeness supplies their endpoints at any finite maximal radius, and the uniform angle bound permits extension by regular local charts. The charts glue through the unique actual embedded germ, so continuation reaches the stated radius. The same argument works over a plane making a sufficiently small angle with the initial tangent plane, with radius $c_g/(2\kappa)$. Choose a once with $10a < c_g$.

The intrinsic constraint has a strict margin in this short-chord regime. Indeed, a minimizing geodesic of length $L \in [a/\kappa_j, 2a/\kappa_j]$ satisfies

$$|\gamma(L) - \gamma(0)| \ge \int_0^L (1 - \kappa_j^2 t^2/2)\, \mathrm{d}t = L - \kappa_j^2 L^3/6 \ge a/(2\kappa_j).$$

The selected chord is much smaller. Thus, if its endpoints are on one component, their intrinsic distance is greater than $2a/\kappa_j$; endpoints on different components retain distance $+\infty$. Minimize distance to the second endpoint on the first endpoint's tangent graph disk of radius $a/(20\kappa_j)$. Its boundary is at distance at least $a/(20\kappa_j) - O(\delta_j)$ from the second endpoint, whereas the initial distance is $O(\delta_j)$. The minimizer is therefore interior and its chord is normal to the first graph. The new endpoint is within ambient distance $O(\delta_j)$ of the initial endpoint, and the small-slope graph joins them by a path of length $O(\delta_j)$. Hence the intrinsic distance to the second endpoint remains greater than a/κ_j. The new pair is admissible, so its distance still lies in $[\delta_j, (1+\eta_j)\delta_j]$. Rotate its chord to the positive e_7 direction.

We next construct the complete thin windows using the actual graph continuation just proved. The small-angle argument also applies to returning branches. Work in a smaller base disk whose prescribed surrounding margin is a sufficiently large fixed multiple of $r = (T/\kappa)^{1/2}$, and let the selected branch be $t = h_1(x)$ with $|Dh_1| \le C\kappa r$ and $|D^2 h_1| \le C\kappa$ there. At a point $Y = (x_0, t_0)$ on another actual branch with $|t_0 - h_1(x_0)| \le CT$, put $a_0 = |\operatorname{proj}_{T_Y\Sigma} e_7|$. If $a_0^2 > C_1\kappa T$ for a sufficiently large fixed C_1, choose a unit tangent vector whose vertical component has magnitude a_0 and points toward the selected branch. By reflecting t if necessary, assume that the initial signed height difference is positive. Along the complete unit-speed geodesic with this initial tangent, the ambient acceleration has norm at most κ. For $\rho = C_2 T/a_0$, with C_2 fixed large enough, Taylor's estimate gives

$$t(\rho) - h_1(x(\rho)) = t_0 - h_1(x_0) - a_0\rho + O(\kappa\rho^2 + \kappa r\rho).$$

The errors are smaller than the leading signed displacement after choosing C_1 large compared with the fixed constants. Also $\rho \le Cr$, so the whole projected geodesic stays inside the supplied larger base disk. Its signed height difference therefore changes sign. By continuity it meets the selected graph. Distinct components cannot meet. For returning branches use the actual embedded zero set directly. Let U be the larger open base disk and $\Gamma_1 = \{(x, h_1(x)) : x \in U\}$. This graph is relatively closed in $\Sigma \cap (U \times \mathbb{R})$ by continuity, and relatively open because at each of

its points it coincides with the unique regular local zero-set germ. The entire projected geodesic remains in U. The set of times at which it lies in Γ_1 is therefore open and closed. The meeting point makes this set nonempty, so it includes the initial time, contrary to the choice of a distinct local branch. This argument does not assume that the other branch already has nonsingular horizontal projection. Hence $a_0 \le C\sqrt{\kappa T}$. For $\kappa T \to 0$ the projection onto the horizontal base is uniformly nonsingular on every retained branch. Completeness and the nested margins continue its graph across the smaller base; the curvature formula then gives $|D^2 h_i| \le C\kappa$. The height drift across that base is $O(\sqrt{\kappa T}\, r + \kappa r^2) = O(T)$, so choosing the larger slab margin first retains every branch meeting the smaller window. Taking $T = B\delta_j$, with B large compared with the square of the working-radius multiplier, gives (3.6). Equivalently one may take slowly diverging outer multipliers before extracting every fixed nested window. This construction concerns the entire zero set, including returning branches; independent graph extensions are never used to create separation.

Distinct zeros on one vertical fiber are intrinsic-off-diagonal. Otherwise completeness supplies a minimizing geodesic of length $< a/\kappa_j$ between them. Starting at the first zero, its projection stays within a/κ_j of that base point. By the tangent-chart radius and the small-angle estimate, the horizontal graph through this zero extends over a disk of radius $2a/\kappa_j$. The same open-and-closed membership argument keeps the whole geodesic on this graph. Its endpoint has the same base coordinate as its starting point, so they coincide, a contradiction. Thus the intrinsic distance is at least a/κ_j, and their separation is at least δ_j. Since $d_j < 2\delta_j$, no third zero lies between the chosen endpoints. Regularity of the zero level makes the crossing signs alternate. Counting at the central fiber gives the stated bound.

Finally $w_j = h_{2,j} - h_{1,j} - \delta_j \ge 0$, $w_j(0) \le \eta_j\delta_j$, and $|D^2 w_j| \le C\kappa_j$. Taylor's inequality in direction $-Dw_j(0)$, optimized on a fixed multiple of ℓ_j, yields

$$|Dw_j(0)| \le C\left(\sqrt{\kappa_j\eta_j\delta_j} + \frac{\eta_j\delta_j}{\ell_j}\right),$$

which is (3.5). □

The condition $d_j - \delta_j = o(1)$, not merely $d_j/\delta_j \to 1$, is essential for exponential force accuracy. In what follows d_j and δ_j can be interchanged in force sizes by a factor $1 + o(1)$; for exact inequalities $h_{i+1,j} - h_{i,j} \ge \delta_j$ we retain δ_j as denominator.

3.2. **Uniform layer and pair-force estimates.** We specify the analytic input before using it, because qualitative convergence $q_s \to q_1$ on fixed intervals does not give relative estimates at a gap tending to infinity. Write

$$\begin{aligned} w_s &= q_s', & a_s(t) &= 3(1 - q_s(t)^2)w_s(t), \\ G_s &= (|D_t|^{2s} + 2)^{-1}, & b_s &= a_s * G_s * a_s, \\ B_s(D) &= \int_D^\infty b_s(r)\, \mathrm{d}r. \end{aligned}$$

Centered uniqueness gives $q_s(t) = -q_s(-t)$ by applying it to $-q_s(-t)$. Hence w_s and a_s are even. Here a_s is a source, not the intrinsic cutoff a. Its exact mass is

$$\int_{\mathbb{R}} a_s(t)\, \mathrm{d}t = 3\int_{-1}^{1}(1 - r^2)\, \mathrm{d}r = 4, \qquad w_s = G_s * a_s. \tag{3.7}$$

The second identity is the differentiated layer equation. The fractional normalization is

$$c_{n,s} = \frac{4^s\Gamma(n/2 + s)}{\pi^{n/2}|\Gamma(-s)|}, \qquad \frac{c_{1,s}}{1 - s} \to 2.$$

Lemma 3.2 (Uniform near-one layer and pair-force estimates)**.** *Uniformly for s near one and large separations, including the transition between the exponential and algebraic tails, one has*

$$1-|q_s(t)| \le C\{e^{-k|t|}+\lambda(1+|t|)^{-p}\}, \tag{3.8}$$

$$|w_s^{(r)}(t)| \le C_r\{e^{-k|t|}+\lambda(1+|t|)^{-p-1}\} \quad (r=0,1,\dots), \tag{3.9}$$

$$a_s(t) \le C\{e^{-\beta|t|}+\lambda^2(1+|t|)^{-2p-1}\}, \quad \beta>k, \tag{3.10}$$

$$0 \le b_s - b_{s,P} \le \alpha_P b_s, \quad \lim_{P\to\infty}\sup_{s \text{ near } 1}\alpha_P = 0, \quad b_{s,P} = a_s\mathbf{1}_{[-P,P]} * G_s * a_s\mathbf{1}_{[-P,P]}. \tag{3.11}$$

Every fixed derivative of the mode has a uniformly bounded and tight first absolute moment. Its fixed normal Sobolev norms and the norms of $tw_s^{(r)}$ are uniform as well. If $p \ge p_* > 3/2$, then $t^2 w_s^{(r)}$ also has a uniform and tight L^2 norm. No second absolute moment of w_s is asserted. For the raw flat-pair force $\mathfrak{F}_s$, with the convention that positive force means attraction,

$$\mathfrak{F}_s(D) = (1+o(1))B_s(D), \quad -\mathfrak{F}_s'(D) = (1+o(1))b_s(D), \quad \mathfrak{F}_s(D) \asymp e^{-kD}+\lambda D^{-p}, \tag{3.12}$$

uniformly as $D\to\infty$, $s\uparrow 1$. The raw force can be replaced by the positive decreasing primitive B_s in an ordered row with total error $o(e^{-k\delta}+\lambda\delta^{-p})$, uniformly in the number of labels whose gaps are at least δ.

Proof. The Stieltjes formula displayed below holds more generally for mass $a \in [a_-,a_+] \Subset (0,\infty)$, with denominator $(r^p-a)^2+2ar^p(1+\cos\pi s)$. Its density is comparable respectively to λr^p, $\lambda/[(r-r_*)^2+\lambda^2]$, and bounded by $C\lambda r^{-p}$ on the intervals separated by $r_*/2$ and $3r_*/2$, where $r_* = a^{1/p}$. The low interval gives $C\lambda d^{-p-1}$. The resonance interval $|r-r_*| \le 1/d$ and its upper wing give $Ce^{-r_* d}$. On the lower wing use the bound $C\lambda d^2$ and integrate the exponential; its cost is at most $C\lambda d e^{-r_* d/2} \le C\lambda d^{-p-1}$. The high interval is smaller. The lower bounds follow by integrating $r \in [1/(2d), 1/d]$ and $|r-r_*| \le \lambda$. If $\lambda d \le 1$, $|r_* - \sqrt{a}|d \le C$; otherwise $e^{-\sqrt{a}d}$ is absorbed by $C\lambda d^{-p-1}$. Thus, including bounded d by positivity and the uniformly integrable Fourier multiplier,

$$G_s^a(d) \asymp e^{-\sqrt{a}|d|}+\lambda(1+|d|)^{-p-1}. \tag{3.13}$$

One density factor r gives the corresponding derivative bound, and integration gives its primitive envelope. These estimates are uniform without a condition on $\lambda \log d$.

The pointwise derivative estimate is used only for $|d| \ge 1$:

$$|(G_s^a)'(d)| \le C\{e^{-\sqrt{a}|d|}+\lambda|d|^{-p-2}\}.$$

At the origin G_s' can be singular when $s<1$; what is uniform there is its integrability. Positivity and monotonicity of the Laplace mixture show $\|(G_s^a)'\|_1 = 2G_s^a(0) \le C$ and $\int |d||(G_s^a)'(d)|\,\mathrm{d}d = 1/a$. Thus none of the following convolution or weighted estimates requires a bounded derivative of G_s at zero.

We derive the faster source envelope instead of assuming an exponential moment for a_s. By monotonicity and local convergence of q_s, choose a fixed T with $3(1-q_s^2) \le \epsilon$ on $|t|>T$. The equation for w_s gives $(|D|^p+2-\epsilon)w_s \le 0$ there. A sufficiently large fixed multiple of $G_s^{2-\epsilon}(t-T)+G_s^{2-\epsilon}(t+T)$ dominates w_s on $[-T,T]$. The massive maximum principle on the complement gives

$$w_s(t) \le C\{e^{-\gamma|t|}+\lambda(1+|t|)^{-p-1}\}, \quad 1-|q_s(t)| \le C\{e^{-\gamma|t|}+\lambda(1+|t|)^{-p}\}, \qquad \gamma=\sqrt{2-\epsilon}.$$

Choose $k<\beta<2\gamma$. Multiplication gives (3.10); for the mixed product use

$$\lambda e^{-\gamma t}(1+t)^{-p} \le Ce^{-\beta t}+C\lambda^2 e^{-(2\gamma-\beta)t}(1+t)^{-2p} \le Ce^{-\beta t}+C\lambda^2(1+t)^{-2p-1}.$$

The comparison is on an unbounded complement, but both functions tend to zero there; exhaustion, or the negative-minimum argument, justifies it.

Write $E_k(t) = e^{-k|t|}$ and $L_q(t) = (1+|t|)^{-q}$, $q = p+1 > 2$. The exponential source tail convolved with E_k is at most $Ce^{-(\beta-k)P}E_k$, and with L_q is at most $Ce^{-cP}L_q$, since $L_q(d-t)/L_q(d) \le (1+|t|)^q$. The polynomial source tail obeys

$$L_{2p+1}(t)\mathbf{1}_{|t|>P} \le (1+P)^{-p}L_{p+1}(t).$$

The uniform convolution inequalities $L_q * L_q \le CL_q$ and $L_q * E_k \le CL_q$ consequently imply

$$(a_s - a_{s,P}) * G_s \le C\{e^{-cP} + \lambda(1+P)^{-p}\}G_s, \qquad a_s * G_s \le CG_s.$$

A fixed core has a uniform positive source mass, so bounded shifts in (3.13) give $a_s * G_s * a_s \ge cG_s$. Expand the difference of the two source convolutions to obtain (3.11), with $\alpha_P \le C(e^{-cP} + \lambda(1+P)^{-p})$. In particular $w_s = a_s * G_s \le CG_s$ proves the mode envelope at order zero, including its sharp exponential rate and algebraic exponent $p+1$. Integrating gives (3.8). To justify all mode derivatives explicitly, fix β with $k < \beta < 2\gamma$, where $\gamma = \sqrt{2-\epsilon} < k$ is the preliminary comparison exponent. For $r \ge 1$ write distributionally

$$w_s^{(r)} = G_s' * a_s^{(r-1)}.$$

Assume the mode envelope is known through order $r-1$. Differentiating $a_s = 3(1-q_s^2)w_s$, the term with no derivative of $1-q_s^2$ is bounded by the preliminary value envelope times the mode envelope. Every other term contains at least two mode factors, since $q_s^{(l)} = w_s^{(l-1)}$ for $l \ge 1$ and $|q_s| \le 1$. Their exponential products have rate at least 2γ and their power products have exponent at least $2p+1$. For mixed products use the Young inequality already displayed in the source-envelope calculation. Consequently

$$|a_s^{(r-1)}(t)| \le C_r\{e^{-\beta|t|} + \lambda^2(1+|t|)^{-2p-1}\}.$$

Split $|G_s'|$ into its restriction to $[-1,1]$, whose L^1 norm is uniform, and its complement, bounded by $C[e^{-k|t|} + \lambda(1+|t|)^{-p-2}]$. The local part preserves the source bound under convolution. For the complement, $E_k * E_\beta \le CE_k$ and the standard uniform polynomial convolution inequalities give

$$|G_s'| * |a_s^{(r-1)}| \le C_r\{E_k + \lambda(1+|t|)^{-p-1}\}.$$

This proves (3.9) at order r and closes the induction. The uniform local smoothness of the layer and these integrable majorants justify each distributional differentiation and identify it with the classical derivative. Finally,

$$\int_{|t|>P} |t||w_s^{(r)}(t)|\,\mathrm{d}t \le C_r\{(P+1)e^{-kP} + P^{1-p_*}\} \longrightarrow 0.$$

The unweighted Sobolev bounds and the first weighted L^2 bounds follow from the same envelope, using one more normal derivative for H^1. For the second weighted L^2 tail, the power integrand is bounded by $C_r\lambda^2 t^{2-2p}$; it is uniformly integrable and tight when $p \ge p_* > 3/2$. These are precisely the moments used in the center-and-width comparisons.

For the raw pair use left-core coordinates $Q_D(t) = q_s(t) - q_s(t-D) - 1$. Its exact cubic residual is

$$R_D(t) = 3(1-q_s(t))(1+q_s(t-D))(q_s(t) - q_s(t-D)).$$

Define its positive projected force by $\mathfrak{F}_s(D) = \int w_s(t)R_D(t)\,\mathrm{d}t$. If $A_s^*(t) = w_s(t)(1-q_s(t))$ and $\tau_s(t) = 1 - q_s(t)$, then $B_s = a_s * \tau_s$ by (3.7) and Tonelli, and exact algebra gives

$$\mathfrak{F}_s(D) = B_s(D) - 3\int A_s^*(t)\tau_s(D-t)^2\,\mathrm{d}t, \qquad \mathfrak{F}_s'(D) = -b_s(D) + 6(A_s^* * A_s^*)(D). \tag{3.14}$$

For $t \ge 0$, $A_s^*(t) \le a_s(t)/3$ and has the fast source envelope; on $t < 0$ it is nonnegative with uniformly bounded integral. Split its convolution at $0, D$ to obtain

$$0 \le 6(A_s^* * A_s^*)(D) \le C\{(1+D)e^{-\beta D} + \lambda^2(1+D)^{-2p-1}\}.$$

Dividing by $b_s(D) \asymp e^{-kD} + \lambda(1+D)^{-p-1}$ gives a quantity tending uniformly to zero as $D \to \infty$. Thus the raw force is decreasing for sufficiently large D, and $-\mathfrak{F}_s' = (1+o(1))b_s$. Its limit at infinity is zero by the explicit residual and the integrable mode. Integrating the derivative estimate from D to infinity proves (3.12). The differentiated remainder is supplied by the exact identity, rather than by differentiation of an asymptotic expansion. Let $\epsilon(\delta) \to 0$ be a uniform

upper bound for the relative raw-force error at every $D \ge \delta$. Index neighbors separately on the two sides of the receiving sheet. On either side, $D_\ell \ge \ell\delta$, and $p \ge p_* > 1$, so

$$\sum_\ell |\mathfrak{F}_s(D_\ell) - B_s(D_\ell)| \le C\epsilon(\delta) \left\{ \sum_{\ell\ge1} e^{-k\ell\delta} + \lambda\delta^{-p} \sum_{\ell\ge1} \ell^{-p} \right\} = o(e^{-k\delta} + \lambda\delta^{-p}).$$

Summing the two sides changes only the constant. This proves the row replacement without a sheet-count factor. □

We now identify the sharp constants in the two possible tail limits. The positive Laplace representation is

$$G_s(t) = \int_0^\infty e^{-r|t|} \rho_s(r)\, \mathrm{d}r, \qquad \rho_s(r) = \frac{\sin(\pi s)}{\pi} \frac{r^p}{(r^p-2)^2 + 4r^p(1+\cos\pi s)}. \tag{3.15}$$

It follows from the Stieltjes inversion of $(z^s+2)^{-1}$ and the one-dimensional kernel of $(-\partial_t^2+r^2)^{-1}$. At low frequency the two source masses give the coefficient $4 = 4^2/2^2$. The kernel and its primitive have different dominance conditions:

$$\begin{aligned} b_{s_j}(D_j) &\sim 4c_{1,s_j} D_j^{-p_j-1} \quad \text{if } e^{-kD_j} = o(\lambda_j D_j^{-p_j-1}), \\ B_{s_j}(D_j) &\sim \frac{4c_{1,s_j}}{p_j} D_j^{-p_j} \quad \text{if } e^{-kD_j} = o(\lambda_j D_j^{-p_j}). \end{aligned} \tag{3.16}$$

In both statements $D_j \to \infty$ and $s_j \uparrow 1$. The stronger condition in the first line is necessary: a resonance contribution can be negligible in the primitive and still dominate the algebraic kernel, since differentiating the power primitive costs a factor D_j^{-1}. On a fixed sufficiently small interval $0 < r < r_0$ the Laplace density is $[\sin(\pi s)/(4\pi)]r^p(1+O(r^p))$, uniformly in s. Substitute $r = z/D_j$, first integrate over a fixed z interval, then exhaust that interval using $Cz^p e^{-z}$ for the kernel and $Cz^{p-1}e^{-z}$ for its primitive. This gives respectively $c_{1,s}D^{-p-1}/4$ and $c_{1,s}D^{-p}/(4p)$ for the low-frequency part of G_s and its primitive. The separated resonance and wing bounds used in (3.13) are negligible under the respective condition above. Apply fixed-source truncation with $m_{s,P} = \int_{-P}^{P} a_s$. Bounded source shifts do not change the low-frequency limits. At fixed P, the two source masses give $m_{s_j,P}^2 \to m_{1,P}^2$ as $j \to \infty$. Then $m_{1,P} \to 4$ as $P \to \infty$. Remove that cutoff using (3.11), also integrated in D for the primitive. This proves both statements without differentiating a primitive asymptotic. In particular the off-contact regime used below satisfies both conditions uniformly.

In the exponential regime

$$\lambda_j d_j^{-p_j} = o(e^{-kd_j}), \tag{3.17}$$

put $r_s = 2^{1/p}$ and restrict (3.15) to $|r - r_s| \le d^{-2}$. Since $\lambda d^N \to 0$ for every fixed N, this interval has mass tending to $1/(2k)$, and $(r-k)d = o(1)$ uniformly on it. The low-frequency part contributes $O(\lambda d^{-p-1})$ to G_s and $O(\lambda d^{-p})$ to its primitive. The near lower wing, with $u = r_s - r$, is bounded on $d^{-2} < u < d^{-1}$ by $Ce^{-r_s d}\lambda d^2$, and on $d^{-1} \le u \le r_s/2$ by $C\lambda d e^{-r_s d/2}$. Both are $o(e^{-kd})$ under (3.17); the upper wing is smaller. Thus, uniformly for bounded z,

$$\frac{G_{s_j}(d_j+z)}{e^{-kd_j}} \to \frac{e^{-kz}}{2k}, \qquad \frac{\int_{d_j+z}^\infty G_{s_j}}{e^{-kd_j}} \to \frac{e^{-kz}}{2k^2}. \tag{3.18}$$

After fixed-source truncation and then (3.11),

$$\int_{\mathbb{R}} a_1(t) e^{kt}\, \mathrm{d}t = 3\int_{-1}^{1} (1+r)^2\, \mathrm{d}r = 8$$

gives the sharp constants

$$\frac{B_{s_j}(d_j+z)}{e^{-kd_j}} \to 16e^{-kz}, \qquad \frac{b_{s_j}(d_j+z)}{e^{-kd_j}} \to 16ke^{-kz}. \tag{3.19}$$

The exact identities (3.14) transfer these constants from B_s to the raw cubic force.

For later use the power statement is needed uniformly off the minimum contact set. Suppose $e^{-kd_j} \le C\lambda_j d_j^{-p_j}$ and $1+\epsilon \le v \le M$. At $D = d_j v$ the resonance and its wings are negligible relative to $\lambda_j d_j^{-p_j}$ for the primitive, and relative to $\lambda_j d_j^{-p_j-1}$ for the kernel. The primitive dominance condition in (3.16) now holds uniformly on this compact v interval. Applying that proved coefficient limit gives

$$\frac{\mathfrak{F}_{s_j}(d_j v)}{(4c_{1,s_j}/p_j)d_j^{-p_j}} \longrightarrow v^{-2} \quad \text{uniformly for } 1+\epsilon \le v \le M. \tag{3.20}$$

The coefficient includes the factor 4^2 after the source cutoff is exhausted, and $\sin(\pi s)\Gamma(p+1)/\pi = c_{1,s}$. The relative raw-force error in (3.14) is already uniform. This off-contact statement, rather than an unproved uniform two-term relative expansion, is what the power limit below uses.

3.3. **Point pins and the finite-window inverse.** The force calculation must use the actual zero graphs, rather than orthogonality centers differing by an uncontrolled amount. On a fixed complete window define

$$b_i(x) = \sqrt{1+|Dh_i(x)|^2}, \quad v_i(x,t) = q_s\left(\frac{t-h_i(x)}{b_i(x)}\right), \quad V = \rho_- + \sum_{i\in\mathcal{I}} \sigma_i(v_i+1) = c_{\mathcal{I}} + \sum_{i\in\mathcal{I}} \sigma_i v_i, \tag{3.21}$$

where $\mathcal{I}$ is the consecutive set of actual ordered labels in the retained window, containing the marked labels $1,2$, $\rho_- \in \{-1,1\}$ is the actual phase below its lowest sheet, and σ_i are its alternating phase jumps, with $\sigma_{\min \mathcal{I}} = -\rho_-$. Thus $c_{\mathcal{I}} = \rho_- + \sum_{i\in\mathcal{I}} \sigma_i$. This convention remains valid when more sheets below the marked pair are retained. Use one compact smooth bump χ with $\chi(0)=1$, supported in $(-1,1)$, and set

$$C(x,t) = -\sum_i V(x,h_i(x))\chi(t-h_i(x)), \qquad Q = V + C. \tag{3.22}$$

Then $Q(x,h_i(x)) = u(x,h_i(x)) = 0$ exactly. Tangential cutoffs are placed only in a larger prepared buffer. The error $e = u - Q$ is point pinned; no L^2 orthogonality to a translation mode is asserted.

At each fixed buffer depth, uniform core preparation and well clearing show $\|e\|_\infty = o(1)$ throughout a smaller complete window. To see the order of choices, first fix a large core depth P. Inside the cores use moving-center convergence; outside them use distance-to-zero clearing and layer tails. Next let $j \to \infty$, and only afterward let $P \to \infty$. This is qualitative preparation on the whole finite window, not a rate.

Lemma 3.3 (Point-pin coercivity)**.** *For s sufficiently near one there is $c > 0$ such that every $z \in H^s(\mathbb{R})$ with continuous trace $z(0) = 0$ satisfies*

$$\langle z, (|D_t|^{2s} - f'(q_s))z\rangle \ge c\|z\|^2_{H^s(\mathbb{R})} \qquad (z(0)=0). \tag{3.23}$$

On a sufficiently separated qualitatively prepared finite stack, the corresponding ambient form has the same coercivity for compact tests supported within the prepared region whose traces vanish on every actual pin.

Proof. Let $\psi_s = w_s/\|w_s\|_2$. The one-dimensional operator is nonnegative by its positive Jacobi field. We justify a uniform L^2 spectral gap on $\psi_s^\perp$. Suppose to the contrary that $s_j \to 1$ and $z_j \perp \psi_{s_j}$ satisfy $\|z_j\|_2 = 1$ and $\langle z_j, (|D|^{2s_j} - f'(q_{s_j}))z_j\rangle \to 0$. The bounded potential gives a uniform H^{s_*} bound for any fixed $1/2 < s_* < s_b$. The mass is tight as follows. Choose a fixed T so that $3q_s^2 - 1 \ge 1$ on $|t| \ge T$, uniformly for s near one; monotonicity and local convergence of q_s supply such a T. Let $\chi_R^2 + \theta_R^2 = 1$, with $\chi_R = 1$ on $[-R,R]$, χ_R supported in $[-2R,2R]$, $\theta_R = 0$ on $[-R,R]$, $\theta_R = 1$ outside $[-2R,2R]$, and with derivatives bounded by C/R. The normalized fractional IMS error is at most $CR^{-2s}\|z\|_2^2$, uniformly for $s \ge s_b$: split its difference integral at distance R, using $|\chi_R(t) - \chi_R(r)| \le C\min(1, |t-r|/R)$, and use $c_{1,s}/(1-s) \le C$. Nonnegativity of the layer operator on the inner part and the positive well potential on the outer part imply, for $R \ge T$,

$$\|\theta_R z_j\|_2^2 \le \langle z_j, (|D|^{2s_j} - f'(q_{s_j}))z_j\rangle + CR^{-2s_b}.$$

Rellich compactness locally and this tightness give, after a subsequence, $z_j \to z$ strongly in $L^2(\mathbb{R})$, with $\|z\|_2 = 1$. The modes also converge strongly in L^2: they converge locally by Lemma 2.7, and

$$\int_{|t|>R} w_s^2 \le \|w_s\|_\infty \int_{|t|>R} w_s = \|w_s\|_\infty \{2 - q_s(R) + q_s(-R)\}$$

is uniformly small after fixing R large and then taking $s \to 1$. Consequently $z \perp q_1'$.

For every fixed Fourier radius M, strong L^2 convergence and uniform convergence of $|\xi|^{2s_j}$ to $|\xi|^2$ on $|\xi| \le M$ give

$$\int_{|\xi|\le M} |\xi|^2 |\widehat{z}|^2 = \lim_j \int_{|\xi|\le M} |\xi|^{2s_j} |\widehat{z}_j|^2.$$

Letting $M \to \infty$ gives the kinetic lower semicontinuity. Strong L^2 convergence, bounded potentials, and local potential convergence give convergence of the potential term. Thus $z \in H^1$ and $\langle z, (-\partial_t^2 - f'(q_1))z\rangle \le 0$. Nonnegativity makes this an equality. The local ground-state identity implies that z is a multiple of q_1', contradicting its orthogonality and unit norm. We have proved a uniform positive L^2 gap on $\psi_s^\perp$. Since

$$\|z_\perp\|_{H^s}^2 \le C\{\langle z_\perp, (|D|^{2s} - f'(q_s))z_\perp\rangle + \|z_\perp\|_2^2\},$$

that gap also controls its full H^s norm.

Write $z = a\psi_s + z_\perp$. The gap and bounded potential control $\|z_\perp\|_{H^s}$. Also $\|\psi_s\|_{H^s} \le C$, since its Jacobi equation and the bounded potential bound its kinetic energy by $C\|\psi_s\|_2^2 = C$. Since $s \ge s_* > 1/2$, the uniform trace estimate

$$|z_\perp(0)|^2 \le C\|z_\perp\|_{H^s}^2, \qquad \int_{\mathbb{R}} (1 + |\xi|^{2s})^{-1}\, \mathrm{d}\xi \le C$$

and $\psi_s(0) \ge c_0 > 0$ control a by the pin. This proves (3.23). Small tilt widths change the potential by $o(1)$, which is absorbed.

The ambient coercivity assertion applies to compact tests supported in the prepared window, with every retained zero-graph trace equal to zero; equivalently it applies to a globally prepared reference and compact tests in that reference. It asserts no coercivity for tests supported in an uncontrolled exterior. In the transmission argument all tested cutoffs have the former support property.

For a stack, use a normal partition with fixed large core depth. Each multiplied test retains its zero trace. The core forms use (3.23); the phase forms have a positive well potential. The normal fractional IMS error is made smaller than the gap by choosing the partition scale large, and preparation errors are then absorbed. The normal partition may depend on x, but its IMS calculation is performed at each fixed x using only the normal form, before integrating in x. Hence no tangential derivative of that partition is required. Since $(|\xi_x|^2 + \xi_t^2)^s \ge |\xi_t|^{2s}$, this supplies ambient L^2 coercivity. Boundedness of the potential upgrades it to ambient H^s coercivity. This argument has no derivative loss in the graph heights. □

Lemma 3.4 (Linear transmission from a prepared buffer)**.** *Suppose a bounded point-pinned error $e \in H^s_{\mathrm{loc}}$ satisfies $He = f + N$ in a prepared ambient window Ω, with $N \in L^1_{\mathrm{loc}}(\Omega)$, where $H = (-\Delta)^s + W$ has the preceding coercivity on point-pinned compact tests, W is bounded, and $|Ne| \le \epsilon|e|^2$ with ϵ at most half the coercivity constant. If $\|f\|_{H^{-s}} \le \mathfrak{r}$ on every fixed ambient cell enlarged to contain the full support of its testing cutoff, then on an inner cell at distance R from Ω^c,*

$$\|e\|_{H^s(\mathrm{cell})} \le C\mathfrak{r} + C\|e\|_\infty \{e^{-cR} + \lambda R^{-p}\}. \tag{3.24}$$

The constant is independent of the number of pins and the exterior geometry.

Proof. We record the powers in the lattice argument, since taking a square root of a near-pair localization error does change its power. Choose a fixed large lattice side r_0, let Q_i be its cubes, and choose cutoffs η_i equal to one on Q_i and supported in a fixed enlargement. Write $M_i = \|e\|_{L^2(Q_i)}$ and $A_i = \|\eta_i e\|_{H^s(\mathbb{R}^7)}$. The equation is tested only at indices whose enlarged cutoff support lies in Ω. Its test $\eta_i^2 e$ is point pinned and compact, so coercivity and

the nonlinear smallness assumption apply. The multiplication estimate $\|\eta_i^2 e\|_{H^s} \le CA_i$ and the exact fractional localization identity give

$$cA_i^2 \le C\mathfrak{r}A_i + Cr_0^{-p} \sum_{|j-i|\le J} M_j^2 + C\lambda r_0^{-p} A_i \sum_{|j-i|>J} (1+|j-i|)^{-7-p} M_j.$$

Here J is fixed, independent of r_0, s, the pins, and the domain. For near pairs, the cutoff difference is bounded by $C\min(1, |X-Y|/r_0)$; integrating its square against the normalized kernel gives Cr_0^{-p} uniformly, since $c_{7,s}/(2-p)$ is bounded. For far pairs one point belongs to the cutoff support; retaining the product of its local error factor and the remote error factor, and then applying Cauchy–Schwarz in each cube, gives the last term. Its coefficient is linear in $c_{7,s} \le C\lambda$. No estimate on derivatives of the remote error has been made.

Solving the quadratic inequality and using $(\sum_{|j-i|\le J} M_j^2)^{1/2} \le \sum_{|j-i|\le J} M_j$ yields

$$M_i \le A_i \le C\mathfrak{r} + (\mathcal{W} * M)_i,$$

where the positive translation-invariant weights may be chosen with finite-range part $Cr_0^{-p/2}\mathbf{1}_{|j|\le J}$ and remote part $C\lambda r_0^{-p}(1+|j|)^{-7-p}\mathbf{1}_{|j|>J}$. Choose r_0 once so large that

$$\|\mathcal{W}\|_{\ell^1} \le \tfrac{1}{2}, \qquad \sum_{|j|\ge N} \mathcal{W}_j \le C\lambda N^{-p} \quad (N \ge 2J).$$

All constants may depend on this fixed r_0.

Put $\mathcal{G} = \sum_{k\ge 0} \mathcal{W}^{*k}$. For $k \le N/(2J)$, a sum of k jumps having magnitude at least N contains a jump of magnitude at least N/k. The union bound thus gives

$$\sum_{|j|\ge N} (\mathcal{W}^{*k})_j \le C\lambda k^{p+1} 2^{1-k} N^{-p}.$$

For larger k the total mass is at most 2^{-k}. Summing the two ranges gives $\sum_{|j|\ge N} \mathcal{G}_j \le C(e^{-cN} + \lambda N^{-p})$, uniformly for $p \ge p_* > 0$.

The global boundedness of e gives a uniform upper bound M_0 for every M_i. Set $b_i = C\mathfrak{r}$ on admissible indices and $b_i = M_0$ on the other indices. Then $M \le b + \mathcal{W} * M$ on all indices: in the interior it was just proved, and on the exterior it follows from $M_i \le M_0$ and positivity. Iteration gives

$$M \le \sum_{k=0}^{n-1} \mathcal{W}^{*k} * b + \mathcal{W}^{*n} * M.$$

The last term is at most $2^{-n} M_0$, so $M \le \mathcal{G} * b$. In the inequality for A_i substitute this comparison to obtain $A_i \le C\mathfrak{r} + (\mathcal{W} * \mathcal{G} * b)_i$; here $\mathcal{W} * \mathcal{G} = \mathcal{G} - \delta_0 \le \mathcal{G}$. An inner cube at physical distance R from Ω^c is separated from those exterior indices by $N \ge cR/r_0 - C$, so convolution with their bounded source contributes at most $C\|e\|_\infty(e^{-cR} + \lambda R^{-p})$. The interior source contributes $C\mathfrak{r}$. The same comparison for A_i therefore proves the stated local H^s estimate. Distances bounded by a fixed constant are absorbed by increasing C. This also shows why the constant does not depend on the pin count or on exterior geometry. □

For the reference residual calculation, use the common ordered graph extension constructed at the start of the proof of Lemma 3.6. Its cutoff is on a fixed multiple of $\ell_j = (\delta_j/\kappa_j)^{1/2}$, so the extended gaps remain at least δ_j, and its global slope and Hessian bounds are $C_B\sqrt{\kappa_j\delta_j}$ and $C_B\kappa_j$, respectively. That construction uses only the geometry of Lemma 3.1; no force-projection estimate is used here. The remote comparison below concerns this reference extension, without asserting preparation of the actual solution outside the supplied window. At a tangential unit cell freeze each h_i at its affine tangent and b_i at the corresponding exact tilt. The frozen profile solves the full ambient equation. Only first parameter derivatives are needed to obtain

$$\|v_i - v_i^{\mathrm{affine}}\|_{H^1(B_3\times\mathbb{R})} \le C\kappa_j.$$

For a remote base point y, the normal L^1 difference is at most $C\kappa_j(|y|^2+|y|)$, whereas the normal L^2 norm of the ambient kernel at tangential radius r is at most $Cc_{7,s}r^{-6-p-1/2}$. Thus the remote residual costs

$$C\lambda_j\kappa_j\int_1^\infty (r^2+r)r^{-p_j-3/2}\,\mathrm{d}r \le C\lambda_j\kappa_j, \tag{3.25}$$

where one fixes $p_j \ge p_* > 3/2$. This gives an all-normal local H^{-s} curvature residual $O(\kappa_j)$ without D^3h_i.

The cubic interaction is $O_B(F_j)$ on each cell, including long chambers. Indeed, write $q_i = \operatorname{sgn}(t-h_i)+\tau_i$ and expand $\sum\sigma_i f(q_i)-f(V)$. The self-square of either chamber-bounding sheet cancels because the step sum is exactly the adjacent well. Every uncancelled square belongs to a sheet at distance at least δ_j; every product from two distinct sheets has total distance at least δ_j. The elementary bound

$$(e^{-kr}+\lambda(1+r)^{-p})(e^{-kt}+\lambda(1+t)^{-p}) \le C\{e^{-k(r+t)}+\lambda(1+r+t)^{-p}\}$$

then proves the assertion using Lemma 3.2. Width inflation is harmless: $b_i-1=O_B(\kappa_j\delta_j)$, and either $\delta_j^2F_j\to 0$, giving an $o(1)$ exponential-argument change, or $F_j\gtrsim\delta_j^{-2}$, making every tilted exponential negligible relative to F_j. The pin correction has local H^1 norm $O_B(F_j)$, including fixed normal derivatives. Consequently the full per-cell weak residual is $O_B(\kappa_j+F_j)$, with all exterior replacements paid separately.

Choose a bounded common exterior reference, equal to the local formula on the prepared buffer. Its bound is independent of the number of retained sheets: an alternating step sum is a well, at most one profile is close to a point, and the remaining tails are summable in $|i|\delta_j$. This prevents a spurious sheet-count factor in the far field. Apply Lemma 3.4 to

$$((-\Delta)^s-f'(Q))e = -((-\Delta)^sQ-f(Q))-3Qe^2-e^3.$$

The force scale is used in its corresponding subsequence regime: $L_j^2=\delta_j/F_{a,j}$ when $F_{a,j}/F_j$ is bounded below, and $L_j^2=F_{e,j}^{-1}$ when $F_{a,j}/F_{e,j}\to 0$. In the contradiction regime $\kappa_j\le CF_j$, these choices satisfy $L_j\le C\ell_j$ and $L_j/\delta_j\to\infty$. Thus their lateral buffers fit the complete thin windows already constructed, with no assertion about the inactive force scale. A normal buffer $B\delta_j$ and lateral buffer comparable with a force scale $L_j\gg\delta_j$ give the following stronger cutoff estimate. Choose a bounded-overlap family of fixed tangential cell cutoffs η_α and a controlled slab cutoff ζ, all with uniformly bounded derivatives and supports in the genuinely prepared region. The inner normal slab has height $O_B(\delta_j)$. Then

$$\beta_{B,j}:=\sup_\alpha\|\eta_\alpha\zeta e\|_{H^{s_j}(\mathbb{R}^7)}\le C_B\sqrt{\delta_j}F_j, \qquad \frac{\beta_{B,j}^2}{F_j}\le C_B\delta_jF_j\to 0. \tag{3.26}$$

The transmission input here is the residual on one fixed *ambient* cell, of size $C_B(\kappa_j+F_j)$, rather than its all-normal norm. In the contradiction regime this is $O_B(F_j)$. Take the outer prepared window sufficiently larger than the support of every testing cutoff. Its normal margin is a fixed large multiple of $B\delta_j$ and its lateral margin a fixed large multiple of L_j. Choose that normal multiplier so that the exponential transmission tail is bounded by $e^{-k\delta_j}$. The algebraic transmission tail is bounded by $C\lambda_j\delta_j^{-p_j}\le CF_j$; the lateral tail is smaller because $L_j/\delta_j\to\infty$. Thus every fixed interior ambient cell has error norm $O_B(F_j)$. For clarity, summing these cells incurs the square root of their count. Decompose a cutoff error on one base cell and its controlled normal slab as $\sum_m e_m$, using a smooth normal partition with bounded overlap and uniformly bounded derivatives. There are $O_B(\delta_j)$ terms, each supported in a fixed ambient cell and each of global H^{s_j} norm $O_B(F_j)$. At any pair of points, at most twice the overlap number of the differences $e_m(X)-e_m(Y)$ are nonzero. Cauchy–Schwarz therefore bounds both the L^2 norm and the full fractional difference integral of the sum by $C\sum_m\|e_m\|_{H^{s_j}}^2$. Consequently its norm is $C_B\sqrt{\delta_j}F_j$, exactly as in (3.26). Every summand is supported in the genuinely prepared region; no global Sobolev norm of the actual error is used.

The factor $\sqrt{\delta_j}$ counts only controlled interior normal cells. No norm of the unknown exterior error is asserted.

3.4. **The weak force equation at the zero pins.** There are two separate operations: transfer the actual equation to the reference, and compute the reference projection. The exact positive self forms are kept until after rescaling.

For one graph h and a positive width b, let $Z = \partial_t v = b^{-1} w_s((t-h)/b)$. Define, for $x \neq y$,

$$K_{s,h,b}(x,y) = c_{7,s} \int_0^1 \iint_{\mathbb{R}^2} \frac{w_{s,b(x)}(T) w_{s,b(y)}(S)}{[|x-y|^2 + (S - T + \theta(h(y)-h(x)))^2]^{(7+p)/2}} \, \mathrm{d}T \, \mathrm{d}S \, \mathrm{d}\theta, \tag{3.27}$$

where $w_{s,b}(t) = b^{-1} w_s(t/b)$, and set

$$\mathcal{E}_{s,h,b}(a,\phi) = \frac{1}{2} \iint (a(x) - a(y))(\phi(x) - \phi(y)) K_{s,h,b}(x,y) \, \mathrm{d}x \, \mathrm{d}y.$$

Lemma 3.5 (Exact self projection and its macroscopic limit)**.** *The kernel* (3.27) *is positive and symmetric, and*

$$\langle (-\Delta)^s v - f(v), Z\phi \rangle = -\mathcal{E}_{s,h,b}(h,\phi). \tag{3.28}$$

For the flat kernel K_s^0, its covariance is

$$\int_{\mathbb{R}^6} z_a z_b K_s^0(z) \, \mathrm{d}z = 2A_s \delta_{ab}, \qquad A_s = s \int_{\mathbb{R}} |\xi|^{2s-2} |\widehat{w}_s(\xi)|^2 \, \mathrm{d}\xi \longrightarrow A_1 = \frac{2\sqrt{2}}{3}. \tag{3.29}$$

Its covariance tail is at most $C\lambda R^{1-p}$ for $R \geq 1$. If $h_j(L_j y)/a_j \to H(y)$ locally in C^1, $L_j \to \infty$, physical slopes and width errors vanish on each fixed rescaled ball, and the extensions have a controlled quadratic rescaled growth and exterior form budget, then

$$\frac{L_j^{2-6}}{a_j} \mathcal{E}_{s_j,h_j,b_j}(h_j, \phi(\cdot/L_j)) \longrightarrow A_1 \int_{\mathbb{R}^6} \nabla H \cdot \nabla \phi \, \mathrm{d}y. \tag{3.30}$$

Proof. Freeze the receiving x, center the normal variable at $h(x)$, and subtract the same profile kept constant in the tangential variable. Its normal residual is $(b(x)^{-p} - 1) f(v)$ and integrates to zero against Z, since $\int_{-1}^1 f = 0$. By the oddness of q_s and evenness of w_s established above, a change of width with no center displacement is odd in the normal variable and likewise integrates to zero. The translation fundamental theorem leaves

$$(h(y) - h(x)) \int_0^1 w_{s,b(y)}(T + r - \theta(h(y) - h(x))) \, \mathrm{d}\theta.$$

Changing the normal integration variable gives (3.27). Exchange x, y and then S, T to verify symmetry. Symmetrization in x, y proves (3.28). Kernel truncation justifies this calculation before passing to the form limit. There are no derivatives of b in the identity.

The flat form has Fourier symbol

$$\int_{\mathbb{R}} \left[(\xi^2 + |\zeta|^2)^s - |\xi|^{2s} \right] |\widehat{w}_s(\xi)|^2 \, \mathrm{d}\xi.$$

Differentiate at $|\zeta|^2 = 0$ to obtain (3.29); this is the direct projection symbol, not a normal Schur complement. Direct normal integration gives the envelope

$$K_s^0(z) \leq C c_{7,s} \big(|z|^{-6-p} \mathbf{1}_{|z| \leq 1} + |z|^{-7-p} \mathbf{1}_{|z|>1} \big),$$

which proves the covariance tail. For slope ν and width error ω, compare the translated denominator and then differentiate the normal density in its width. The absolute kernel difference has a radial majorant J with

$$\int |z|^2 J(z) \, \mathrm{d}z \leq C(\nu + \omega), \qquad \int_{|z|>R} |z|^2 J(z) \, \mathrm{d}z \leq C(\nu + \omega) \lambda R^{1-p}.$$

Use the L^2 translation estimate $\|f(\cdot + z) - f\|_2 \leq |z| \|\nabla f\|_2$ on a fixed rescaled ball. This removes the slope and width errors there. The covariance identity then gives the local Dirichlet form by smooth approximation and tightness of the second moments. The remaining exterior is removed using the specified quadratic growth and covariance tail. In particular, globally vanishing slopes are unnecessary; a fixed small global extension slope is enough outside each working ball. An uncontrolled exterior cannot be removed by this local computation. □

For $e = u - Q$ and $H_Q = (-\Delta)^s - f'(Q)$, the actual equation is exactly

$$\langle R_Q, \Psi_i\rangle = -\langle e, H_Q\Psi_i\rangle - \langle 3Qe^2 + e^3, \Psi_i\rangle, \qquad R_Q = (-\Delta)^s Q - f(Q). \tag{3.31}$$

For mode truncation choose a separate even cutoff $\vartheta \in C_c^\infty(\mathbb{R})$ with $0 \le \vartheta \le 1$, $\vartheta = 1$ on $[-1,1]$, and support in $[-2,2]$. The pin bump χ remains the fixed microscopic bump; its value $\chi(0) = 1$ alone is insufficient for a tail-sized mode defect.

Use the compact modes

$$\Psi_{i,B}(x,t) = b_i^{-1} w_s((t-h_i)/b_i)\vartheta((t-h_i)/(B\delta_j b_i))\phi(x/L_j),$$

with the larger normal and tangential buffers containing their full supports. The appropriate norm for the first pairing is ambient H^{s_j} on the retained slab. A merely fiberwise normal H^{s_j} bound would not pair with a general ambient H^{-s_j} distribution.

Lemma 3.6 (Localized adjoint and reference force passage)**.** *For the common bounded reference with genuine nested buffers and the ordered extensions constructed in the proof, fix B and a test ϕ. Let ζ be the controlled cutoff equal to one near the mode support, and let $\{\eta_\alpha\}$ be the bounded-overlap base partition on its support, as constructed in the proof. Then*

$$|\langle \zeta z, H_Q\Psi_{i,B}\rangle| \le C_{B,\phi} L_j^6 \left(\kappa_j + \sqrt{\delta_j} F_j + \frac{\nu_{B,j}}{L_j} + L_j^{-2}\right) \sup_\alpha \|\eta_\alpha \zeta z\|_{H^{s_j}(\mathbb{R}^7)}. \tag{3.32}$$

The pin correction and reference-mode de-truncation obey the same paired estimates with their respective $O_B(F_j)$ and tail norms. The full reference projection has

$$L_j^{-6}|X_{i,j}(\phi)| = o_{j,B}(F_j), \tag{3.33}$$

$$P_{i,j}(x) = \sum_{k<i} \sigma_k \mathfrak{F}_{s_j}(h_i - h_k) - \sum_{k>i} \sigma_k \mathfrak{F}_{s_j}(h_k - h_i) + r_{i,j}(x), \tag{3.34}$$

$$L_j^{-6}\|r_{i,j}\|_{L^1(L_jK)} = o_{j,B}(F_j),$$

where $X_{i,j} = \sum_{k\neq i} \sigma_k \langle R_{v_k}, \Psi_i\rangle$ and $P_{i,j} = \int Z_i(\sum_k \sigma_k f(v_k) - f(V))\,\mathrm{d}t$. Any replacement outside the controlled reference is instead charged to (3.42) below, with a constant independent of B.

Proof. Use the pin bump and plateau cutoff fixed above, and write $\Psi_i = \Psi_{i,B}$. We construct one common reference and estimate its projection on genuine nested buffers.

Fix the test $\phi \in C_c^\infty(\mathbb{R}^6)$ and all the working multipliers before taking $j \to \infty$. Write $\delta = \delta_j$, $\kappa = \kappa_j$, $p = 2s_j$, $\lambda = 1 - s_j$, $F = e^{-k\delta} + \lambda\delta^{-p}$, and $\ell = (\delta/\kappa)^{1/2}$. We work in the contradiction regime $\kappa \le CF$, $\kappa\delta \to 0$, with $p \ge p_* > 3/2$. Let L be either of the two force scales; in both cases $L \le C\ell$. Take the common graph-extension cutoff on a disk of radius $C_*\ell$, with a fixed larger disk supplied by Lemma 3.1. Choose the outer normal slab multiplier after this disk multiplier, large enough to contain the height drift of all retained branches and the normal/lateral support buffers below. Every graph needed in the smaller window is retained; omitted branches lie outside the larger slab. The actual central-height labels form a finite family $M \le C_B$.

Set

$$\widetilde{h}_i(x) = h_i(0) + \chi_*(x/\ell)(h_i(x) - h_i(0)), \qquad b_i = (1 + |D\widetilde{h}_i|^2)^{1/2}.$$

Here χ_* is one common cutoff, equal to one on the supplied working disk, and vanishing inside the larger supplied disk. The gap between two extended heights is the convex combination of their actual and central gaps, so remains at least δ. On these extensions,

$$|D\widetilde{h}_i| \le C_B\sqrt{\kappa\delta} =: \nu, \qquad |D^2\widetilde{h}_i| \le C_B\kappa, \qquad |Db_i| \le C_B\nu\kappa, \qquad 1 \le b_i \le 1 + C_B\nu^2.$$

Indeed the cutoff terms are controlled by $\nu/\ell = O_B(\kappa)$ and $\kappa\ell^2/\ell^2 = \kappa$. Using a cutoff at the smaller force scale without checking these terms would not give this Hessian bound. The extensions are reference objects only; all actual error estimates below use compact supports in the genuinely prepared larger window. Suppress tildes in the notation from now on.

Define the bounded alternating reference $Q = V + C$ as before, and

$$Z_i = b_i^{-1} w_s((t-h_i)/b_i), \qquad \Psi_i = Z_i\vartheta((t-h_i)/(B\delta b_i))\phi(x/L).$$

At a fixed B, choose the prepared outer slab and disk so that the support of Ψ_i has normal distance at least $B\delta$ and lateral distance at least L from the complement of an intermediate controlled cutoff region. Further fixed enlargement supplies the support of that cutoff. These choices are possible because the complete thin windows contain every prescribed fixed multiple of ℓ and because $L \le C\ell$. The constants attached to these regions are fixed before $j \to \infty$.

For a unit base cube U, enlarged by a fixed amount when necessary, define its all-normal dual norm by testing against global $H^s(\mathbb{R}^7)$ functions supported in $U \times \mathbb{R}$. In particular it is not a fiberwise $H^{-s}(\mathbb{R})$ norm. We claim that

$$\sup_{\substack{\operatorname{supp} z \subset U\times\mathbb{R} \\ \|z\|_{H^s}\le 1}} |\langle z, H_Q\Psi_i\rangle| \le C_{B,\phi}\{\kappa + \sqrt{\delta}F + \nu/L + L^{-2}\}, \qquad H_Q = (-\Delta)^s - f'(Q). \tag{3.35}$$

The supremum may first be taken over smooth compact tests; density extends the result to the displayed class.

Affine comparison in the all-normal norm. Freeze h_i at a point x_0 within a fixed distance of U: $h_i^0(x) = h_i(x_0) + Dh_i(x_0)\cdot(x - x_0)$ and $b_i^0 = (1 + |Dh_i(x_0)|^2)^{1/2}$. Let v_i^0 and Z_i^0 be the exact tilted layer and its vertical derivative. Both solve their full ambient equation and Jacobi equation, respectively. For a fixed base cutoff η equal to one on an enlargement of U,

$$\|\eta(v_i - v_i^0)\|_{H^1(\mathbb{R}^7)} + \|\eta(Z_i - Z_i^0)\|_{H^1(\mathbb{R}^7)} \le C_B\kappa. \tag{3.36}$$

Taylor's theorem gives $|h_i - h_i^0| \le C\kappa$ and $|Dh_i - Dh_i(x_0)| \le C\kappa$ on this fixed base. Also $|b_i - b_i^0| + |Db_i| \le C_B\nu\kappa$. Differentiate the smooth profile with respect to its center and width before differentiating in x or t. The resulting normal factors are fixed derivatives of q_s, w_s, and their first polynomial moments; their norms are uniformly bounded by Lemma 3.2. A tangential derivative of Z_i uses only Dh_i and $Db_i = b_i^{-1}Dh_i \cdot D^2h_i$, so no third height derivative occurs. This proves both terms in (3.36). In particular, the normal derivative comparison is proved directly, rather than obtained by differentiating an H^{-s} residual estimate.

The mode estimates used here are the sharper bounds following from $w_s = G_s * a_s$: for each fixed r, $|w_s^{(r)}(t)| \le C_r\{e^{-k|t|} + \lambda(1 + |t|)^{-p-1}\}$. In particular their first absolute moments, the L^2 norms of $tw_s^{(r)}$, and (since $p_* > 3/2$) the L^2 norms of $t^2w_s^{(r)}$ are uniformly bounded and tight. These weighted norms justify differentiating widths, including the normal-mode tail and its tangential width derivative; a second *absolute* moment is neither claimed nor required.

For a remote base point y, with $r = |y - x_0| \ge 1$, the same parameter identities and the uniform first moments give

$$\|v_i(y,\cdot) - v_i^0(y,\cdot)\|_{L^1(\mathbb{R})} + \|Z_i(y,\cdot) - Z_i^0(y,\cdot)\|_{L^1(\mathbb{R})} \le C_B\kappa(r^2 + r). \tag{3.37}$$

Large parameter changes cause no problem: the difference of two translated layers has L^1 norm at most 2 times the center change, and the mode differences can be bounded by the integral of their parameter derivatives. All widths remain in a fixed compact interval.

For $x \in U$, the normal L^2 norm of the remote ambient kernel is

$$\left\| c_{7,s}\{r^2 + (t - \cdot)^2\}^{-(7+p)/2} \right\|_{L^2_t} \le C\lambda r^{-6-p-1/2}.$$

Minkowski's inequality, (3.37), and the six-dimensional radial shell therefore bound the remote field by

$$C_B\lambda\kappa \int_1^\infty (r^2 + r) r^{-p-3/2}\, \mathrm{d}r \le C_B\lambda\kappa$$

in $L^2(U \times \mathbb{R})$, uniformly for $p \ge p_* > 3/2$. Local differences are controlled in H^{-s} by their H^1 norms, because $s \le 1$. Consequently, replacing $Z_i\phi(x/L)$ by $Z_i^0\phi(x/L)$ costs $C_{B,\phi}\kappa$ in this all-normal dual norm. The reaction part of the replacement is controlled by the bounded potential and the local L^2 part of (3.36). The same argument without a normal derivative proves

$$\|(-\Delta)^s Q - f(Q)\|_{H^{-s}(U\times\mathbb{R})} \le C_B(\kappa + \sqrt{\delta}F), \tag{3.38}$$

once the elementary cubic estimate below and the H^1 pin-bump bound are included.

The flat tilted commutator. Let n be the unit normal of v_i^0, so $|n'| \le C_B\nu$ and n_7 is bounded away from zero. The Fourier frequencies of its normal mode, multiplied by $\phi(x/L)$, have the form $\xi n + (\zeta, 0)$. Since the mode is an exact Jacobi field, only

$$m(\xi,\zeta) = |\xi n + (\zeta,0)|^p - |\xi|^p$$

appears in the commutator. Uniformly for $p \ge p_* > 3/2$,

$$\|m(\cdot,\zeta)\widehat{w}_s\|_{L^2_\xi} \le C_B(\nu|\zeta| + |\zeta|^2). \tag{3.39}$$

For $|\xi| \ge 2|\zeta|$, Taylor expansion gives the linear term $p|\xi|^{p-2}\xi n' \cdot \zeta$ and a remainder bounded by $C|\zeta|^2|\xi|^{p-2}$. Its normal L^2 integral is uniformly finite near zero because $2p-4 > -1$; at infinity the fixed derivative bounds of w_s apply. On $|\xi| < 2|\zeta|$, boundedness of $\widehat{w}_s$ gives a remainder norm $C|\zeta|^{p+1/2}$, which is at most $C|\zeta|^2$ for $|\zeta| \le 1$. For $|\zeta| \ge 1$, the same claim follows from $p \le 2$ and the fixed normal Sobolev norms. The linear term has norm $C|n'||\zeta|$. This proves (3.39). Fourier inversion in the tangential variable, Plancherel in t, and Minkowski then give a bound uniform at every x:

$$\|[(-\Delta)^s, \phi(x/L)]Z_i^0\|_{L^2_t} \le C_{B,\phi}(\nu/L + L^{-2}),$$

because $\widehat{\phi(\cdot/L)}(\zeta) = L^6\widehat{\phi}(L\zeta)$. Thus this commutator has the same bound in the local ambient dual norm.

Potential mismatch and the normal cutoff. On each normal fiber, the ordered heights are separated by δ; their number and their maximum vertical height span are bounded by C_B and $C_B\delta$, respectively. The layer envelopes and the alternating step sum give

$$\|[f'(v_i) - f'(Q)]Z_i\|_{L^2_t} \le C_B\sqrt{\delta}F. \tag{3.40}$$

For detail, on the span of the stack expand each profile as its well step plus its tail. Near core i, the other tails are $O_B(F)$. Between two consecutive cores the product of the two bounding tails is at most C_BF; every other tail is already $O_B(F)$. Multiplying the potential difference by the selected mode gives the same pointwise bound. Near another core the selected mode is $O_B(F)$. These regions have total length $O_B(\delta)$, giving the stated L^2 bound. Outside the stack, either i is the outermost core, in which case the mismatch comes from remote tails of size $O_B(F)$ multiplying an integrable mode, or the selected mode is itself a remote tail; the bounds give an additional $O_B(F)$ normal L^2 norm. The small width inflation does not change these estimates. Put $\omega = C_B\kappa\delta$. If $\delta^2F \to 0$ along a subsequence, $\omega\delta \to 0$. Along a complementary subsequence $F \ge c\delta^{-2}$, every exponential with rate $k/(1+C\omega)$ is negligible compared with F. This proves the requisite bound and the later relative width errors without a relation between λ and κ. The comparison to the affine mismatch changes (3.40) by $O_B(\kappa)$ using (3.36).

Let $E_i = Z_i(\vartheta((t-h_i)/(B\delta b_i)) - 1)\phi(x/L)$. The plateau means this is a genuine mode tail at normal distance at least $B\delta$. On a fixed base cell,

$$\|E_i\|_{H^1} \le C_{B,\phi}\sqrt{\delta}\{e^{-cB\delta} + \lambda(B\delta)^{-p}\},$$

and on every normal fiber its L^1 norm is at most $C_\phi\{e^{-cB\delta} + \lambda(B\delta)^{-p}\}$. These bounds follow from the mode and fixed derivative envelopes. The tangential width derivative has coefficient $Db_i = O_B(\nu\kappa)$; its first normal moment on the tail is harmless since $\kappa\nu B\delta \to 0$ and $p_* > 3/2$. The remote kernel estimate above, now without the r^2 factor, shows that applying H_Q to this tail costs the same local dual bound. Choose the fixed normal cutoff and slab multipliers large enough that $cB > k$. The bracket is then $O_B(F)$, and the cutoff contributes $C_{B,\phi}\sqrt{\delta}F$. Affine comparison, the commutator bound, potential mismatch, and this tail bound prove (3.35).

Actual transfer on nested buffers. Choose a smooth controlled cutoff ζ, equal to one on a neighborhood of $\operatorname{supp}\Psi_i$, with its transition at normal distance at least $B\delta$ and lateral distance at least L from that support. Additional fixed enlargement bounds its support. The numerical distance constants may be chosen independently of B; only the supplied window size and its preparation constants grow with B. A fixed-base bounded-overlap partition of unity $\sum_\alpha \eta_\alpha = 1$ on $\operatorname{supp}\zeta$ gives global ambient H^s tests $\eta_\alpha\zeta e$, where $e = u - Q$. Their multiplier norms are

uniform because all cutoff derivatives are bounded, and every support is in the prepared region. The transmission bound gives $\|\eta_\alpha \zeta e\|_{H^s} \le C_B\sqrt{\delta}F$. There are $O_\phi(L^6)$ base cells. Hence

$$\frac{|\langle \zeta e, H_Q\Psi_i\rangle|}{FL^6} \le C_{B,\phi}\left\{\kappa\sqrt{\delta} + \delta F + \frac{\nu\sqrt{\delta}}{L} + \frac{\sqrt{\delta}}{L^2}\right\} \longrightarrow 0. \tag{3.41}$$

Every term tends to zero on either force scale, using $\kappa \le CF$ and $\delta F \to 0$. Also $|\langle 3Qe^2 + e^3, \Psi_i\rangle|/(FL^6) \le C_B\delta F \to 0$ by the local L^2 bound and boundedness of e.

For the separated exterior, $\Psi_i(Y) = 0$ wherever $1 - \zeta(Y) \neq 0$. The globally bounded e, $\|\Psi_i\|_1 \le C_\phi L^6$, and the ambient kernel tail give

$$L^{-6}|\langle (1-\zeta)e, H_Q\Psi_i\rangle| \le C_\phi\lambda\{(B\delta)^{-p} + L^{-p}\}. \tag{3.42}$$

The normal part divided by F is $O(B^{-p_*})$; it must be exhausted after the j limit. The lateral normalized part is $(F_a/F)(\delta/L)^p \to 0$. This is the entire contribution of the uncontrolled actual exterior. No exterior Sobolev norm is used.

Pin correction and reference de-truncation. The pin amplitudes and their first tangential derivatives are $O_B(F)$. Indeed the remote value tails at a pin are $O_B(F)$ and differentiation uses only $Dh_i - Dh_k$, Db_k, and $D = h_i - h_k = O_B(\delta)$. The disjoint fixed-width pin bumps therefore have cell H^1 norm and fiber L^1 norm $O_B(F)$. Using

$$\langle R_Q - R_V, \Psi_i\rangle = \langle C, H_V\Psi_i\rangle + \langle 3VC^2 + C^3, \Psi_i\rangle, \tag{3.43}$$

and (3.35) applied to ζC, the inner normalized contribution tends to zero; the quadratic part is $O_B(F)$. Here the estimate for H_Q applies to H_V as well: $(H_V - H_Q)\Psi_i = (f'(Q) - f'(V))\Psi_i$ and $\|f'(Q) - f'(V)\|_\infty = O_B(F)$. Its pairing with ζC costs $O_B(F^2L^6)$, hence $O_B(F)$ after normalization. For $(1-\zeta)C$, use the separated exterior kernel estimate and $\|C\|_\infty = O_B(F)$. After division by FL^6 its contribution is at most $C_B\lambda\{(B\delta)^{-p} + L^{-p}\} \to 0$. Thus the cancellation is justified for the global bounded correction, without assigning it a finite global Sobolev norm. The $O(F)$ bump residual is not itself declared an $o(F)$ quantity. For the reference only, pair the all-normal residual (3.38) with the discarded mode tail. After division by F, its cell bound is $C_B(\kappa\sqrt{\delta} + \delta F) \to 0$. Thus the reference projection may be de-truncated on the full normal fiber. The actual equation is still tested only by the compact mode, and its exterior remains accounted for by the separated estimate.

Separated cross-curvature. For $k \neq i$, localize $v_k - v_k^0$ in a fixed enlarged base cell. Its H^1 norm divided by κ is bounded and its normal tails, centered at sheet k, tend uniformly to zero as a fixed depth tends to infinity. This follows from the parameter identities in (3.36), their fixed normal derivatives, and the uniform first moments. The cell-localized selected mode has tight H^1 tails centered at sheet i. At fixed normal depth the two retained supports separate by at least $\delta - O(1)$: the heights have small slopes on a fixed base cell and their gaps are at least δ. Their reaction pairing is zero, and their fractional bilinear pairing is a separated-kernel integral bounded by $o_j(\kappa)$. The discarded terms are at most $\kappa\epsilon(P)$ by the ambient H^s–H^{-s} dual inequality, with $\epsilon(P) \to 0$. The remote tangential part is $O_B(\lambda\kappa)$ by (3.37). First let $j \to \infty$, then $P \to \infty$. Summing the fixed finite family and $O(L^6)$ cells gives

$$L^{-6}\left|\sum_{k\neq i}\sigma_k\langle R_{v_k}, Z_i\phi(x/L)\rangle\right| = o_B(\kappa) = o_B(F).$$

This uses tight tails of the profile difference, not an unjustified tight-tail assertion for the distribution R_{v_k}/κ.

The cubic interaction projection. Fix a normal fiber and put $q_k = v_k(x,t)$, $\epsilon_k = \operatorname{sgn}(t - h_k)$, $r_k = q_k - \epsilon_k$. On the interval $I_i = (h_{i-1}, h_{i+1})$, interpreting a missing endpoint as infinity, the alternating step identity gives $c_{\mathcal{I}} + \sum_{k\neq i}\sigma_k\epsilon_k = 0$, where every sum is over the retained actual labels. Therefore $V = \sigma_i q_i + R$, with $R = \sum_{k\neq i}\sigma_k r_k$. Using $f(\epsilon + r) = -2r - 3\epsilon r^2 - r^3$ and

expanding the cubic exactly, one obtains

$$\begin{aligned} Z_i\{\textstyle\sum_k \sigma_k f(q_k) - f(V)\} = & -a_{s,b_i}(t-h_i)\sum_{k\neq i}\sigma_k r_k \\ & + 3Z_i\sum_{k\neq i}(\sigma_i q_i - \sigma_k\epsilon_k)r_k^2 \\ & + 6\sigma_i q_i Z_i \sum_{k<l\,;\,k,l\neq i}\sigma_k\sigma_l r_k r_l \\ & + 3Z_i\sum_{k\neq l\,;\,k,l\neq i}\sigma_l r_k^2 r_l + 6Z_i\sum_{k<l<a\,;\,k,l,a\neq i}\sigma_k\sigma_l\sigma_a r_k r_l r_a. \end{aligned}$$

Here $a_{s,b_i} = 3(1-q_i^2)Z_i$. The remainder has absolute value $C_B F^2$ on the finite span of I_i. For a nonadjacent k, $|r_k| \le C_B F$ throughout I_i. For adjacent $k = i-1$, its square coefficient is $\sigma_i(q_i+1)$; for $k = i+1$, it is $\sigma_i(q_i-1)$. On the corresponding side of core i, the selected mode times this coefficient is bounded by the square of its own tail. Thus each adjacent square is bounded by the square of a two-tail product across a gap $D \ge \delta$, and hence by $C_B F^2$. For a mixed adjacent product, one tail is at least δ away on either side of core i, while the selected mode times the other tail is a two-tail product across a gap at least δ. This also gives $C_B F^2$. Products involving a nonadjacent tail are easier, and the remaining extra factors in cubic terms are bounded. These estimates use the uniform elementary two-tail inequality from the preceding residual calculation, with the already controlled width inflation. The finite span has length $O_B(\delta)$; an unbounded end of I_i has all other tails at distance at least δ and an integrable selected mode. The integrated remainder is consequently $O_B(\delta F^2) = o_B(F)$.

Outside I_i, the unprojected cubic interaction is $O_B(F)$ by the alternating step expansion. The selected mode has total mass $O_B(F)$ there, since its distance from its center is at least δ. Thus that projected part is $O_B(F^2)$. The omitted part of the proposed leading source term is also $o_B(F)$: the faster source envelope gives

$$\int_{|t-h_i|\ge\delta} a_{s,b_i}(t-h_i)\,\mathrm{d}t \le C_B\{e^{-\beta'\delta} + \lambda^2\delta^{-2p}\} = o(F), \qquad \beta' > k.$$

Hence the full projection equals, up to $o_B(F)$,

$$\sum_{k<i}\sigma_k\int a_{s,b_i}(r)\tau_s((h_i-h_k+r)/b_k)\,\mathrm{d}r - \sum_{k>i}\sigma_k\int a_{s,b_i}(r)\tau_s((h_k-h_i-r)/b_k)\,\mathrm{d}r.$$

Evenness of the source changes $+r$ to $-r$ in the first integrals. The source is even by the layer symmetry established in Section 3.2.

To check the estimate directly, write the mixed-width primitive as

$$J_s(b_i,b_k;D) = \int_{\mathbb{R}} a_s(t)\tau_s((D-b_i t)/b_k)\,\mathrm{d}t.$$

Its two width derivatives have absolute values bounded by

$$C\int_{\mathbb{R}} a_s(t)(D+1+|t|)w_s((D-b_i t)/b_k)\,\mathrm{d}t.$$

The profiles are smooth, and the faster source envelope and the mode first moment justify differentiation under the integral. Choose the fixed source exponent $\beta > k$ and take j large enough that $\beta > kb_i/b_k$ for all retained widths. The product of their exponential envelopes integrates to $C(1+D)e^{-kD/b_k}$: split at $t=0$ and $t = D/b_i$ and integrate the three resulting geometric exponentials. A source exponential with the mode power tail contributes at most $C\lambda D^{-p}$, by splitting at $|t| = D/(2b_i)$. A source power tail with the mode exponential is at most $C\lambda^2 D^{-2p}$, by splitting at $|D-b_i t| = D/2$. The product of the two power tails is at most $C\lambda^3 D^{-p}$, using that either $|t|$ or $|D-b_i t|$ is a fixed multiple of D and integrating the other factor with its first moment. These constants are uniform for widths in a fixed neighborhood of one and $p \ge p_* > 1$. Since $\lambda \le 1$ and $D \ge 1$, the last two terms are absorbed by $C\lambda D^{-p}$. This proves the stated mixed-width bound for the actual source convolution.

To remove the widths, use the derivative estimate just proved along the segment from (b_i, b_k) to $(1,1)$. Since $b_i, b_k \le 1 + C_B\omega$, that estimate is bounded by

$$C_B\{(1+D)e^{-kD/(1+C_B\omega)} + \lambda D^{-p}\}, \qquad \omega = C_B\kappa\delta \to 0, \qquad \delta \le D \le C_B\delta.$$

Multiply by the width change $O_B(\omega)$. The algebraic error is $o_B(F)$. If $\delta^2 F \to 0$, then $\omega\delta \to 0$ and the exponential error is $o_B(F)$ as well. On any remaining subsequence $F \ge c\delta^{-2}$, the exponential error divided by F is bounded by a fixed polynomial in δ times $e^{-(k-o(1))\delta}$ and therefore tends to zero. This exhausts possible subsequences. The unscaled integral is $B_s(D)$; the uniform raw-force identity in Lemma 3.2 replaces $B_s(D)$ by $\mathfrak{F}_s(D)$ with another $o_B(F)$ error. We have proved

$$P_i(x) = \sum_{k<i} \sigma_k \mathfrak{F}_s(h_i - h_k) - \sum_{k>i} \sigma_k \mathfrak{F}_s(h_k - h_i) + o_B(F),$$

uniformly on the retained base. Integrating against $\phi(x/L)$ gives the required $o_B(FL^6)$ interaction remainder.

Exterior self-form budget. For the common-cutoff extensions used here, the exterior self-form budget follows directly for the heights whose limits are required. In the power case it applies to every retained height, with $a_j = \delta_j$; in the exponential case it is needed, and is asserted, only for the two selected heights, with $a_j = 1$. Apply the self-form limit to $h_i - h_i(0)$; forms are unchanged by this subtraction. In particular the upper exponential-scale height is centered at its own value d_j, which differs from δ_j by $o(1)$. Taylor's estimate and the common-cutoff construction give globally

$$|\widetilde{h}_j(L_j y) - \widetilde{h}_j(0)| \le C\{L_j|Dh_j(0)||y| + \kappa_j L_j^2|y|^2\} \le C_R a_j(1+|y|^2).$$

Indeed $\kappa_j L_j^2/a_j$ is bounded on both force scales. In the power case, $L_j|Dh_j(0)|/\delta_j \le C_B\sqrt{\kappa_j/F_{a,j}}$. In the exponential case $Dh_{1,j}(0) = 0$, whereas (3.5), $\kappa_j \le CF_{e,j}$, and $\eta_j\delta_j \to 0$ give

$$L_j|Dh_{2,j}(0)| \le C\sqrt{\delta_j}(\sqrt{\eta_j} + \eta_j) \longrightarrow 0.$$

The common cutoff preserves these quadratic bounds and all its extensions have vanishing global slope at each fixed buffer. The weaker general slope bound $O_B(\sqrt{\kappa_j\delta_j})$ does not imply a bounded exponential-scale slope for an arbitrary other height; no self-form limit for such a height is used.

For widths uniformly close to one and this small slope, the denominator comparison in (3.27), followed by normal integration, gives

$$K_j(x,y) \le C\lambda_j|x-y|^{-7-p_j} \qquad (|x-y| \ge 1).$$

Fix ϕ supported in B_R, and take $T > 2R+2$. In the self form tested by $\phi(x/L_j)$, pairs with one point in L_jB_R and the other outside L_jB_T have, after normalization by L_j^{-4}/a_j, absolute contribution at most

$$C_R\lambda_j L_j^{-4} L_j^6 \int_{L_j(T-R)}^{\infty} \{1 + (r/L_j)^2\} r^{-7-p_j} r^5 \, \mathrm{d}r \le C_R\lambda_j\{L_j(T-R)\}^{1-p_j}.$$

Here the harmless bounded part is absorbed by the quadratic part since $r/L_j \ge T - R > 1$. Because $p_j \ge p_* > 1$ and $L_j \to \infty$, this tends to zero for every fixed outer multiplier. On bounded rescaled sets the small-slope/width covariance comparison and the C^1 height convergence give the local form limit. This verifies the asserted exterior budget for the actual extensions in both first-force cases. The cell estimates, separated cross-curvature, and cubic projection prove (3.32)–(3.34). The actual exterior is charged only to (3.42) and is exhausted after the j limit. No third derivative of a graph height or global Sobolev norm of the actual error has been used. □

Proposition 3.7 (Actual signed weak force system)**.** *On every retained complete window, then after buffer exhaustion,*

$$-\mathcal{E}_i(h_i, \phi(\cdot/L_j)) = \int G_{i,j}(x)\phi(x/L_j)\,\mathrm{d}x + o(F_jL_j^6), \tag{3.44}$$

where

$$G_{i,j} = \sum_{k<i} (-1)^{i-k+1} \mathfrak{F}_{s_j}(h_i - h_k) - \sum_{k>i} (-1)^{k-i+1} \mathfrak{F}_{s_j}(h_k - h_i). \tag{3.45}$$

Proof. The reference decomposition is exact:

$$\langle R_V, Z_i \phi_L \rangle = -\sigma_i \mathcal{E}_i(h_i, \phi_L) + \int P_{i,j} \phi_L + X_{i,j}(\phi).$$

Combine it with (3.31), (3.43), and the verified error budgets; then multiply by $-\sigma_i$. The sign agrees with $A_1 \Delta h_i = G_i$ only after the macroscopic limit. Different prelimit sheets carry different exact self forms and are not silently assigned the same Laplacian. □

Lemma 3.8 (The full alternating row gives the scalar upper sign)**.** *For any positive decreasing force f and any adjacent gap $D = h_{i+1} - h_i$, in a finite ordered family or an infinite family with absolutely convergent rows, the signed rows satisfy*

$$G_{i+1}[f] - G_i[f] = 2f(D) - \mathcal{N}_i[f], \qquad \mathcal{N}_i[f] \geq 0. \tag{3.46}$$

Proof. A label $i - a$ to the left contributes $(-1)^a \{f(r_a) + f(r_a + D)\}$, where $r_a = h_i - h_{i-a}$ increases with a. The positive magnitudes decrease, and the alternating sum starts negative. Pair successive terms; if the list is finite its unpaired last term is also negative. The right labels have the same calculation. Absolute convergence of the algebraic and exponential tails justifies infinite lists. Convexity is unnecessary here; monotonicity is enough. □

3.5. **Stability and the first force.** The next bridge is about the actual solution, not stability of a superposed profile. Put

$$g = \partial_t u, \quad W = 3(1 - u^2), \quad A = (-\Delta)^s + 2.$$

Fix a large normal depth P and a bounded convex complete window Ω containing all support buffers. Let K_P be the union of depth-P vertical cores of every actual graph supplied on Ω. Retain the inner boundary strip as well as the exterior by setting

$$\Omega^- = \{X \in \Omega : \operatorname{dist}(X, \Omega^c) > 2P\}, \qquad C = (\Omega^-)^c \cup (K_P \cap \Omega^-),$$

$$V = W \mathbf{1}_{\Omega^- \setminus K_P}, \qquad G_V = (A - V)^{-1}.$$

For large j the graph slopes are at most one. If $X \in \Omega^-$ is within $P/2$ of a zero, that zero lies in Ω, and its vertical graph distance from X is at most $(1 + \nu)P/2 \leq P$. Thus $X \in K_P$. Every point of C^c is therefore at distance at least $P/2$ from the whole zero set. Clearing gives $0 \leq V \leq \epsilon_P < 2$, with $\epsilon_P \to 0$ as $P \to \infty$, after the j limit. All active amplitudes are supported in the deep working region and are zero on $(\Omega^-)^c$. Cores and this retained exterior are disjoint. The inner parallel window Ω^- is convex; from this point onward, the exterior-field estimates are applied to Ω^-. Its boundary distance loses only the fixed amount $2P$, leaving all normal and lateral buffer limits unchanged. Stability is $A - W \geq 0$, hence its Schur complement gives

$$T = \sqrt{W \mathbf{1}_C}\, G_V \sqrt{W \mathbf{1}_C} \leq I.$$

The Jacobi relation $Ag = Wg$ gives $T(\sqrt{W} g|_C) = \sqrt{W} g|_C$ also for this bounded, possibly non-L^2 field. The bounded inversion argument in Lemma 2.1, with e^{-2t} in place of e^{-t}, gives $g = A^{-1}(Wg)$. Decompose $W = V + W\mathbf{1}_C$ and iterate

$$g = A^{-1}(Vg) + A^{-1}(W \mathbf{1}_C g).$$

Since $\|A^{-1}V\|_{L^\infty \to L^\infty} \leq \epsilon_P/2 < 1$, its positive Neumann series converges on bounded functions and yields $g = G_V(W\mathbf{1}_C g)$. Multiplication by $\sqrt{W}$ on C proves the identity. Every compact-amplitude pairing below is absolutely convergent: one endpoint is in a compact set, the fields are bounded, and the positive massive comparison gives an integrable kernel row.

For arbitrary compact retained amplitudes ζ, write $v = \sqrt{W} g|_C$. Expanding against this exact signed ground state gives

$$0 \leq \langle v\zeta, (I - T) v\zeta \rangle = \frac{1}{2} \iint_{C \times C} W(X) g(X) G_V(X, Y) W(Y) g(Y) (\zeta(X) - \zeta(Y))^2 \, \mathrm{d}X \, \mathrm{d}Y. \tag{3.47}$$

On core i, g has sign σ_i, and $S_i = W|g| = (1+o_j(1))a_{s_j}(t)$ relatively on each fixed normal core. Thus the internal cross row has sign $(-1)^{i-k}$. The exterior remains signed and is not assigned a fictitious sheet decomposition.

For a selected small-slope graph core and a tangential amplitude, positivity of the resolvent gives $G_V \le ((-\Delta)^s + 2 - \epsilon_P)^{-1}$. Flatten its graph by the elementary distance comparison

$$|(x-y, h_i(x) - h_i(y) + r - t)| \ge (1-\nu)\sqrt{|x-y|^2 + |r-t|^2}.$$

Fourier differentiation of the resulting positive flat majorant yields

$$E_{ii}(\zeta_i) \le (1+o_j(1)) \left(\frac{2}{2-\epsilon_P}\right)^2 (1-\nu)^{-7-p} A_s \int |\nabla \zeta_i|^2. \tag{3.48}$$

To track the dilation, put $b = 1-\nu$ and $a = 2-\epsilon_P$. The free ambient kernel satisfies $G^a_{7,s}(bZ) = b^{p-7} G^{ab^p}_{7,s}(Z)$. The quadratic gradient coefficient of the positive flat majorant is therefore at most

$$b^{p-7} s \int_{\mathbb{R}} |\xi|^{p-2} \frac{|\widehat{a}_s(\xi)|^2}{(|\xi|^p + ab^p)^2}\, \mathrm{d}\xi \le \left(\frac{2}{a}\right)^2 b^{-7-p} A_s,$$

because $\widehat{a}_s = (|\xi|^p + 2)\widehat{w}_s$ and $(z+2)/(z+ab^p) \le 2/(ab^p)$. The symbol at tangential frequency ζ is bounded by this coefficient times $|\zeta|^2$: its derivative in $|\zeta|^2$ decreases, since $s \le 1$ and the massive denominator increases. This proves the displayed self budget for arbitrary compact amplitudes. There is no normal taper cost in (3.47).

Lemma 3.9 (Sharp actual row comparisons and signed exterior)**.** *Fix γ_0 with $k/2 < \gamma_0 < 1$. For each sufficiently large fixed P, on the supplied actual graph windows the same-orientation upper rows have algebraic coefficient at most $(1+o(1))\mu_P^2 c_{1,s}/(2-\epsilon_P)^2$ and a summable $C_P e^{-\gamma_0 D}$ remainder. The opposite-orientation lower rows have coefficient at least $(1-o(1))\mu_P^2 c_{1,s}/4$, where $\mu_P = \int_{-P}^{P} a_1 \to 4$. The relative errors vanish for $D \to \infty$, $j \to \infty$, and vanishing slopes, with the fixed normal-core shifts included. Their cross tangential second moments are bounded by*

$$C_{P,B}\{\mathrm{poly}(D) e^{-cD} + \lambda D^{1-p}\} = o(1). \tag{3.49}$$

Finally, with $F_0 = 3u - u^3$, the signed exterior field satisfies

$$\left| \int_{(\Omega^-)^c} G_V(X,Y) \partial_t F_0(Y)\, \mathrm{d}Y \right| \le C\{e^{-ch} + \lambda(1+h)^{-p-1}\}, \qquad h = \mathrm{dist}(X, \partial\Omega^-), \tag{3.50}$$

uniformly in the window aspect ratio and number of cores.

Proof. For the algebraic lower coefficient, the positive Stieltjes formula for $((-\Delta)^s + 2)^{-1}$ has density

$$\frac{\sin(\pi s)}{\pi} \frac{t^s}{t^{2s} + 4t^s \cos(\pi s) + 4}.$$

On $0 < t < 1$ its denominator is at most 4 for s sufficiently near one. The off-diagonal fractional-kernel representation consequently gives

$$G^2_{7,s}(Z) \ge (1 - o_{|Z|}(1)) \frac{c_{7,s}}{4} |Z|^{-7-p}.$$

Tangential integration and the two fixed-core masses give the lower row coefficient $\mu_P^2 c_{1,s}/4$. The matching upper coefficient comes from the mass $2-\epsilon_P$ majorant; its positive exponential remainder is used only for same-orientation distances, which are at least twice the minimum gap. This reserve is essential: a fixed small error in the mass cannot be compared relatively at the selected exponential distance.

The signed exterior mechanism is an integration by parts, not an unsigned packing estimate. For the free massive kernel $G = G^2_{7,s}$,

$$G * (\mathbf{1}_{(\Omega^-)^c} \partial_t F_0)$$

is a gradient-volume integral plus a boundary integral against the bounded function F_0. Convexity gives $\mathcal{H}^6(\partial\Omega^- \cap B_R(X)) \le CR^6$ uniformly in aspect ratio: partition by a coordinate

direction in which the unit normal has magnitude at least $1/\sqrt{7}$, project each of the fourteen pieces, and use that a line meets a convex set in one interval. Massive kernel and gradient bounds therefore give $C\{e^{-ch} + \lambda(1+h)^{-p-1}\}$, gaining one algebraic power over an unsigned volume tail. To retain all phase visits, let $\beta(h) = e^{-ch} + \lambda(1+h)^{-p-1}$. Coarea on the convex inner parallel sets and one-dimensional convolution estimates give

$$\int_{\Omega^-} G(X-Y)\beta(\operatorname{dist}(Y,\partial\Omega^-))\,\mathrm{d}Y \le C_0\beta(\operatorname{dist}(X,\partial\Omega^-)).$$

Choosing P so that $\epsilon_P C_0 < 1$, the resolvent series is absorbed in this weighted norm. Boundedness of F_0 replaces any exterior area or absolute gradient-mass assumption.

For completeness, the uniform free-kernel estimates used in this argument can be obtained directly from the same positive density. In one dimension, for $a \in [1.9, 2]$ and $d \ge 1$, put

$$A_a(r) = [r^{2p} + 2a\cos(\pi s)r^p + a^2]^{-1}.$$

On $0 \le r \le 1$ the denominator is uniformly positive and $A_a(r) = a^{-2} + O(r^p)$. Integration of $e^{-dr}r^p$ gives

$$G_s^a(d) \le \frac{c_{1,s}}{a^2}d^{-p-1}(1 + Cd^{-p}) + Ce^{-d}. \tag{3.51}$$

For $r \ge 1$ use $e^{-dr} \le e^{-d}$ and the uniform total density mass $G_s^a(0)$. This proves the sharp upper row coefficient after normal-core integration. For the lower coefficient, the low-mass Stieltjes integral omitted at $t \ge 1$ is bounded, relative to $c_{7,s}|Z|^{-7-p}$, by a fixed polynomial in $|Z|$ times $e^{-|Z|/2}$; scale $\sqrt{t}|Z|$ in the Yukawa heat formula to see this. The ratio $c_{7,s}/\sin(\pi s)$ stays bounded above and below. This justifies the uniform $o_{|Z|}(1)$ in the lower bound.

The exact tangential second-moment identity for normal separation D is

$$\int_{\mathbb{R}^6} |z|^2 G_{7,s}^a(z,D)\,\mathrm{d}z = 6\int_0^\infty \rho_s^a(r)e^{-r|D|}\left(\frac{|D|}{r} + \frac{1}{r^2}\right)\,\mathrm{d}r.$$

It follows either by differentiating the classical one-dimensional resolvent in its mass before integrating the positive mixture, or from the Gaussian heat-time moment. The density split used in (3.13) bounds it by $C\{(1+D)e^{-\sqrt{a}D} + \lambda D^{1-p}\}$. Flattening a receiving graph replaces a by $a(1-\nu)^p$ and introduces a factor tending to one. For sufficiently large fixed P and then sufficiently large j, the positive upper-row masses remain in $[1.9, 2]$. Formula (3.51) then has exponential remainder $C_P e^{-(1-\nu)D}$; because $\nu \to 0$, this is bounded by $C_P e^{-\gamma_0 D}$. Fixed normal depths shift D by at most $2P$, which is absorbed in C_P. This proves the row estimates and the individual second-moment bound (3.49). Those second moments are not summed over all labels: the algebraic bound D^{1-p} is not summable for $p < 2$. Only finitely many active-active pairs require that moment estimate. Inactive-label tails are instead bounded by their zeroth row masses $C_P\{e^{-\gamma_0 D} + \lambda D^{-p-1}\}$, which are summable over ordered distances independently of the number of sheets.

The ambient kernel and gradient bounds needed for the exterior are

$$G_{7,s}^2(Z) \le C\big(|Z|^{p-7}\mathbf{1}_{|Z|\le 1} + e^{-c_0|Z|} + \lambda(1+|Z|)^{-7-p}\big),$$

$$|\nabla G_{7,s}^2(Z)| \le C\big(|Z|^{p-8}\mathbf{1}_{|Z|\le 1} + e^{-c_0|Z|} + \lambda(1+|Z|)^{-8-p}\big).$$

Near zero these follow by scaling the heat integral and integrating the high-mass density; away from zero the low-density and resonance splits above give the displayed terms. Differentiation is also justified by the dimension-raising identity $-(G_{n,s}^2)'(r) = 2\pi r G_{n+2,s}^2(r)$, obtained by differentiating the heat kernel. The local gradient is integrable uniformly for $p > 1$. Dyadic annuli and the convex-surface estimate therefore bound both the boundary integral and the gradient-volume integral by $C\beta(h)$. At infinity their boundary term is $O(e^{-cR} + \lambda R^{-p-1})$ and vanishes. For nonsmooth convex cylinders use Gauss–Green or smooth convex approximations; the same surface bound controls the limit without corner terms.

We spell out the weighted convolution step so that repeated phase visits do not become an unproved assertion. The distance function $h(Y) = \operatorname{dist}(Y, (\Omega^-)^c)$ is 1-Lipschitz and $|\nabla h| = 1$ almost everywhere inside Ω^-. Its inner parallel sets $(\Omega^-)_t = \{h > t\}$ are convex, and

$\operatorname{dist}(X,\partial(\Omega^-)_t) \ge |h(X)-t|$. The surface estimate applied to $\partial(\Omega^-)_t$ bounds its kernel integral by $C[e^{-2c|h(X)-t|}+\lambda(1+|h(X)-t|)^{-p-1}]$ after decreasing c. Coarea and the elementary inequalities, uniform for $p \ge p_* > 1$,

$$e^{-2c|\cdot|} * e^{-c|\cdot|} \le Ce^{-c|\cdot|}, \quad e^{-2c|\cdot|} * (1+|\cdot|)^{-p-1} \le C(1+|\cdot|)^{-p-1},$$
$$(1+|\cdot|)^{-p-1} * (1+|\cdot|)^{-p-1} \le C(1+|\cdot|)^{-p-1}$$

prove the weighted Schur inequality. For the last two bounds split where either convolution argument has magnitude at least half the output magnitude; the other factor has uniformly bounded integral. The first is direct integration. As $\lambda^2 \le \lambda$, the constant C_0 is independent of the window size and aspect ratio.

Writing the exterior field as $Z = B + GVZ$, its weighted supremum is finite for each bounded window because $\beta(h)$ has a positive minimum there. Absorb $\epsilon_P C_0 < 1$ and obtain $|Z| \le C\|F_0\|_\infty \beta(h)/(1-\epsilon_P C_0)$. Here $\|F_0\|_\infty \le 2$. This proves (3.50) through every allowed phase visit, and completes the massive-row and exterior estimates. □

Proposition 3.10 (Actual power matrix and its adjacent two-site test)**.** *Suppose*

$$L_j^2 = \frac{\delta_j}{F_{a,j}}, \quad F_j \asymp F_{a,j}, \qquad H_{i,j}(y) = \frac{h_{i,j}(L_j y)}{\delta_j} \to H_i(y)$$

locally in C^1 on every finite-height label. Then $H_{i+1} - H_i \ge 1$ and, for finitely many compact amplitudes,

$$0 \le A_1 \sum_i \int |\nabla\zeta_i|^2 + 4\Lambda \sum_{i<k} (-1)^{i-k} \int \frac{(\zeta_i-\zeta_k)^2}{(H_k-H_i)^3}, \qquad \Lambda = \lim \frac{c_{1,s_j}}{\lambda_j} = 2. \tag{3.52}$$

For any adjacent gap $v = H_{i+1} - H_i$, regardless of whether it remains minimal away from the origin,

$$8\Lambda \int v^{-3}\eta^2 \le A_1 \int |\nabla\eta|^2. \tag{3.53}$$

Proof. Divide (3.47) by L_j^{6-2}. The row normalization is

$$L_j^2 c_{1,s_j} \delta_j^{-p_j-1} = c_{1,s_j}/\lambda_j \to \Lambda.$$

For positive same-orientation rows use the upper massive comparison; for negative rows use the lower comparison, discarding any extra opposite-orientation exponential interaction only with its favorable sign. The positive exponential rows vanish since

$$L_j^2 e^{-2\gamma_0\delta_j} \le C\delta_j e^{-(2\gamma_0-k)\delta_j} \to 0.$$

The algebraic lower comparison is initially integrated only over the actual donor patch. For each fixed pair, choose a larger fixed rescaled lateral buffer, so that its donor patch contains a tangential ball of radius cL_j about every receiving point in the amplitude support. Completing this algebraic row to $\mathbb{R}^6$ loses at most

$$C_P c_{1,s_j} \int_{|z|\ge cL_j} |z|^{-7-p_j}\, \mathrm{d}z \le C_P c_{1,s_j} L_j^{-p_j-1}.$$

After stability normalization this is bounded by $C_P L_j^2 c_{1,s_j} L_j^{-p_j-1} \le C_P \lambda_j L_j^{1-p_j} = o(1)$. Apply this completion first to a fixed finite set of labels; the remaining labels are exhausted afterward using the summable zeroth row masses. Thus neither this loss nor a tangential second moment is multiplied by the total number of sheets.

Only pairs whose two amplitudes are nonzero require replacing a receiving tangential foot, and there are finitely many such pairs. If $\zeta_k \equiv 0$, the cross term is already $\zeta_i(x)^2$: its receiving variable is integrated directly using the summable zeroth row mass, without a second-moment error. For each active-active pair use

$$\int |\zeta((x+z)/L_j) - \zeta(x/L_j)|^2\, \mathrm{d}x \le L_j^4 |z|^2 \int |\nabla\zeta|^2.$$

For a positive row apply $(a+b)^2 \le (1+\tau)a^2 + C_\tau b^2$; for a negative row apply $-(a+b)^2 \le -(1-\tau)a^2 + C_\tau b^2$. The moment (3.49) removes the error. On negative rows this is done after replacing by the algebraic lower kernel, so a potentially much larger exponential mass never enters that error. The exterior is at most

$$C_P L_j^2 \{e^{-cB\delta_j} + \lambda_j (B\delta_j)^{-p_j-1}\} \le o_j(1) + C_P B^{-p_*-1}.$$

A label beyond N intervening sheets contributes at most $C\sum_{k>N} k^{-p_j-1} = O(N^{-p_*})$. First pass to the limit with fixed buffers and labels, then exhaust them, let $P \to \infty$, and finally let $\tau \downarrow 0$ in the receiving-foot comparison. The self coefficient in (3.48) tends to A_1; both row coefficients tend to 4Λ. This proves (3.52).

For (3.53), choose $\zeta_i = \eta$, $\zeta_{i+1} = -\eta$ and every other amplitude zero. With $b(r) = 4\Lambda r^{-3}$, the selected pair is $-4b(v)\eta^2$. The two active interactions of label $i-a$ to the left sum to

$$(-1)^a \{b(r_a) - b(r_a+v)\}\eta^2.$$

The bracket is positive and decreases with r_a, since b is decreasing and convex. Its alternating sum starts negative and is nonpositive. The right side is identical. Thus the total form is bounded above by $2A_1 \int |\nabla\eta|^2 - 16\Lambda \int v^{-3}\eta^2$. Divide by two. The full signed matrix gives the coefficient 8Λ; a premature nearest-row estimate would lose the needed sharp ratio. □

For the exponentially dominated case the actual selected row has a sharper lower limit. Let the selected vertical gap be $\delta_j + z_j(x/L_j)$ with $z_j \to z$ locally uniformly, and take $L_j^2 = e^{k\delta_j}$. The first-force contradiction gives

$$\nu_{B,j}\sqrt{\delta_j} \le C_B \delta_j \sqrt{\kappa_j} \le C_B \delta_j e^{-k\delta_j/2} \to 0, \qquad \kappa_j\delta_j \to 0, \qquad L_j/\sqrt{\delta_j} \to \infty. \tag{3.54}$$

The ambient positive mixture is

$$G^{;2}_{7,s}(X) = \int_0^\infty 2r\rho_s(r) Y_{7,r}(X)\, \mathrm{d}r,$$

where $Y_{7,r}$ is the classical Yukawa kernel. Its heat integral gives

$$Y_{7,r}(X) = \frac{1}{2r}\left(\frac{r}{2\pi}\right)^3 |X|^{-3} e^{-r|X|}(1 + O(|X|^{-1})).$$

Restrict to $|r - r_s| \le \delta_j^{-2}$ and to receiving feet $x' - x = \sqrt{\delta_j}\, a$, $|a| \le K$. The graph correction to the normal separation is at most $K\nu_{B,j}\sqrt{\delta_j} + CK^2\kappa_j\delta_j = o(1)$. The normalized Yukawa kernel converges to

$$\frac{1}{2k}\left(\frac{k}{2\pi}\right)^3 e^{-k(z+r-t+|a|^2/2)}.$$

Integrate the actual fixed-core sources, then let $K, P \to \infty$. Their limiting exponential moments are 8, so the selected actual cross row, divided by $e^{-k\delta_j}$, has lower limit $16ke^{-kz}$. Positivity permits every restriction just used.

Proposition 3.11 (Actual exponential two-site stability)**.** *Under* (3.17) *and the selected additive compactness just stated,*

$$32k \int e^{-kz}\eta^2 \le A_1 \int |\nabla\eta|^2. \tag{3.55}$$

Proof. Use $\eta, -\eta$ on the selected adjacent cores in (3.47). Their self budget is $2A_1 \int |\nabla\eta|^2$. Retain only the selected negative resonance interaction and discard other opposite rows. Its full factor four is justified by

$$[\eta(x/L_j) + \eta(x'/L_j)]^2 \ge 4(1-\tau)\eta(x/L_j)^2 - C_\tau[\eta(x'/L_j) - \eta(x/L_j)]^2.$$

On the restricted receiving disk the error, after normalization, is $O_{P,K,\tau}(\delta_j e^{-k\delta_j}) = o(1)$. Positive same-orientation rows vanish because

$$L_j^2 e^{-2\gamma_0\delta_j} = e^{-(2\gamma_0-k)\delta_j} \to 0, \qquad L_j^2\lambda_j\delta_j^{-p_j-1} = \frac{F_{a,j}}{\delta_j F_{e,j}} \to 0.$$

The signed exterior vanishes after choosing B with $cB > k$; its algebraic part has the same bound. Let the truncations tend to infinity, then $\tau \downarrow 0$. Nonnegativity of the exact form gives $0 \le 2A_1 \int |\nabla\eta|^2 - 64k \int e^{-kz}\eta^2$, which is (3.55). □

3.6. **Limiting gap equations and the reciprocal test.**

Lemma 3.12 (The six-dimensional power reciprocal test)**.** *There is no positive $v \in C^{1,1}_{\mathrm{loc}}(\mathbb{R}^6)$ satisfying*

$$\Delta v \le a v^{-2}, \qquad 2a \int v^{-3}\varphi^2 \le \int |\nabla\varphi|^2 \quad (\varphi \in C_c^\infty(\mathbb{R}^6)), \qquad a > 0. \tag{3.56}$$

Proof. Fix $\alpha = 7/2$. With

$$I = \int v^{-3-2\alpha}\eta^2, \quad J = \int v^{-2\alpha-2}|\nabla v|^2\eta^2, \quad D = \int v^{-2\alpha}|\nabla\eta|^2,$$

test the distributional upper equation by $v^{-2\alpha-1}\eta^2$. This gives $(2\alpha+1)J \le aI + 2\sqrt{JD}$. Test stability by $v^{-\alpha}\eta$ to get $2aI \le \alpha^2 J + 2\alpha\sqrt{JD} + D$. For any sufficiently small fixed $\epsilon > 0$, Young's inequality gives

$$J \le \frac{a}{2\alpha + 1 - \epsilon} I + C_\epsilon D, \qquad 2aI \le (\alpha^2 + \epsilon)J + C_\epsilon D.$$

Since $\alpha^2 = 49/4 < 2(2\alpha+1) = 16$, absorption yields $I \le CD$. These tests are legitimate by approximation on compact sets, where v has a positive minimum.

Take $\eta = \chi_R^t$, where $\chi_R = 1$ on B_R, vanishes outside B_{2R}, $|\nabla\chi_R| \le C/R$, and $t \ge (3+2\alpha)/3$. Hölder gives

$$I \le CR^{-2} I^{2\alpha/(3+2\alpha)} R^{18/(3+2\alpha)}.$$

Consequently

$$\int_{B_R} v^{-3-2\alpha} \le CR^{6-2-4\alpha/3} = CR^{-2/3} \longrightarrow 0. \tag{3.57}$$

A fixed ball has strictly positive integral, a contradiction. In the usual notation $\Delta v \le a v^{-p}$, $bp \int v^{-p-1}\varphi^2 \le \int |\nabla\varphi|^2$, the threshold is $b/a > \alpha_c^2/[p(2\alpha_c+1)]$ with $\alpha_c = (m-2)(p+1)/4$. Here $p = 2, m = 6$ gives $9/14 < 1$. The word "Hardy" describes the resulting inverse-power potential; no unproved comparison of v with $|x|^{2/3}$ has been used. □

Lemma 3.13 (The six-dimensional exponential reciprocal test)**.** *There is no $z \in C^{1,1}_{\mathrm{loc}}(\mathbb{R}^6)$ satisfying*

$$\Delta z \le a e^{-kz}, \qquad ak \int e^{-kz}\varphi^2 \le \int |\nabla\varphi|^2, \qquad a, k > 0. \tag{3.58}$$

Proof. Choose $\alpha = 3k/2$. Test the upper equation by $e^{-2\alpha z}\eta^2$ and stability by $e^{-\alpha z}\eta$. With $I = \int e^{-(k+2\alpha)z}\eta^2$, $J = \int e^{-2\alpha z}|\nabla z|^2\eta^2$, and $D = \int e^{-2\alpha z}|\nabla\eta|^2$, the same calculation gives

$$(2\alpha - \epsilon)J \le aI + C_\epsilon D, \qquad akI \le (\alpha^2 + \epsilon)J + C_\epsilon D.$$

Absorption is possible because $\alpha < 2k$. Thus $I \le CD$. Using $\eta = \chi_R^t$ with $t \ge (k+2\alpha)/k$, Hölder gives

$$\int_{B_R} e^{-(k+2\alpha)z} \le CR^{6-2-4\alpha/k} = CR^{-2} \longrightarrow 0,$$

again impossible. In general the strict range is $k(m-2)/4 < \alpha < 2k$; it is nonempty for $m = 6$. □

Theorem 3.14 (First-force exclusion in stable ambient dimension seven)**.** *Under the preparation preceding Lemma 2.10,*

$$e^{-\sqrt{2}\delta_j} + (1 - s_j)\delta_j^{-2s_j} = o(\kappa_j). \tag{3.59}$$

In particular the statement concerns the actual intrinsic-off-diagonal infimum and the actual positive curvature supremum. It is not a classification or a curvature-retention theorem by itself.

Proof. The infinite-gap and curvature-scale-isolation cases were disposed of immediately after (3.1). If the conclusion fails, pass to (3.2), select the actual consecutive pins by Lemma 3.1, and pass to a subsequence for which

$$\theta_j = F_{a,j}/F_j \longrightarrow \theta \in [0,1].$$

All constants in a working window are fixed before $j \to \infty$. In applying the scalar alternating-row inequality, we may replace $\mathfrak{F}_{s_j}$ by the positive decreasing function B_{s_j}: Lemma 3.2 bounds the total error in each ordered row by $o(F_j)$, including its summable label tail. This replacement does not change either limiting upper inequality.

Suppose first $\theta > 0$. Put $L_j^2 = \delta_j/F_{a,j}$. Then $L_j \le C\ell_j$, $L_j/\delta_j \to \infty$, and the complete thin windows contain every fixed rescaled ball and its necessary buffer. For every finite-height label,

$$H_{i,j}(y) = h_{i,j}(L_j y)/\delta_j, \qquad |D^2 H_{i,j}| \le C\kappa_j/F_{a,j} \le C.$$

The marked slope estimates and central heights bound their remaining local $C^{1,1}$ norms. A diagonal subsequence supplies all labels whose normalized central heights stay finite. A label with $|h_{i,j}(0)|/\delta_j \to \infty$ also escapes uniformly on every fixed rescaled compact base: otherwise it would have height $O(\delta_j)$ at a sequence of points in that base, and the complete thin-window construction would continue it to the central fiber with height $O(\delta_j)$, a contradiction. The ordered zeroth-row tails are summable uniformly in j. First truncate the labels, then use this compact-uniform escape for the finite collection, and finally remove the label truncation. Hence escaping labels contribute vanishing normalized rows. For the selected gap

$$v_j = (h_{2,j} - h_{1,j})(L_j\cdot)/\delta_j \ge 1, \qquad v_j(0) \to 1,$$

one obtains $v \in C^{1,1}_{\mathrm{loc}}(\mathbb{R}^6)$ with $v \ge 1$, $v(0) = 1$. The same almost-minimum Taylor estimate as in (3.5) gives $Dv_j(0) \to 0$.

On each compact subset of $\{v > 1\}$ the selected force is algebraic: if $v_j \ge 1+\epsilon$, the exponential contribution, normalized by $F_{a,j}$, is $O(e^{-c\epsilon\delta_j})$. The low-frequency coefficient (3.16), the exact form limit (3.30), and the signed upper row (3.46) imply

$$A_1 \Delta v \le 4\Lambda v^{-2} \quad \text{on } \{v > 1\}. \tag{3.60}$$

Neighboring exponential contacts have favorable negative signs in this inequality; they are not asserted small componentwise. Nor may the selected positive contact be dropped merely because its limiting set has empty interior.

Instead use the local $C^{1,1}$ regularity of v established above directly from $|D^2 h_{i,j}| \le C\kappa_j$ and $\kappa_j/F_{a,j} \le C$. On $E = \{v = 1\}$, $Dv = 0$ pointwise because v is C^1 and attains its minimum there. Each $\partial_a v$ is locally Lipschitz, and its weak derivative vanishes almost everywhere on its zero set. Consequently $D^2 v = 0$ almost everywhere on E. Since $v \in C^{1,1}_{\mathrm{loc}}$, its distributional Hessian is an L^∞ function and has no singular contact part. Thus (3.60) holds on all of $\mathbb{R}^6$. The actual matrix supplies (3.53), so with $a = 4\Lambda/A_1$ the pair of inequalities is exactly (3.56). Lemma 3.12 is a contradiction. This contact argument would be invalid for a merely Lipschitz limit: $1 + |x_1|$ has a positive singular Laplacian on its minimum set.

Suppose next $\theta = 0$. Put $L_j^2 = F_{e,j}^{-1}$. Then $L_j/\ell_j \le C/\sqrt{\delta_j} \to 0$. Only the two selected additive heights are needed:

$$Z_{1,j}(y) = h_{1,j}(L_j y), \qquad Z_{2,j}(y) = h_{2,j}(L_j y) - \delta_j.$$

Their Hessians are bounded by $C\kappa_j/F_{e,j} \le C$. The first marked slope is zero, while the second has scaled size at most

$$CL_j\sqrt{\kappa_j\delta_j}(\sqrt{\eta_j} + \eta_j) \le C\sqrt{\kappa_j/F_{e,j}}\sqrt{\eta_j\delta_j}(1 + \sqrt{\eta_j}) \to 0.$$

Their central values remain bounded because $d_j - \delta_j = o(1)$. Hence their difference $z_j = Z_{2,j} - Z_{1,j} \ge 0$ converges locally in $C^{1,\alpha}$, $\alpha < 1$, to $z \in C^{1,1}_{\mathrm{loc}}$ with $z(0) = 0$. There is no additive compactness assertion for the other sheets.

The selected exponential force limit (3.19), the transferred actual rows and the common self-form limit now give

$$A_1 \Delta z \le 32 e^{-kz}.$$

Every exterior neighboring force is discarded only after retaining its favorable sign. The transport precision (3.54) gives the compatible actual stability (3.55). With $a = 32/A_1$, this is exactly (3.58). Lemma 3.13 gives the second contradiction. The two cases exhaust the subsequences, proving (3.59) with the preceding estimates applied in the stated order of limits. □

Consequences of the first-force estimate. The established geometric output is $\delta_j \to \infty$, with genuine normal tubes and complete thin-window actual graph families. The analytic conclusion is $F_j = o(\kappa_j)$ by Theorem 3.14. The pin estimate used here is $\beta_{B,j} = O_B(\sqrt{\delta_j} F_j)$ only in the contradiction regime $F_j \gtrsim \kappa_j$. It is neither the later $O(\kappa_j)$ natural-window preparation nor permission to differentiate a value-controlled correction. The subsequent curvature-scale normal compactness, minimality, and whole-carrier stability arguments remain separate steps.

4. Quantitative preparation of the transition region

4.1. **Natural windows and chamber estimates.** We now work in the stable end dimension $n = 7$, with sheet dimension $m = 6$. Suppress the sequence index unless a limit is being taken. Thus

$$\begin{gathered} p = 2s \longrightarrow 2, \qquad \lambda = 1 - s \longrightarrow 0, \qquad \kappa \longrightarrow 0, \qquad \delta \longrightarrow \infty, \\ F = e^{-k\delta} + \lambda \delta^{-p} = o(\kappa), \quad k = \sqrt{2}. \end{gathered} \tag{4.1}$$

Here $\kappa = \sup_\Sigma |A| > 0$, $\Sigma = \{u = 0\}$ is the regular, complete, ambiently closed embedded zero hypersurface, and δ is an intrinsic-off-diagonal separation lower bound with one fixed small intrinsic cutoff a_{geom}/κ. Distances between different components are infinite in the intrinsic metric. The actual equations and stability are

$$(-\Delta)^s u = u - u^3, \quad |u| \le 1, \qquad \int |\xi|^{2s} |\widehat{\zeta}(\xi)|^2 \, \mathrm{d}\xi + \int (3u^2 - 1)\zeta^2 \, \mathrm{d}X \ge 0. \tag{4.2}$$

The Fourier normalization is used throughout. In particular $c_{n,s} \asymp \lambda$, while $c_{n,s}/(2-p)$ remains bounded above and below. We take $p \ge p_* > 3/2$ after discarding finitely many indices. Qualitative convergence to a centered layer at every moving zero, uniform well clearing, and the hypotheses in (4.1) have already been supplied. No rate relation between λ and κ is added.

The first-force estimate (4.1) permits quantitative preparation on growing windows. We first establish zero-trace coercivity from the qualitative moving-center limit and then control the chambers. This order provides the higher signed-distance derivatives needed for the signed capacity estimates. The source-to-phase inverse will be developed in Section 6.

Lemma 4.1 (Uniform layer and massive-kernel estimates)**.** *For the centered increasing layer q_s of Lemma 2.7, put $w_s = q_s'$ and $W_s = 3(1 - q_s^2)$. Uniformly near one,*

$$q_s(0) = 0, \quad q_s(\pm\infty) = \pm 1, \quad w_s > 0, \quad \int_{\mathbb{R}} w_s = 2, \quad w_s(0) \ge c > 0, \tag{4.3}$$

$$1 - |q_s(t)| + |w_s(t)| + |w_s'(t)| \le C\{e^{-k|t|} + \lambda(1 + |t|)^{-p}\}, \tag{4.4}$$

$$H_s := |D_t|^p + 2 - W_s, \qquad H_s w_s = 0, \qquad \langle v, H_s v \rangle \ge \gamma \|v\|_2^2 \quad (v \perp w_s). \tag{4.5}$$

All fixed derivatives of q_s are uniformly bounded. For the positive massive resolvent $G_s^\mu = ((-\Delta)^s + \mu)^{-1}$, with μ in a fixed compact subset of $(0, \infty)$, one has mass $1/\mu$ and

$$G_s^\mu(Z) \le C_\mu\{e^{-c_\mu |Z|} + \lambda |Z|^{-n-p}\} \quad (|Z| \ge 1). \tag{4.6}$$

Fixed scaled radial derivatives have the same type of bound. When μ is sufficiently close to 2, the exponential rate can be chosen strictly greater than $k/2$. The one-dimensional massive kernel obeys

$$g_s^{\mu_\kappa}(D) \le C\{e^{-(k - C\kappa)D} + \lambda(1 + D)^{-p-1}\}, \qquad |\mu_\kappa - 2| \le C_0 \kappa, \quad D \ge 0. \tag{4.7}$$

Proof. The centered layer is the smooth increasing solution of $|D_t|^p q_s = q_s - q_s^3$ normalized by $q_s(0) = 0$ and $q_s(-\infty) = -1$, $q_s(+\infty) = 1$, recalled in Section 1; smoothness follows from Proposition 2.2. Its derivative $w_s = q_s'$ is exactly the positive Jacobi field used in Lemma 3.3. Thus $w_s > 0$, and the fundamental theorem of calculus followed by the two end limits gives $\int_{\mathbb{R}} w_s = 2$. Lemma 2.7 gives $w_s(0) \to 1/\sqrt{2}$, hence a uniform positive lower bound. Differentiating the

layer equation gives $(|D_t|^p + 3q_s^2 - 1)w_s = 0$, which is $H_s w_s = 0$. The tails are Lemma 3.2; its sharper mode bounds imply (4.4). Since $H_s = |D_t|^p - f'(q_s)$, the uniform orthogonal spectral gap is exactly the gap established in the proof of Lemma 3.3. Thus no new layer or spectral-gap argument is required.

Positivity and mass $1/\mu$ follow from the heat resolvent. The scalar Stieltjes representation used in Lemma 3.9, with its mass parameter replaced by μ, gives, for $s < 1$,

$$\begin{aligned} G^{\mu}_{7,s}(Z) &= \int_0^\infty 2r\, \rho^{\mu}_s(r) Y_{7,r}(Z)\, \mathrm{d}r, \\ \rho^{\mu}_s(r) &= \frac{\sin(\pi s)}{\pi} \frac{r^p}{(r^p-\mu)^2 + 2\mu r^p(1+\cos(\pi s))}. \end{aligned} \tag{4.8}$$

Here $Y_{7,r}$ is the free Yukawa kernel of mass r^2. For μ in a fixed compact set $\mathcal{K} \Subset (0,\infty)$, the density is $O(\lambda r^p)$ below a fixed threshold smaller than all resonance locations $r_* = \mu^{1/p}$, has bounded total mass near the resonance, and is $O(\lambda r^{-p})$ above a fixed large threshold. These are the mass-uniform density estimates proved in Section 3.2.

We retain the additional scaled-radial calculation needed here. In dimension seven,

$$Y_{7,r}(R) = C_7 R^{-5} e^{-rR}\{1 + rR + (rR)^2/3\}.$$

Each fixed derivative $(R\partial_R)^a Y_{7,r}$ is bounded by $C_a R^{-5}(1 + (rR)^{a+2})e^{-rR}$. For $R \le 1$, the high-frequency part of (4.8) therefore costs at most $C_a \lambda \Gamma(2-p) R^{p-7}$ for the zeroth polynomial term. Positive powers of rR have bounded Gamma factors and the same power of R. The product $\lambda\Gamma(2-p)$ remains bounded. The resonance part costs $CR^{-5} \le CR^{p-7}$. For $R \ge 1$, the low-density contribution, including the factor $2r$, is bounded by $C_a \lambda R^{-7-p}$. Choose $0 < c < \inf_{\mu\in\mathcal{K}} \sqrt{\mu}$ and a fixed threshold between c and all r_* for orders sufficiently near one. On a compact interval away from zero the mixing mass is uniformly bounded. On the unbounded high-frequency interval retain the factor e^{-rR} while integrating $C\lambda r^{1-p}$; the remaining exponential absorbs every fixed power of rR after the rate is decreased to c. These two contributions are at most $C_{a,c} e^{-cR}$. The full mixing measure has infinite mass when $p < 2$, so a total-mass bound is used only on the compact interval. Consequently

$$\mathcal{H}_s(Z) = C_a \begin{cases} |Z|^{p-7}, & 0 < |Z| \le 1, \\ e^{-c|Z|} + \lambda |Z|^{-7-p}, & |Z| \ge 1 \end{cases}$$

controls the free kernels and the finitely many scaled radial derivatives needed below. At $s = 1$ use $Y_{7,\sqrt{\mu}}$ directly. Small fixed constant metric changes and their parameter derivatives have the same majorant after slightly decreasing c.

Whenever the rate $c > k/2$ is required, restrict the mass set to a sufficiently small fixed neighborhood of 2. Choose $c > k/2$ with a fixed margin first, and then choose the metric neighborhood so the decreased rate still satisfies $2c > k$. No such fast rate is asserted for an arbitrary compact mass set.

Finally, the one-dimensional mass-uniform estimate (3.13) gives $g^{\mu_\kappa}_s(D) \le C\{e^{-\sqrt{\mu_\kappa} D} + \lambda(1+D)^{-p-1}\}$. Since $\sqrt{\mu_\kappa} \ge k - C\kappa$, this proves (4.7) without a relation between λ and κ. □

4.1.1. *Uniform physical smoothness and zero-trace coercivity.* Proposition 2.2 already gives the absolute physical derivative bounds, independently of chamber preparation:

$$\|u\|_{C^k(\mathbb{R}^7)} \le C_k \qquad (k \text{ fixed}). \tag{4.9}$$

Write $\mathcal{E}_s(v) = \int_{\mathbb{R}^7} |\xi|^p |\widehat{v}(\xi)|^2 \, \mathrm{d}\xi$. The trace on Σ is understood in its regular local graph charts.

Lemma 4.2 (A gap before quantitative preparation)**.** *Under (4.5), qualitative moving-center preparation and well clearing imply, eventually,*

$$\mathcal{E}_s(v) + \int (3u^2 - 1)v^2 \ge c\|v\|^2_{H^s(\mathbb{R}^7)}, \qquad v \in H^s(\mathbb{R}^7), \quad \mathrm{Tr}_\Sigma\, v = 0, \tag{4.10}$$

with one uniform $c > 0$.

Proof. The point-pin estimate (3.23), applied fiberwise, and $(|\xi_x|^2+\tau^2)^s \geq |\tau|^{2s}$ give a uniform flat ambient L^2 gap. Bounded potential upgrades it to an H^s gap. We pass from this model to the entire actual zero set.

First choose D_0 so that well clearing gives $3u^2-1 \geq 1$ whenever $\operatorname{dist}(X,\Sigma) \geq D_0$, for all sufficiently large indices. Choose a large fixed $R_0 \geq D_0$. Take translates, on the R_0-lattice, of a fixed nonnegative smooth bump whose translates cover space, and divide each by the square root of their sum of squares. This gives a smooth square partition $\sum_a \eta_a^2 = 1$ with uniformly bounded overlap, support diameter CR_0, and $|D\eta_a| \leq C/R_0$. In particular $\sum_a |D\eta_a|^2 \leq C/R_0^2$. If a support is not entirely at distance at least D_0 from Σ, choose a nearby zero. The support lies in an enlarged ball of radius $C'R_0$ about that zero. Section 2 supplies the single actual graph and qualitative layer comparison on that ball. Center its tangent frame at the chosen zero, so $h(0)=0$ and $Dh(0)=0$. The curvature bound on this fixed small-slope graph gives $|D^2h| \leq C\kappa$, and hence $|Dh| \leq C\kappa R_0$ and $|h| \leq C\kappa R_0^2$ on the required enlargement. Multiply h by a tangential cutoff with derivative bounded by C/R_0, equal to one on a neighborhood of the projected test support. The cutoff derivative term is at most $(C/R_0)(C\kappa R_0^2)$, so the extended height satisfies $\|Dh_{\mathrm{ext}}\|_\infty \leq C\kappa R_0$. The global map $(x,t) \mapsto (x, t+h_{\mathrm{ext}}(x))$ has determinant one and changes all pairwise distances by relative $O(\kappa R_0)$. For a localized global test $v_a = \eta_a v$, put $f_a = v_a \circ \Phi$, where $\Phi(x,t) = (x, t+h_{\mathrm{ext}}(x))$. Writing $L = \|Dh_{\mathrm{ext}}\|_\infty \leq C\kappa R_0$, one has for every pair $P, Q \in \mathbb{R}^7$

$$(1-L)|P-Q| \leq |\Phi(P)-\Phi(Q)| \leq (1+L)|P-Q|.$$

The determinant-one change of variables in the entire energy gives

$$(1+L)^{-7-p}\mathcal{E}_s(f_a) \leq \mathcal{E}_s(v_a) \leq (1-L)^{-7-p}\mathcal{E}_s(f_a).$$

This controls every fractional interaction pair, including pairs with an exterior endpoint. The actual and extended pins agree on the test support, while the localized test vanishes elsewhere, so f_a has zero flat trace. Fiberwise and hypersurface traces agree by the uniform one-dimensional trace inequality and density. The zero extension here is the already global smooth localization $\eta_a v$, not a characteristic extension across a new boundary.

The potential comparison is also uniform in the chosen zero. Indeed the C^1 preparation gives $\nabla u(0) = w_s(0)e + o(1)$ for its approximate direction e. Since $w_s(0) \geq c$, the phase normal $\nu = \nabla u(0)/|\nabla u(0)|$ satisfies $e-\nu = o(1)$. Rotate ν to the last coordinate. On this fixed enlarged ball, $|e\cdot X - t| = o_{R_0}(1)$; the uniform Lipschitz bound on q_s therefore changes $q_s(e\cdot X)$ to $q_s(t)$ by $o_{R_0}(1)$. After the shear, the additional height shift is at most $C\kappa R_0^2$. Thus

$$\sup_{\operatorname{supp} f_a} |u(\Phi(x,t)) - q_s(t)| = o_{R_0}(1),$$

and the two potentials differ by $o_{R_0}(1)$ there. The flat H^s gap absorbs $O(\kappa R_0)\mathcal{E}_s(f_a) + o_{R_0}(1)\|f_a\|_2^2$ when the sequence index is large. Supports entirely beyond D_0 have the positive well gap. For this scaled partition, the fundamental theorem of calculus, Cauchy–Schwarz, and $\sum_a |D\eta_a|^2 \leq C/R_0^2$ give

$$\sum_a |\eta_a(X) - \eta_a(Y)|^2 \leq C\min\{1, |X-Y|^2/R_0^2\}.$$

The bound by a constant follows directly from $\sum_a \eta_a^2 = 1$. The square-partition IMS identity and $|v(X)v(Y)| \leq (v(X)^2+v(Y)^2)/2$ consequently bound its error by

$$Cc_{7,s}\left(\frac{1}{2-p}+\frac{1}{p}\right)R_0^{-p}\|v\|_2^2 \leq CR_0^{-p}\|v\|_2^2.$$

This is uniform near one. Choose R_0 first to make this smaller than the local gap, and only then take the index large. This proves the global L^2 lower bound. Boundedness of $3u^2-1$ upgrades it to (4.10). The local point-pin estimate and pullbacks apply to H^s tests directly; compact exhaustion justifies the same localization for every zero-trace H^s function. □

4.1.2. *Complete natural windows and a same-zero reference.* Set $R_* = (\delta/\kappa)^{1/2}$ when $\kappa\delta \le a_0$, where a_0 will be chosen after all fixed window enlargement factors. At a marked zero, rotate its tangent plane to $t = 0$. The marked graph on a fixed large multiple of R_* has height $O(\kappa R_*^2) = O(\delta)$ and slope $O(\kappa R_*) = O(\sqrt{\kappa\delta})$.

Use the tangent-chart continuation established in the proof of Lemma 3.1; this fact uses only completeness and the curvature bound. The actual graph through a point extends to radius c_{geom}/κ, and contains its intrinsic ball of radius a_{geom}/κ after fixing a_{geom} sufficiently small. Normal variation along a lifted path is bounded by curvature times its intrinsic length. Completeness supplies every finite endpoint. An actual graph is open and closed relative to $\Sigma \cap (U \times \mathbb{R})$, by continuity and the unique embedded local germ. Consequently a surface path projecting inside U and meeting that graph remains on it.

For another zero Y of height $O(\delta)$, choose a tangent unit vector with maximal vertical component $v_t \ge 0$. Along the complete unit-speed geodesic with this initial vector, $|\gamma''| \le \kappa$. At parameters $\pm CR_*$ the vertical displacement is $\pm CR_* v_t + O(\kappa R_*^2)$, with C fixed. The marked graph has height and variation $O(\delta)$ on the larger supplied base. If $v_t > C'\delta/R_*$, one of these geodesic segments crosses it. The preceding graph-membership argument contradicts a distinct branch, including a returning branch. Thus every such zero has tilt $O(\sqrt{\kappa\delta})$. Its horizontal graph continues over the larger common base: curvature maintains transversality and ambient closedness rules out a finite terminal point. Every zero in the desired inner prism continues to the central fiber at height $O(\delta)$. To verify separation before counting, choose a_0 after all fixed enlargements so the entire working base has diameter much smaller than a_{geom}/κ. Each small-tilt branch has a larger horizontal curvature chart containing that working base and its intrinsic ball of radius a_{geom}/κ. If points on two distinct retained branches could be joined by a shorter minimizing geodesic, it would remain in this larger chart. The horizontal graph and its continuation over the working base identify its endpoint with the starting branch, a contradiction. Thus arbitrary points on distinct retained branches are intrinsically off diagonal and have ambient distance at least δ. Select all central-fiber zeros in a sufficiently large closed interval $[-C\delta, C\delta]$. They are δ-separated, so their number is bounded by $2C + 1$. An omitted zero between two selected graphs would continue to a central intercept between their intercepts and thus would already have been selected. We have obtained a complete consecutive finite stack with

$$M \le C, \qquad |h_i| \le C\delta, \qquad |Dh_i| \le C\sqrt{\kappa\delta}, \qquad |D^2 h_i| \le C\kappa. \tag{4.11}$$

All constants depend on the prescribed finite enlargements, never on chamber length. The proof uses the actual surface throughout; independent extensions of the graphs are not used to infer separation.

Choose fixed constants $r_{\mathrm{eval}} < r_{\mathrm{mid}} < r_{\mathrm{out}}$ and $H_{\mathrm{eval}} < H_{\mathrm{mid}} < H_{\mathrm{out}}$ and put

$$\Omega_\ell = B^6_{r_\ell R_*} \times (-H_\ell\delta, H_\ell\delta), \qquad \ell \in \{\mathrm{eval}, \mathrm{mid}, \mathrm{out}\}.$$

Take each consecutive radius difference at least K and each consecutive height difference at least B. The previously constructed actual graphs are continued over a larger common base and the finite stack is chosen to contain every actual zero in the closed δ-neighborhood of Ω_{out}, together with all intermediate graphs. This requires only a fixed increase in the height and base multipliers, since $\delta/R_* = \sqrt{\kappa\delta}$ is bounded; the finite sheet bound remains uniform. In particular, for each fixed A and all sufficiently large indices, any point of Ω_{out} within distance A of the full zero set has every nearest zero on a retained branch. Any additional zero in the auxiliary domain lies at distance at least δ from Ω_{out}. The fixed height constants can be increased to include the marked graph on the working base. A prepared region means such a complete-stack region, equipped with the qualitative moving-zero layer and well-clearing estimates of Proposition 2.8. Its phase cells are the connected components of its complement of the actual zero set. The actual phase of a cell is the constant sign of u there. A transmission cell, when used below, is instead one of the fixed-size ambient localization neighborhoods in Lemma 3.4; it may meet a zero graph. The lateral and vertical margins between consecutive prepared regions are at least KR_* and $B\delta$, respectively. All constants, including the larger bases and the auxiliary domains below, are fixed before a_0 is decreased.

Choose larger auxiliary prisms of height $O(R_*)$ and base radius $O(R_*)$, with a further lateral buffer in the actual common graph domain. They may contain zeros outside the retained finite stack; preparation and phase matching are asserted only in the prepared regions. Here ρ_i denotes distance to the individual retained actual graph, with its upward sign, rather than the minimum of the distances to all zero graphs. All graph points used as feet remain in the actual larger base.

For $X = (x, t)$ in an auxiliary prism the foot equation is

$$y = x + (t - h_i(y))Dh_i(y). \tag{4.12}$$

Its right-hand side moves x by at most $CR_*\sqrt{\kappa\delta} = C\delta$, and its derivative in y has norm at most $C(\kappa\delta + \sqrt{\kappa\delta})$. The larger base is chosen with this margin, so the map is a contraction on a closed ball about x contained in that base. The squared-distance function on the base has Hessian

$$2\{I + Dh_i \otimes Dh_i - (t - h_i)D^2h_i\} \geq I$$

after decreasing a_0. Its critical point is therefore the unique minimizer. Choose the further lateral buffer larger than the fixed auxiliary height bound; comparison with the vertical graph point then excludes a minimizing point on the edge of the graph domain. Thus this foot and its distance are genuine.

Here is the differentiation using only second derivatives of h_i. At the foot write

$$b = Dh_i(y), \quad H = D^2h_i(y), \quad a = t - h_i(y), \quad A = I + b \otimes b - aH, \quad w = (1 + |b|^2)^{1/2}.$$

Since $A \geq \frac{1}{2}I$, differentiation of the foot equation gives

$$A\, D_X y = (I_6 \;\; b), \qquad |D_X y| \leq C.$$

With $Y = (y, h_i(y))$ and $N = (-b, 1)/w$, we have $X - Y = awN$, so $\rho_i = aw$. Differentiating this vector identity and taking its scalar product with N gives

$$d\rho_i = N \cdot dX,$$

because $N \cdot dY = N \cdot dN = 0$. This calculation also holds at $a = 0$, without division by a. Therefore

$$D\rho_i = N(y), \qquad |D\rho_i| = 1, \qquad D^2\rho_i = D_y N\, D_X y, \qquad |D^2\rho_i| \leq C|H| \leq C\kappa.$$

Indeed $D_y N[v] = (-Hv, 0)/w - N(b \cdot Hv)/w^2$, whose norm is at most $C|H||v|$. No third derivative of a graph has entered.

The domains and one fixed $c > 0$ are chosen so that, for every $X \in \Omega_{\text{out}}$, the closed ball $B_{2cR_*}(X)$ is inside the common genuine-distance domain with a positive foot margin. The required Taylor segments have the following precise meaning. For each graph take an orthogonal map O with $Oe_7 = D\rho_i(X)$. For $|(z, t)| \leq cR_*$ they are

$$\{X + \theta tOe_7 : 0 \leq \theta \leq 1\}, \qquad \{X + tOe_7 + \theta O(z, 0) : 0 \leq \theta \leq 1\}.$$

Both lie in $\overline{B}_{cR_*}(X)$, hence in the larger genuine domain, and their feet remain on the actual graph. These are the normal-axis and tangential segments used in the profile comparison.

Order the graphs upward. Let $a_0^{\text{ph}} = \rho_-$ be the actual phase immediately below the first graph and let a_i^{ph} be the phase immediately above graph i. These phase symbols are unrelated to the small geometric constant a_0. Regularity of the zero set and completeness of the consecutive stack give $a_i^{\text{ph}} = -a_{i-1}^{\text{ph}}$. Define

$$\sigma_i = \frac{a_i^{\text{ph}} - a_{i-1}^{\text{ph}}}{2} = -a_{i-1}^{\text{ph}} \in \{-1, 1\}, \qquad c_M = \rho_- + \sum_{i=1}^{M} \sigma_i.$$

Thus $2\sigma_i$ is the actual jump; the coefficients σ_i alternate and the phase normal on graph i is $\sigma_i N_i$. The saturated stack is the piecewise constant function

$$Q_0(X) = c_M + \sum_i \sigma_i \operatorname{sgn} \rho_i(X).$$

In the cell above precisely the first j graphs it equals $\rho_- + 2\sum_{i\le j}\sigma_i = a_j^{\mathrm{ph}}$. There are no omitted zeros in a prepared region, so this is exactly its actual phase, including in its outer phase cells.

Choose a fixed $w_0 > 0$ and a fixed $\chi_0 \in C_c^\infty((-w_0, w_0))$ with $\chi_0(0) = 1$. Choose $\eta(x) = \eta_*(x/R_*)$, where $\eta_* \in C_c^\infty(\mathbb{R}^6)$ is fixed, equals one on $\overline{B}_{r_{\mathrm{out}}}$, and has support in a fixed larger scaled base. Thus $\eta = 1$ on the entire prepared base and has support strictly inside the larger actual base. For every fixed integer $j \ge 0$,

$$\|D^j\eta\|_\infty \le C_j R_*^{-j}.$$

On the common genuine-distance domain set

$$\begin{aligned} Q_{\mathrm{raw}} &= c_M + \sum_i \sigma_i q_s(\rho_i), & A_i(x) &= Q_{\mathrm{raw}}(x, h_i(x)), \\ C(X) &= \eta(x)\sum_i A_i(x)\chi_0(t - h_i(x)), & Q &= Q_{\mathrm{raw}} - C. \end{aligned} \tag{4.13}$$

The symbol A_i in this display denotes a scalar pin coefficient. Choose a single smooth ambient cutoff χ supported strictly inside the auxiliary domain, equal to one on the support of C and on $\bigcup_{X\in\Omega_{\mathrm{out}}} \overline{B}_{2cR_*}(X)$. The fixed enlargements permit this choice with $\|D^j\chi\|_\infty \le C_j R_*^{-j}$. Globally use $Q = \chi Q_{\mathrm{raw}} - C$, with each compactly supported term extended by zero. On the stated neighborhoods this is the displayed reference, and $\|Q\|_\infty \le C_M$.

For every x in the larger base, separation of distinct actual branches gives $|h_i(x) - h_j(x)| \ge \delta$ when $i \ne j$. For large indices $\delta > 2w_0$, so the vertical bump supports are disjoint, and in particular $\chi_0(h_i(x) - h_j(x)) = 0$ for $i \ne j$. At each actual zero in Ω_{out} we therefore have $C(x, h_i(x)) = A_i(x)$ and $Q(x, h_i(x)) = 0$.

The saturated value with its ith sign omitted is zero at graph i, since $a_{i-1}^{\mathrm{ph}} + \sigma_i = 0$. Consequently

$$A_i(x) = \sum_{j\ne i} \sigma_j\{q_s(\rho_j(x, h_i(x))) - \mathrm{sgn}(i - j)\}.$$

The pin and the other-sheet foot lie on distinct actual branches, so $|\rho_j(x, h_i(x))| \ge \delta$. Its first two derivatives as a function of x are bounded uniformly: the chain rule uses only $D\rho_j, D^2\rho_j, Dh_i, D^2h_i$, already controlled above. The layer tail and derivative envelopes imply

$$\max_{0\le j\le 2} \|D^j A_i\|_\infty \le C_M F.$$

Differentiating the displayed formula for C now gives $\|C\|_{C^{1,1}(\mathbb{R}^7)} \le C_M F$: derivatives of η are uniformly bounded since $R_* \to \infty$; derivatives through order two of $t - h_i(x)$ are uniformly bounded; and the bump has fixed derivative bounds. Compact support inside the genuine domain justifies the zero extension. Finally the symmetric-difference formula, split at radius one, gives

$$\|(-\Delta)^s C\|_\infty \le C c_{n,s}\left\{\frac{\|D^2 C\|_\infty}{2 - p} + \frac{\|C\|_\infty}{p}\right\} \le C_M F. \tag{4.14}$$

This proves the claimed global correction bounds using only graph derivatives of order at most two.

If $\kappa\delta \ge a_0$, choose a fixed $b = b(a_0) > 0$ sufficiently small and put $R_{\mathrm{iso}} = b/\kappa$. The tangent-chart constants and Lemma 2.10 allow b to be chosen so that $B_{64R_{\mathrm{iso}}}$ contains only the marked actual graph and $32R_{\mathrm{iso}}$ is smaller than the supplied lower bound for $\mathrm{reach}(\Sigma)$. Indeed $\delta \ge a_0/\kappa$, while that reach is at least $c\min(1, a_0)/\kappa$; a zero off the marked tangent chart is intrinsically off diagonal from the marked zero and hence at distance at least δ. Choose $64b$ smaller than these fixed chart and separation constants. On the actual graph domain one has $h(0) = Dh(0) = 0$, $|D^2h| \le C\kappa$, $|Dh(x)| \le C\kappa|x|$, and $|h(x)| \le C\kappa|x|^2$.

Use $B_{R_{\mathrm{iso}}}$, $B_{2R_{\mathrm{iso}}}$, and $B_{4R_{\mathrm{iso}}}$ as the evaluation, middle, and outer prepared regions. For $X \in B_{16R_{\mathrm{iso}}}$, the inequality $\mathrm{dist}(X, \Sigma) \le |X|$ places every nearest point in $B_{32R_{\mathrm{iso}}}$. The reach bound gives uniqueness there, and the one-sheet assertion places that foot on the marked actual graph. The upward signed distance ρ is therefore genuine on $B_{16R_{\mathrm{iso}}}$; the foot calculation above gives $|D\rho| = 1$ and $|D^2\rho| \le C\kappa$ there. In particular each point of $B_{4R_{\mathrm{iso}}}$ has a radius-$2R_{\mathrm{iso}}$

genuine neighborhood, containing all radius-R_{iso} normal-axis and tangential Taylor segments with actual feet in the larger chart.

Let $\sigma \in \{-1, 1\}$ be the sign of the upward normal derivative of u on this graph. It is constant along the graph, and σN is its phase normal. Take a smooth χ_{iso} equal to one on $B_{8R_{\mathrm{iso}}}$ and supported in $B_{12R_{\mathrm{iso}}}$, with $\|D^j \chi_{\mathrm{iso}}\|_\infty \le C_j R_{\mathrm{iso}}^{-j}$, and set

$$Q_{\mathrm{iso}}(X) = \chi_{\mathrm{iso}}(X)\, \sigma q_s(\rho(X)) \quad \text{on } B_{16R_{\mathrm{iso}}}, \qquad Q_{\mathrm{iso}} = 0 \quad \text{outside}.$$

This bounded global reference equals the single phase-oriented profile on every required neighborhood, vanishes on the actual zero set in the outer prepared ball, and has saturated value $\sigma \operatorname{sgn} \rho$, the actual phase of each of its two cells. Its constant term and pin correction are zero. Thus the same geometric construction, including the buffered Taylor domain, is available in this regime with $R_{\mathrm{iso}} \asymp \kappa^{-1}$.

Normal-axis comparison. The centered layer is odd: $-q_s(-t)$ is another centered increasing layer with the same end states and equation, so the previously established uniqueness of the layer gives $q_s(-t) = -q_s(t)$. Let ρ be any one of the genuine signed distances just constructed, with either orientation. Fix an evaluation point X, and let $L \ge 1$ be such that its radius-$2L$ buffered neighborhood is in the genuine distance domain, all the required normal and tangential segments are retained, and $|D^2\rho| \le A\kappa$ there. Here A is fixed. Put

$$\tau = \rho(X), \qquad \nu = D\rho(X), \qquad P_X(Y) = q_s(\tau + \nu \cdot Y).$$

Uniformly in X, τ, L and $p \in [p_*, 2]$, for $0 < R \le L$,

$$\sup_{|Y|<R} |q_s(\rho(X+Y)) - P_X(Y)| \le C\kappa R^2, \qquad \|q_s(\rho(X+\cdot)) - P_X\|_{L^2(B_R)} \le C\kappa R^{n/2+2}, \tag{4.15}$$

and, for $p \in [p_*, 2)$,

$$c_{n,s} \int_{0<|Y|<L} \frac{|q_s(\rho(X+Y)) - P_X(Y)|}{|Y|^{n+p}}\, \mathrm{d}Y \le C\kappa. \tag{4.16}$$

In particular the last estimate holds on any smaller measurable integration region, including $2 < |Y| < r_*$ for $r_* \le L$. The constants depend only on A, n, p_* and the uniform layer bounds, not on the normal offset τ.

Proof of the comparison. Use orthonormal coordinates $Y = (z, t)$ with $\nu = e_t$. Differentiating $|D\rho|^2 = 1$ gives $D^2\rho\, D\rho = 0$. Its gradient trajectories are therefore straight unit normal lines. The retained normal segments consequently have

$$\rho(X + te_t) = \tau + t, \qquad D\rho(X + te_t) = e_t.$$

Taylor's formula along the retained segment from $X + te_t$ to $X + z + te_t$ gives

$$|\rho(X + z + te_t) - \tau - t| \le E(r) := \tfrac{1}{2} A\kappa r^2, \qquad r = |z|.$$

The uniform Lipschitz bound for q_s proves (4.15), including its stated L^2 factor.

For (4.16), first integrate over $0 < r < 1/2$. The same Lipschitz bound, followed by integration of a positive majorant over the whole normal line, gives

$$C\kappa c_{n,s} \int_0^{1/2} r^{n-2} r^2 \int_{\mathbb{R}} (r^2 + t^2)^{-(n+p)/2}\, \mathrm{d}t\, \mathrm{d}r \le C\kappa c_{n,s} \int_0^{1/2} r^{1-p}\, \mathrm{d}r \le C\kappa.$$

Here the one-dimensional integral has a uniformly bounded scale constant, and $c_{n,s}/(2-p)$ is uniformly bounded. This includes the thin axial region with small r and large $|t|$.

For $r \ge 1/2$, monotonicity gives the nonnegative bound

$$|q_s(\rho(X + z + te_t)) - q_s(\tau + t)| \le q_s(\tau + t + E(r)) - q_s(\tau + t - E(r)).$$

Since $\int_{\mathbb{R}} q_s' = 2$, Tonelli gives exactly

$$\int_{\mathbb{R}} \{q_s(\tau + t + E(r)) - q_s(\tau + t - E(r))\}\, \mathrm{d}t = 4E(r).$$

Bounding the kernel by r^{-n-p} thus bounds this part by

$$C\kappa c_{n,s}\int_{1/2}^{L} r^{n-2+2-n-p}\,\mathrm{d}r \le C\kappa c_{n,s}\int_{1/2}^{\infty} r^{-p}\,\mathrm{d}r \le C\kappa,$$

because $p \ge p_* > 1$. Only positive planar majorants were extended to the whole normal line; no value of ρ outside its genuine domain has been used. Both estimates are independent of τ. □

Lemma 4.3 (The full-cell residual uses only second derivatives)**.** *The reference above satisfies throughout* Ω_{out}*,*

$$\|(-\Delta)^s Q - (Q - Q^3)\|_\infty \le C_M(\kappa + F + \lambda R_*^{-p}), \qquad \|u - Q\|_\infty = o(1). \tag{4.17}$$

In an isolation ball the same estimate holds with the auxiliary radius $R_{\mathrm{iso}} \asymp \kappa^{-1}$ *in place of* R_* *and a single profile; the interaction and pin-correction terms then vanish.*

Proof. Write $f(v) = v - v^3$. The integral comparison below is for $s < 1$. At $s = 1$, its conclusion for each distance profile follows directly from

$$-\Delta(q_1(\rho)) - f(q_1(\rho)) = -q_1'(\rho)\Delta\rho = O(\kappa),$$

using $|D\rho| = 1$; the interaction, pinning, and qualitative arguments are the same. Thus it suffices to present the fractional calculation. For the following global linear calculation take $Q_{\mathrm{raw}} = Q + C$ also outside the auxiliary domain. This is a bounded extension of its displayed genuine-distance formula, with a bound depending only on M. At $X \in \Omega_{\mathrm{out}}$ put $L = cR_*$, $\tau_i = \rho_i(X)$ and $\nu_i = D\rho_i(X)$, and define the bounded planar stack

$$P_X(Y) = c_M + \sum_i \sigma_i q_s(\tau_i + \nu_i \cdot Y).$$

One has $P_X(0) = Q_{\mathrm{raw}}(X)$ and $(-\Delta)^s P_X(0) = \sum_i \sigma_i f(q_s(\tau_i))$. On B_L, the normalized absolute comparison (4.16) bounds the difference of these operators by $C_M\kappa$. On its complement, boundedness and $c_{n,s} \asymp \lambda$ give

$$c_{n,s}\int_{|Y|\ge L} \frac{|Q_{\mathrm{raw}}(X+Y) - P_X(Y)|}{|Y|^{n+p}}\,\mathrm{d}Y \le C_M\lambda L^{-p}.$$

The local difference is absolutely integrable, so subtraction of the principal values is justified. Consequently, throughout the prepared region,

$$\left|(-\Delta)^s Q_{\mathrm{raw}}(X) - \sum_i \sigma_i f(q_s(\rho_i(X)))\right| \le C_M(\kappa + \lambda R_*^{-p}).$$

Here is the interaction estimate, with its uniform tail product made explicit. For $a \ge 0$ set

$$T(a) = e^{-ka} + \lambda(1+a)^{-p}, \qquad k = \sqrt{2}.$$

For $a, b \ge 0$, $p \le 2$ and $0 \le \lambda \le 1$,

$$e^{-ka}(1+b)^{-p} \le C_k(1+a+b)^{-p}, \qquad C_k = \sup_{r\ge 0} e^{-kr}(1+r)^2 < \infty,$$

because $(1+a+b)/(1+b) \le 1+a$. The same estimate holds with a, b interchanged. Moreover,

$$(1+a)^{-p}(1+b)^{-p} \le (1+a+b)^{-p}.$$

Expanding the product and using $\lambda^2 \le \lambda$ therefore gives

$$T(a)T(b) \le e^{-k(a+b)} + (2C_k+1)\lambda(1+a+b)^{-p} \le CT(a+b). \tag{4.18}$$

All constants are uniform as $p \uparrow 2$.

Off the zero graphs, write $q_i = q_s(\rho_i)$, $s_i = \operatorname{sgn}\rho_i$, $Q_0 = c_M + \sum_i \sigma_i s_i \in \{-1, 1\}$ and $\alpha_i = \sigma_i(q_i - s_i)$. The saturated value Q_0 is the actual phase of the cell. Expanding the cubic

gives the exact identity

$$\sum_i \sigma_i f(q_i) - f(Q_{\mathrm{raw}}) = 3\sum_i (Q_0 - \sigma_i s_i)\alpha_i^2 + 6Q_0 \sum_{i<j} \alpha_i\alpha_j + 3\sum_{i\neq j} \alpha_i^2\alpha_j + 6\sum_{i<j<\ell} \alpha_i\alpha_j\alpha_\ell.$$

For either bounding sheet of a cell, including the single bounding sheet of an outer cell, $Q_0 = \sigma_i s_i$, so its self-square coefficient is zero. If i is not bounding, the segment to its genuine foot crosses an intervening actual sheet; separation of that crossing from the foot gives $|\rho_i| \geq \delta$. The separation of two genuine feet also gives $|\rho_i| + |\rho_j| \geq \delta$ for $i \neq j$. The layer tail bound gives $|\alpha_i| \leq CT(|\rho_i|)$, while $|\alpha_i| \leq 2$. Since T is decreasing and $T(\delta) \leq CF$, (4.18) bounds each mixed pair by CF. Every mixed cubic monomial is such a pair times a factor bounded by 2, and every remaining self-square has one factor bounded by CF and the other by 2. There are only $O_M(1)$ terms. Thus the interaction is $O_M(F)$. Continuity of the unexpanded cubic expression extends this estimate across the zero graphs.

It remains to account for the nonlinear effect of pinning. The raw reference, C, and $Q = Q_{\mathrm{raw}} - C$ are uniformly bounded by a fixed B_M, after discarding finitely many indices. The mean value theorem, with $\sup_{|v|\leq B_M} |f'(v)| \leq 1 + 3B_M^2$, gives

$$|f(Q_{\mathrm{raw}}) - f(Q_{\mathrm{raw}} - C)| \leq (1 + 3B_M^2)|C| \leq C_M F.$$

Using

$$(-\Delta)^s Q - f(Q) = (-\Delta)^s Q_{\mathrm{raw}} - f(Q_{\mathrm{raw}}) - (-\Delta)^s C + f(Q_{\mathrm{raw}}) - f(Q)$$

and (4.14) proves the residual bound.

For qualitative smallness, restore the sequence index j and fix $A \geq 1$. Set $d_j(X) = \mathrm{dist}(X, \Sigma_j)$ and $\nu_{\mathrm{ph}}(y) = \nabla u_j(y)/|\nabla u_j(y)|$. Uniform moving-center preparation in C^1, together with $q'_{s_j}(0) \geq c > 0$, identifies its approximate layer direction with $\nu_{\mathrm{ph}}(y)$ up to an error tending uniformly to zero: for a preparing direction $e_{j,y}$, $\nabla u_j(y) = q'_{s_j}(0)e_{j,y} + o(1)$ uniformly in y, and normalizing gives $e_{j,y} - \nu_{\mathrm{ph}}(y) = o(1)$. The uniform Lipschitz bound for q_{s_j} therefore implies

$$\varepsilon_j(A) := \sup_{y\in\Sigma_j} \|u_j(y + \cdot) - q_{s_j}(\nu_{\mathrm{ph}}(y)\,\cdot)\|_{L^\infty(B_{2A})} \longrightarrow 0.$$

This uses only the previously established qualitative moving-center preparation, not a quantitative chamber or jet conclusion.

If $X \in \Omega_{\mathrm{out},j}$ and $d_j(X) \leq A$, closedness supplies a nearest zero y. For large j, the retained-zero buffer includes y in the actual family, say on sheet i, and its fixed-radius neighborhood lies in the genuine domain. The upward orientation and actual half-jump satisfy $\nu_{\mathrm{ph}}(y) = \sigma_i D\rho_i(y)$. By the oddness established in the normal-axis comparison, $\sigma_i q_{s_j}(\rho_i) = q_{s_j}(\sigma_i\rho_i)$. Apply (4.15) at y to the phase-oriented distance $\sigma_i\rho_i$. It gives

$$|u_j(X) - \sigma_i q_{s_j}(\rho_i(X))| \leq \varepsilon_j(A) + C_A\kappa_j.$$

The genuine foot on sheet i is y, so $|\rho_i(X)| = d_j(X) \leq A$. The other-foot separation gives $|\rho_\ell(X)| \geq \delta_j - A$ for $\ell \neq i$. The actual saturated phases on the two sides of sheet i give $c_M + \sum_{\ell\neq i} \sigma_\ell \operatorname{sgn}\rho_\ell(X) = 0$ on this neighborhood. Hence the other profiles change $\sigma_i q_{s_j}(\rho_i(X))$ by at most $C_M T_j(\delta_j - A)$, and the pin changes it by at most $C_M F_j$. For fixed A these both tend to zero, uniformly in X. Thus

$$\sup_{\substack{X\in\Omega_{\mathrm{out},j}\\ d_j(X)\leq A}} |u_j(X) - Q_j(X)| \longrightarrow 0 \quad \text{for every fixed } A.$$

On the other hand, if $d_j(X) > A$, every retained genuine distance has $|\rho_i(X)| \geq A$ and the saturated reference equals $\operatorname{sgn} u_j(X)$. Therefore

$$|Q_j(X) - \operatorname{sgn} u_j(X)| \leq C_M T_j(A) + C_M F_j.$$

Define

$$\omega(A) = \limsup_{j\to\infty} \sup_{\{X:\, d_j(X)>A\}} |u_j(X) - \operatorname{sgn} u_j(X)|,$$

with the empty supremum equal to zero. Uniform well clearing gives $\omega(A) \to 0$ as $A \to \infty$. Combining the two regions yields

$$\limsup_{j\to\infty} \|u_j - Q_j\|_{L^\infty(\Omega_{\mathrm{out},j})} \le \omega(A) + C_M\{e^{-kA} + (1+A)^{-p_*}\}.$$

Let $A \to \infty$ after the sequence limit. This proves the required qualitative smallness on the whole prepared region, including its boundary neighborhoods.

Finally, use the already constructed isolated reference on its nested balls, with auxiliary radius $R_{\mathrm{iso}} \asymp \kappa^{-1}$, $Q = \sigma q_s(\rho)$ on the genuine buffered domain and $C = 0$. Apply the same normal-axis comparison with $L = cR_{\mathrm{iso}}$ to the phase-oriented distance $\sigma\rho$. There is only one profile, so there is no cubic interaction, and no pin term. The bounded exterior costs $C\lambda R_{\mathrm{iso}}^{-p}$, giving $|(-\Delta)^s Q - f(Q)| \le C(\kappa + \lambda R_{\mathrm{iso}}^{-p})$. For each fixed A, all nearest zeros at distance at most A from the prepared balls belong to this single actual sheet for large j, by the isolation buffer. The preceding near-zero argument then has no remote profiles or correction, and the far-zero argument uses the actual phases on its two sides. The same two-region limit proves $\|u - Q\|_\infty = o(1)$ there. □

4.1.3. *Linear transmission and the quantitative chamber bound.* Use Lemma 3.4 with $N = 0$. For a bounded error with zero trace on all actual pins in a prepared region Ω, bounded coefficient, and the zero-trace coercivity, the exact localization step from its proof is

$$c\|\eta_i e\|_{H^s(\mathbb{R}^7)}^2 \le C\|f\|_{H^{-s}(\operatorname{supp}\eta_i)}\|\eta_i e\|_{H^s(\mathbb{R}^7)} + \frac{c_{n,s}}{2}\iint |e(X)e(Y)|\frac{(\eta_i(X)-\eta_i(Y))^2}{|X-Y|^{n+p}}\,\mathrm{d}X\,\mathrm{d}Y. \tag{4.19}$$

Here η_i is the fixed enlarged ambient-cell cutoff, and the source norm is the dual norm on global tests supported in its full enlargement. The lattice iteration already proved there keeps the far-error product linear in $c_{n,s}$. If β bounds these local source norms and the cell is at distance D from the complement of the prepared region, it gives

$$\|\eta_i e\|_{H^s(\mathbb{R}^7)} \le C\beta + C\|e\|_\infty(e^{-cD} + \lambda D^{-p}). \tag{4.20}$$

This is the global cutoff norm established by that proof. The distance is to the whole region of equation, pins, and coercivity.

Apply this to $e = u - Q$ and the exact secant equation

$$[(-\Delta)^s + u^2 + uQ + Q^2 - 1]e = -\{(-\Delta)^s Q - Q + Q^3\}. \tag{4.21}$$

The coefficient differs from $3u^2 - 1$ by $O(|e|)$. The spectral antecedent of Lemma 4.2 is precisely the orthogonal layer gap (4.5), supplied by Lemma 4.1; moving-center preparation and clearing are part of the standing hypotheses. Thus that coercivity lemma applies, and the qualitative smallness of e on the prepared region absorbs the secant perturbation. In particular the coefficient and coercivity constants there are uniform after the sequence index is large. The transmission decay rate is independent of the fixed buffer multipliers; the bounded exterior extension affects only the factor $\|e\|_\infty$. For a fixed full cell enlargement $\mathcal{C}_i$ and a global H^s test ϕ supported there,

$$|\langle f, \phi\rangle| \le |\mathcal{C}_i|^{1/2}\|f\|_{L^\infty(\mathcal{C}_i)}\|\phi\|_{L^2} \le C\|f\|_{L^\infty(\mathcal{C}_i)}\|\phi\|_{H^s}.$$

The cell volumes are fixed, so the constant is uniform in location and window size. This verifies the required local H^{-s} source bound with $\beta \le C_M(\kappa + F + \lambda R_*^{-p})$ in the natural regime; in isolation replace R_* by R_{iso} and omit F. On an inner natural prism use fixed normal buffers $B\delta$ and lateral buffers KR_*. Equations (4.17)–(4.20) give

$$\begin{aligned}\|\eta_i e\|_{H^s(\mathbb{R}^7)} &\le C_M(\kappa + F + \lambda R_*^{-p}) \\ &\quad + C\|e\|_\infty\{e^{-cB\delta} + e^{-cKR_*} + \lambda(B\delta)^{-p} + \lambda(KR_*)^{-p}\}.\end{aligned} \tag{4.22}$$

Choose $cB > k$ and then a_0 so $cK/\sqrt{a_0} > k$. The identity

$$\frac{\lambda R_*^{-p}}{\kappa} = \frac{\lambda\delta^{-p}}{\kappa}(\kappa\delta)^{p/2} = o(1) \tag{4.23}$$

shows that all terms apart from $C_M\kappa$ are $o(\kappa)$. In the isolation regime use $D = c/\kappa$; the exterior terms are $e^{-c'/\kappa} + \lambda\kappa^p = o(\kappa)$.

Lemma 4.4 (Normalized interior estimates)**.** *Let $v \in H^s_{\mathrm{loc}}(B_2) \cap L^2(B_2)$ be measurable on $\mathbb{R}^n$ and suppose its normalized tail*

$$T_s(v) = c_{n,s} \int_{\mathbb{R}^n \setminus B_2} \frac{|v(Y)|}{|Y|^{n+p}}\, \mathrm{d}Y$$

is finite. Suppose $c, h \in L^\infty(B_2)$, $\|c\|_\infty \le C$, and $[(-\Delta)^s + c]v = h$ weakly on B_2. Uniformly for s near one,

$$\|v\|_{L^\infty(B_1)} \le C' \left(\|v\|_{L^2(B_2)} + \|h\|_{L^\infty(B_2)} + c_{n,s} \int_{\mathbb{R}^n \setminus B_2} \frac{|v(Y)|}{|Y|^{n+p}}\, \mathrm{d}Y \right).$$

For the homogeneous equation, each fixed j and $0 < r < 2$ also satisfy

$$\|v\|_{C^j(B_r)} \le C_{j,r}\{\|v\|_{L^2(B_2)} + T_s(v)\},$$

provided the finitely many coefficient derivatives used below are bounded. The constants may depend on these derivative bounds, j, r, and the fixed near-one interval, but are uniform in s.

Proof. Put $L_s = (-\Delta)^s$ and, for $s < 1$, write

$$\mathcal{B}_s(a,b) = \frac{c_{n,s}}{2} \iint \frac{(a(X) - a(Y))(b(X) - b(Y))}{|X - Y|^{n+p}}\, dX\, dY, \qquad \mathcal{E}_s(a) = \mathcal{B}_s(a,a).$$

Only the nonvacuous case $\|v\|_{L^2(B_2)} + \|h\|_{L^\infty(B_2)} < \infty$ needs proof. All tests below have compact support in B_2. For such a support, choose a larger relatively compact open subset of B_2. On pairs inside this larger set the bilinear form is continuous by local H^s regularity and Cauchy–Schwarz. Pairs with the other endpoint outside it are separated from the test support; their absolute integrals are controlled by local L^2 and the displayed weighted L^1 tail. Thus the weak equation extends from smooth tests to compactly supported H^s tests, and distributional subsolution inequalities extend to nonnegative such tests. For the latter assertion, nonnegative mollifications of the zero-extended test preserve positivity, converge in H^s, and have support in one fixed compact subset of B_2. No global H^s norm of v is used in this extension.

Here is the weak Kato step in this class. Set $j_\varepsilon(a) = \sqrt{a^2 + \varepsilon^2} - \varepsilon$. For $\phi \in C_c^\infty(B_2)$, $\phi \ge 0$, the test $j'_\varepsilon(v)\phi$ is admissible: j'_ε is bounded and Lipschitz for each $\varepsilon > 0$. Convexity gives, for all $a, b \in \mathbb{R}$ and $A, B \ge 0$,

$$(j_\varepsilon(a) - j_\varepsilon(b))(A - B) \le (a - b)\{Aj'_\varepsilon(a) - Bj'_\varepsilon(b)\}.$$

Indeed the difference of the right side and left side is

$$A\{(a-b)j'_\varepsilon(a) - j_\varepsilon(a) + j_\varepsilon(b)\} + B\{j_\varepsilon(a) - j_\varepsilon(b) - (a-b)j'_\varepsilon(b)\},$$

whose two braces are nonnegative. Integrating with $A = \phi(X)$, $B = \phi(Y)$ and using the weak equation yields

$$\mathcal{B}_s(j_\varepsilon(v), \phi) \le \int_{B_2} (h - cv) j'_\varepsilon(v)\phi \le \int_{B_2} (|h| + \|c\|_\infty |v|)\phi.$$

The left side tends to $\mathcal{B}_s(|v|, \phi)$. On local pairs this follows by dominated convergence using $|j_\varepsilon(a) - j_\varepsilon(b)| \le |a - b|$ and the local Cauchy–Schwarz majorant; on separated pairs use $0 \le j_\varepsilon(v) \le |v|$ and the weighted L^1 tail. Consequently $z = |v|$ satisfies

$$\mathcal{B}_s(z, \phi) \le \int_{B_2} (\|c\|_\infty z + \|h\|_\infty)\phi \qquad (0 \le \phi \in H^s,\ \operatorname{supp}\phi \Subset B_2).$$

Let

$$T = c_{n,s} \int_{\mathbb{R}^n \setminus B_2} \frac{|v(Y)|}{|Y|^{n+p}}\, dY, \quad K = \|v\|_{L^2(B_2)} + \|h\|_\infty + T + \epsilon, \quad w = z + K,$$

where $\epsilon > 0$ will finally decrease to zero. Since constants have zero bilinear form, $w \geq K > 0$ is a weak subsolution with right side at most $(1 + \|c\|_\infty)w$. For $q \geq 2$ and $M > 0$ define the linearly truncated power and its energy variable by

$$b_M(a) = a \min(a, M)^{q-2}, \qquad W_M(a) = a \min(a, M)^{(q-2)/2} \quad (a \geq 0).$$

Both are Lipschitz for fixed q, M, and $W_M(a)^2 = a b_M(a) \leq a^q$. Thus $\phi = \eta^2 b_M(w)$ is an admissible nonnegative compactly supported H^s test whenever $\eta \in C_c^\infty(B_2)$. This truncation is linear, rather than bounded, above M; its bounded slope and local L^2 integrability are precisely what is needed for admissibility.

We record the cutoff power inequality, including its dependence on q. Put $c_q = 4(q-1)/q^2$. Almost everywhere on $(0, \infty)$, $|W_M'|^2 \leq c_q^{-1} b_M'$, so Cauchy–Schwarz on the interval between a and b gives

$$E := (a-b)(b_M(a) - b_M(b)) \geq c_q |W_M(a) - W_M(b)|^2.$$

Moreover, since $b_M(a)/a$ is nondecreasing,

$$|a-b|\{b_M(a) + b_M(b)\} \leq 2\sqrt{E}\sqrt{W_M(a)^2 + W_M(b)^2}.$$

For example, if $a > b$, then $b_M(a) - b_M(b) \geq (a-b) b_M(a)/a$ and $b_M(a) + b_M(b) \leq 2 b_M(a)$, which prove this estimate; the other cases follow by symmetry or a limit. Writing $S = W_M(a)^2 + W_M(b)^2$ and using Young's inequality gives

$$\begin{aligned}
&(a-b)\{A^2 b_M(a) - B^2 b_M(b)\} \\
&\quad \geq \tfrac{1}{4}(A^2 + B^2)E - 2(A-B)^2 S \\
&\quad \geq \frac{1}{4q}|A W_M(a) - B W_M(b)|^2 - 3(A-B)^2 S \qquad (A, B \geq 0).
\end{aligned}$$

For the last step use $2/q \leq c_q \leq 1$ and $|A W_M(a) - B W_M(b)|^2 \leq 2(A^2 + B^2)|W_M(a) - W_M(b)|^2 + 2(A-B)^2 S$.

Now let $1 \leq r < R \leq 2$, $d = R - r$, and choose $0 \leq \eta \leq 1$ equal to one on B_r, supported in $B_{(R+r)/2}$, with $|D\eta| \leq C/d$. Write $U = W_M(w)$ and $F = \eta U$. For $X \in \operatorname{supp} \eta$ the exterior quantity

$$J_R(X) = c_{n,s} \int_{B_R^c} \frac{w(Y)}{|X - Y|^{n+p}}\, dY$$

is bounded by $C d^{-N} K$, with one fixed $N = N(n)$, uniformly near one. In fact, the constant part K costs CKd^{-p}; the part of z in $B_2 \setminus B_R$ costs $C d^{-n-p} \|v\|_{L^2(B_2)}$ by separation and Cauchy–Schwarz. For $2 \leq |Y| \leq 4$ use $|X - Y| \geq d/2$ and $|Y| \leq 4$, and for $|Y| > 4$ use $|X - Y| \geq |Y|/2$, to bound the remaining part by $C d^{-N} T$. This retains the factor $c_{n,s}$ in T throughout.

Decompose the tested form exactly as

$$\mathcal{B}_s(w, \eta^2 b_M(w)) = I_R + c_{n,s} \int_{B_R} \eta^2 b_M(w(X)) \int_{B_R^c} \frac{w(X) - w(Y)}{|X - Y|^{n+p}}\, dY\, dX,$$

where I_R is the bilinear integral over $B_R \times B_R$, including its factor $c_{n,s}/2$. The second term is at least $-C d^{-N} K \int \eta^2 b_M(w) \geq -C d^{-N} \int_{B_R} U^2$, since $K \leq w$. The subsolution inequality therefore gives

$$I_R \leq C d^{-N} \int_{B_R} U^2.$$

Integrating the cutoff power inequality over $B_R \times B_R$ gives

$$\frac{1}{4q} \mathcal{E}_s^{B_R \times B_R}(F) \leq I_R + C d^{-p} \int_{B_R} U^2.$$

Here the cutoff error is uniform because

$$c_{n,s} \int_{\mathbb{R}^n} \frac{\min(1, |H|^2/d^2)}{|H|^{n+p}}\, dH = |\mathbb{S}^{n-1}| c_{n,s} \left(\frac{1}{2-p} + \frac{1}{p} \right) d^{-p} \leq C d^{-p}.$$

Finally, the zero-extended F has the exact full energy

$$\mathcal{E}_s(F) = \mathcal{E}_s^{B_R\times B_R}(F) + c_{n,s}\int_{B_R} F(X)^2 \int_{B_R^c} |X-Y|^{-n-p}\,dY\,dX.$$

The last term is at most $Cd^{-p}\int_{B_R} U^2$, since the support of F is separated from B_R^c by $d/2$. We have proved the uniform global cutoff estimate

$$\|\eta W_M(w)\|^2_{H^s(\mathbb{R}^n)} \le Cqd^{-N}\int_{B_R} W_M(w)^2 \le Cqd^{-N}\int_{B_R} w^q.$$

This estimate requires $w \in L^q(B_R)$ only on its right side; the truncated tests themselves were admissible using just the original local H^s and L^2 hypotheses.

Fix $0 < t < \min(s_b, n/2)$ and put $\chi = n/(n-2t) > 1$. The fixed H^t Sobolev inequality and $\|F\|_{H^t} \le C\|F\|_{H^s}$, followed by monotone convergence $M \to \infty$, give

$$\|w\|_{L^{q\chi}(B_r)} \le [Cqd^{-N}]^{1/q}\|w\|_{L^q(B_R)}.$$

Start with $q_0 = 2$, take $q_j = 2\chi^j$ and $R_j = 1 + 2^{-j}$, and apply this estimate with $R = R_j$, $r = R_{j+1}$. The logarithm of the product of iteration constants is bounded by a constant times $\sum_j (1 + j + \log q_j)/q_j < \infty$. Thus

$$\|v\|_{L^\infty(B_1)} \le \|w\|_{L^\infty(B_1)} \le C\{\|v\|_{L^2(B_2)} + \|h\|_\infty + T + \epsilon\}.$$

Let $\epsilon \downarrow 0$.

If $s = 1$ is included, omit the exterior tail and use $\mathcal{B}_1(a,b) = \int \nabla a \cdot \nabla b$. The same convex test proves Kato by the nonnegative term $j_\varepsilon''(v)|\nabla v|^2\phi$. For the truncated power test, $b_M^2/b_M' \le W_M^2$ and $|W_M'|^2 \le c_q^{-1} b_M'$ give directly

$$\nabla w \cdot \nabla(\eta^2 b_M(w)) \ge \frac{c_q}{4}|\nabla(\eta W_M(w))|^2 - 3|\nabla\eta|^2 W_M(w)^2.$$

This yields the same energy bound with d^{-2} and the same iteration.

For higher derivatives put $L_s = |D|^p$ and $A = \|v\|_{L^2(B_2)} + T_s(v)$. Choose smooth cutoffs χ, θ compactly supported in B_2, with $\theta = 1$ near $\operatorname{supp}\chi$, and set $f = \theta v$. The exact localized identity is

$$(1 + L_s)(\chi v) = \chi(1-c)f + [L_s, \chi]f + F, \qquad F = -\chi L_s((1-\theta)v).$$

The input defining F is separated from $\operatorname{supp}\chi$. Differentiating its kernel gives $\|F\|_{C^a} \le C_a A$ for each fixed a. The local annulus uses the $L^2(B_2)$ norm; the exterior uses exactly $T_s(v)$, retaining $c_{n,s}$.

For $p \in [p_*, 2]$ one has uniformly

$$\big||\xi|^p - |\zeta|^p\big| \le C|\xi - \zeta|\{|\zeta|^{p-1} + |\xi - \zeta|^{p-1}\}.$$

The Fourier convolution formula for the commutator, Peetre's weight inequality, and the rapid decay of $\widehat{\chi}$ therefore give, for every fixed $r \ge 0$,

$$\|[L_s, \chi]f\|_{H^{r-p+1}} \le C_{r,\chi}\|f\|_{H^r}.$$

Indeed transfer the weight $\langle\xi\rangle^{r-p+1}$ to $\langle\zeta\rangle^r$; all remaining powers of $\xi - \zeta$ are absorbed by an integrable Schwartz seminorm of $\widehat{\chi}$, and apply Young's convolution inequality. This argument does not differentiate $|\xi|^p$ at the origin. The multiplier $(1 + |\xi|^p)^{-1}$ maps H^a to H^{a+p} uniformly. Since $p \ge p_* > 1$, the localized identity yields

$$\|\chi v\|_{H^{r+1}} \le C_r\{\|\theta v\|_{H^r} + A\},$$

using the fixed coefficient derivative bounds for multiplication. Start at $r = 0$ and iterate over finitely many nested cutoffs until the Sobolev order exceeds $j + n/2$. Sobolev embedding proves the claimed C^j estimate. No derivative or unnormalized norm of the exterior data is required. □

Use the three nested prepared regions fixed above. Every cell in Ω_{mid} has the transmission margins to Ω_{out} and hence local L^2 norm $O_M(\kappa)$. Here cells meeting the middle-region boundary are also allowed: their fixed enlargements still lie in the outer region, and subtracting a fixed cell diameter from the growing buffers changes only constants. Fix an evaluation point X. Cover Ω_{mid} by the fixed lattice cells $\mathcal{C}_\ell$, with centers a_ℓ, and write $A_\ell = \mathcal{C}_\ell \cap \Omega_{\mathrm{mid}} \cap \{|Y - X| > 2\}$. The uniform cell diameter implies

$$\sup_{Y\in A_\ell} |X-Y|^{-n-p} \le C(1+|a_\ell - X|)^{-n-p}, \qquad \sum_\ell (1+|a_\ell - X|)^{-n-p} \le C.$$

The first assertion uses $|X-Y| > 2$ for the finitely many nearby cells and $|X-Y| \ge |a_\ell - X|/2$ for the distant cells. For the second, the number of centers in the annulus of radius 2^j is at most $C2^{jn}$, giving a geometric sum bounded by $C\sum_{j\ge 0} 2^{-jp_*}$, uniformly in X and window size. Cellwise Cauchy–Schwarz now gives

$$c_{n,s}\sum_\ell \int_{A_\ell} \frac{|e(Y)|}{|X-Y|^{n+p}}\,\mathrm{d}Y \le Cc_{n,s}\sum_\ell (1+|a_\ell - X|)^{-n-p}\|e\|_{L^2(\mathcal{C}_\ell)} \le C_M\kappa.$$

Thus the controlled region contributes $O_M(\kappa)$ to the normalized tail, with no factor depending on the number of cells. The complement of Ω_{mid} is separated from Ω_{eval} by normal and lateral buffers $B\delta$ and KR_*. Global boundedness charges it by $C_M\lambda\{(B\delta)^{-p} + (KR_*)^{-p}\} = o(\kappa)$. The analogous nested isolation balls give the same estimate there. Lemma 4.4 therefore gives

$$|u - Q| \le C_M\kappa \tag{4.24}$$

on every prescribed smaller natural prism or isolation ball, including its long chambers.

4.1.4. *Relative physical jets and full genuine-distance jets.* For $a \ge 0$ define, with the sequence parameters s, p, λ,

$$T(a) = T_s(a) := e^{-ka} + \lambda(1+a)^{-p}, \qquad k = \sqrt{2}. \tag{4.25}$$

The layer theorem gives $1 - |q_s(t)| + |q_s'(t)| + |q_s''(t)| \le CT(|t|)$. At a marked zero, translated to the origin, choose $t = \nu_{\mathrm{ph}} \cdot Y$, where $\nu_{\mathrm{ph}} = \nabla u(0)/|\nabla u(0)|$, and put $r_* = c\min(\delta, \kappa^{-1})$ with a fixed sufficiently small c. Use the genuine distance to the own sheet with this orientation; denote it by ρ_0, so that $\rho_0(0) = 0$ and $\nabla\rho_0(0) = \nu_{\mathrm{ph}}$. If the reference uses the upward distance ρ_i and coefficient $\sigma_i \in \{-1, 1\}$, this means $\rho_0 = \sigma_i\rho_i$. By the oddness recalled in the normal-axis comparison, the own profile is $q_s(\rho_0)$ in either convention. On a larger controlled window write $u = q_s(\rho_0) + e +$ remote tails $- C$. On every fixed microscopic ball the last two terms are $O_R(F) = o(\kappa)$ by the layer bound and separation. Shrink c once so that B_{2r_*} and its required segments remain in the own sheet's genuine distance domain, with actual feet in the larger patch. The natural-reference lemma supplies $|D^2\rho_0| \le C\kappa$ there. Apply the comparison (4.15)–(4.16) at this zero with $L = r_*$, $\tau = 0$, and $\nu = \nu_{\mathrm{ph}}$. Its explicit conclusions give

$$\|q_s(\rho_0) - q_s(t)\|_{L^2(B_2)} \le C\kappa, \qquad c_{n,s}\int_{2<|Y|<r_*} \frac{|q_s(\rho_0(Y)) - q_s(t)|}{|Y|^{n+p}}\,\mathrm{d}Y \le C\kappa.$$

For every other fixed ball B_R, its local conclusion is $\|q_s(\rho_0) - q_s(t)\|_{L^2(B_R)} \le C_R\kappa$. At $s = 1$ use its local comparison clause and omit all nonlocal tail terms, as in the local interior estimate.

The controlled e tail is $O(\kappa)$ by cell summation. The remaining remote profiles on this isolation region have size $CT(c'\delta)$; their normalized contribution is $O(\lambda T(c'\delta)) = O(\lambda\delta^{-p}) = o(\kappa)$, since $e^{-kc'\delta} \le C\delta^{-p}$ and $\lambda \le 1$. The exterior past r_* costs $C\lambda(\delta^{-p} + \kappa^p) = o(\kappa)$. The difference $v = u - q_s(t)$ satisfies the homogeneous equation

$$(-\Delta)^s v = [1 - u^2 - uq_s(t) - q_s(t)^2]v.$$

Its coefficients have the absolute jets (4.9). The same comparison works on every fixed B_{2R+2} about the zero: chamber values and the distance-versus-tangent-plane bound give an $O_R(\kappa)$ local L^2 error, and the normalized tail calculation changes only by a fixed R-dependent constant. Apply Lemma 4.4 to this homogeneous equation. Uniformly at every moving zero and for every fixed R, k,

$$\|u - q_s(t)\|_{C^k(B_R)} \le C_{R,k}\kappa, \qquad |D^j h| \le C_j\kappa\ (j \ge 2), \qquad |\nabla_\Sigma^j A| \le C_j\kappa. \tag{4.26}$$

The last conclusions follow by implicit differentiation using $w_s(0) \geq c$. This order of proof never differentiates distance three times to obtain the initial chamber estimate.

We make the passage from these graph jets to ambient distance jets explicit. At each possible actual foot choose a fixed orthonormal ambient frame and a fixed-radius graph chart $Y(a) = (a, h(a))$, shrinking its radius once so that $|Dh| \leq 1$. The frame is kept fixed when differentiating within this chart. The preceding estimates, uniformly at moving zeros, give

$$|D_a^\ell h| \leq C_\ell \kappa \quad (\ell \geq 2).$$

Write $E = D_a Y$, $g = E^T E$, and let $N(a)$ be the unit normal defining the signed distance. With its fixed choice of sign,

$$N(a) = \pm \frac{(-Dh(a), 1)}{\sqrt{1 + |Dh(a)|^2}}, \qquad S(Y(a)) = -(D_a N) g^{-1} E^T. \tag{*}$$

The last formula extends the shape operator by zero on the normal line: it satisfies $SE = -D_a N$ and $SN = 0$. All denominators in these expressions are uniformly bounded away from zero. Each positive-order derivative of N contains a derivative of h of order at least two, and each derivative of S contains such a factor, including its order-zero expression. Leibniz and chain rules in this fixed chart therefore give

$$|D_a^\ell N| \leq C_\ell \kappa \quad (\ell \geq 1), \qquad |D_a^\ell (S \circ Y)| \leq C_\ell \kappa \quad (\ell \geq 0). \tag{4.27}$$

These are derivatives of ambient vector and matrix components, not a tacit identification with covariant derivatives. For comparison with the curvature notation, the graph metric has $\Gamma^c_{ab} = g^{cd} h_d h_{ab}$ and the second fundamental form has components $\pm h_{ab}/\sqrt{1 + |Dh|^2}$. Their differentiated coordinate formulas and the formula for covariant differentiation express $\nabla^j_\Sigma A$ as finite sums of these graph derivatives. Every such sum retains a curvature factor and is $O_j(\kappa)$. In particular the graph bounds already proved supply precisely the ambient-component bounds needed below.

Let $X = y + r\nu(y)$ belong to a genuine buffered tube, with $r = \rho(X)$, actual foot $y = y(X)$ in the larger patch, and $\kappa|r| \leq c < 1$. Here ν is the normal of this signed distance; for the upward distance it is the geometric normal, whereas the phase normal differs by the fixed sheet sign. For clarity put

$$\begin{aligned} \mathcal{N}(X) &= \nu(y(X)), & \mathcal{S}(X) &= S(y(X)), \\ P(X) &= I - \mathcal{N}(X) \otimes \mathcal{N}(X), & M(X) &= (I - r(X)\mathcal{S}(X))^{-1}. \end{aligned}$$

Thus $M = M_r$ and $P = P_T$ in the statement. Since $\|S\| \leq |A| \leq \kappa$, one has $\|M\| \leq (1 - c)^{-1}$. Differentiating the normal-coordinate identity gives $\mathrm{d}X = (I - rS)\,\mathrm{d}y + \nu\,\mathrm{d}r$, with $\mathrm{d}y$ tangent. Its normal and tangential parts yield

$$D_X y = MP, \qquad Dr = \mathcal{N}, \qquad D^2 r = -\mathcal{S}M. \tag{4.28}$$

Indeed $D_X \mathcal{N} = -\mathcal{S} D_X y = -\mathcal{S}MP = -\mathcal{S}M$, since $\mathcal{S}\mathcal{N} = 0$ and $M\mathcal{N} = \mathcal{N}$. Matrix differentiation, without any commutation assumption on $D\mathcal{S}$, gives

$$DM = M\big(\mathcal{S}\,Dr + r\,D\mathcal{S}\big)M, \qquad D(\mathcal{S}M) = M(D\mathcal{S})M + (\mathcal{S}M)^2 Dr.$$

Here the identity $I + r\mathcal{S}M = M$ combines the two $D\mathcal{S}$ terms. The Gauss formula applied to the ambient matrix S, extended by zero on ν, gives $|D_\tau S| \leq C(|\nabla_\Sigma A| + |A|^2)|\tau|$ for tangent τ: its tangential part is the covariant derivative, and differentiating the tangent frame or the equation $S\nu = 0$ produces two shape-operator factors. Together with $|Dy| \leq C_c$ this proves

$$|D^2 \rho| \leq C_c \kappa, \qquad |D^3 \rho| \leq C_c(|\nabla_\Sigma A| + |A|^2) \leq C_c \kappa. \tag{4.29}$$

Here is a closed induction for every higher order; in particular it tracks the derivatives of r, ν, and P_T used when differentiating the inverse. For $j \geq 1$ take as induction hypothesis

$$|D^\ell y| \leq C_{c,j} \ (1 \leq \ell \leq j), \qquad |Dr| = 1, \qquad |D^\ell r| \leq C_{c,j}\kappa \ (2 \leq \ell \leq j). \tag{I_j}$$

The last range is empty for $j = 1$, and (I_1) follows from the first two identities above. In the fixed foot chart the coordinate map $a(X)$ is the linear coordinate projection of $y(X)$. Thus its

derivatives through order j are bounded by (I_j). Applying the ordinary higher chain rule to (4.27), and then the product rule for P, gives

$$|D^\ell \mathcal{N}| + |D^\ell P| \le C_{c,j}\kappa \quad (1 \le \ell \le j), \qquad |D^\ell \mathcal{S}| \le C_{c,j}\kappa \quad (0 \le \ell \le j). \tag{II_j}$$

Set $B = I - r\mathcal{S}$. For a multi-index β with $1 \le |\beta| \le j$, its differentiated entries are

$$\partial^\beta B = -r\,\partial^\beta \mathcal{S} - \sum_{0<\gamma\le\beta} \binom{\beta}{\gamma} (\partial^\gamma r)(\partial^{\beta-\gamma}\mathcal{S}).$$

The first term is bounded because $|r|\kappa \le c$; every other term is bounded by (I_j) and (II_j) (in fact it is $O_{c,j}(\kappa)$). Hence $|D^\ell B| \le C_{c,j}$ for $1 \le \ell \le j$. Differentiating $BM = I$ gives the explicit recursion

$$\partial^\alpha M = -M \sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} (\partial^\beta B)(\partial^{\alpha-\beta} M).$$

Induction on $|\alpha|$, starting from $|M| \le (1-c)^{-1}$, therefore bounds all derivatives of M through order j by $C_{c,j}$. Finally, differentiating $Dy = MP$ exactly j times gives $|D^{j+1}y| \le C_{c,j+1}$, while differentiating $Dr = \mathcal{N}$ exactly j times gives $|D^{j+1}r| \le C_{c,j+1}\kappa$. This proves (I_{j+1}) and closes the induction. Equivalently, differentiating $D^2\rho = -\mathcal{S}M$ now retains an $O(\kappa)$ factor in every product. All fixed physical derivatives $D^j\rho$, $j \ge 2$, are consequently $O_{c,j}(\kappa)$.

All these computations take place in a neighborhood whose actual feet stay in the larger genuine patch. At each point the chart may be recentered at its foot, with the same constants; neither the absolute location of y nor a bound for r independent of κ^{-1} is used. The prescribed condition $\kappa|r| \le c < 1$ and the genuine-foot condition are retained. A formal graph extension does not supply an additional distance domain.

4.2. **Smooth cores and positive interaction rows.** Choose the fixed constants in the following order. First choose a preliminary depth P_0 so that the free comparison mass is in a neighborhood of 2 admitting a uniform exponential rate strictly greater than $k/2$. Reserve a smaller rate, still greater than $k/2$, for the polynomial losses in radial and annular integrations. Choose the logarithmic cutoff K_0 and its normal buffer multiplier K_1 using this reserved rate. Finally increase the fixed depth to $P \ge P_0$ so that the one-dimensional insertion norm below is less than $1/2$ for the resulting finite sheet bound $M(K_0, K_1)$. Increasing P only improves the comparison mass. All these choices precede the sequence limit. Fix an even cutoff $\psi \in C_c^\infty((-2P, 2P))$, with $0 \le \psi \le 1$ and equal to one on $[-P, P]$. Keep P fixed while the sequence tends to its limit. Complete small-slope curvature charts are used with fixed buffer radius $H = \rho/\kappa$ and strictly smaller evaluation regions. The graph construction above also gives such charts: use scale $d_* = \rho/\kappa$, lateral radius $\sqrt{\rho}/\kappa$, and a fixed height multiplier. Taking ρ small makes all slopes $O(\sqrt{\rho})$. All returning branches are retained. The number of branches need not be bounded on a curvature chart. The fixed-depth version of (4.26), with $R > 2P+2$, covers the full depth-$2P$ cores. By Lemma 2.7, $q_s' \to q_1'$ locally uniformly, and $\min_{|t|\le 2P+2} q_1'(t) > 0$. Thus q_s' has a uniform positive minimum on this fixed interval. The relative C^1 estimate then gives $g \ge c_P$ throughout the cores. In the phase tangent frame, $u - q_s(t) = O_P(\kappa)$ in C^2; the planar rank-one Hessian cancels in differentiation of the normalized gradient. Thus $|D(\nabla u/g)| \le C_P\kappa$ throughout the cores.

For an upward graph $Y_i(x) = (x, h_i(x))$, set

$$\nu_i(x) = \frac{(-Dh_i(x), 1)}{\sqrt{1 + |Dh_i(x)|^2}}, \qquad X_i(x, r) = Y_i(x) + r\nu_i(x).$$

These are the genuine normal fibers in the buffered fixed-depth tubes, and $\rho_i(X_i(x, r)) = r$. Write $J_i(x, r) = |\det DX_i(x, r)|$, so physical measure is $J_i(x, r)\,\mathrm{d}x\,\mathrm{d}r$. For a physical field, coordinate notation such as $N_i(x, r)$ means $N_i(X_i(x, r))$. On the zero sheet, $\nu_i = \sigma_i\nabla u/|\nabla u|$. Let $\mathcal{C}_{\mathrm{eval}}$ be the union of the receiving fixed-depth cores on which the row estimates will be evaluated. Fix $b_0 > 0$ and choose the window multipliers so that its closed $2b_0H$ ambient neighborhood lies in the supplied smaller window, with a further $3P$ margin. This is possible with b_0 fixed and small, since $H = \rho/\kappa \to \infty$ and all window enlargements are fixed before the sequence limit. In

the following construction the *evaluation buffer* is the closed b_0H neighborhood of $\mathcal{C}_{\mathrm{eval}}$. Choose a smooth core-preserving window η which is one on the evaluation buffer and constant along each retained depth-$2P$ fiber. Here is a construction with uniform derivatives. The fixed ambient margins allow a cutoff $\alpha_\kappa \in C_c^\infty$, $0 \le \alpha_\kappa \le 1$, equal to one on a neighborhood of the closed evaluation buffer and all its points at distance at most $3P$, supported strictly inside the supplied smaller window with a further $3P$ margin, and satisfying

$$\|D^j \alpha_\kappa\|_\infty \le C_j \kappa^j \qquad (j \ge 1).$$

For example, smooth a slightly enlarged evaluation buffer at a fixed small multiple of H; the reserved b_0H margins contain both the smoothing region and its support. Equivalently, one may write $\alpha_\kappa(X) = \alpha(\kappa(X - X_c))$, where the curvature-coordinate cutoffs α have uniformly bounded derivatives; their shapes may depend on the buffered window. Set

$$\eta_0 = \alpha_\kappa^2, \qquad a_i(x) = \alpha_\kappa(Y_i(x)).$$

Then a_i has compact support in the larger genuine graph base, its depth-$3P$ sweep lies inside the supplied smaller window, and $a_i = 1$ at every foot of an evaluation-buffer point of depth at most $3P$. The fixed lateral buffers give a collar where both $a_i(x) = 0$ and $\eta_0(X_i(x,r)) = 0$ for $|r| \le 3P$. Take $0 \le \theta \le 1$, with $\theta = 1$ on $[-2P, 2P]$ and $\operatorname{supp}\theta \subset (-3P, 3P)$, and define

$$\eta(X_i(x,r)) = (1 - \theta(r))\eta_0(X_i(x,r)) + \theta(r)a_i(x)^2$$

on the retained tubes, with $\eta = \eta_0$ elsewhere. For large indices these fixed-depth tubes are disjoint by Lemma 2.10. The lateral collar and the normal cutoff therefore give smooth gluing and $0 \le \eta \le 1$. In a tube let $\pi_i(X)$ denote the genuine foot. Since $|\pi_i(X) - X| \le 3P$, the distance and foot jets already proved above give, for every fixed j,

$$\|\eta_0 \circ \pi_i - \eta_0\|_{C^j(\{|\rho_i| \le 3P\})} \le C_{P,j}\kappa.$$

For order zero use the displacement bound and $\|D\eta_0\|_\infty = O(\kappa)$; for every positive order the chain rule contains a derivative of η_0, while all fixed foot jets are bounded. Applying this to $\eta = \eta_0 + \theta(\rho_i)(\eta_0 \circ \pi_i - \eta_0)$ yields

$$\|D^j\eta\|_\infty + \sup_i \|D_x^j a_i\|_\infty \le C_{P,j}\kappa \qquad (j \ge 1).$$

Disjointness of the tubes makes these constants independent of the number of retained branches. In particular the cutoffs used below have uniform physical derivatives through every fixed order.

At an evaluation-buffer point in a tube both η_0 and a_i are one, so the interpolation equals one; outside the tubes it is $\eta_0 = 1$. The prescribed foot supports and normal margins keep the support inside the supplied smaller window. The feet and all transition regions lie inside the larger chart. In particular the constructed plateau satisfies

$$\operatorname{dist}\big(\mathcal{C}_{\mathrm{eval}}, \operatorname{supp}(1 - \eta)\big) \ge b_0H.$$

Thus the later unsigned exterior estimate at radius b_0H applies to $W_{\mathrm{ext}} = (1 - \eta)W$ uniformly on every receiving evaluation core. This spatial separation concerns the entire ambient plateau, including the phase regions between its fibers. Retain every actual zero whose depth-$3P$ tube meets the window support. Thus if $X \in \operatorname{supp}\eta$ has distance less than P from the entire zero set, its unique nearest zero belongs to a retained branch and one core cutoff equals one. Set

$$W = 3(1-u^2), \qquad W_i = \eta\psi(\rho_i)^2 W, \qquad V = \eta\Big(1 - \sum_i \psi(\rho_i)^2\Big)W, \qquad W_{\mathrm{ext}} = (1-\eta)W. \tag{4.30}$$

These are globally smooth nonnegative functions after zero extension of the tube cutoffs, and

$$W = V + \sum_i W_i + W_{\mathrm{ext}}, \qquad 0 \le V \le \delta_P < 2, \qquad G_V = ((-\Delta)^s + 2 - V)^{-1}. \tag{4.31}$$

The bound $\delta_P \to 0$ follows by clearing, with P chosen first and the sequence index sufficiently large. It gives $G_s^2 \le G_V \le G_s^{2-\delta_P}$ by the positive Neumann series. No transition-annulus remainder has been omitted.

On each core define

$$\begin{aligned} &g = |\nabla u|, \qquad N_i = \sigma_i \nabla u / g, \qquad S_i = W_i g, \\ &\mathcal{J}_{ij}(Z, Z') = J_i(Z) S_i(X_i Z) G_V(X_i Z, X_j Z') S_j(X_j Z') J_j(Z'), \end{aligned} \tag{4.32}$$

where $Z = (x, r)$. The signs alternate with the crossings so the N_i point in the same graph direction. They obey $|DN_i| \le C_P \kappa$ in both normal and tangential physical directions, by (4.26). Put

$$K_{ij}(x, y) = \int_C \int_C \mathcal{J}_{ij}(x, r; y, q)\, \mathrm{d}r\, \mathrm{d}q, \qquad B_{ij}(x) = \int K_{ij}(x, y)\, \mathrm{d}y, \qquad C = [-2P, 2P]. \tag{4.33}$$

Every coordinate Jacobian occurs exactly once at each integrated endpoint. These are actual positive rows before the orientation signs are inserted. At tapered or clipped fibers no positive lower mass is claimed. All lower bounds and point-depth comparisons below concern full fibers on a window plateau.

4.2.1. *Unsigned exterior control.* Let $r_0 = \min(\delta, \kappa^{-1})$. Lemma 2.10 already gives reach at least $c r_0$. For $R \ge P$, sweep $\Sigma \cap B_{R+P}(X)$ through normal distances $b = c \min(R, r_0)$, with c smaller than the reach constant. The map is injective, its Jacobian is bounded above and below, and its image lies in $B_{CR}(X)$. The area formula therefore gives

$$\mathcal{H}^6(\Sigma \cap B_{R+P}(X)) \le C R^7 / \min(R, r_0) \le C(R^6 + R^7 / r_0).$$

Sweeping this surface piece through depth P covers the desired P-tube inside $B_R(X)$ and gives

$$|\{\mathrm{dist}(\cdot, \Sigma) < P\} \cap B_R(X)| \le C_P(R^6 + R^7 / r_0) \qquad (R \ge P). \tag{4.34}$$

We supply the norm and whole-space comparison steps. Write $v = \nabla u$, $a = 3u^2 - 1$ and $j_\varepsilon(v) = (|v|^2 + \varepsilon^2)^{1/2}$. Convexity gives pointwise, and hence distributionally,

$$(-\Delta)^s j_\varepsilon(v) \le \frac{v}{j_\varepsilon(v)} \cdot (-\Delta)^s v = -a \frac{|v|^2}{j_\varepsilon(v)}.$$

For example, this follows first for the truncated singular integrals from $j_\varepsilon(v(X)) - j_\varepsilon(v(Y)) \le Dj_\varepsilon(v(X)) \cdot (v(X) - v(Y))$ and then by the symmetric principal-value limit. The uniform smoothness of v and its boundedness justify this limit and passage to distributions. Letting $\varepsilon \downarrow 0$ yields $[(-\Delta)^s + a] g \le 0$, also at zeros of v. Choose a separate fixed clearing depth $P_c \ge P$ sufficiently large that clearing gives $a \ge \mu > 0$ outside $E_c = \{\mathrm{dist}(\cdot, \Sigma) < P_c\}$. Then the uniform bound for g gives

$$[(-\Delta)^s + \mu] g \le C \mathbf{1}_{E_c}, \qquad g \le C G_s^\mu * \mathbf{1}_{E_c}.$$

Here is a bounded-subsolution justification of the second inequality. The convolution on the right is bounded, since $\int G_s^\mu = 1/\mu$, and solves its resolvent equation distributionally. Their difference h therefore satisfies $[(-\Delta)^s + \mu] h \le 0$ on all of $\mathbb{R}^7$. Mollify h with a nonnegative compactly supported mollifier, so the same inequality holds for a bounded smooth h_ϵ. Choose $0 < \beta < \min(1, p_*)$ and $b(X) = (1 + |X|^2)^{\beta/2}$. The bounded second derivatives of b, together with $|b(X + Z) - b(X)| \le C|Z|^\beta$ for $|Z| \ge 1$, give $\|(-\Delta)^s b\|_\infty \le C$ uniformly near one by splitting at $|Z| = 1$. The function $h_\epsilon - \tau b$ attains a maximum. At its maximum point X_τ the integral definition gives $(-\Delta)^s h_\epsilon(X_\tau) \ge -C\tau$, and hence $\mu h_\epsilon(X_\tau) \le C\tau$. For every fixed X, $h_\epsilon(X) \le h_\epsilon(X_\tau) + \tau b(X)$. Letting $\tau \downarrow 0$ and then $\epsilon \downarrow 0$ proves $h \le 0$. For $s = 1$ the same argument uses $-\Delta$.

The preceding packing calculation with depth P_c and Tonelli give, for $R \ge P_c$,

$$\int_{B_R(X)} g(Y)\, \mathrm{d}Y \le C \int G_s^\mu(Z) |E_c \cap B_R(X - Z)|\, \mathrm{d}Z \le C_{P_c}(R^6 + R^7 / r_0).$$

For $P \le R < P_c$, the bound $g \le C$ gives the same estimate after increasing the fixed constant, since $R^7 \le P_c R^6$. Thus the gradient-mass estimate holds for the prescribed $R \ge P$, without requiring that this P itself be the clearing depth.

On the dyadic annulus of radius $2^\ell R$, multiply this ball bound by $c_{n,s}(2^\ell R)^{-7-p}$. The resulting two geometric series have exponents $-p-1$ and $-p$, with uniform sums for $p \geq p_*$. Consequently

$$c_{n,s}\int_{|Y-X|>R}\frac{g(Y)}{|X-Y|^{n+p}}\,\mathrm{d}Y \leq C\lambda(R^{-p-1}+R^{-p}/r_0). \tag{4.35}$$

At $R=\rho/\kappa$ this is $o(\kappa^2)$. If $\delta \geq \kappa^{-1}$, division by κ^2 gives $C\lambda\kappa^{p-1}\to 0$. Otherwise, with $\alpha=\lambda\delta^{-p}/\kappa\to 0$, the exact identity is

$$\frac{\lambda\kappa^{p-2}}{\delta}=\alpha^{1/p}\lambda^{1-1/p}\kappa^{(p-1)^2/p}\longrightarrow 0. \tag{4.36}$$

For the massive exterior, the prior choice of the fixed core depth ensures $\delta_P \leq 1$, so $2-\delta_P \in [1,2]$. Use $G_V \leq G_s^{2-\delta_P}$ from (4.31), $0 \leq W \leq 3$, and the uniform massive bound (4.6). Its algebraic part is bounded by the same dyadic gradient-mass calculation, up to a fixed constant. Its exponential part on $|Y-X|>\rho/\kappa$ is bounded using $g \leq C$ and is smaller than every fixed power of κ. Consequently

$$\int_{|Y-X|>\rho/\kappa} G_V(X,Y)W(Y)g(Y)\,\mathrm{d}Y = o(\kappa^2).$$

This accounts for all components and all omitted spatial pieces without a global sheet count.

4.2.2. *Positive row estimates and the gap-excess angle.* Here are the conventions for a retained row and a controlled pair. Fix the receiving base point P_i and use one orthonormal frame with origin P_i and horizontal plane $T_{P_i}\Sigma$ throughout its calculation. The corresponding common graph coordinates are denoted by z; the receiver coordinate is $z=0$. Orient the vertical axis so the donor is above the receiver; a pair below it is treated by simultaneously reversing the vertical orientation. On this frame write the actual ordered graphs as h_j, so $h_i(0)=0$ and $Dh_i(0)=0$. Set

$$H=\rho/\kappa, \qquad R=\sqrt{\rho}/\kappa.$$

Retain the complete consecutive family through the height-$O(H)$, base-radius-$O(R)$ inner window and every intermediate branch. Its actual branches, including returning branches, are continued over the larger common disk B_{2LR} as justified below. Indices count consecutive branches in this local vertical ordering, not global connected components. The number of branches may depend on the index of the sequence; all constants below are independent of it.

The donor core cutoff is supported on a strictly smaller base inside B_{LR}; all its normal fibers have depth at most $2P$. The complete buffered donor portion used for distance minimization is

$$\Gamma_k^{\mathrm{buf}}=\{(z,h_k(z)) : z\in\overline{B}_{LR}\}, \qquad \ell_{ik}=\min_{Z\in\Gamma_k^{\mathrm{buf}}}|Z-P_i|.$$

Thus this minimum is over the actual enlarged donor graph, including its buffer, whether or not its cutoff is equal to one at the minimizer. Every endpoint appearing in the retained donor row lies on a bounded normal fiber of this portion. The definition makes no identification of two different returning branches of one component. As in the core construction, the actual row and its Jacobian factors are expressed in these common coordinates. Translating the receiving coordinate from 0 to x gives the notation $|x-y|$ in the row moments below.

In this fixed frame define

$$D_{ik}(z)=h_k(z)-h_i(z), \quad D=D_{ik}(0), \quad d=|k-i|\delta, \quad t=D-d.$$

A controlled pair is such a retained pair with $0<D\leq H$ at its receiving point. Derivatives DD_{ik} mean derivatives of this same-base graph difference with the frame held fixed. We prove

$$|DD_{ik}(0)|^2 \leq C\kappa(D-d) \tag{4.37}$$

and the distance and angle comparisons in this convention. The construction is repeated uniformly at any receiving evaluation point; the frame is not varied while taking this derivative.

We first establish the initial small angle at this general curvature scale. The marked tangent graph h_i is defined on a fixed multiple of R and has $|h_i| \leq C\kappa R^2 = CH$ and $|Dh_i| \leq C\sqrt{\rho}$. Let Y be another zero in the fixed inner prism. Choose a unit tangent vector at Y with maximal

upward component $v_t \geq 0$, and let γ be the complete unit-speed geodesic with that initial vector. Fix a constant $A > 0$ so that the horizontal projections of $\gamma([-AR, AR])$ remain strictly inside the supplied marked-graph base. The bound $|\gamma''| \leq \kappa$ gives

$$\gamma_t(\pm AR) = Y_t \pm ARv_t + O(A^2 H).$$

The height of the marked graph at the two projected endpoints is also $O_A(H)$. Therefore, if $v_t > C_A H/R = C_A\sqrt{\rho}$, the two values of $\gamma_t - h_i(\gamma_x)$ have opposite signs. The geodesic would meet the marked graph while its entire projection remains in the interior of that graph's base. An actual graph is both open and closed relative to the zero set over this base: it is open by the unique embedded local germ and closed by continuity of its graph function. The whole connected geodesic segment would therefore lie on the marked graph, contrary to Y being on a distinct branch. Points of the marked graph already have the same angle bound. Since $v_t = (1 - |\nu(Y) \cdot e_t|^2)^{1/2}$, every such zero has upward normal angle $O(\sqrt{\rho})$ from e_t.

The horizontal projection lifts now continue each branch over a fixed initial base, with slope at most $C_0\sqrt{\rho}$; all initial evaluation fibers and intermediate branches have this bound, with C_0 independent of the subsequent enlargement. Every zero needed in the inner prism continues to its central fiber, with intercept $O(H)$, so selecting all intervening central intercepts gives a complete family. Choose the fixed enlargement L larger than the evaluation-base multiplier and the Taylor-step multiplier below. Continue the actual branches over B_{2LR}. Along such a projection lift the normal turns by at most $CL\sqrt{\rho}$. Choose ρ after L, so this stays below the fixed transversality and tangent-chart thresholds. Completeness supplies every finite endpoint and ambient closedness identifies the limiting actual zero, so no lift terminates at a finite interior point. Their slopes are bounded by an arbitrarily small fixed ε, and $|D^2 h_j| \leq C_h\kappa$, with C_h uniform for slopes below this threshold. Two such actual graph continuations cannot cross: if they meet they have the same embedded germ, and continuation on the connected common base identifies the two graphs.

Ordering is not yet the intrinsic separation assertion. To verify the precise assertion needed here, let Q, Q' lie on two distinct retained branches at the same horizontal coordinate in the enlarged working disk. If they were joined intrinsically at distance $\ell < a_{\mathrm{geom}}/\kappa$, completeness would supply a minimizing unit-speed geodesic γ between them. Write π for horizontal projection. Since the upward normal at Q has slope at most ε, its initial tangent velocity obeys $|\pi\gamma'(0)| \geq (1 - \varepsilon^2)^{1/2}$. The global curvature bound gives $|\gamma''| \leq \kappa$, and hence

$$\left|\pi(Q' - Q) - \ell\pi\gamma'(0)\right| \leq \int_0^\ell \kappa r \, \mathrm{d}r = \tfrac{1}{2}\kappa\ell^2.$$

But $\pi(Q' - Q) = 0$, so this would imply $(1 - \varepsilon^2)^{1/2} \leq \kappa\ell/2 < a_{\mathrm{geom}}/2$, impossible for the fixed small intrinsic cutoff and the chosen small slope threshold. Different components already have infinite intrinsic distance. Thus distinct same-fiber branches are intrinsically off diagonal at every point of the enlarged base. Each consecutive same-fiber gap is at least δ, and summing these gaps gives $D_{ik}(z) \geq |k - i|\delta$ on the whole Taylor segment. This argument uses the original intrinsic cutoff and does not infer intrinsic separation merely from preserved ordering.

Consequently, for the pair under consideration,

$$D \geq d = |i - k|\delta, \qquad |D^2 D| \leq C_h\kappa$$

throughout that base. Write $a = DD(x)$. The segment from x in direction $-a$ of length $|a|/(C_h\kappa)$ is contained in the larger base: the small-slope bound gives $|a| \leq C\sqrt{\rho}$, and L was fixed larger than C/C_h. Taylor's formula at its endpoint gives

$$d \leq D(x) - \frac{|a|^2}{C_h\kappa} + \frac{|a|^2}{2C_h\kappa}, \qquad |DD(x)|^2 \leq 2C_h\kappa(D(x) - d).$$

Thus the derivative estimate uses actual separation on an actual larger base. A missing Taylor segment is not a spatial exterior contribution.

In the fixed receiving tangent frame, write

$$h_k(z) = D + a \cdot z + O(\kappa|z|^2), \qquad a = Dh_k(0) = DD_{ik}(0), \qquad |a|^2 \leq C\kappa t.$$

The point $(0, D)$ belongs to Γ_k^{buf}, so $\ell_{ik} \le D$. A minimizing point $(z_*, h_k(z_*))$ exists by compactness and satisfies $|z_*| \le D$. As $D \le H \ll LR$, it is in the interior of the actual buffered donor disk, irrespective of the smaller core cutoff support. Consequently its stationarity equation and height bound are

$$z_* = -h_k(z_*)Dh_k(z_*), \qquad |h_k(z_*)| \le D.$$

The Hessian bound gives $|Dh_k(z_*)| \le |a| + C\kappa|z_*|$. Absorbing $C\kappa D \le C\rho$ yields $|z_*| \le CD|a|$. Taylor's formula therefore gives

$$\ell_{ik} \ge h_k(z_*) \ge D - CD|a|^2 \ge D - C\kappa Dt \ge d + \tfrac{1}{2}t, \tag{4.38}$$

after decreasing ρ. At the common base coordinate 0, the receiving upward normal is e_t and the donor upward normal is $(-a, 1)/(1+|a|^2)^{1/2}$, so their difference is at most $C|a|$. The fixed-depth normal-variation estimate from the relative jets then gives, for $|r|, |q| \le 2P$,

$$|N_i(0, r) - N_k(0, q)| \le C_P(\sqrt{\kappa}\sqrt{t} + \kappa). \tag{4.39}$$

Here the N_j are the common upward geometric normals used in the row construction, equivalently the phase normals multiplied by their half-jump signs. Normal offsets of these depths change any endpoint distance by at most $4P$.

Let x be the receiving coordinate in this fixed frame, put $r_0 = |x-y|$ and $S = (D_{ik}(x)^2+r_0^2)^{1/2}$, and write $P_j(z) = (z, h_j(z))$. The donor slope bound gives

$$|h_k(y) - h_k(x)| \le \varepsilon r_0, \qquad |P_k(y) - P_i(x)| \ge S - \varepsilon r_0 \ge (1-\varepsilon)S.$$

For the actual normal coordinates, $X_j(z, r) = P_j(z) + r\nu_j(z)$, this implies

$$|X_k(y, q) - X_i(x, r)| \ge (1-\varepsilon)S - 4P, \qquad |r|, |q| \le 2P.$$

All donor coordinates in the retained row lie in the actual common base. Its bounded graph and tubular Jacobians and the bounded core endpoint factors are absorbed into C_P. Since $D \ge \delta \to \infty$, the physical distance is also at least $S/2$ for large indices. Choose a free massive exponential rate $\gamma > k/2$, available after the fixed large-depth choice. Choose the slope threshold so small that $\gamma(1-\varepsilon) > k/2$, and reserve

$$k/2 < c < \gamma(1-\varepsilon).$$

The positive massive comparison therefore bounds the kernel of a retained row by

$$C_P\left\{e^{-\gamma(1-\varepsilon)S} + \lambda S^{-7-p}\right\}.$$

The additive $4P$ costs only a fixed exponential constant. The remaining tangential integration uses the strict gap between c and $\gamma(1-\varepsilon)$; thus the common-coordinate comparison preserves an exponential rate strictly larger than $k/2$.

The rough positive row estimates needed below are

$$B_{ik}(x) \le C_P\{(1+D_{ik})^L e^{-cD_{ik}} + \lambda D_{ik}^{-p-1}\}, \tag{4.40}$$

$$\int |x-y|K_{ik}(x, y)\,\mathrm{d}y \le C_P\{(1+D_{ik})^L e^{-cD_{ik}} + \lambda D_{ik}^{-p}\}, \tag{4.41}$$

$$\int |x-y|^2 K_{ik}(x, y)\,\mathrm{d}y \le C_P\{(1+D_{ik})^L e^{-cD_{ik}} + \lambda D_{ik}^{1-p}\}. \tag{4.42}$$

Here $c > k/2$ after choosing P large and slopes small. Fix $0 < c_1 < c$ and a logarithmic cutoff K_0 so large that $c_1 K_0 > 2$; the later normal buffers are chosen after K_0. They follow by integrating the massive majorant over a fixed-depth bounded-slope graph tube, using the radial integrals in six tangential variables. For clarity, after the physical-to-graph distance comparison, the algebraic integrals for $a = 0, 1, 2$ are exactly bounded by

$$C_P\lambda \int_0^\infty \frac{r^{5+a}}{(D^2+r^2)^{(7+p)/2}}\,\mathrm{d}r = C_P\lambda D^{a-p-1}\int_0^\infty \frac{z^{5+a}}{(1+z^2)^{(7+p)/2}}\,\mathrm{d}z.$$

The last integral is uniform for $p \in [p_*, 2]$: near zero its integrand is $O(z^{5+a})$, and at infinity it is $O(z^{a-p-2})$, whose exponent is at most $-p_* < -1$. The corresponding exponential integrals are bounded by a polynomial in D times e^{-cD}, with the reserved $c > k/2$. The same estimates

hold at a fixed receiving core point with its bounded endpoint factor instead of receiving normal integration.

On the logarithmic range $\delta \le D \le K_0 \log(1/\kappa)$ one needs the sharper row

$$B_{ik}(x) \le C_{P,K_0}\{e^{-k\ell_{ik}} + \lambda D^{-p-1}\}. \tag{4.43}$$

Here is a reduction accounting for all phase visits. Put $L_0 = \log(1/\kappa)$, $R = \kappa^{-1/2}/L_0$, and normal buffer $H_1 = K_1 L_0$. In the receiver tangent frame take

$$\mathcal{B} = \{|z| < R,\ -H_1 < t < D + H_1\}, \qquad V_0 = \mathbf{1}_{\mathcal{B}} V.$$

The receiving core is within $O_P(1)$ of $(0,0)$, and the retained donor core is within $O_P(1)$ of height D on the lateral box. Thus omitted insertions have lateral distance at least $R - O_P(1)$ or normal distance at least $H_1 - O_P(1)$ from the receiving core. The fixed shifts are absorbed in the constants below. Since $e^{-k\delta} = o(\kappa)$, we have $\delta \ge L_0/(2k)$ eventually. Complete natural-window geometry therefore gives only a fixed number of sheets in this box, with slopes $O(\sqrt{\kappa L_0})$ and graph oscillation $O(L_0^{-1/2})$ on its base. On the nonempty logarithmic range, $L_0/(2k) \le \delta \le D \le K_0 L_0$. Let $a_i = h_i(0)$. The signed-distance conversion therefore gives, uniformly in this box,

$$|\rho_i(x,t) - (t - a_i)| \le C\{L_0^{-1/2} + \kappa L_0^2\} =: \varepsilon_\kappa \longrightarrow 0.$$

For $P > 2$ and sufficiently small κ, the box-restricted phase potential is bounded globally by $C\kappa + v(t)$, where one may take

$$v(t) = C \sum_i \mathbf{1}_{\{|t-a_i| \ge P-1\}}\{e^{-k|t-a_i|} + \lambda(1 + |t - a_i|)^{-p}\}.$$

Indeed $\psi = 1$ when $|\rho_i| \le P$, so the actual phase potential vanishes in those cores; outside them the chamber estimate bounds the squared-phase error by $C\kappa$ plus the displayed tails. Hence

$$\|v\|_1 \le C_M\{e^{-k(P-1)} + \lambda(P-1)^{1-p}/(p-1)\}, \qquad \|v\|_\infty \longrightarrow 0 \quad (P \to \infty).$$

The constant $C\kappa$ is absorbed into the massive coefficient, and positivity permits domination by this planar resolvent. Endpoints beyond the lateral box contribute separately at most $C_P(e^{-cR} + \lambda R^{-p-1})$, which is covered by the row remainder.

The one-dimensional massive kernel has a positive Laplace representation $g(r) = \int_0^\infty e^{-ur}\rho(\,\mathrm{d}u)$. Nonnegative covariance of the two decreasing functions e^{-Au} and e^{-Bu} proves

$$g(A)g(B) \le g(0)g(A+B).$$

The kernel is decreasing, and $|a - z| + |z - b| \ge |a - b|$. Hence

$$g(|a-z|)g(|z-b|) \le g(0)g(|a-b|).$$

Integrating this inequality against $v(z)\,\mathrm{d}z$ shows that every insertion costs at most $q = g(0)\|v\|_1$. Choose P so $q < 1/2$. Summing the positive series bounds the full one-dimensional planar resolvent by $(1-q)^{-1}g$. To apply it to the actual curved cores, use the signed-distance conversion above: for large indices their physical source densities are bounded by C_P and supported in the fixed flat bands $|t - a_i| \le 3P$. Dominate the donor density by $C_P \mathbf{1}_{\{|t-a_k| \le 3P\}}\,\mathrm{d}X$, and similarly bound the receiving core integral. These planar bands are independent of tangential variables. Tonelli then reduces every positive ambient resolvent term exactly to its one-dimensional term. The fixed band shifts are absorbed in C_P. Equation (4.7) gives $C_P\{e^{-kD} + \lambda D^{-p-1}\}$ because $\kappa D \to 0$. Since $\ell_{ik} \le D$, this implies (4.43) for the box-restricted resolvent.

Exterior phase visits in the logarithmic comparison. Let $\mu_k = S_k\,\mathrm{d}X$ be the donor source restricted to its supplied complete bounded-slope graph disk, including its full fixed-depth fibers. For $R \ge 1$ it obeys the translation-uniform bound $\mu_k(B_R(Z)) \le C_P R^6$. If G_s^μ is a free massive kernel, the positive measure with density $F_k = G_s^\mu * \mu_k$ satisfies the identical ball bound: Tonelli gives

$$\int_{B_R(Z)} F_k(X)\,\mathrm{d}X = \int G_s^\mu(T)\mu_k(B_R(Z-T))\,\mathrm{d}T \le (C_P/\mu)R^6.$$

For $H \geq 1$, annular integration against the massive upper bound therefore gives

$$\int_{|Z-X|>H} G_s^{\mu}(X-Z)F_k(Z)\,\mathrm{d}Z \leq C_P\{e^{-cH} + \lambda H^{-p-1}\}.$$

This is the estimate for an omitted insertion, rather than just an omitted endpoint. Indeed the exact identity $G_V - G_{V_0} = G_{V_0}(V-V_0)G_V$ and the two positive comparisons by $G_s^{2-\delta_P}$ bound its donor-integrated row by this integral, times the bounded insertion and receiving source factors. For a normal buffer $H_1 = K_1\log(1/\kappa)$ and the lateral buffer $R = \kappa^{-1/2}/\log(1/\kappa)$, both distances exceed the retained gap $D \leq K_0\log(1/\kappa)$, after K_1 is chosen large and the index is large. Their algebraic errors are therefore $O_P(\lambda D^{-p-1})$. Choose K_1 so $cK_1 > \sqrt{2}K_0$; then the normal exponential error is bounded by $Ce^{-\sqrt{2}D}$, and the lateral exponential is smaller still. Choose $cK_1 > 2$ as well to make the normal *exponential* insertion error $o(\kappa^2)$. The algebraic error remains in the $O_P(\lambda D^{-p-1})$ row remainder; a logarithmic buffer does not by itself give it an $o(\kappa^2)$ bound. These choices precede the sequence limit. This supplies the sharp row comparison for the full positive resolvent, with every visit included.

The sum of the second moments up to curvature distance is $o(1)$. Fix $0 < \epsilon_0 < p_* - 1$ and let $L_* = C/(\kappa\delta) \geq 1$. Then

$$\sum_{1\leq\ell\leq L_*} \ell^{1-p} \leq C_{\epsilon_0} L_*^{2-p+\epsilon_0},$$

so its algebraic contribution is bounded by $C\lambda\kappa^{p-2-\epsilon_0}\delta^{-1-\epsilon_0}$. Writing $\alpha = \lambda\delta^{-p}/\kappa$, this equals

$$C\alpha^{(1+\epsilon_0)/p}\lambda^{1-(1+\epsilon_0)/p}\kappa^{(p-1)(p-1-\epsilon_0)/p} = o(1). \tag{4.44}$$

All powers are uniformly positive by the choice of ϵ_0. The exponential part of the moment sum also vanishes: after absorbing the fixed polynomial into a slightly smaller positive rate, $D_{ik} \geq |i-k|\delta$ gives

$$\sum_{k\neq i}(1+D_{ik})^L e^{-cD_{ik}} \leq C\sum_{m\geq 1} e^{-c'm\delta} = \frac{Ce^{-c'\delta}}{1-e^{-c'\delta}} = o(1).$$

If $\kappa\delta$ is bounded below, only boundedly many retained labels occur and the conclusion is immediate. This proof does not lose a factor $(2-p)^{-1}$ in a near-two harmonic sum. We now spell out the angular summation on the buffered evaluation cores. The fixed-depth derivative bound and (4.39) give, with $M_{2,ik} = \int |x-y|^2 K_{ik}(x,y)\,\mathrm{d}y$, the upper budget

$$C_P\sum_{k\neq i}\{(\kappa t_{ik}+\kappa^2)B_{ik} + \kappa^2 M_{2,ik}\},$$

plus the unsigned geometric exterior. Bounded fixed endpoint factors give the same estimate at a fixed receiving normal point. The truncated moment sum is $o(1)$, so tangential displacement contributes $o(\kappa^2)$. The same ordered exponential sum and $\sum_{m\geq 1}\lambda(m\delta)^{-p-1} \leq C\lambda\delta^{-p-1}$ give $\sum_{k\neq i} B_{ik} = o(1)$, so the $\kappa^2 B_{ik}$ terms are negligible. On logarithmic gaps the remaining same-foot terms satisfy

$$\kappa t_{ik}e^{-k\ell_{ik}} \leq C\kappa e^{-kd_{ik}}, \qquad \kappa t_{ik}\lambda D_{ik}^{-p-1} \leq C\kappa\lambda d_{ik}^{-p}.$$

Summing the geometric and ℓ^{-p} series gives $C_P\kappa F = o(\kappa^2)$. For $D > K_0\log(1/\kappa)$, the algebraic term has the same $\kappa\lambda d_{ik}^{-p}$ bound. Absorb the polynomial in the rough exponential into $e^{-c_1 D}$. On the retained range $\kappa t_{ik} \leq \kappa D_{ik} \leq \rho$. The same-base gaps increase by at least δ along either side of the ordered stack. Hence the far exponential sum is bounded by

$$C_P\sum_{D_{ik}>K_0L_0} e^{-c_1 D_{ik}} \leq \frac{2C_P e^{-c_1K_0L_0}}{1-e^{-c_1\delta}} = o(\kappa^2).$$

Pairs beyond curvature distance are in the unsigned exterior: the uniform graph slope bound gives ambient endpoint separation at least $c\rho/\kappa$, with fixed depth shifts absorbed. Equation (4.35) and the positive massive comparison bound all these pieces by $o(\kappa^2)$ without a

global label count. Combining the budgets proves

$$\sup_{i,x,r:\ X_i(x,r)\in\mathcal{C}_{\mathrm{eval}}} \sum_{k\neq i} \int \mathcal{J}_{ik}(x,r;y,q)|N_i(x,r)-N_k(y,q)|^2\,\mathrm{d}y\,\mathrm{d}q \le C_P\kappa F + o(\kappa^2) = o(\kappa^2). \quad (4.45)$$

Multiplication of the row's endpoint factors by uniformly bounded functions preserves this upper estimate. Here this is the meaning of a bounded normalization; the unit-normal difference itself is kept unchanged. For the self row, paths on the small-slope graph and its fixed-depth fibers give $|N_i(X)-N_i(Y)| \le C_P\kappa|X-Y|$ inside the common buffer. On $|X-Y|<1$, the massive majorant is $C|X-Y|^{p-7}$, so its product with $|X-Y|^2$ has a uniformly finite integral in seven physical dimensions. On the remaining fixed-depth tube, the exponential second moment is uniformly finite, and the algebraic second moment is bounded by $C_P\lambda\int_1^\infty r^{-p}\,\mathrm{d}r \le C_P\lambda/(p_*-1)$. The bounded endpoint densities and Jacobians therefore give a self angular row of $O(\kappa^2)$. Any omitted geometric part is already covered by the unsigned exterior estimate.

5. Capacities and the normal equation

5.1. **Adjacent-row capacity from signed stability.** Use a complete plateau chart with the spatial margins specified below. All active fibers are full, and the two active rows use the same consecutive retained label set. On its common tangential base put

$$I_i = \tfrac{1}{2}(B_{i,i+1}+B_{i+1,i}), \qquad T_i = \sum_{k\neq i,i+1}\{(-1)^{i-k}B_{ik} + (-1)^{i+1-k}B_{i+1,k}\}. \quad (5.1)$$

Temporarily the sum includes every core of the common smooth partition; its later restriction to central labels is paid by the unsigned exterior estimate.

Write $A_V = (-\Delta)^s + 2 - V$. The exact partition and stability give $A_V - \sum_j W_j = H + W_{\mathrm{ext}} \ge 0$. On the direct sum of the core L^2 spaces let $U(f_j) = \sum_j \sqrt{W_j} f_j$, extended by zero outside the cores. The form inequality says $\|A_V^{-1/2}U\| \le 1$, and hence $I - U^*G_VU \ge 0$. This specifies the endpoint multiplication and its square roots; no normal cutoff is differentiated here. Let $Z = G_V(W_{\mathrm{ext}}\nabla u)$. Applying the bounded massive inverse to the differentiated equation and using $\nabla u = \sigma_j g N_j$ on core j gives

$$W_i g^2 = \sum_k \sigma_i\sigma_k \int S_i(X)G_V(X,Y)S_k(Y)N_i(X)\cdot N_k(Y)\,\mathrm{d}Y + \sigma_i S_i(X)N_i(X)\cdot Z(X). \quad (5.2)$$

The bounded inverse and the exterior pairing are justified by the positive massive comparison and (4.35).

Lemma 5.1 (Exact two-site inequality)**.** *For full plateau fibers over the test support,*

$$4\int I_i\eta^2 \le C\int |D\eta|^2 + \int T_i\eta^2 + C\kappa^2\int \eta^2. \quad (5.3)$$

Proof. For compact tangential amplitudes ζ_j, apply the nonnegative form to $f_j = \sigma_j\sqrt{W_j}g\zeta_j$. Substitute (5.2) in its diagonal terms and symmetrize. Its exact value is

$$\frac{1}{2}\sum_{j,k}\sigma_j\sigma_k \iint K_{jk}(x,y)(\zeta_j(x)-\zeta_k(y))^2\,\mathrm{d}x\,\mathrm{d}y + \mathcal{A}(\zeta) + \mathcal{X}(\zeta),$$

where, with physical normal coordinates understood,

$$\mathcal{A} = -\frac{1}{4}\sum_{j,k}\sigma_j\sigma_k \iint \mathcal{J}_{jk}(Z,Z')|N_j(Z)-N_k(Z')|^2(\zeta_j(x)^2+\zeta_k(y)^2)\,\mathrm{d}Z\,\mathrm{d}Z',$$

$$\mathcal{X} = \sum_j \sigma_j \int S_j(X)N_j(X)\cdot Z(X)\zeta_j(x)^2\,\mathrm{d}X.$$

Take $\zeta_i = \eta$, $\zeta_{i+1} = -\eta$, and all other amplitudes zero. The angular estimate (4.45), its self-row counterpart, and the unsigned exterior bound give $|\mathcal{A}| + |\mathcal{X}| \le C\kappa^2\int\eta^2$. The self terms cost

$C\int|D\eta|^2$ by the self-row moment estimate already established in Section 4.2. The selected unordered pair contributes exactly

$$-\iint K_{i,i+1}(x,y)(\eta(x)+\eta(y))^2\,\mathrm{d}x\,\mathrm{d}y=-4\int I_i\eta^2+\iint K_{i,i+1}(x,y)(\eta(x)-\eta(y))^2\,\mathrm{d}x\,\mathrm{d}y.$$

This uses $K_{i+1,i}(y,x)=K_{i,i+1}(x,y)$ and is an exact identity, without replacing either endpoint by the other foot. The positive term is at most $C\int|D\eta|^2$. Indeed the translation-invariant majorant from the proof of (4.42) has uniformly bounded second moment for this single pair, since its gap is at least δ; integrate $|\eta(x+z)-\eta(x)|^2\le|z|^2\int_0^1|D\eta(x+tz)|^2\,\mathrm{d}t$ against that majorant. Every active–inactive unordered pair contributes its signed row mass times η^2, and their sum is exactly $\int T_i\eta^2$. Nonnegativity of the original form proves the inequality. □

Throughout this subsection all labels are expressed in one fixed common graph frame, and $D_{ab}(x)=|h_a(x)-h_b(x)|$. Thus for $a<b$, $D_{ab}=\sum_{j=a}^{b-1}d_j$ exactly; this is the gap convention used in the algebraic model and its alternating cancellation. Write $d_j=h_{j+1}-h_j\ge\delta$, $q=p+1$, and

$$A_j=e^{-kd_j}+\lambda d_j^{-q},\qquad \mu_{\chi,s}=\int\psi(r)^2W_s(r)w_s(r)\,\mathrm{d}r,\qquad b_s(D)=\frac{c_{1,s}\mu_{\chi,s}^2}{4}D^{-q}=c_s\lambda D^{-q}.$$

Here c_s is common to every row compared in this chart. Its upper and positive lower bounds may be chosen independently of $P\ge1$: $\psi=1$ on $[-1,1]$, local layer convergence gives a positive depth-one mass, and

$$\int_{\mathbb{R}}W_sw_s=\int_{-1}^{1}3(1-v^2)\,\mathrm{d}v=4.$$

Also $c_{1,s}/\lambda$ has positive finite bounds near one.

We use the sharp free massive coefficients already proved in Lemma 3.9, in particular (3.51), and the normalized mixture (4.8). Only their application to the present smooth cores remains to be checked. On a full plateau fiber, the relative jet estimate gives, for fixed P,

$$J_j(x,r)S_j(X_j(x,r))=\sqrt{1+|Dh_j(x)|^2}\,\psi(r)^2W_s(r)w_s(r)(1+o(1)+O(\kappa P)).$$

This is a relative statement on the fixed interval after cancelling the common factor ψ^2; it requires no division by the taper. For upper bounds a tapered endpoint only decreases the mass. The squared-slope factor is $1+O(\varepsilon^2)$ if the graph slopes are at most ε, uniformly in P.

For a fixed receiving base point and a donor point over y, put $R=(D_{ab}^2+|x-y|^2)^{1/2}$. The genuine graph-distance comparison from Section 4.2 and the fixed normal depths bound the physical distance between $(1-C\varepsilon)R-4P$ and $(1+C\varepsilon)R+4P$. Since $D_{ab}\ge\delta\to\infty$, these are multiplicative changes $1+O(\varepsilon+P/\delta)$. Tangential integration of the algebraic kernel gives exactly

$$c_{7,s}\int_{\mathbb{R}^6}(|z|^2+D^2)^{-(7+p)/2}\,\mathrm{d}z=c_{1,s}D^{-q}.$$

Equivalently, for the free upper kernel the dilation identity $G^{\mu}_{7,s}(bZ)=b^{p-7}G^{\mu b^p}_{7,s}(Z)$ followed by (3.51) gives coefficient $c_{1,s}\mu^{-2}b^{-7-p}$. Use $G_s^2\le G_V\le G_s^{2-\delta_P}$. The relative errors in the common coefficient are therefore $O(\varepsilon+\delta_P+P/\delta)+o(1)$, as well as the vanishing free-kernel asymptotic error. Choose these errors below any prescribed fixed $\omega>0$ in the order stated below. The free upper exponential rate can still be chosen $c>k/2$. On the common donor plateau this proves

$$\begin{aligned}B_{ab}&\ge(1-\omega)b_s(D_{ab})-r_{ab}^{-}&&(a-b\text{ odd}),\\ B_{ab}&\le(1+\omega)b_s(D_{ab})+C_Pe^{-cD_{ab}}+r_{ab}^{+}&&(a-b\text{ even},\ a\ne b).\end{aligned}\tag{5.4}$$

The lower truncations and all omitted physical pieces are the $r^{\pm}$ estimated below. Positivity alone does not permit replacing a finite donor disk by the whole plane in a lower bound.

We record one depth-one lower argument for use both here and in chart transfer. Let P_* and Q_* be actual receiver and donor points at distance $d\ge\delta$, with full depth-one fibers and a donor plateau containing an actual tangent disk of radius c_0d. Assume κd is below the fixed curvature threshold. On a smaller such disk all endpoint distances are at most Cd. The mass-two algebraic

lower kernel and the uniform positive depth-one source densities give a row lower bound $c\lambda d^{-q}$: the donor tangential volume is cd^6. All constants in this argument are independent of the larger P.

For the exponential term it suffices to treat $d \le K\log(1/\kappa)$ and $\lambda d^{-q} \le e^{-kd}$; otherwise the algebraic bound or an $O(\kappa^2)$ remainder suffices, with $kK > 2$. Parametrize the donor at the actual point Q_* by tangent vectors v, with remainder $O(\kappa|v|^2)$. Restrict to

$$|v| \le \sqrt{d}, \qquad (Q_* - P_*)\cdot v \le 0.$$

This half-ball has volume cd^3, and its points, including depth-one normal shifts at both ends, have distance at most $d + C + C\kappa d = d + O(1)$ from the receiver. The favorable linear term is taken at the donor, not at an unrelated graph foot. In (4.8) with mass 2, let $r_s = 2^{1/p}$. On its resonance interval, $2r\rho_s^2(r) \ge c\lambda/((r-r_s)^2+\lambda^2)$. The assumption gives $\lambda d^2 \le d^{q+2}e^{-kd} \to 0$, so the band $|r - r_s| \le d^{-2}$ has mixing mass bounded below. Moreover $(r_s - k)d = O(\lambda d) = o(1)$. The explicit seven-dimensional Yukawa kernel from the proof of Lemma 4.1 is consequently at least $cd^{-3}e^{-kd}$ on the selected endpoint pairs. Multiplication by the tangential volume and by the fixed depth-one masses proves the exponential row lower bound. At the local endpoint the same argument uses $Y_{7,k}$ directly.

Apply this construction to the actual same-base adjacent pair, whose distance is d_j. The common plateau disk is large enough as specified below, and $I_j \ge B_{j,j+1}/2$. Thus

$$A_j \le C_0 I_j + C_0\kappa^2, \tag{5.5}$$

where C_0 can be chosen independently of the larger core depth.

The spatial and chart-transfer accounting. Fix a small curvature-chart parameter H and supply complete actual graphs and their genuine foot buffers over a window containing $B_{8H/\kappa}$. Its slopes are $O(\sqrt{H})$, by the construction in Section 4.2. Evaluation tests lie in $B'_{r/\kappa}$, with $r \ll H$, and the active pair lies in the inner central band. Choose the common consecutive labels $\mathcal{K}$ by central intercepts in $[-4H/\kappa, 4H/\kappa]$, increasing the fixed outer multipliers if necessary. All returning and intermediate branches are retained. Choose the smooth window equal to one on the depth-$2P$ fibers of these graphs over the common donor disk $B'_{2H/\kappa}$ and on the receiving buffer; its taper has the outer margins above. Thus for each receiving $x \in B'_{r/\kappa}$ the donor disk $B'_{H/\kappa}(x)$ really consists of full plateau fibers for all labels in $\mathcal{K}$. Sources outside this common family, or outside its supplied lateral region, have distance at least cH/κ from the active receiver cores. Indeed a central-intercept discrepancy of order H/κ cannot be cancelled by the $O(\sqrt{H})$ slope on these bases; lateral omissions are already that far away. The fixed-depth offsets are absorbed for large indices. Equation (4.35) and the massive majorant give total $o(\kappa^2)$ for these unsigned physical pieces, without a count of exterior labels. Both active rows use the same $\mathcal{K}$.

For a lower algebraic comparison integrate only the existing donor disk $B'_{H/\kappa}(x)$. The tail of the complete flat algebraic row outside it is at most $C\lambda(H/\kappa)^{-q}$ per label. Since same-fiber gaps are at least δ, $\#\mathcal{K} \le 1 + CH/(\kappa\delta)$. Hence the total deficit is

$$C_H\{\lambda\kappa^{p+1} + \lambda\kappa^p/\delta\} = o(\kappa^2). \tag{5.6}$$

The first ratio to κ^2 is $\lambda\kappa^{p-1} \to 0$; the second is (4.36). This proves that the incident remainders in (5.4), including the common-label omissions, sum to $o(\kappa^2)$.

Here is the scalar chart transfer with a constant independent of P. Fix $d_0 \ll r \ll H$. For an adjacent edge j in the old common chart put $\theta_j(x) = \chi(\kappa d_j(x)/d_0)$, where $0 \le \chi \le 1$, $\chi = 1$ on $[0,1]$, and $\chi = 0$ on $[2,\infty)$. On $1 - \theta_j^2 \ne 0$ we have $d_j \ge d_0/\kappa$, and hence $A_j \le C_{d_0}\kappa^2$. Also $|D\theta_j| \le C\kappa/d_0$ by the small-slope graph bounds. Cover the support of the old test by a bounded-overlap square partition $\sum_\alpha \zeta_\alpha^2 = 1$ with disks of radius cr/κ and $|D\zeta_\alpha| \le C\kappa/r$. At an actual point of the old sheet j over each disk center, make a fresh tangent chart. The coordinate map sends an old point of that same sheet to its horizontal coordinate in the new frame. It is the restriction of an orthogonal projection to a small-slope graph, so its Jacobian and inverse derivative have universal upper and positive lower bounds. The source sheet $j+1$ remains in the fresh central band on $\theta_j \ne 0$, since $d_j \le 2d_0/\kappa \ll r/\kappa$. Old completeness on the larger buffer

shows that these same two actual sheets remain adjacent. No new intermediate zero can occur in their old zero-free chamber.

Transport $\theta_j\zeta_\alpha\eta$ by this coordinate map and call the resulting test $\eta_{j,\alpha}$. With

$$\mathcal{E}^*_\kappa(\eta) = \int |D\eta|^2 + (\kappa/d_0)^2 \int \eta^2,$$

the change of variables, product rule, bounded overlap, and $d_0 < r$ give

$$\sum_\alpha \mathcal{E}^*_\kappa(\eta_{j,\alpha}) \le C_{\mathrm{tr}}\mathcal{E}^*_\kappa(\eta).$$

The constant is independent of j, P, and the sequence index; it is uniform for the fixed small-slope threshold and the chosen fixed partition geometry.

At a transported receiver point use the actual old adjacent source point as Q_* in the depth-one lower construction above. Its physical separation is exactly the old vertical gap $d_j(x)$. The fresh donor plateau contains the required radius-$c_0 d_j$ disk because $d_0 \ll r$, and it contains the logarithmic radius-$\sqrt{d_j}$ disk for large indices. Consequently that same lower construction, with no comparison of phase resolvents, gives the old potential $A_j(x)$ bounded by a universal constant times the fresh positive adjacent row, plus $C\kappa^2$. After change of variables and summation,

$$\int A_j\eta^2 \le C'_{\mathrm{tr}} \sum_\alpha \int I^{\mathrm{fresh}}_{j,\alpha}\eta^2_{j,\alpha} + C\mathcal{E}^*_\kappa(\eta). \tag{5.7}$$

The coefficient C'_{tr} is independent of P, since the positive source lower bounds used only depth-one fibers. For the exponential term the same actual path and favorable donor half-ball give the old exponent e^{-kd_j}; a multiplicative comparison of old and new gaps is not used.

Choose the target ω using these universal transfer and depth-one constants. Next choose H so the slopes are sufficiently small, then fix r, d_0 as above, then choose P so δ_P is small, and finally take the sequence index large for relative preparation, P/δ, and all vanishing errors. This order keeps the coefficient used in the final absorption independent of any later P-dependent additive constant.

Theorem 5.2 (Simultaneous actual adjacent capacity)**.** *The actual rows on these plateau charts obey, uniformly in the marked point and adjacent label,*

$$\int I_i\eta^2 \le C\left(\int |D\eta|^2 + \kappa^2 \int \eta^2\right). \tag{5.8}$$

The same bound holds for each A_i after changing its constant. For $y = \kappa x$, every admissible translated rescaled ball satisfies

$$\int_{B_r(z)} \kappa^{-2} I_i(y/\kappa)\,\mathrm{d}y \le C(r^{m-2} + r^m) = C(r^4 + r^6), \qquad 0 < r \le 1. \tag{5.9}$$

The ball and its fixed enlargement must lie in the admissible chart.

Proof. For $q = p + 1$ and positive consecutive gaps, convexity gives

$$(d_1 + \cdots + d_\ell)^{-q} \le \ell^{-q-1} \sum_{j=1}^{\ell} d_j^{-q}.$$

An edge at label distance n from an active site occurs only in paths of length at least n; its resulting algebraic weight is at most $C\sum_{\ell \ge \max(1,n)} \ell^{-q-1}$. These weights are summable over edges, uniformly for $p \ge p_*$. For positive same-orientation rows, ℓ is even and at least two. Jensen's inequality for the exponential gives

$$e^{-c\sum_{j=1}^{\ell} d_j} \le e^{-(c\ell-k)\delta} \frac{1}{\ell} \sum_{j=1}^{\ell} e^{-kd_j}.$$

Summed over incident rows, its total edge weight is at most $C_P \sum_{\ell \text{ even}}^{\ell \ge 2} e^{-(c\ell-k)\delta} = o_\delta(1)$, since $2c > k$.

At each base point the common algebraic model has a favorable exact sign. On the left of the selected edge its contribution to T_i is

$$\sum_{\ell=1}^{N_-}(-1)^{\ell}\{b_s(r_\ell)-b_s(r_\ell+d_i)\},\qquad r_\ell=h_i-h_{i-\ell}.$$

Since b_s is positive, decreasing, and convex, the alternating-difference calculation in the proof of Proposition 3.10 applies without change and makes this sum nonpositive; the right side has the same sign. That calculation applies to finite consecutive lists, including an odd last negative term, and requires no symmetry of adjacent gaps. Subtract this nonpositive model from (5.4). The relative errors cost ω times the absolute model rows; positive exponential rows have the weights just estimated; all incident remainders have the truncation/exterior budget above. Therefore there are nonnegative edge weights t_n such that

$$T_i\le\sum_j t_{j-i}A_j+C\kappa^2,\qquad \sum_n t_n\le C\omega+o_\delta(1).$$

The coefficient C here is independent of the chosen larger P: the common algebraic mass is at most 4^2, and the only P-dependent exponential coefficient is in $o_\delta(1)$ after P has been fixed.

Let M_κ be the supremum of $\int I_i\eta^2/\mathcal{E}^*_\kappa(\eta)$ over all marked plateau charts, their admissible adjacent pairs, and nonzero compact tests. It is finite for each sufficiently large fixed sequence index: the positive massive comparison and bounded fixed-depth endpoint densities give $I_i\le C_P$, uniformly in these charts, whence $M_\kappa\le C_Pd_0^2\kappa^{-2}$. Apply (5.7) separately to every A_j. Uniformity of its constants and the summability of t_n give

$$\int T_i\eta^2\le\{C_*(C\omega+o_\delta(1))M_\kappa+C\}\mathcal{E}^*_\kappa(\eta),$$

where C_* is independent of P. First choose ω so that $C_*C\omega<1$, and then take the sequence index large enough that the full coefficient is at most two. Insert this bound into (5.3) and take the supremum: $4M_\kappa\le C+2M_\kappa$. Its finiteness justifies subtraction, giving a uniform bound for M_κ. Since d_0 is fixed, this is (5.8); the chart transfer and (5.5) give the same bound for A_i.

Finally put $\eta(x)=\phi(\kappa x)$ and multiply the form bound by κ^{m-2}. It becomes

$$\int\kappa^{-2}I_i(y/\kappa)\phi(y)^2\,\mathrm{d}y\le C\int(|D\phi|^2+\phi^2)\,\mathrm{d}y.$$

A cutoff equal to one on $B_r(z)$, supported on its admissible fixed enlargement, and with derivative $O(r^{-1})$ gives (5.9) for $m=6$. □

This proof uses a fixed small relative error in rows, never an unproved $o(\kappa^2)$ absolute matching error. It remains valid when I_i/κ^2 is unbounded pointwise. The selected gap need not be a global minimum. Capacity here refers to a positive density consequence of the signed stability calculation; it is not positivity of the signed coupled operator.

5.2. **From row capacities to the full-normal source.**

5.2.1. *Uniform point-depth domination.* All point-depth estimates in this subsection are restricted to a smaller receiving plateau: $a_i(x)=1$, the whole depth-$2P$ fiber is retained, and a fixed neighborhood of that fiber lies in the larger genuine chart. The final tangential cutoff is supported inside this plateau. No point-depth conclusion is asserted at tapered or clipped receiving fibers. On this plateau, the fixed-depth preparation gives $g\ge c_P$, $W\ge c_P$ for $|r|\le P$, and bounded positive Jacobians. Consequently

$$\int_{-P}^{P}S_i(X_i(x,r))J_i(x,r)\,\mathrm{d}r\ge c_P>0.$$

Without the plateau assumption the left side could be arbitrarily small, and the unweighted point-depth potential cannot be controlled by the integrated row with a uniform constant.

An integrated-row capacity estimate must be transferred to each fixed normal depth before it can be applied to a full-normal source. We record a uniform proof. On $B_1 \subset \mathbb{R}^7$ the fractional Poisson and Dirichlet Green kernels are

$$P_s(z,y) = \pi_{7,s} \frac{(1-|z|^2)^s}{(|y|^2-1)^s |z-y|^7}, \qquad \pi_{7,s} = \frac{\Gamma(7/2)\sin(\pi s)}{\pi^{9/2}},$$

$$\mathcal{G}_s(x,z) = k_{7,s}|x-z|^{2s-7} \int_0^{(1-|x|^2)(1-|z|^2)/|x-z|^2} \frac{t^{s-1}}{(1+t)^{7/2}}\,\mathrm{d}t, \quad k_{7,s} = \frac{\Gamma(7/2)}{2^{2s}\pi^{7/2}\Gamma(s)^2}.$$

For $x \in B_{1/2}$, $y \notin \overline{B}_1$, these formulas imply

$$\int_{B_1} \mathcal{G}_s(x,z) P_s(z,y)\,\mathrm{d}z \le C P_s(x,y). \tag{5.10}$$

To check uniformity, in $|z| \le 3/4$ use $\mathcal{G}_s(x,z) \le C|x-z|^{2s-7}$ and a bounded Poisson ratio. For $|z| > 3/4$, write $d(z) = 1-|z|$; the Green formula improves to $\mathcal{G}_s(x,z) \le C d(z)^s$. The Poisson ratio is at most $C d(z)^s |x-y|^7/|z-y|^7$. If $1 < |y| < 2$, then $d(z) \le |z-y|$ and the product is at most $C|z-y|^{2s-7}$, whose integral is uniformly bounded. For $|y| \ge 2$ the ratio is bounded directly. Scaling adds a factor r^{2s} on B_r. These exact formulas are given, with their normalizations, in [4, Theorems 2.10 and 3.1–3.3]. Only these explicit formulas, not a fixed-order Harnack constant, are used here.

If $v \ge 0$ globally and $[(-\Delta)^s + c]v = 0$ on B_r, with $0 \le c \le \Lambda$, let h be its positive fractional-harmonic Poisson lift. Then $0 \le h - v = \mathcal{G}_s(cv) \le \Lambda \mathcal{G}_s h$. Equation (5.10) gives $h - v \le C\Lambda r^{2s} h$. Choose a fixed r with $C\Lambda r^{2s_*} \le 1/2$. Thus $h/2 \le v \le h$ on the half-ball. The explicit Poisson ratio bounds $\sup h / \inf h$ there uniformly in s. A fixed finite chain therefore gives a uniform Harnack estimate.

The preceding Poisson and Green representations apply to bounded nonnegative solutions with bounded forcing on the ball, which is the class used here. At $s = 1$ use the classical ball Poisson and Green representations: the same interior/boundary split proves the conditional-Green estimate and hence the same Harnack conclusion. For $s < 1$ all constants above are uniform for $s \ge s_* > 1/2$.

Apply the estimate to $v_k = G_V S_k \ge 0$ along a receiving plateau fiber, $k \ne i$. Its fixed-radius neighborhoods are donor-free because $\delta \to \infty$, and there $[(-\Delta)^s + 2 - V]v_k = 0$, with $0 \le 2 - V \le 2$. A fixed finite chain through the depth-$2P$ fiber yields

$$\sup_{|r| \le 2P} v_k(X_i(x,r)) \le C_P \inf_{|r| \le P} v_k(X_i(x,r)).$$

Multiplication by the positive receiving density and integration over $[-P,P]$ therefore give

$$\sup_{|r| \le 2P} (G_V S_k)(X_i(x,r)) \le C_P B_{ik}(x). \tag{5.11}$$

No division by the edge cutoff ψ occurs. For each fixed receiving base point x and fixed donor-base center z_0, apply the same argument to $G_V(|y-z_0|S_k)$, with y the donor base coordinate. First truncate the nonnegative weight; its source is bounded and still donor-free near the receiving fiber. Monotone convergence, and the established finite first-row moment, then give

$$\sup_{|r| \le 2P} \int G_V(X_i(x,r),Y)|y-z_0|S_k(Y)\,\mathrm{d}Y \le C_P \int |y-z_0|K_{ik}(x,y)\,\mathrm{d}y.$$

Uniformly bounded receiving factors preserve these value bounds. Whenever their derivatives are used below, the factors also have uniformly bounded physical derivatives of every required fixed order.

We distinguish the fixed chart coordinates used for integration from the receiver tangent frame used by the sharp angle estimate. Let $Y_i(x) = (x, h_i(x))$ and let $\nu_i(x)$ be its upward geometric normal. For a retained branch j, let

$$Q_j(x) = Y_i(x) + \tau_j(x)\nu_i(x) \in \Sigma_j, \qquad \Phi_j(x) = \pi Q_j(x), \qquad \tau_i = 0,$$

where π is the fixed chart's horizontal projection. The larger genuine buffers retain these intersections and every intermediate branch, with a fixed horizontal margin exceeding the normal-ray horizontal displacement, the fresh-chart test radius, and the depth-$2P$ offsets. The fixed parameters are chosen so these margins lie inside the supplied complete graph window. Work first on one side of i, with $0 < \tau_{i+1} < \cdots$ and $\tau_k \le \rho/\kappa$; the other side is identical with the reversed unit normal. The small slopes and the normal transversality give uniqueness and smoothness of each intersection within this buffer. Its normal equation differentiates to

$$D\tau_j = -\frac{\nu_j(Q_j)\cdot(DY_i + \tau_j D\nu_i)}{\nu_j(Q_j)\cdot\nu_i}.$$

If all fixed-chart slopes are at most ε, then $|D\tau_j| \le C(\varepsilon+\rho)$, because DY_i is tangent to ν_i, $|D\nu_i| \le C\kappa$, and the denominator is bounded below. Since

$$D\Phi_j = I + \nu_i^{\mathrm{hor}} \otimes D\tau_j + \tau_j D\nu_i^{\mathrm{hor}}, \qquad \|D\Phi_j - I\| \le C(\varepsilon^2+\rho),$$

choose the fixed slope and curvature thresholds so the last bound is less than $1/2$. These maps and their inverses are uniformly bi-Lipschitz on the smaller common base, with bounded Jacobians. Their rescaled versions have the same bounds. This construction is localized when a target intersection approaches the curvature-distance boundary. Choose a smooth ζ equal to one on $[0,1]$ and zero on $[2,\infty)$, and for a fixed target k use

$$\chi_k(x) = \zeta(\kappa\tau_k(x)/\rho).$$

The larger genuine buffer retains the intersections up to $3\rho/\kappa$; thus χ_k extends by zero before any intersection domain ends, and no formal surface extension is used. On its support all intermediate intersections exist, with $0 \le \tau_j \le 2\rho/\kappa$; use the preceding derivative bounds with 2ρ in place of ρ. In particular $|D\chi_k| \le C\kappa$, with a fixed constant depending on ρ. The bi-Lipschitz claim also holds if this localized set is not convex: compare the two intersection equations at x, x'. The Lipschitz bounds of the actual graphs and $|\tau_j| \le 2\rho/\kappa$ give $|\tau_j(x)-\tau_j(x')| \le C(\varepsilon+\rho)|x-x'|$, and then

$$|\Phi_j(x) - \Phi_j(x') - (x-x')| \le C(\varepsilon^2+\rho)|x-x'|.$$

This uses graph values in the larger convex common base, not an unprovided intersection at every point of the connecting segment. All transfers below use tests multiplied by χ_k. Only rows with $D_{|k-i|} \le \rho/\kappa$, where $\chi_k = 1$, enter the retained estimates. Complementary actual rows are part of the single unsigned $o(\kappa^2)$ spatial remainder.

In this subsection set

$$D_\ell(x) = \tau_{i+\ell}(x), \qquad \widehat{d}_j(x) = \tau_{j+1}(x) - \tau_j(x), \qquad D_\ell = \sum_{j=i}^{i+\ell-1} \widehat{d}_j, \qquad d_\ell = \ell\delta, \quad t_\ell = D_\ell - d_\ell.$$

Here is the intrinsic-separation check in this rotated frame. The endpoints Q_j, Q_{j+1} have equal projection onto $\nu_i(x)^\perp$. If a unit-speed minimizing intrinsic geodesic of length $L < a_{\mathrm{geom}}/\kappa$ joined them, its initial tangent would have projection of norm at least $(1-C\varepsilon^2)^{1/2}$, because the initial sheet normal is within $C\varepsilon$ of ν_i. The curvature bound gives $|\gamma''| \le \kappa$, and hence

$$L(1-C\varepsilon^2)^{1/2} \le \tfrac{1}{2}\kappa L^2,$$

contradicting the fixed small choice of a_{geom}. Completeness supplies such a minimizing geodesic if the intrinsic distance were below that threshold; distinct components have infinite intrinsic distance. Thus these endpoints are intrinsically off diagonal. Their Euclidean separation is exactly $\widehat{d}_j$, so the definition of δ gives $\widehat{d}_j \ge \delta$. Thus D_ℓ is the height in the tangent frame of its receiver; it is not identified with the vertical height in the old fixed chart. The Section 4 sharp and rough estimates apply to this D_ℓ. At the donor, its same-base point is $Q_k(x)$, with old coordinate $\Phi_k(x)$. The source displacement weight is accordingly $|y - \Phi_k(x)|$. Projection onto the receiver tangent plane gives a uniformly bi-Lipschitz coordinate on the donor graph centered at $Q_k(x)$, so this weight is comparable to its receiver-frame tangential radius. The established first-moment estimate and its point-depth version therefore hold with this weight.

In the physical-value and later source splits below, "same-base" and "displacement" refer to these points and this weight.

We justify the scalar capacity transfer needed below. For each intermediate pair, $Q_j(x)Q_{j+1}(x)$ is an actual segment of length $\widehat{d}_j(x)$. Transfer to a fresh marked chart on sheet j, using Φ_j and the fixed finite chart partition from the capacity proof. On the support of the intermediate-gap cutoff specified below, the depth-one positive lower-row construction at the transported receiver gives

$$e^{-k\widehat{d}_j} + \lambda \widehat{d}_j^{-p-1} \le C I_j^{\mathrm{fresh}}(Q_j) + C\kappa^2.$$

The fresh row is evaluated in its own tangential coordinate. For the exponential part use the actual source point Q_{j+1} at distance $\widehat{d}_j$: on the tangent half-disk of radius $\sqrt{\widehat{d}_j}$ favorable to the receiver, endpoint distance does not increase by more than $O(1) + O(\kappa\widehat{d}_j)$. The favorable linear term may shorten the distance by more than this amount, which only improves the positive kernel lower bound. On logarithmic gaps $\kappa\widehat{d}_j \to 0$, so the resonant lower calculation in (5.5) applies. For larger gaps that exponential is $O(\kappa^2)$. The algebraic lower bound uses a disk at scale $\widehat{d}_j$ and bounded geometric distortion. If an intermediate gap exceeds the small fresh-chart scale d_0/κ, its whole scalar potential is already $O(\kappa^2)$. To implement this split, multiply the transferred test also by $\theta_j = \chi(\kappa\widehat{d}_j/d_0)$ as in the earlier chart transfer. Its derivative is $O(\kappa/d_0)$, since $|D\widehat{d}_j| \le C(\varepsilon + \rho)$; the complementary potential is $O(\kappa^2)$. Thus the fresh donor disk is required only where it lies inside the guaranteed plateau. The fresh pair remains adjacent in the same zero-free chamber. This argument does not replace a gap by a fixed multiplicative approximation inside its exponential.

Apply the fresh form capacity to the transported tests $\theta_j\chi_k\eta$ times the fixed chart partition, and bound the complementary scalar potential by $C\kappa^2$. The derivative bound for χ_k adds at most $C\kappa^2\int\eta^2$ to the physical energy. Uniform coordinate Jacobians and distortion therefore give, after a ball cutoff,

$$\int_{B_r(z)} \chi_k(y/\kappa)^2\kappa^{-2}\{e^{-k\widehat{d}_j(y/\kappa)} + \lambda\widehat{d}_j(y/\kappa)^{-p-1}\}\,\mathrm{d}y \le C(r^{m-2} + r^m) \tag{5.12}$$

on every smaller admissible rescaled ball and for every intermediate gap j of this target. Only the support of χ_k is transported; the maps need not exist on the whole test ball. Explicitly, its image under each rescaled coordinate map is contained in the image of a uniformly controlled local chart, and the image of a radius-r ball under the rescaled map lies in a radius-Cr ball; change variables with its bounded Jacobian and apply the fresh capacity bound. For the largest admissible radii use the same fixed finite chart cover. The $C\kappa^2$ comparison error contributes Cr^m. All constants are independent of the intermediate label. In the nonadjacent-row capacity calculation below, Jensen and the exponential product inequality use precisely these increments $\widehat{d}_j$ and these localized capacities. Scalar envelopes used below are restricted to the same retained set $D_{|k-i|} \le \rho/\kappa$. The composite target potentials on $D_{|k-i|} \ge \rho/\kappa$ are $e^{-kD_{|k-i|}} + \lambda D_{|k-i|}^{-p-1} = O(\kappa^2)$, but no such smallness is asserted for a single intermediate gap there. They are not summed as per-label errors: the complementary actual source was already placed in the one unsigned exterior remainder.

5.2.2. *Physical value and smooth jets of the entire source.* Let $\mathcal{R}_i$ denote the full signed cross difference, with a uniformly bounded receiving multiplier b_i having the uniform fixed-order smooth jets required below:

$$\mathcal{R}_i(X) = b_i(X)\sum_{k\ne i}\sigma_i\sigma_k\int G_V(X,Y)S_k(Y)[N_i(X) - N_k(Y)]\,\mathrm{d}Y. \tag{5.13}$$

The final fixed-profile choice will be specified exactly in (5.41); it is not an arbitrary change of source convention. Using (4.39), split the normal difference into its same-base part and

$C_P\kappa|y-\Phi_k(x)|$. In the logarithmic range

$$\sqrt{t}\,e^{-k(d+t/2)} \le Ce^{-kd}, \qquad \sqrt{D-d}\,D^{-p-1} \le Cd^{-p-1/2}.$$

Above that range choose its fixed cutoff sufficiently large and use (4.40); its exponential sum is $o(\kappa^{3/2})$. Summing all labels and retaining the spatial exterior gives

$$\begin{aligned}\|\mathcal{R}\|_\infty &\le C_P\sqrt{\kappa}\{e^{-k\delta}+\lambda\delta^{-p-1/2}\} \\ &\quad + C_P\kappa\{(1+\delta)^{L'}e^{-c\delta}+\lambda\delta^{-p}\}+o(\kappa^{3/2})=o(\kappa^{3/2}).\end{aligned} \tag{5.14}$$

For the last exponential term use $\delta \ge k^{-1}\log(1/\kappa)$ eventually and $c/k > 1/2$; the strict rate margin absorbs every fixed polynomial.

The required high derivatives concern the entire smooth source, not its individual positive majorants. If $h, V \in C_b^N$ and $0 \le V \le 2-\mu$, then $v = G_V h$ satisfies

$$A_V(\tau_z v - v) = \tau_z h - h + (\tau_z V - V)\tau_z v, \qquad \|G_V\|_{\infty\to\infty} \le \mu^{-1}. \tag{5.15}$$

For the differentiability argument take $h, V \in C_b^{N+1}$, with $0 \le V \le 2-\mu$ and fixed positive μ. Their derivatives through order N are uniformly continuous. First the undivided identity (5.15) gives

$$\|\tau_z v - v\|_\infty \le \mu^{-1}\big(\|\tau_z h - h\|_\infty + \|\tau_z V - V\|_\infty \|v\|_\infty\big) \longrightarrow 0.$$

Thus v is uniformly continuous before any derivative is asserted. For $z = te_j$, divide the identity by t. Its right side converges uniformly to $\partial_j h + (\partial_j V)v$, since $\tau_{te_j} v \to v$ uniformly. The bounded operator G_V therefore shows uniform convergence of the difference quotients of v. Their limit is its classical bounded continuous derivative. Repeat the argument on the differentiated equations; at each order the right side is uniformly continuous by the preceding orders and the extra derivative of h, V. This proves

$$A_V D^\alpha v = D^\alpha h + \sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} (D^\beta V) D^{\alpha-\beta} v, \qquad \|v\|_{C_b^N} \le C_{N,\mu,V}\|h\|_{C_b^N}. \tag{5.16}$$

The constant depends on the bounded derivatives of V through order N, but not on s. The additional derivative is used for uniform continuity, not in the displayed quantitative bound. The actual smooth cutoffs and tube sources satisfy these assumptions for every fixed N. In particular, the final multiplier $-f/g$ has uniform fixed-order jets on the full receiving plateau because $g \ge c_P$ there and f has uniform fixed-profile jets.

Disjoint smooth tube supports imply

$$\left\|\sum_k \sigma_k S_k\right\|_{C_b^N} + \left\|\sum_k \sigma_k S_k N_k\right\|_{C_b^N} \le C_{P,N},$$

independently of the number of labels. The latter ambient source is smooth through the exterior critical set because each term is supported where $g > 0$. Put $H_{\neq i} = \sum_{k\neq i}\sigma_k S_k$ and $Z_{\neq i} = \sum_{k\neq i}\sigma_k S_k N_k$. Then

$$\mathcal{R}_i = b_i\sigma_i\{N_i G_V H_{\neq i} - G_V Z_{\neq i}\}.$$

Equations (5.16) and the product and pullback rules prove $\|\mathcal{R}_i \circ X_i\|_{C^N_{x,r}} \le C_{P,N}$. The exact exterior vector source is $G_V(W_{\mathrm{ext}}\nabla u)$; it is globally smooth and has the same jets. One never differentiates a characteristic function or extends $\nabla u/|\nabla u|$ through a critical point. A bounded smooth fiber projection preserves these estimates by differentiating its finite-interval defining integral.

5.2.3. *A direct capacity-to-source estimate.* Let e be a compactly supported density in a fixed rescaled chart, with values in a Banach space $\mathbb{B}$; below $\mathbb{B} = \mathbb{R}^7$, $H = L^2(C;\mathbb{R}^7)$, or $X = C(C;\mathbb{R}^7)$. Suppose $v \ge 0$ is a locally integrable density with capacity constant K on admissible balls. Localize strictly inside an admissible chart, with a fixed positive margin to its boundary. The local capacity estimates imply, after zero extension of a slightly larger cutoff of the density, that

$$\int_{B_r(z)} v\,\mathrm{d}y \le K(r^{m-2}+r^m), \quad 0<r\le 1, \qquad \int v\,\mathrm{d}y \le CK, \tag{5.17}$$

for all centers z, with a fixed change of K. Indeed a small ball meeting the support is contained in a twice-larger admissible ball centered on that support. Balls of radius above a fixed fraction of the margin, and the total mass, are controlled by a fixed finite cover. Thus no capacity estimate outside its allowed chart is assumed.

Suppose $A, a, h, K > 0$, $\alpha \in [\alpha_*, 1]$ with $\alpha_* > 1/2$, and

$$\|e(y)\|_{\mathbb{B}} \le \frac{A}{h} \min\left\{1, \left(\frac{h^2 v(y)}{a}\right)^{\alpha}\right\}. \tag{5.18}$$

For a fixed inhomogeneous Littlewood–Paley partition define

$$\|e\|_{\mathcal{D}^{-1}(\mathbb{B})} = \|S_{-1}e\|_{L^\infty \mathbb{B}} + \sum_{j\ge 0} 2^{-j} \|\Delta_j e\|_{L^\infty \mathbb{B}}.$$

Put $r = 2^{-j}$. The absolute Schwartz kernel of Δ_j has bounded mass; its shell decomposition and (5.17) give $\||K_j| * v\|_\infty \le CKr^{-2}$. For shells above radius one use $\int v \le CK$ and rapid kernel decay. Jensen's inequality for the concave function t^α, and the triangle inequality for the Banach norm, yield

$$r\|\Delta_j e\|_{L^\infty \mathbb{B}} \le C \min\left\{\frac{Ar}{h}, Aa^{-\alpha} h^{2\alpha-1} K^\alpha r^{1-2\alpha}\right\}.$$

The two bounds meet at $r_0 = h\sqrt{K/a}$, with common value $A\sqrt{K/a}$. If $r_0 \le 1$, sum the amplitude bound for $r \le r_0$ and the capacity bound for $r \ge r_0$; both are geometric series with constants uniform for $\alpha \ge \alpha_*$. If $r_0 > 1$, the amplitude bound alone suffices. For the low block, the same mass and Jensen argument at scale one gives

$$\|S_{-1}e\|_{L^\infty \mathbb{B}} \le C(A/h) \min\{1, r_0^{2\alpha}\} \le C(A/h) r_0.$$

Consequently

$$\|e\|_{\mathcal{D}^{-1}(\mathbb{B})} \le CA\sqrt{K/a}. \tag{5.19}$$

All localization factors and margins are fixed in rescaled variables. A larger cutoff of v, equal to one on the support of the cutoff of e, preserves (5.18). No derivative of e or of its subsequent measurable pieces is used.

Throughout the following capacity and package calculations, each retained row and its vector-source pieces are restricted to $D_\ell \le \rho/\kappa$ and extended by zero outside that set. Their complementary actual source is included in the single unsigned spatial remainder already specified.

For label distance $\ell \ge 1$, use the receiver-frame gaps D_ℓ, $d_\ell = \ell\delta$, and $t_\ell = D_\ell - d_\ell$ defined above, and put

$$a_\ell^e = \kappa^{-1} e^{-k d_\ell}, \qquad a_\ell^a = \kappa^{-1} \lambda d_\ell^{-p}. \tag{5.20}$$

The density $\kappa^{-2} B_\ell(y/\kappa)$ has capacity constant

$$\widehat{K}_1 \le C, \qquad \widehat{K}_\ell \le C\{\ell^{-p-1} + e^{-\gamma(\ell-1)\delta}\} \quad (\ell \ge 2). \tag{5.21}$$

For the adjacent row this follows from $B_{i,i+1} \le 2I_i$ and Theorem 5.2. For $\ell \ge 2$, use the rough row, absorb its polynomial into $e^{-c'D}$ with $k/2 < c' < \min(c, k)$, and apply

$$D_\ell^{-p-1} \le \ell^{-p-2} \sum_{j=i}^{i+\ell-1} \widehat{d}_j^{-p-1}, \qquad e^{-c' \sum_{j=i}^{i+\ell-1} \widehat{d}_j} \le e^{-(c'\ell - k)\delta} \frac{1}{\ell} \sum_{j=i}^{i+\ell-1} e^{-k\widehat{d}_j}.$$

The receiver-frame scalar capacities (5.12) (applied with the target cutoff, which is one on the retained set) yield (5.21) with $\gamma = 2c' - k > 0$. Summed spatial truncation terms remain in the total $o(\kappa^2)$ exterior, rather than becoming uncontrolled per-label remainders.

There are two distinct uses of capacity. For actual-row pieces one uses the capacity of $\kappa^{-2}B_\ell$. For displacement envelopes one needs scalar-potential capacities, which do not follow merely from domination of the actual row by an envelope. We record them directly from (5.12), whose chart transfer was justified above. Write $D_\ell(x) = \sum_{j=i}^{i+\ell-1} \widehat{d}_j(x)$ and set, on the retained rescaled set $E_\ell = \{y : D_\ell(y/\kappa) \le \rho/\kappa\}$,

$$v_\ell^a(y) = \kappa^{-2} \lambda D_\ell(y/\kappa)^{-p-1}, \qquad v_\ell^e(y) = \kappa^{-2} e^{-k D_\ell(y/\kappa)}.$$

Extend these densities by zero off E_ℓ and then apply the fixed tangential localization. In the following pointwise inequalities the right sides are multiplied by $\chi_{i+\ell}(y/\kappa)^2$, which is one on E_ℓ; hence all intermediate quantities are used only on their genuine domains. The convex-power inequality gives

$$v_\ell^a \le \ell^{-p-2} \sum_{j=i}^{i+\ell-1} \kappa^{-2} \lambda \widehat{d}_j^{-p-1}.$$

Also, for every one of these adjacent gaps, $e^{-kD_\ell} \le e^{-k(\ell-1)\delta} e^{-k\widehat{d}_j}$; averaging yields

$$v_\ell^e \le \frac{e^{-k(\ell-1)\delta}}{\ell} \sum_{j=i}^{i+\ell-1} \kappa^{-2} e^{-k\widehat{d}_j}.$$

Thus their capacity constants are respectively

$$K_\ell^a \le C\ell^{-p-1}, \qquad K_\ell^e \le Ce^{-k(\ell-1)\delta}. \tag{5.22}$$

Arguments of $\widehat{d}_j$ in the right sides are y/κ. All inequalities hold on the same admissible rescaled support; the preceding cutoff procedure supplies their global localized form. These are capacities of the scalar comparison potentials themselves. They will be used for the two first-moment families only.

We specify how scalar row bounds decompose the actual vector source. Fix the receiving label i and a retained donor k, and write $Y = X_k(y,q)$. In the integral defining its contribution to $\mathcal{R}_i$, split exactly

$$N_i(x,r) - N_k(y,q) = [N_i(x,r) - N_k(\Phi_k(x),q)] + [N_k(\Phi_k(x),q) - N_k(y,q)].$$

Denote the resulting actual vector contributions, including the signs and receiving multiplier, by F_{ik}^{base} and F_{ik}^{disp}. All coordinates used here belong to the common genuine base; omitted spatial pieces are retained separately in the single unsigned exterior remainder. The point-depth bounds give

$$\|F_{ik}^{\mathrm{base}}(x,\cdot)\|_X \le C_P(\sqrt{\kappa}\sqrt{t_\ell} + \kappa) B_\ell(x), \qquad \|F_{ik}^{\mathrm{disp}}(x,\cdot)\|_X \le C_P \kappa \{m_\ell^e(x) + m_\ell^a(x)\},$$

where

$$m_\ell^e = (1+D_\ell)^L e^{-cD_\ell}, \qquad m_\ell^a = \lambda D_\ell^{-p}.$$

After defining $B_\ell^e, B_\ell^a, B_\ell^{\mathrm{rem}}$ below, define the corresponding base-source pieces by

$$F_{ik}^{\mathrm{base},\tau} = \frac{B_\ell^\tau}{B_\ell} F_{ik}^{\mathrm{base}}, \quad \tau = e, a, \mathrm{rem}.$$

Set them to zero where $B_\ell = 0$; the point-depth estimate implies $F_{ik}^{\mathrm{base}} = 0$ there. The three pieces sum exactly to the base source and have the same bounds with B_ℓ replaced by B_ℓ^τ. For the displacement source set

$$F_{ik}^{\mathrm{disp},\tau} = \frac{m_\ell^\tau}{m_\ell^e + m_\ell^a} F_{ik}^{\mathrm{disp}}, \quad \tau = e, a,$$

with the same zero convention. These sum exactly to the displacement source and have bounds $C_P \kappa m_\ell^\tau$. The weights depend measurably on x and are scalar in the normal variable, so the pieces take values in X and also in H. No derivative of a weight or piece is asserted. The smooth-jet argument applies to their entire sum by its original resolvent formula, independently of this measurable decomposition.

Use the actual upper row on its logarithmic range to make a measurable split

$$B_\ell^e = \min\{B_\ell, C_P e^{-k(d_\ell + t_\ell/2)}\}, \qquad B_\ell^a = B_\ell - B_\ell^e \le C_P \lambda D_\ell^{-p-1}. \tag{5.23}$$

The two-way exponential/algebraic split is made only when $D_\ell \le K_0 L_0$. On its complement define

$$B_\ell^e = 0, \qquad B_\ell^a = \min\{B_\ell, C_P \lambda D_\ell^{-p-1}\}, \qquad B_\ell^{\mathrm{rem}} = B_\ell - B_\ell^a.$$

On the logarithmic range put $B_\ell^{\mathrm{rem}} = 0$ and retain the previous split. Every B_ℓ^a is bounded by the actual row and the algebraic envelope. Choose $c_1 \in (k/2, c)$ and absorb the fixed polynomial

in the rough exponential into $e^{-c_1 D}$. Increase the fixed K_0, if needed, so that $c_1 K_0 > 2$, and make the already specified normal-buffer and fixed-depth choices in that order. Along either side of the ordered stack, consecutive values of D_ℓ differ by at least δ. Hence

$$\sum_\ell B_\ell^{\mathrm{rem}} \le \frac{C_P e^{-c_1 K_0 L_0}}{1 - e^{-c_1 \delta}} = o(\kappa^2).$$

The unit-normal differences and receiving multipliers are bounded, so the sum of the base-source remainder pieces is $o(\kappa^2)$ in X and H. Combine it with the unsigned spatial exterior into one bounded physical remainder. Its compactly localized version, multiplied by κ^{-2}, has $\mathcal{D}^{-1}$ norm $o(1)$ by the amplitude estimate and the geometric sum $\sum_{j\ge 0} 2^{-j} < \infty$.

The retained base-source pieces inherit actual-row capacity. If $\lambda = 0$, all algebraic pieces below vanish and are omitted; no division by λ or a zero package parameter is made. For the exponential same-base source, put $u = B_\ell^e/(\kappa a_\ell^e)$. The upper envelope gives $u \le Ce^{-kt_\ell/2}$ and hence, for any fixed $\alpha \in (1/2, 1)$,

$$u\sqrt{1+t_\ell} \le C \min(1, u^\alpha).$$

The normalized source $\kappa^{-3/2} B_\ell^e \sqrt{1+t_\ell}$ satisfies (5.18) with $A = C_P a_\ell^e$, $a = a_\ell^e$, $h_\ell = \sqrt{\kappa}$ and $v = \kappa^{-2} B_\ell^e$. Its budget is $C\sqrt{a_\ell^e \widehat{K}_\ell}$. For the power part, set $u = B_\ell^a d_\ell^{p+1}/\lambda$. Then

$$\frac{\kappa^{-3/2} B_\ell^a \sqrt{D_\ell}}{a_\ell^a/\sqrt{\kappa d_\ell}} = u\sqrt{D_\ell/d_\ell} \le C \min(1, u^\beta), \qquad \beta = \frac{p+1/2}{p+1} > \frac{1}{2}.$$

This gives $A = C_P a_\ell^a$, $a = a_\ell^a$, $h_\ell = \sqrt{\kappa d_\ell}$, $v = \kappa^{-2} B_\ell^a$, and budget $C\sqrt{a_\ell^a \widehat{K}_\ell}$. The harmless κB_ℓ normal-depth term is absorbed in the same-base bound.

The displacement first moments require two further families of estimates. For the power term, after rescaling it is $C a_\ell^a (d_\ell/D_\ell)^p$. Up to curvature distance $h_\ell^2 = \kappa d_\ell \le C$, so it satisfies (5.18) with $A = C a_\ell^a$, $a = a_\ell^a$, $\alpha = p/(p+1) > 1/2$, and density $v = \kappa^{-2} \lambda D_\ell^{-p-1}$. For the exponential first moment, choose $1/2 < \alpha < \min(c/k, 1)$ and use

$$\kappa^{-1}(1+D_\ell)^L e^{-cD_\ell} \le C\kappa^{-1}(1+d_\ell)^L e^{-cd_\ell} e^{-\alpha k t_\ell}.$$

Set $h_\ell = \sqrt{\kappa}$, $a = a_\ell^e$, $A = C\kappa^{-1/2}(1+d_\ell)^L e^{-cd_\ell}$, and $v = \kappa^{-2} e^{-kD_\ell}$. Its scalar comparison capacity constant is K_ℓ^e from (5.22), at most $Ce^{-k(\ell-1)\delta}$. The budget is

$$C(1+d_\ell)^L e^{-(c-k/2)d_\ell} e^{-k(\ell-1)\delta/2}.$$

These estimates are justified pointwise through the receiving fiber by (5.11) and its weighted version. Bounded receiving normalizations and fixed fiber projections preserve them.

Theorem 5.3 (Vanishing label-summable direct budget)**.** *For the compactly localized normalized full-normal source $e(y) = \kappa^{-2}\mathcal{R}(y/\kappa)$, all retained label pieces satisfy estimates (5.17)–(5.18), plus a summed $o(1)$ bounded compactly supported remainder, such that*

$$\sum_\nu A_\nu \sqrt{K_\nu/a_\nu} \le C\left\{\sqrt{\kappa^{-1}e^{-k\delta}} + \sqrt{\kappa^{-1}\lambda\delta^{-p}} + (1+\delta)^L e^{-(c-k/2)\delta}\right\} + o(1) = o(1). \tag{5.24}$$

All capacity estimates are translation uniform on smaller admissible charts. The same conclusion holds in H and X and after bounded smooth receiving multipliers.

Proof. The two base-source families use the actual subrows and hence $K \le \widehat{K}_\ell$; the two displacement families use (5.22), not the actual-row capacity. For the algebraic displacement family the estimate with $A = Ca_\ell^a$ follows because $h_\ell = \sqrt{\kappa d_\ell}$ is bounded above on the retained curvature range; its bound is absorbed into the fixed constant in A. For clarity, the four summed budgets

are bounded as follows:

$$\sum_{\ell\ge1}\sqrt{a^e_\ell\widehat{K}_\ell}\le C\sqrt{\kappa^{-1}e^{-k\delta}},$$
$$\sum_{\ell\ge1}\sqrt{a^a_\ell\widehat{K}_\ell}\le C\sqrt{\kappa^{-1}\lambda\delta^{-p}},$$
$$\sum_{\ell\ge1}\sqrt{a^a_\ell K^a_\ell}\le C\sqrt{\kappa^{-1}\lambda\delta^{-p}}\sum_{\ell\ge1}\ell^{-p-1/2},$$
$$\sum_{\ell\ge1}(1+d_\ell)^L e^{-(c-k/2)d_\ell}e^{-k(\ell-1)\delta/2}\le C(1+\delta)^L e^{-(c-k/2)\delta}.$$

The second line uses the same polynomial series plus a geometric series from the exponential part of $\widehat{K}_\ell$. All constants are uniform for $p\ge p_*>3/2$. The first two right sides vanish by $F=o(\kappa)$, and the last vanishes since $c>k/2$ and $\delta\to\infty$. The measurable source decompositions above make this an estimate for the actual signed vector source, not a scalar replacement of it. The direct lemma and the single bounded remainder therefore prove (5.24), in both H and X. Only bounded receiving multipliers are needed for this value-and-capacity assertion; their uniform smooth jets are additionally required for the next assertion. □

At physical frequencies $t=2^l\ge1$, the full smooth source of amplitude $A_\kappa=o(\kappa^{3/2})$ obeys

$$\kappa^{-1}\sum_{l\ge0}2^l\min\{A_\kappa,C_N2^{-Nl}\}\le C_N\kappa^{-1}A_\kappa^{1-1/N}=o(1),\qquad N>3. \tag{5.25}$$

Here the tangential cutoff is smooth, supported strictly inside the receiving plateau, and has the form $\chi(\kappa x)$. The entire localized source extends by zero and retains both its small amplitude $A_\kappa=o(\kappa^{3/2})$ and its bounded fixed-order physical jets. Schwartz-kernel cancellation consequently gives, in X or H, $\|\Delta_l^{\mathrm{phys}}\mathcal{R}\|_\infty\le C\min(A_\kappa,C_N2^{-Nl})$. Splitting the series at $2^l\simeq(C_N/A_\kappa)^{1/N}$ proves (5.25). More explicitly,

$$\kappa^{-1}A_\kappa^{1-1/N}=\left(\frac{A_\kappa}{\kappa^{3/2}}\right)^{1-1/N}\kappa^{(N-3)/(2N)}=o(1),\qquad N>3.$$

No derivative bound is used for a separate measurable label piece. All smooth derivatives concern the actual full source before its capacity decomposition.

Conversion of the direct budget to a variable-coefficient phase-gradient estimate additionally requires the same-gauge inverse and Schur mapping estimates in the subsequent operator argument. The scalar capacity calculation by itself does not provide those operator bounds.

5.3. **Self comparison and the normal equation.**

5.3.1. *Actual potential comparison on the full insertion region.* Fix the self sheet 0. Its actual smooth phase potential on the plateau is $V_a=(1-\sum_i\psi(\rho_i)^2)W$ and its true-distance one-sheet comparison is $V_d=(1-\psi(\rho_0)^2)W_s(\rho_0)$. The same reference used for (4.24) gives

$$|W-W_s(\rho_0)|\le C_M\kappa+C_M\sum_{i\ne0}T(\mathrm{dist}(X,\Sigma_i)),\quad T(t)=e^{-kt}+\lambda(1+t)^{-p}. \tag{5.26}$$

This is a comparison of squared phases. In chambers adjacent to sheet 0, the saturated constants of all other profiles cancel and leave $\pm q_s(\rho_0)$. In another chamber, both squared expressions are close to one; a segment to the true self foot crosses an intervening sheet, whose distance is at most $|\rho_0|$. Thus its tail also bounds the own-profile tail there. This proves (5.26) from boundedness and $|u-Q|\le C_M\kappa$. The exact cutoff algebra is

$$V_a-V_d=(1-\sum_i\psi(\rho_i)^2)(W-W_s(\rho_0))-\sum_{i\ne0}\psi(\rho_i)^2W_s(\rho_0).$$

On a remote cutoff support its own tail is bounded below by a positive P-dependent constant, so the same right-hand side as in (5.26) bounds $|V_a - V_d|$. The remote cutoff discrepancy is therefore retained.

For the actual restricted graphs,

$$d_i^{\mathrm{set}} = \operatorname{dist}(\Sigma_0, \Sigma_i) \geq \max\{\delta, c_*|i|\delta\}. \tag{5.27}$$

The second bound follows by minimizing $|x-y|^2 + (|i|\delta - \theta|x-y|)_+^2$ on a common small-slope base. For the exact first bound, use the all-pair separation argument preceding (4.11). It also applies to these restricted curvature graphs: choose ρ after the fixed buffer multipliers so that the common working-base diameter is much smaller than a_{geom}/κ. The tangent-chart construction there, using the uniform small tilt, gives a larger horizontal graph through each retained point, defined over the common base and containing its intrinsic ball of radius a_{geom}/κ. A shorter joining geodesic would remain in this larger graph. Unique continuation of the embedded graph germ on the connected common base then identifies its endpoint with the starting branch, contradicting distinctness of the two retained branches. Thus every such pair is intrinsically off diagonal, and (2.5) gives its ambient separation at least δ. This keeps the exact exponential rate. These set distances are never asserted for independent extensions.

Enlarge δ_P, still with $\delta_P \to 0$, to bound both the actual phase potential and the one-sheet potential. Extend the latter boundedly outside its genuine domain, with values in $[0, \delta_P]$; differences outside the controlled insertion region are charged to (5.29) below. For the corresponding positive resolvents,

$$R_a - R_d = R_a(V_a - V_d)R_d, \qquad 0 \leq R_a, R_d \leq G_s^{2-\delta_P}.$$

Both endpoints return to the same fixed-depth self core C_0. Use the weighted row norm

$$\|E\|_{1,\mathrm{row}} = \sup_{X \in C_0} \int_{C_0} (1 + |X - Y|)|E(X,Y)| \, \mathrm{d}Y.$$

Write $G = G_s^{2-\delta_P}$ and, for $a = 0, 1$, put

$$F_a(Z) = \int_{C_0} |Z - Y|^a G(Z - Y) \, \mathrm{d}Y.$$

At true distance t from the self sheet, the codimension-one ball bound for the core gives

$$F_0(Z) \leq C_P\{e^{-ct} + \lambda(1+t)^{-p-1}\}, \qquad F_1(Z) \leq C_P\{e^{-ct} + \lambda(1+t)^{-p}\}.$$

On the genuine distance levels $\Gamma_t^{\pm}$, their uniform area bound gives the same two estimates for $\int_{\Gamma_t^{\pm}} G(X - Z) \, \mathrm{d}\mathcal{H}^6(Z)$ and $\int_{\Gamma_t^{\pm}} |X - Z| G(X - Z) \, \mathrm{d}\mathcal{H}^6(Z)$, respectively. The fixed endpoint depths are absorbed in C_P, and the exponential rate is reserved so that $2c > k$. These level estimates are used only in the retained genuine domain; their nonnegative majorants may be integrated over all $t \geq 0$.

Splitting $|X - Y| \leq |X - Z| + |Z - Y|$ and applying coarea therefore gives the returning insertion majorant

$$K_P(t) \leq C_P\{e^{-2ct} + \lambda e^{-ct}(1+t)^{-p} + \lambda^2(1+t)^{-1-2p}\}. \tag{5.28}$$

For bounded t, use the bounded ambient convolution and first moment; no singular pointwise assertion on a distance level is required. Since $\operatorname{dist}(Z, \Sigma_i) \geq (d_i^{\mathrm{set}} - t)_+$,

$$\int_0^\infty K_P(t) T((d-t)_+) \, \mathrm{d}t \leq C_P\{e^{-kd} + \lambda(1+d)^{-p}\}.$$

Indeed the purely exponential product integrates to Ce^{-kd} because $2c > k$. Every other term contains a factor λ; splitting at $d/2$ and d, exponential decay on the first half and the displayed algebraic tail on the second give $C_P\lambda(1+d)^{-p}$. All constants are uniform for $p \geq p_* > 1$. Summation using (5.27) costs $C_P F$. A bulk discrepancy $O(\kappa)$ costs $O(\kappa)$ by the finite ambient first moment of the two resolvents.

An arbitrary bounded insertion outside distance H from the receiving self core has weighted row norm at most

$$C_P\{e^{-cH} + \lambda H^{-p}\}. \tag{5.29}$$

Indeed F_0 has translation-uniform ball mass CR^6, F_1 is bounded, and the row is bounded by $C\int_{|Z-X|>H} G_s^{\mu}(X-Z)[(1+|X-Z|)F_0(Z)+F_1(Z)]\,\mathrm{d}Z$. This gives (5.29) by annular integration. Consequently the actual-to-distance-reference self kernel has weighted first norm $O_P(\kappa)$, with all visits retained.

Only a fixed neighboring block is needed for the chamber comparison. Choose L with $2cL > k$ and retain enough actual labels to cover the entire strip $|\rho_0| \le L\delta$ with another fixed transmission buffer. Omitted normal-strip insertions have norm

$$C_P\{e^{-2cL\delta} + \lambda^2(L\delta)^{-2p}\} = o(\kappa).$$

For lateral natural-window omissions, (5.29) and (4.23) give $o(\kappa)$. Omitted self endpoints are different: boundedness of the normals and the codimension-one one-leg tail give

$$\frac{\lambda R_*^{-p-1}}{\kappa^2} = \frac{\lambda\delta^{-p}}{\kappa}(\kappa\delta)^{(p-1)/2} = o(1). \tag{5.30}$$

Choose the fixed threshold a_0 after all enlargements so the associated exponentials are $o(\kappa^2)$. If the strip reaches curvature distance, use isolation and the corresponding curvature exterior estimates. Thus no chamber comparison uniform in a growing number of labels is being assumed.

5.3.2. *Genuine normal gauge and all-visit geometric freezing.* Fix a small-slope extension h of the actual self graph, agreeing on the entire smaller curvature chart and satisfying $|Dh| \le \theta$ and $|D^j h| \le C_j\kappa$ for fixed $j \ge 2$. Subtract the tangent affine part before cutting off the remainder. The height and slope remainders on a radius-R chart are $O(\kappa R^2)$ and $O(\kappa R)$, so the extension keeps these derivative bounds. Put

$$Y(x) = (x, h(x)), \qquad \nu(x) = \frac{(-Dh(x), 1)}{\sqrt{1+|Dh(x)|^2}}.$$

Here is an explicit extension of its Fermi map. Let φ be a smooth compactly supported mollifier of mass one, put $a(r) = (P^2+r^2)^{1/2}$, and choose an even smooth ζ equal to zero on $[-2P, 2P]$ and to one outside $(-3P, 3P)$. Set

$$\bar\nu(x,r) = \nu(x) + \zeta(r)\{(\varphi_{a(r)} * \nu)(x) - \nu(x)\}.$$

For $|r| \ge 3P$ and $|\alpha| + b \ge 1$, mollifier cancellation and $|D\nu| \le C\kappa$ give

$$|D_x^\alpha \partial_r^b \bar\nu(x,r)| \le C_{\alpha,b}\kappa|r|^{1-|\alpha|-b}.$$

On the fixed transition strip the corresponding derivatives are $O_{P,\alpha,b}(\kappa)$, by the already proved physical jets. Choose an even χ equal to one on $[-1,1]$ and to zero outside $(-2,2)$, and define

$$\Phi(x,r) = Y(x) + r\{e_7 + \chi(\kappa r/\rho)(\bar\nu(x,r) - e_7)\}.$$

For large sequence index this is the exact Fermi map on $|r| \le 2P$. The displayed derivative bounds and the cutoff scale give

$$\|D\Phi - I\|_\infty \le C(\theta+\rho) < \tfrac14, \qquad \|D^j\Phi\|_\infty \le C_{P,j,\rho}\kappa \quad (j \ge 2). \tag{5.31}$$

The first inequality makes $\Phi - I$ a contraction perturbation of the identity, proving global invertibility and both Lipschitz bounds.

For $|r| \le \rho/\kappa$ the normal cutoff is one and $|\bar\nu - \nu| \le C\kappa|r|$. Hence $|\Phi(x,r) - Y(x) - r\nu(x)| \le C\kappa r^2$. Choose the fixed ρ inside the genuine normal buffer; its distance function is Lipschitz, so $|\rho_0(\Phi(x,r)) - r| \le C\kappa r^2$ there. The transition $\rho/\kappa \le |r| \le 2\rho/\kappa$ and its exterior are at fixed curvature distance from the self core and may be charged directly to (5.29). For $v_s(r) = (1-\psi(r)^2)W_s(r)$, the potential difference on the controlled part is bounded by

$$C_P\kappa(1+|r|)^2\{e^{-c|r|} + \lambda(1+|r|)^{-p}\}.$$

Here ρ is small enough that the two normal arguments are comparable; the derivative tail and the fixed-depth cutoff give the estimate by the mean value theorem. Multiplication by the two-leg majorant is integrable uniformly: its slowest product is $C_P\kappa\lambda^3(1+|r|)^{1-3p}$. Thus the weighted returning-row difference is $O_P(\kappa)$. The remaining curvature-distance insertion bound is $O_P(e^{-c/\kappa} + \lambda\kappa^p) = o(\kappa)$. No global $O(\kappa)$ supremum for the potential difference is used.

Take a single nonnegative radial majorant H_* for the free kernels under all the allowed linear and bi-Lipschitz changes. It can be chosen independent of s and κ, with

$$H_*(Z) \le C \begin{cases} |Z|^{p_*-7}, & |Z| \le 1, \\ e^{-c|Z|} + |Z|^{-7-p_*}, & |Z| > 1. \end{cases}$$

For each fixed derivative order the required scaled radial derivatives have the same majorant up to a fixed constant. Writing

$$M_0(t) = \int_{\mathbb{R}^6} H_*(z,t)\, \mathrm{d}z, \qquad \widetilde{M}_2(t) = \int_{\mathbb{R}^6} (|z|^2 + t^2) H_*(z,t)\, \mathrm{d}z,$$

one has $\|M_0\|_1 + \|M_0\|_\infty + \|\widetilde{M}_2\|_\infty \le C_0$. The last bound uses $p_* > 1$; it does not assert a finite ambient second moment.

For a path with ℓ free legs, fixed initial normal endpoint $r \in C$, and final normal endpoint in C, let $L_{\mathrm{full}} = \sum_{\alpha=1}^{\ell} |Z_\alpha - Z_{\alpha-1}|$. Then

$$\int (1 + L_{\mathrm{full}}^2) \prod_{\alpha=1}^{\ell} H_*(Z_\alpha - Z_{\alpha-1}) \le C_P \{\|M_0\|_\infty + \ell^2 \|\widetilde{M}_2\|_\infty\} \|M_0\|_1^{\ell-1}. \tag{5.32}$$

The integral includes all intermediate points and the final tangential and normal endpoints. To prove it, use $L_{\mathrm{full}}^2 \le \ell \sum_\alpha |Z_\alpha - Z_{\alpha-1}|^2$. For each marked leg use the bounded section $\widetilde{M}_2$, and use the L^1 sections M_0 on all the other legs; then integrate the final normal endpoint over C. The unweighted term uses M_0 on the marked leg.

The unitary free kernel in $Z = (x, r)$ coordinates is $J_\Phi(Z)^{1/2} G_s^2(\Phi Z - \Phi Z') J_\Phi(Z')^{1/2}$. Freeze the full differential at $Z_c = ((x+y)/2, 0)$, where x, y are the two tangential endpoints. Every point of the path is at distance at most $C(L_{\mathrm{full}} + P)$ from Z_c, so (5.31) gives

$$|\Phi Z_\alpha - \Phi Z_{\alpha-1} - D\Phi(Z_c)(Z_\alpha - Z_{\alpha-1})| \le C_P \kappa |Z_\alpha - Z_{\alpha-1}| (L_{\mathrm{full}} + P).$$

This includes all intermediate normal excursions. The actual leg and its frozen linear leg both lie within one quarter of the original leg, by $\|D\Phi - I\| < 1/4$. Their interpolating segment therefore stays a fixed relative distance from zero, so the scaled radial derivative bounds justify the kernel mean-value estimate even for long paths. Telescope the leg kernels and Jacobians in the positive Neumann series for $v_s(r)$. A changed factor costs $C_P \kappa (L_{\mathrm{full}} + P)$, and the endpoint first weight costs at most $C_P(1 + L_{\mathrm{full}})$. Thus (5.32) bounds the ℓ-leg error by $C_P \kappa \ell^3 C_0^{\ell-1} \delta_P^{\ell-1}$, after enlarging C_0. The per-leg constant C_0 is fixed before P; only the prefactor C_P uses the length of the endpoint interval. Choose P so that $C_0 \delta_P < 1/2$. The weighted first-row difference is then $O_P(\kappa)$. Tangential translated derivatives mark only finitely many factors, and add a fixed polynomial in ℓ. Derivative constants occur only on the marked factors, so the same choice of P works for each fixed derivative order. These are translated tangential and metric-parameter derivative estimates. Full physical jets of the resulting smooth fields follow from (5.16) and the smooth coordinate and source extensions. Acting on a known normal difference bounded by $C_P \kappa |Z - Z'|$ gives an $O_P(\kappa^2)$ error.

For later localization this calculation gives a common envelope, not merely individual moment bounds. For all allowed b, s,

$$\|K_b(z)\|_{C(C) \to C(C)} \le h(z) := C_P \sum_{\ell \ge 1} \delta_P^{\ell-1} \int_{-4P}^{4P} H_*^{*\ell}(z,t)\, \mathrm{d}t, \qquad \int_{\mathbb{R}^6} (1 + |z|^2) h(z)\, \mathrm{d}z < \infty.$$

Indeed shift the initial normal endpoint to zero and discard the bounded insertion factors; the final normal difference lies in $[-4P, 4P]$. Equation (5.32) sums the moments of this majorant. Consequently $\int_{|z|>R} |z|^2 h(z)\, \mathrm{d}z \to 0$ uniformly for the model family. A finite number of marked derivative factors gives the corresponding parameter-derivative moment bounds in exactly the same way.

The actual physical normal must stay inside the difference. The symmetric pulled-back physical flux kernel is exactly

$$\mathcal{J}_a(Z, Z') = J_\Phi(Z) S_0(\Phi Z) G_V(\Phi Z, \Phi Z') S_0(\Phi Z') J_\Phi(Z'). \tag{5.33}$$

The original pulled-back row is $J_\Phi(Z)^{-1}$ times this flux row. Equivalently its endpoint against the unitary resolvent is $\sqrt{J_\Phi}S_0$. Relative core preparation changes that endpoint from its profile value by $O_P(\kappa)$ after its leading tangent density is retained. Multiplying a massive kernel by one endpoint error has weighted first norm $O_P(\kappa)$. The preceding potential, coordinate, endpoint, and freezing estimates therefore prove

$$\|\{\mathcal{L}_a^{\mathrm{self}} - \mathcal{L}_{\mathrm{F,mid}}\}N\|_\infty \le C_P\kappa^2 \tag{5.34}$$

in the physical flux convention.

These estimates retain the cutoff factors at the two endpoints. More precisely, on the controlled plateau fibers one may factor $\psi(r)^2\psi(q)^2$ from every physical flux comparison: the remaining endpoint coefficients involve only J_Φ, W, g and their profile counterparts, and are uniformly bounded. Relative preparation holds for those coefficients after the leading tangent density is retained. Replacing its endpoint factor $\sqrt{\beta(x)\beta(y)}$ by $\beta((x+y)/2)$ costs $C\kappa|x-y|$; the same path second moment and the retained normal difference give $O_P(\kappa^2)$ action. Multiplication by the receiving factor f/f_a^2 below cancels only one power of $\psi(r)$. The resulting normalized comparisons therefore still have a bounded receiving factor $\psi(r)$ and the same $O_P(\kappa^2)$ action, uniformly up to the fiber edge. In particular the argument does not divide an unweighted error estimate by a vanishing cutoff.

All uncontrolled endpoints and insertions carry their separate budgets above. Local natural-window comparisons can be pulled back to the single fixed curvature-chart extension: the extensions agree on a larger nested near region, and their differences elsewhere are exactly the lateral insertion and endpoint terms already estimated. No donor-dependent metric is left in the final equation.

5.3.3. *Fixed-profile normalization and the receiving diagonal.* Let $g_{\tan}$ be the first fundamental form of the chosen self chart, and now use the unambiguous model parameter

$$b(x) = g_{\tan}(x)^{-1}, \qquad \beta(x) = \sqrt{\det g_{\tan}(x)}. \tag{5.35}$$

Here b is a matrix and β is its density; neither is the regular solution-gradient magnitude $g = |\nabla u|$. On $C = [-2P, 2P]$ set

$$\begin{aligned} &U_s(r) = \psi(r)\sqrt{W_s(r)}, \qquad f(r) = U_s(r)w_s(r), \\ &H = L^2(C;\mathbb{R}^7), \qquad X = C(C;\mathbb{R}^7), \qquad j(x,r) = f(r)N_0(x,r). \end{aligned} \tag{5.36}$$

Here $D_r^2 = -\partial_r^2$, so $(D_r^2 + \xi^T b\xi)^s$ denotes the nonnegative Fourier multiplier $(\tau^2 + \xi^T b\xi)^s$ in the normal variable. The frozen family, with this fixed profile normalization, is

$$\begin{aligned} &\widehat{K}_b(\xi) = U_s[(D_r^2 + \xi^T b\xi)^s + 2 - v_s(r)]^{-1}U_s, \qquad v_s = (1-\psi^2)W_s, \\ &B[b]j = j - \int K_{b(x)}(z)j(x-z)\,\mathrm{d}z. \end{aligned} \tag{5.37}$$

The fiber is genuine normal distance, so there is no vertical normal Fourier shear. The equality $H_s w_s = 0$ gives $\widehat{K}_b(0)f = f$. Thus $B[b](fN)$ has the exact frozen row- difference representation. Its left quantization is even in z. The fixed coefficient map, applied componentwise, is $P_f j = (\int_C f(r)j(r)\,\mathrm{d}r)/(\int_C f(r)^2\,\mathrm{d}r) \in \mathbb{R}^7$; the corresponding projections are fP_f and $Q_f = I - fP_f$.

Let

$$\widehat{R}_a(Z,Z') = \sqrt{J_\Phi(Z)}\,G_V(\Phi Z, \Phi Z')\sqrt{J_\Phi(Z')}, \qquad U_a(Z) = \sqrt{W_0(\Phi Z)}, \qquad T_a = U_a\widehat{R}_a U_a.$$

The smooth square root U_a includes the spatial donor taper. It is multiplication in unitary coordinates, whose Jacobian factors already occur in $\widehat{R}_a$. On a receiving plateau fiber, $U_a = \psi(r)\sqrt{W(\Phi Z)}$. The phase field and comparison multipliers there are

$$f_a = \sqrt{J_\Phi}\,U_a g, \qquad M = f_a/f, \qquad \widetilde{M} = f_a/(\sqrt{\beta}f).$$

Cancel the common ψ before forming the quotients. On the fixed plateau fiber, w_s, W_s, g are bounded below, so M, M^{-1} extend smoothly across its edge and $\widetilde{M} = 1 + O_P(\kappa)$. The factor

$\sqrt{\beta}$ is retained exactly. At donor points $M(fN)$ means f_aN, with its actual taper and zero extension; no inverse of a donor taper is used. The exact kernel identity is

$$f_a(Z)T_a(Z,Z')f_a(Z') = \mathcal{J}_a(Z,Z').$$

Define

$$d_a = (T_af_a)/f_a, \qquad B_a^{\mathrm{row}} = d_a - T_a$$

on the receiving plateau. The quotient and the following row identities are understood there after the same cutoff cancellation.

$$M^{-1}B_a^{\mathrm{row}}M(fN) = \frac{f}{f_a^2}\int \mathcal{J}_a(Z,Z')[N(Z)-N(Z')]\,\mathrm{d}Z'. \tag{5.38}$$

The right side is smooth at the edge after cancellation. This is not an identity-diagonal operator: exactly

$$M^{-1}(I-T_a)M = M^{-1}B_a^{\mathrm{row}}M + (1-d_a). \tag{5.39}$$

The last term cannot be declared quadratic by self comparison. Its cross receiving diagonal remains in the full signed normal identity.

At a constant frozen metric the physical coordinate flux equals $\beta f(r)K_b(z;r,q)f(q)$. Dividing its receiving row by $\beta f(r)$ gives (5.37) on $j = fN$. The receiving factor in (5.38) relative to this normalization is

$$\frac{\beta f^2}{f_a^2} = \widetilde{M}^{-2} = 1 + O_P(\kappa).$$

The frozen normalized action on fN is $O_P(\kappa)$ by the finite first moment and $DN = O(\kappa)$, so this factor contributes $O_P(\kappa^2)$. Midpoint-to-receiving replacement of β or b has the same quadratic action: its parameter difference is $O(\kappa|z|)$ and the retained normal difference is $O(\kappa(|z|+|r-q|))$. The corresponding parameter-derivative second moments follow from the same fixed-endpoint path calculation. This converts (5.34) to the left operator (5.37) in the specified fixed-profile gauge.

5.3.4. *Exact source assembly and localization.* The differentiated equation and its scalar product give, before any self/cross splitting,

$$\sum_k \sigma_i\sigma_k \int J_{ik}(X,Y)[N_i(X)-N_k(Y)]\,\mathrm{d}Y = \frac{N_i(X)}{2}\sum_k \sigma_i\sigma_k \int J_{ik}(X,Y)|N_i(X)-N_k(Y)|^2\,\mathrm{d}Y + \sigma_i S_i(X)\{Z(X) - N_i(X)[N_i(X)\cdot Z(X)]\}, \tag{5.40}$$

where here $J_{ik}(X,Y) = S_i(X)G_V(X,Y)S_k(Y)$ is the ambient flux kernel and $Z = G_V(W_{\mathrm{ext}}\nabla u)$. This notation is distinct from the Jacobian J_i in (4.33). Subtract the complete cross difference from both sides and apply the same receiving normalization as in (5.38). The resulting cross source on the plateau is exactly

$$R_i(x,r) = -\frac{f(r)}{g(X_i(x,r))}\sum_{k\neq i}\sigma_i\sigma_k \int G_V(X_i(x,r),Y)S_k(Y)[N_i(x,r)-N_k(Y)]\,\mathrm{d}Y. \tag{5.41}$$

Indeed $(f/f_a^2)J_iS_i = f/g$. This specifies the receiving multiplier in (5.13) and cancels the edge cutoff exactly. It includes both N_i and $-N_k$ terms, hence the receiving diagonal. No positive inverse is being assumed for this signed cross operator. The angular estimates are used with the receiving endpoint removed. At a fixed receiving core point X, set $\mathrm{d}\mu_k(Y) = G_V(X,Y)S_k(Y)\,\mathrm{d}Y$. The positive point-depth row and moment majorants used in (4.40)–(4.42) hold for this measure: their proofs integrate the donor source against the massive kernel and use only an upper bound for a receiving density. Set that density equal to one from the outset. The same gap-excess decomposition and ordered sums proving (4.45) therefore give

$$\sup_{i,X}\sum_{k\neq i}\int |N_i(X)-N_k(Y)|^2\,\mathrm{d}\mu_k(Y) = o(\kappa^2).$$

For the self row use $|N_i(X) - N_i(Y)| \le C_P\kappa|X - Y|$ on the buffered core. The local massive singularity has a finite weighted second integral, and the codimension-one far integral is bounded by $C_P\lambda \int_1^\infty t^{-p}\,dt$. The unsigned exterior handles the omitted part, giving $\int |N_i(X) - N_i(Y)|^2\,d\mu_i(Y) \le C_P\kappa^2$. The receiving normalization is now harmless: $(f/f_a^2)J_iS_i = f/g$, which is smooth and bounded on the whole fixed fiber, including its cutoff edge. It gives the asserted $O_P(\kappa^2)$ self angular term and $o(\kappa^2)$ cross angular term. The normalized exterior term is $\sigma_i(f/g)\{Z - N_i(N_i \cdot Z)\}$ and is $o(\kappa^2)$ by (4.35). Add these terms to the factored same-gauge self comparison and the model-conversion errors, and call the resulting sum E.

Finally take $\eta_\kappa(x) = \eta_0(\kappa x)$ and extend the known field with bounded physical jets and $\|D_x j\|_{L_x^\infty X} \le C_P\kappa$. Only this tangential Lipschitz bound is needed; the fixed profile $f(r)$ need not have a small normal derivative. The commutator is

$$[B[b], \eta_\kappa]j(x) = \int K_{b(x)}(z)\{\eta_\kappa(x) - \eta_\kappa(x - z)\}j(x - z)\,dz.$$

In the first part replace $j(x - z)$ by $j(x - z) - j(x)$; the two increments cost $C_P\kappa^2|z|^2$ in X. In the remaining part subtract the linear Taylor term of η_κ; it integrates to zero by the evenness of the operator-valued frozen kernel. Its remainder has the same bound. The common envelope h above proves that localization costs $C_P\kappa^2$. Changes beyond a fixed curvature-distance buffer cost

$$C \int_{|z|>c/\kappa} h(z)\,dz \le C\kappa^2 \int_{|z|>c/\kappa} |z|^2 h(z)\,dz = o(\kappa^2).$$

The source R is multiplied by the same tangential cutoff; the commutator and the displayed exterior changes enter E. All physical jets remain bounded by the smooth extensions, (5.16), and the finite-order model kernel calculus.

Theorem 5.4 (Prepared same-gauge source equation)**.** *Starting from (4.1), the stated moving-center qualitative preparation and clearing, and the near-one flat analytic estimates proved above, the preceding construction gives the following uniform conclusions on every smaller moving curvature chart:*

(1) There is a same-zero, same-reference finite stack on each prescribed nested natural window (or a single-sheet isolation window) with chamber error (4.24). All relative physical jets (4.26) and genuine full-buffer distance jets (4.29) hold.

(2) Actual smooth positive adjacent rows obey (5.8) and (5.9), with the signed stability calculation and all coordinate measures retained.

(3) In the gauge (5.36) and the model family (5.37), a controlled extension and localization give

$$B[b]j = R + E, \qquad \|R\|_{L_x^\infty X} = o(\kappa^{3/2}), \qquad \|E\|_{L_x^\infty X} \le C\kappa^2, \tag{5.42}$$

with R exactly the localized source (5.41), including its signed receiving diagonal. For every fixed finite N,

$$\|R\|_{C_x^N X} + \|E\|_{C_x^N X} + \|j\|_{C_x^N X} + \|b\|_{C_x^N} \le C_{P,N}. \tag{5.43}$$

The localized density $\kappa^{-2}R(y/\kappa)$ has translation-uniform direct capacity bounds (5.18) with total budget (5.24) tending to zero, plus one full $o(\kappa^2)$ physical remainder.

The conclusion does not include a scalar phase inverse, a normal-complement reconstruction theorem, or curvature-scale C^1 compactness. Those are subsequent operator conclusions and must use this exact gauge, source, and diagonal.

Proof. The preparation follows from the zero-trace gap, complete reference, linear cell transmission, and normalized interior estimates, in that order. The physical jets then justify the smooth core construction. The signed two-site inequality and simultaneous finite-form-norm absorption give capacity. The exact vector identity (5.40), the self comparison (5.34), and the algebraic normalization (5.38) give (5.42). Its cross amplitude is (5.14); its capacity budget is Theorem 5.3. The smooth differentiated-resolvent argument and the model path estimates give the full jets. All extensions and localization terms were included in E. In particular no implication $o(\kappa^{3/2}) = o(\kappa^2)$ is used. □

6. The finite-core operator and curvature retention

Throughout this section the stable equation is posed in $\mathbb{R}^7$, its zero hypersurfaces have dimension $m = 6$, and

$$s_j \uparrow 1, \quad p_j = 2s_j \geq p_* > 3/2, \quad \lambda_j = 1 - s_j, \quad \kappa_j \downarrow 0, \quad \delta_j \to \infty, \qquad e^{-\sqrt{2}\delta_j} + \lambda_j \delta_j^{-p_j} = o(\kappa_j).$$

All constants below are uniform in j at each fixed normal depth P. The preceding preparation is used on strictly smaller buffered charts, with the same reference and the same row normalization throughout. In particular we start with the *actual on-solution* equation

$$B[b_j]j_j = R_j + E_j, \qquad j_j(x,r) = f_j(r)N_j(x,r), \tag{6.1}$$

The equation and budgets are the conclusion of Theorem 5.4, in its fixed-profile similarity gauge. All fiber operators act componentwise on the seven normal coordinates. Here $x \in \mathbb{R}^6$, $r \in C_P = [-2P, 2P]$, and

$$\|R_j\|_{L^\infty_x \mathcal{X}} = o(\kappa_j^{3/2}), \qquad \|E_j\|_{L^\infty_x \mathcal{X}} = O(\kappa_j^2), \qquad \|R_j\|_{C^N_x \mathcal{X}} + \|E_j\|_{C^N_x \mathcal{X}} \leq C_N, \quad N > 3. \tag{6.2}$$

Here $\mathcal{H} = L^2(C_P)$, $\mathcal{X} = C(C_P)$, $f_j = \psi_P \sqrt{3(1 - q_{s_j}^2)} q'_{s_j}$, and N_j denotes the controlled extension of the consistently oriented normal field. It equals the actual unit normal on the evaluation plateau; its extension is only required to satisfy the stated bounds. The smooth cutoff is nonnegative and equals one on its inner plateau. Put $E_P = \{r \in (-2P, 2P) : \psi_P(r) > 0\}$. Division by f_j will be restricted to compact subsets of E_P, where profile convergence gives a uniform positive lower bound for all sufficiently large j. We require $f_j(0) \geq c_P > 0$ and $\|f_j\|_2 \geq c_P > 0$. The equation includes controlled extension and localization costs. The *whole signed cross difference*, including the receiving cross diagonal, belongs to R_j. The symbol E_j includes the self comparison, gauge, angular reaction and exterior errors; it is not obtained by simply declaring a self Jacobi diagonal negligible.

Write

$$\mathsf{P}_j v = \frac{\langle f_j, v \rangle}{\mathfrak{m}_j}, \quad \mathfrak{m}_j = \|f_j\|_2^2, \quad \mathsf{Q}_j = I - f_j \mathsf{P}_j, \qquad a_j = \mathsf{P}_j j_j.$$

Physical preparation gives $\mathrm{Lip}_{x,r} N_j \leq C_P \kappa_j$, $\|D_x b_j\|_\infty \leq C\kappa_j$, $\|b_j - I\|_\infty \leq \eta_0$, and bounded absolute physical jets at every fixed order used below. No relative smallness of the differentiated source in (6.2) is assumed.

6.1. **The frozen operator and its scalar symbol.** Fix temporarily s, P and put $w = q'_s$, $W_q = 3(1 - q_s^2)$, $U = \psi_P \sqrt{W_q}$, $f = Uw$, $c = U^2 w$, and $M = \|f\|_2^2$. The phase potential and its resolvent are

$$V_P = (1 - \psi_P^2) W_q, \qquad R_{V,s}(\tau) = \big((D_r^2 + \tau)^s + 2 - V_P\big)^{-1}, \quad D_r^2 = -\partial_r^2.$$

Take P large enough that $0 \leq V_P \leq \epsilon_P < 2$ uniformly near one. The genuine-normal-fiber core operator is

$$B_s(\tau) = I - U R_{V,s}(\tau) U, \qquad H_s(\tau) = (D_r^2 + \tau)^s + 2 - W_q. \tag{6.3}$$

Functions on the core are extended by zero when multiplied by U. Since $H_s(0)w = 0$, one has $B_s(0)f = 0$. A constant inverse tangent metric b replaces τ by $\xi^T b \xi$; it does not tilt the normal fiber or change f. We retain the fixed-profile similarity gauge, so the diagonal normalization is $d_0 = 1$ throughout this section.

For $\tau > 0$, $H_s(\tau)$ is strictly positive and invertible, as verified below. The resolvent identity gives

$$B_s(\tau)^{-1} = I + U H_s(\tau)^{-1} U.$$

For a scalar a, minimization subject to $\langle f, v \rangle = Ma$ gives

$$\min \langle v, B_s(\tau) v \rangle = \frac{M^2}{\langle f, B_s(\tau)^{-1} f \rangle} |a|^2.$$

Consequently the exact scalar Schur symbol is

$$\Sigma_{s,P}(\tau) = \frac{M^2}{M + \langle c, H_s(\tau)^{-1} c\rangle}. \tag{6.4}$$

This uses the unnormalized scalar pairing $\langle f, \cdot\rangle$; using P instead divides the symbol and scalar source by M.

The whole-line operator $H_s(0)$ has the uniform orthogonal gap already proved in Lemma 3.3 and recalled in (4.5). Put

$$A_s = s \int_{\mathbb{R}} |\xi|^{2s-2} |\widehat{w}(\xi)|^2 \, \mathrm{d}\xi,$$

with the Plancherel Fourier measure. The profile estimates and (3.29) give $0 < c \le A_s \le C$ near one. The uniform small-frequency block calculation below yields

$$\langle c, H_s(\tau)^{-1} c\rangle = \frac{M^2}{A_s \tau}\big(1 + O(\tau^\theta)\big) + O(1), \qquad \frac{\Sigma_{s,P}(\tau)}{\tau} \longrightarrow A_s \quad (\tau \downarrow 0),$$

where one may take $\theta = p_*/2 - 1/2 > 0$. The error and convergence are uniform near one at fixed P. At high frequencies one has

$$\Sigma(\tau) \asymp \min(\tau, 1), \quad \Sigma(0) = 0, \quad \Sigma(\infty) = M, \quad \tau^a \partial_\tau^a (\Sigma(\tau) - M) = O_a(\tau^{-s}) \quad (\tau \ge 1). \tag{6.5}$$

The scalar function in the denominator of (6.4), divided by M^2, is a Stieltjes function. Its reciprocal is therefore a complete Bernstein function, by the Stieltjes/complete-Bernstein characterization.[1] The hypotheses are checked below. Its integral representation, together with the two endpoint limits in (6.5), gives

$$\Sigma(\tau) = \int_{(0,\infty)} \frac{\tau}{\tau + t} \, \mu_{s,P}(\mathrm{d}t), \qquad \mu_{s,P}((0,\infty)) = M, \qquad \int_{(0,\infty)} t^{-1} \mu_{s,P}(\mathrm{d}t) = A_s. \tag{6.6}$$

There is no constant or linear term. In particular, for every integer $a \ge 1$,

$$|\tau^a \Sigma^{(a)}(\tau)| \le a! \Sigma(\tau), \qquad \tau > 0.$$

If Y_t is the kernel of $(-\Delta_x + t)^{-1}$ on $\mathbb{R}^6$, the even nonnegative jump kernel is $k_0(z) = \int_{(0,\infty)} t Y_t(z) \mu_{s,P}(\mathrm{d}t)$, and

$$\int_{\mathbb{R}^6} k_0 = M, \qquad \int_{\mathbb{R}^6} |z|^2 k_0(z) \, \mathrm{d}z = 12 A_s. \tag{6.7}$$

Thus the finite-core scalar reduction has bounded high-frequency order, while its low-frequency coefficient is A_s.

Lemma 6.1 (Uniform spectral gap and all-visit moment estimates)**.** *The flat spectral and kernel assertions above hold for the smooth finite-core family, under the retained uniform near-one profile bounds.*

Proof. We use the whole-line gap (4.5); its proof is not repeated. Write $A_0 = |D_r|^{2s} + 2 - V_P$ and $H_0 = A_0 - U^2$. The kernel correspondence between $B_s(0) = I - U A_0^{-1} U$ and H_0 sends w to $f = Uw$; the kernel is therefore exactly $\mathbb{R} f$. Nonnegativity follows from $H_0 \ge 0$ by the Birman–Schwinger congruence. For $F \perp f$, one has $UF \perp w$. Set $v = H_0^\dagger UF$, where the inverse on $w^\perp$ has norm at most γ^{-1}. Then $A_0 v = U(F + Uv)$, so $B_s(0)(F + Uv) = F$. With Q the orthogonal projection onto $f^\perp$, this proves

$$[B_s(0)|_{f^\perp}]^{-1} = \mathsf{Q}[I + U H_0^\dagger U]\mathsf{Q}, \qquad \|[B_s(0)|_{f^\perp}]^{-1}\| \le 1 + 3/\gamma.$$

Thus its complementary gap is at least $(1 + 3/\gamma)^{-1}$, uniformly near one and for the retained depths. Monotonicity of $(D_r^2 + \tau)^s$ gives $B_s(\tau) \ge B_s(0)$ for $\tau \ge 0$, preserving this gap.

For the small-frequency calculation put $n_s = \|w\|_2$, $e = w/n_s$, and let Q_e project onto $e^\perp$ on the whole line. Both n_s and its reciprocal are uniformly bounded by the profile bounds. Define

$$D_\tau = (D_r^2 + \tau)^s - |D_r|^{2s}, \quad \alpha_\tau = \langle e, D_\tau e\rangle, \quad v_\tau = Q_e D_\tau e, \quad C_\tau = Q_e H_s(\tau) Q_e.$$

[1] R. L. Schilling, R. Song and Z. Vondraček, *Bernstein Functions: Theory and Applications*, second edition, De Gruyter, 2012, Chapters 6–7.

The positive increment gives $C_\tau \geq \gamma$. Since $s \geq p_*/2 > 3/4$,

$$\|D_\tau e\|_2 \leq \tau \|s|\xi|^{2s-2}\widehat{e}(\xi)\|_2 \leq C\tau.$$

The integrability at zero uses the uniform L^1 profile bound and $4s-4 > -1$; at infinity the multiplier is bounded and $e \in L^2$. Splitting the scalar Fourier integral at $|\xi| = \sqrt{\tau}$ and $|\xi| = 1$ gives

$$\alpha_\tau = \frac{A_s}{n_s^2}\tau + O(\tau^{1+\theta}), \qquad \theta = p_*/2 - 1/2 > 0, \qquad 0 < \tau \leq 1.$$

Indeed the small interval contributes at most $C\tau \int_0^{\sqrt{\tau}} \xi^{2s-2}\,\mathrm{d}\xi$ to the remainder; on the other intervals the second-order Taylor bound contributes $C\tau^2 \int_{\sqrt{\tau}}^1 \xi^{2s-4}\,\mathrm{d}\xi + C\tau^2$. These bounds are uniform and at most $C\tau^{1+\theta}$. The scalar Schur denominator is therefore

$$d_\tau = \alpha_\tau - \langle v_\tau, C_\tau^{-1} v_\tau\rangle = \frac{A_s}{n_s^2}\tau + O(\tau^{1+\theta}) + O(\tau^2).$$

It is bounded below by $c\tau$ for uniformly small positive τ. The block inverse then makes $H_s(\tau)$ positive and invertible in this range. Monotonicity extends invertibility to every $\tau > 0$, with a uniform lower bound on any fixed interval bounded away from zero.

Write $c = (M/n_s)e + c_\perp$. The block inverse gives exactly

$$\langle c, H_s(\tau)^{-1} c\rangle = \frac{|M/n_s - \langle v_\tau, C_\tau^{-1} c_\perp\rangle|^2}{d_\tau} + \langle c_\perp, C_\tau^{-1} c_\perp\rangle.$$

The mixed numerator correction is $O(\tau)$ and the final term is $O(1)$, proving the announced expansion and $\Sigma(\tau)/\tau \to A_s$ uniformly. For large τ, $H_s(\tau) \geq \tau^s - 1$, while

$$\|\partial_\tau^a H_s(\tau)\| \leq C_a \tau^{s-a}, \qquad a \geq 1.$$

Differentiating the inverse yields $\|\partial_\tau^a H_s(\tau)^{-1}\| \leq C_a \tau^{-s-a}$. Substitution in (6.4) proves the high-frequency estimates. The remaining compact frequency interval is controlled by the preceding uniform positivity and monotonicity.

For the complex representation take the principal branch on $\mathcal{D} = \mathbb{C} \setminus (-\infty, 0]$. The free operator $F_s(z) = (D_r^2 + z)^s + 2$ is invertible there. Multiplication by the bounded decaying W_q, followed by $F_s(z)^{-1}$, is compact: local compactness follows from its H^{2s} gain and the tail from $W_q \to 0$. Thus $H_s(z) = F_s(z) - W_q$ is Fredholm of index zero. For $\operatorname{Im} z > 0$ its imaginary form is strictly positive on every nonzero vector. This excludes a kernel; the lower half-plane follows by conjugation, and the positive real axis was handled above. The resulting inverse is analytic on $\mathcal{D}$. If $h(z) = \langle c, H_s(z)^{-1} c\rangle$, then

$$\operatorname{Im} h(z) = -\langle H_s(z)^{-1} c, (\operatorname{Im} H_s(z)) H_s(z)^{-1} c\rangle < 0 \qquad (\operatorname{Im} z > 0).$$

The function $(M + h(z))/M^2$ is positive on $(0, \infty)$, has limit $+\infty$ at zero, and has the opposite-half-plane mapping property. These are precisely the hypotheses of the Stieltjes characterization cited above. Its reciprocal is a complete Bernstein function by the same characterization. The resulting integral representation has zero constant term because $\Sigma(0+) = 0$, and zero linear term because Σ is bounded. Monotone convergence as $\tau \to \infty$ gives the mass M in (6.6); as $\tau \downarrow 0$ it gives the inverse first moment A_s. Differentiating its positive integrand gives the displayed derivative bound. Finally $\int Y_t = t^{-1}$ and $\int |z|^2 Y_t(z)\,\mathrm{d}z = 12t^{-2}$, by the Gaussian heat resolvent in six dimensions. Tonelli proves (6.7).

The all-visit fixed-endpoint moment estimate is already (5.32). Its common envelope, established immediately after that calculation, controls the free kernels, the finitely many marked parameter derivatives, and all allowed constant metric changes. Apply it to the present potential V_P, with each intermediate phase factor bounded by ϵ_P, and choose P so that $C_0\epsilon_P < 1/2$. The endpoint multipliers U are uniformly bounded and supported in C_P. The resulting positive series is dominated by the same common envelope with finite tangential second moment and a tight second-moment tail. Finite paths converge as $s \to 1$ by the common massive majorants; the geometric series is uniformly summable. Consequently the kernel family and every fixed marked parameter derivative have the required uniform moments and continuity up to $s = 1$. No ambient second moment is asserted. □

6.2. **Weighted complementary inversion and point-normal envelopes.** We give the model inversion argument explicitly. On $\mathsf{Q}\mathcal{H}$ let

$$C_b = \mathsf{Q}B_b\mathsf{Q}, \quad D_b = \mathsf{Q}B_b f, \quad E_b v = \langle f, B_b\mathsf{Q}v\rangle, \quad \Gamma_b = -C_b^{-1}D_b.$$

We use the actual fixed-profile similarity gauge, so $d_0 = 1$. The two-moment path estimate (5.32), its marked parameter-derivative versions, and the common envelope proved after it already supply the required all-visit bounds. Applied also to the transpose fiber kernel, their row estimates and the Schur test give

$$\int_{\mathbb{R}^6} (1+|z|^2)\|\partial_b^\alpha K_{s,b}(z)\|_{\mathcal{H}\to\mathcal{H}}\,\mathrm{d}z \le C_{P,\alpha}.$$

They give continuity at the local endpoint and uniformly tight weighted tails. We do not repeat the phase-path proof here. All subsequent inverse statements concern this same normal-fiber space and this same gauge.

For a Hilbert space H, let $\mathcal{A}_2(H)$ be the unitized algebra of operator-valued convolution kernels with norm

$$\|L\|_{1,2} = \int_{\mathbb{R}^6} (1+|z|^2)\|L(z)\|_{H\to H}\,\mathrm{d}z.$$

If $I - \widehat{L}(\xi)$ is invertible at every frequency and L lies in this algebra, then its inverse is $I+T$ with $T \in \mathcal{A}_2(H)$. We record the operator-valued Wiener argument. Let $L \in L^1(\mathbb{R}^6; \mathcal{B}(H))$ and suppose $A(\xi) = I - \widehat{L}(\xi)$ is invertible for every ξ. At a fixed ξ_0, choose a scalar smooth cutoff equal to one near ξ_0, of frequency width ϵ. The inverse Fourier kernel of

$$\chi_\epsilon(\xi)\{\widehat{L}(\xi) - \widehat{L}(\xi_0)\}$$

has operator-valued L^1 norm tending to zero as $\epsilon \downarrow 0$. Indeed its norm is at most the integral of $\|L(y)\|$ times the L^1 difference of a modulated cutoff kernel and its appropriately phased translate. For each fixed y this difference tends to zero by continuity of translation after dilation; it is bounded by twice the fixed cutoff-kernel norm. Dominated convergence applies. Consequently

$$A(\xi_0) - \chi_\epsilon(\xi)\{\widehat{L}(\xi) - \widehat{L}(\xi_0)\}$$

has an inverse in the unitized operator-valued convolution algebra, by factoring the constant inverse of $A(\xi_0)$ and using a Neumann series. On a neighborhood of ξ_0 it agrees with A^{-1}.

For the exterior construction choose a scalar approximate identity ϕ_ϵ whose Fourier transform has compact support. Then $\|L - L*\phi_\epsilon\|_1 \to 0$, so $I - (L - L*\phi_\epsilon)$ has a global algebra inverse. Its Fourier transform agrees with A^{-1} outside a large compact set. Cover the remaining compact frequency set by finitely many of the preceding neighborhoods and patch by scalar smooth cutoffs, including one exterior cutoff. Scalar cutoffs are central, so the patched element is a two-sided inverse even though the operator coefficients need not commute. The exterior piece has constant term I, and all remaining pieces are L^1 kernels; the inverse is $I+T$ with $T \in L^1$. Injectivity of the distributional Fourier transform identifies the convolution inverse.

The inverse-derivative identities are

$$\partial_i(I-\widehat{L})^{-1} = (I-\widehat{L})^{-1}(\partial_i\widehat{L})(I-\widehat{L})^{-1},$$
$$\partial_i^2(I-\widehat{L})^{-1} = (I-\widehat{L})^{-1}(\partial_i^2\widehat{L})(I-\widehat{L})^{-1} + 2(I-\widehat{L})^{-1}(\partial_i\widehat{L})(I-\widehat{L})^{-1}(\partial_i\widehat{L})(I-\widehat{L})^{-1}.$$

If $\int(1+|z|^2)\|L(z)\|\,\mathrm{d}z < \infty$, each Fourier derivative of $\widehat{L}$ of order at most two is the transform of the corresponding integrable weighted kernel. Every product on the right of the inverse-derivative identities is therefore the Fourier transform of an L^1 kernel: the inverse factors are $I+T$, and at least one differentiated factor is an L^1 kernel. By distributional uniqueness, z_iT and z_i^2T belong to L^1 for every coordinate i. Summing the second-coordinate weights proves $T \in \mathcal{A}_2$. The weight $(1+|z|)^2$ is submultiplicative and equivalent to $1+|z|^2$, so this is a Banach algebra. Inversion is continuous there, and its parameter derivatives follow by differentiating the algebra identity.

Apply this argument to $\mathsf{Q}K_b\mathsf{Q}$. The flat gap makes every $C_b(\xi)$ invertible and yields

$$C_b^{-1} = I + L_b, \qquad \|\partial_b^\alpha L_b\|_{1,2} + \|\partial_b^\alpha \Gamma_b\|_{1,2} \le C_{P,\alpha}, \qquad \int \Gamma_b = 0. \tag{6.8}$$

The last assertion is exact: $B_b(0)f = 0$, hence $D_b(0) = 0$. All these kernels are even in z. Adjoin the identity on span f_s when applying the algebra inverse, so all operators act on the same ambient Hilbert space. For the sequence it suffices to use the compact order set $\{s_j : j \geq j_0\} \cup \{1\}$ and a slightly larger compact metric neighborhood. The prior kernel continuity and the established flat gap make the inverse and each fixed metric derivative a compact family in weighted L^1. Thus (6.8) is uniform.

For each fixed order s and multi-index α, choose versions of the parameter kernels on a rectangular metric neighborhood. Iterated one-dimensional Sobolev inequalities bound their pointwise parameter supremum by integrals of finitely many further mixed metric derivatives. Tonelli gives

$$\int (1+|z|^2) \sup_b \|\partial_b^\alpha L_{s,b}(z)\| \, \mathrm{d}z \leq C_{P,\alpha}. \tag{6.9}$$

Apply the same inequality on $|z| > R$. Compactness of each derivative family in weighted L^1 makes its weighted tails uniformly small; integration over the fixed parameter box then gives

$$\sup_s \int_{|z|>R} (1+|z|^2) \sup_b \|\partial_b^\alpha L_{s,b}(z)\| \, \mathrm{d}z \longrightarrow 0.$$

The same conclusions hold for Γ and the other frozen kernels. The supremum over b is taken at a fixed s; no differentiability in s or integrable pointwise supremum over s is being presumed. These envelopes, with uniform mass and tails, suffice for every variable-parameter estimate below.

Evaluation at a normal point needs a different mapping estimate. By Cauchy–Schwarz in the donor fiber,

$$\|K_b(z)\|_{\mathcal{H}\to\mathcal{X}} \leq C_P \sup_r \left(\int_{C_P} G^{\mu}_{7,s}(cz, c(r-t))^2 \, \mathrm{d}t \right)^{1/2} \leq C_P(1+|z|^{p-13/2}) \quad (|z| \leq 1).$$

This is tangentially integrable for $p > 1/2$. At infinity it has the same finite second moment as above. Fixed metric derivatives obey the same estimate. The identity

$$\Gamma_b = -D_b + \mathsf{Q}K_b\mathsf{Q}\Gamma_b$$

then upgrades Γ_b and its parameter derivatives from $\mathbb{R} \to \mathcal{H}$ to $\mathbb{R} \to \mathcal{X}$ second-moment kernels. Convolution preserves two moments because $1+|z+w|^2 \leq 2(1+|z|^2)(1+|w|^2)$. Approximation of the massive integral by kernels away from its integrable singularity gives continuous normal representatives. Thus this estimate really defines the value at $r = 0$; an L^∞ equivalence class alone would not do so.

The same argument upgrades the nonlocal part of the inverse itself. Writing $K_b^Q = \mathsf{Q}K_b\mathsf{Q}$, the frozen inverse identity gives

$$L_b = K_b^Q + K_b^Q * L_b.$$

Here Q is bounded on $\mathcal{X}$ as well as on $\mathcal{H}$, uniformly in s. Thus the $\mathcal{H} \to \mathcal{X}$ kernel bound for K_b and the $\mathcal{H} \to \mathcal{H}$ two-moment bound for L_b give an $\mathcal{H} \to \mathcal{X}$ two-moment bound for L_b. Differentiate the identity to obtain the same conclusion for fixed metric derivatives. The local identity part of C_b^{-1} acts on $\mathcal{X}$ directly. Consequently the later point-normal reconstruction does not infer point evaluation from a Hilbert-space inverse alone. These point-normal kernels also have the required uniform tails. Cutting off a neighborhood of the integrable singularity and the far tail reduces their continuity to convergence on a compact set away from the singularity. The common massive majorants make both discarded weighted norms uniformly small. Hence K_b is continuous in weighted $L^1(\mathcal{B}(\mathcal{H},\mathcal{X}))$ along the retained family. The convolution identities just proved give the corresponding continuity for L_b and Γ_b and their metric derivatives. Compactness and the same parameter Sobolev argument then give point-normal common envelopes with uniformly vanishing weighted tails.

The same weighted tails hold for the scalar jump kernels used below. Let $J_b(z) = \langle f, K_b(z)f\rangle$. The frozen Schur identity and $C_b^{-1} = I + L_b$ give

$$k_b = J_b + E_b * D_b + E_b * L_b * D_b.$$

Indeed the scalar Schur operator is $MI - k_b*$; Fourier uniqueness identifies this kernel with the Bernstein jump kernel having symbol $\Sigma(\xi^T b\xi)$. Every factor on the right and every fixed metric derivative is continuous in the weighted convolution algebra on the compact order/metric family. Their products are therefore compact in weighted L^1. The parameter Sobolev argument above then gives common envelopes and uniformly vanishing second-moment tails for k_b and its derivatives. This supplies tight tails, which a uniform bound on (6.7) alone would not imply.

6.3. **Exact variable elimination and the row-preserving midpoint.** For a kernel family write $K[b]v(x) = \int K_{b(x)}(z)v(x-z)\,\mathrm{d}z$. If $\int H_b(w)\,\mathrm{d}w = 0$, exact expansion gives

$$\begin{aligned}&(K[b]H[b] - (K_b * H_b)[b])a(x)\\ &\quad = \iint K_{b(x)}(z)[H_{b(x-z)}(w) - H_{b(x)}(w)][a(x-z-w) - a(x-z)]\,\mathrm{d}w\,\mathrm{d}z. \end{aligned}\tag{6.10}$$

The parameter difference costs $C\kappa|z|$ and the retained phase difference costs $C\kappa|w|$. Two first-moment envelopes therefore make (6.10) $O(\kappa^2)$. Without the zero-mass cancellation, the composition defect still has a useful positive majorant. Put $P[b] = I + L[b]$, the frozen inverse quantized at the receiver, and let k, ℓ, ℓ_1, k_1 be positive scalar majorants for K_b^Q, L_b and their first metric derivatives. Their zeroth and first moments are finite, uniformly in s. In the similarity gauge $C[b] = I - K^Q[b]$, exact expansion gives

$$C[b]P[b] = I + U_R, \qquad P[b]C[b] = I + U_L,$$

where, componentwise in norm,

$$|U_R F| \le u_R * |F|, \quad u_R = C\kappa\,(|z|k(z)) * \ell_1, \qquad |U_L F| \le u_L * |F|, \quad u_L = C\kappa\,(|z|\ell(z)) * k_1.$$

Both positive kernels have mass $O(\kappa)$. For large indices their Neumann series converge on $L^\infty_x\mathcal{H}$, and the left and right inverses coincide. Hence

$$C[b]^{-1} = P[b](I + U_R)^{-1}, \qquad |(C[b]^{-1} - P[b])F| \le \rho_\kappa * |F|,$$

where one may take

$$\rho_\kappa = (\delta_0 + \ell(z)\,\mathrm{d}z) * \sum_{n\ge 1} u_R^{*n}, \qquad \rho_\kappa(\mathbb{R}^6) = O(\kappa).$$

These are positive finite measures controlling operator norms, not positivity assertions about the inverse. Only their mass is needed; no extra third moment is inferred from this construction.

Let $q = j - fa$ and define the *exact* variable Schur operator and source map by

$$\begin{aligned} S[b] &= \langle f, B[b]f\,\cdot\rangle - E[b]C[b]^{-1}D[b],\\ \mathfrak{S}_b F &= \langle f, F\rangle - E[b]C[b]^{-1}\mathsf{Q}F. \end{aligned}\tag{6.11}$$

Thus (6.1) implies, with no frozen-quantization identification,

$$S[b]a = \mathfrak{S}_b(R+E), \qquad q = -C[b]^{-1}D[b]a + C[b]^{-1}\mathsf{Q}(R+E).\tag{6.12}$$

Set $q_0 = \Gamma[b]a$. The zero-mass composition formula gives

$$C[b]q_0 + D[b]a = O(\kappa^2), \qquad S[b]a = (\Sigma(\xi^T b\xi))[b]a + O(\kappa^2).$$

These are estimates of the action on the actual slowly varying phase, not equality of two quantizations.

For later source estimates define the frozen left source map

$$\mathcal{T}^{\mathrm{fr}}[b]F = \langle f, F\rangle - (E_b * C_b^{-1})[b]\mathsf{Q}F.$$

It has an identity pairing and an integrable kernel dominated by a positive finite convolution measure of uniformly bounded mass. Besides $C[b]^{-1} - P[b]$, the difference from the exact source map contains the composition defect $E[b]P[b] - (E_b * C_b^{-1})[b]$. Its majorant is $C\kappa(|z|e(z)) * \ell_1$, where e dominates the kernel of E_b. It follows that

$$|(\mathfrak{S}_b - \mathcal{T}^{\mathrm{fr}}[b])F| \le \omega_\kappa * \|F\|_{\mathcal{H}}, \qquad \omega_\kappa(\mathbb{R}^6) = O(\kappa).\tag{6.13}$$

This accounts for quantization of the product as well as correction of the inverse. An inverse defect alone would not account for both.

All source maps preserve bounded fixed-order physical jets of the entire source. To justify this without differentiating a label piece, write $C[b]v = F$. A translated difference satisfies the same bounded inverse equation with right side the translated difference of F plus the parameter-kernel difference applied to the translated v. The argument used for (5.16) first proves uniform translation continuity and then convergence of difference quotients. Induction gives

$$\|C[b]^{-1}F\|_{C_x^N\mathcal{H}} \le C_N\|F\|_{C_x^N\mathcal{H}},$$

with constants depending on the stated absolute jets of b and kernel-parameter derivatives. For $F \in C_x^N\mathcal{X}$, the identity $v = F + K^Q[b]v$ and the $\mathcal{H} \to \mathcal{X}$ kernel bound also give bounded $C_x^N\mathcal{X}$ jets. The fixed pairing and $E[b]$ give the corresponding scalar source jets. The same reasoning applies to the small midpoint perturbation. No smallness of differentiated sources or of differentiated comparison errors is claimed.

We next compare with the full symmetric midpoint model. In the actual gauge the frozen diagonal is 1. Writing $K_b(z;r,t) = U(r)\mathcal{G}_b(z;r,t)U(t)$, the midpoint diagonal is chosen to kill f exactly and is interpreted on the closed fiber as

$$d^{\mathrm{mid}}(x,r) = \frac{1}{w(r)}\iint \mathcal{G}_{b(x-z/2)}(z;r,t)U(t)f(t)\,\mathrm{d}t\,\mathrm{d}z.$$

This cancels U before division and gives the continuous value at the cutoff edge. The uniform positive lower bound of w on the fixed fiber and the point-normal parameter envelopes show that $d^{\mathrm{mid}} - 1 = O(\kappa)$ in multiplication-operator norm. Thus $B^{\mathrm{mid}} - B[b]$, including its diagonal, has a positive operator-norm convolution majorant of mass $O(\kappa)$. The already constructed inverse and another Neumann series give a bounded inverse of its complementary block. This perturbation also preserves the $O(\kappa)$ source-map-majorant conclusion.

On the actual phase, $\operatorname{Lip} a = O(\kappa)$ and the zero mass of Γ_b give $q_0 = \Gamma[b]a = O(\kappa)$, in both $\mathcal{H}$ and $\mathcal{X}$. The row-killed midpoint-minus-left difference on fa is

$$\iint [K_{b(x-z/2)} - K_{b(x)}](z;r,t)f(t)[a(x) - a(x-z)]\,\mathrm{d}t\,\mathrm{d}z = O(\kappa^2),$$

by its coefficient and phase increments. Its action on q_0 is $O(\kappa^2)$ by the operator norm just established. Thus q_0 has a complementary residual $O(\kappa^2)$ for either operator. Correcting this residual with the corresponding bounded inverse shows that both exact Schur actions agree with the frozen left scalar action up to $O(\kappa^2)$.

Finally let k_b be the scalar jump kernel with symbol $\Sigma(\xi^T b\xi)$. Its parameter-derivative two-moment envelopes give directly

$$\left|\int [k_{b(x-z/2)}(z) - k_{b(x)}(z)][a(x) - a(x-z)]\,\mathrm{d}z\right| \le C\kappa^2\int |z|^2 m(z)\,\mathrm{d}z.$$

This is precisely the difference between the row-killed scalar midpoint and left actions. It proves the claimed midpoint comparison without identifying two quantizations. The scalar diagonal is the integral of its own midpoint jump row, consistently with the full fiber diagonal above. All physical-jet assertions are absolute bounds and follow by the kernel and inverse differentiation argument; no relative smallness of their higher derivatives is needed.

Localization by $\eta_\kappa(x) = \eta(\kappa x)$ is quadratic as well. In its commutator subtract $j(x)$ from $j(x-z)$, paying $C\kappa^2|z|^2$. For the remaining term subtract the linear Taylor term of η_κ, whose integral vanishes because the frozen kernel is even. Its remainder has the same budget. Every extension outside the actual buffer must still satisfy the exterior estimate assumed in (6.1).

6.4. **Direct source capacity and the two frequency ranges.** Define an inhomogeneous dyadic norm in curvature variables $y = \kappa x$:

$$\|F\|_{Y_\kappa} = \|\Delta_{-1}F\|_\infty + \sum_{l\ge 0}\max(2^{-l}, \kappa^2 2^l)\|\Delta_l F\|_\infty. \tag{6.14}$$

A bounded family $F_j \in Y_{\kappa_j}$ is called tight if

$$\lim_{L\to\infty} \limsup_{j\to\infty} \sum_{l\ge L} \max(2^{-l}, \kappa_j^2 2^l)\|\Delta_l F_j\|_\infty = 0.$$

We use the direct-capacity lemma (5.19), already proved in the preceding section. Recall its hypotheses: for a Banach-valued piece e, a nonnegative density v, and $\alpha \in [\alpha_*, 1]$, $\alpha_* > 1/2$, one has

$$\|e(y)\| \le \frac{A}{h}\min\{1, (h^2 v(y)/a)^\alpha\}, \qquad \int_{B_r(z)} v \le K(r^4 + r^6) \quad (0 < r \le 1), \qquad \int v \le CK.$$

All parameters A, h, a, K here are positive package parameters. The lemma gives the block estimate

$$r\|\Delta_l e\|_\infty \le C\min\{Ar/h, Aa^{-\alpha}h^{2\alpha-1}K^\alpha r^{1-2\alpha}\}, \qquad r = 2^{-l}, \tag{6.15}$$

and its summed conclusion

$$\|\Delta_{-1} e\|_\infty + \sum_{l\ge 0} 2^{-l}\|\Delta_l e\|_\infty \le CA\sqrt{K/a}. \tag{6.16}$$

We only have to show that the actual source map preserves these packages; it need not commute with dyadic blocks.

Let T be any of the variable source maps with a common positive convolution majorant μ, including a possible atom for its local part, and write $M = \mu(\mathbb{R}^6)$. If $M = 0$, its contribution is zero. Otherwise let μ_κ be the pushforward under $z \mapsto \kappa z$ and $\nu_\kappa = M^{-1}\mu_\kappa$, a probability measure. For a physical source $F(x) = \kappa^2 e(\kappa x)$, pointwise domination gives

$$\|\kappa^{-2} TF(y/\kappa)\| \le M \int \|e(y - z)\|\, \nu_\kappa(\,\mathrm{d}z).$$

If e has the package with density v, set $v' = \nu_\kappa * v$. The function $t \mapsto \min(1, t^\alpha)$ is concave for $0 < \alpha \le 1$, so Jensen gives the new pointwise package

$$\|\kappa^{-2} TF(y/\kappa)\| \le \frac{MA}{h}\min\{1, (h^2 v'(y)/a)^\alpha\}.$$

Tonelli and translation-uniform capacity give

$$\int_{B_r(z)} v' \le K(r^4 + r^6), \qquad \int v' \le CK.$$

Thus the transformed budget is at most $CMA\sqrt{K/a}$. The convolution need not have compact support: the proof of the direct lemma uses precisely these global ball and total-mass bounds, which were supplied by compact localization before convolution. No spatial translation of the measure depends on the evaluation point; a merely point-dependent majorant with bounded mass would not justify this argument. Here the common parameter envelopes provide the required fixed convolution majorant despite variable coefficients.

Apply this construction separately to every original measurable label package, and then sum their budgets. The exact linear output is the sum of these transformed pieces, so the budget still tends to zero. In particular it applies to $\mathcal{T}^{\mathrm{fr}}[b]$. The normalized physical remainder remains $o(1)$ in the low-frequency norm by its amplitude and the bounded L^∞ source map. Equation (6.13) additionally gives

$$\|(\mathfrak{S}_b - \mathcal{T}^{\mathrm{fr}}[b])R\|_\infty = o(\kappa^{5/2}),$$

whose normalized low-frequency norm is $o(\kappa^{1/2})$. Only zeroth-order domination is used in this package transfer; no smoothness of an individual measurable label piece is required.

For the high frequencies use the entire-source estimate (5.25), with the source-map jet bounds just proved. If $\mathfrak{F}(y) = \kappa^{-2}F(y/\kappa)$, physical frequency t corresponds to curvature frequency t/κ. For $t \ge 1$, the weight in (6.14) is κt; thus the contribution is bounded by

$$C\kappa^{-1} \sum_{t\ge 1 \text{ dyadic}} t\min(A, C_N t^{-N}) \le C_N \kappa^{-1} A^{1-1/N}. \tag{6.17}$$

A dyadic grid shifted by a factor in $[1,2]$ gives the same bound, so no dyadic assumption on κ is needed. For the exact source from R, its amplitude remains $o(\kappa^{3/2})$ and its absolute physical jets are bounded; hence this contribution is $o(1)$ for $N > 3$. For a source of size $O(\kappa^2)$ with the same jet bounds, the high contribution is $O(\kappa^{1-2/N}) = o(1)$. Its physical-low tail above curvature frequency 2^L is bounded by $C\sum_{l\ge L} 2^{-l} \le C2^{-L}$. Consequently the exact normalized scalar source from R tends to zero in Y_κ, while the sources from E and the Schur, midpoint, and localization errors are bounded and tight in that norm. Only the full smooth physical output is used at high frequencies.

6.5. **The midpoint cancellation and Lipschitz-coefficient absorption.** The following annular bounds have an explicit measure proof. Its derivative hypotheses follow from the positive measure representation rather than constitute an additional input. For $r \ge 1$ that representation gives

$$|\Sigma^{(r)}(\tau)| = r! \int \frac{t}{(\tau+t)^{r+1}}\,\mu(\mathrm{d}t) \le r!A_s\tau^{1-r},$$
$$\tau^r|\Sigma^{(r)}(\tau)| \le r! \int \frac{t}{\tau+t}\,\mu(\mathrm{d}t) = r![M-\Sigma(\tau)].$$

The first inequality uses $t^2\tau^{r-1} \le (\tau+t)^{r+1}$ and $\int t^{-1}\mu = A_s$. The chain rule for $\Sigma(|\zeta|^2)$ therefore gives all low annular derivative bounds $C_N|\zeta|^{2-N}$ and all high bounds $C_N|\zeta|^{-p-N}$ for the mass-subtracted symbol. The same estimates hold for each metric expansion coefficient: differentiating $\Sigma(\zeta^T b\zeta)$ in b only adds powers of $|\zeta|^2$ and derivatives of Σ, and the analytic coefficient extraction supplies its stated C_ν.

Let $\psi_\nu(\zeta) = \int[1-\cos(\zeta\cdot z)]k_\nu(z)\,\mathrm{d}z$ be an even signed jump symbol with finite absolute second moment and annular derivative bounds

$$|\psi_\nu(\zeta)| \le C_\nu \min(|\zeta|^2, 1), \qquad |D\psi_\nu(\zeta)| \le C_\nu \begin{cases} |\zeta|, & |\zeta| \le 1, \\ |\zeta|^{-2s-1}, & |\zeta| \ge 1. \end{cases}$$

Higher rescaled annular derivatives obey the corresponding bounds. The scalar metric family has such an expansion: write

$$k_{I+B}(z) = \det(I+B)^{-1/2}k_0((I+B)^{-1/2}z)$$

and expand analytically in the small entries of B. The retained similarity gauge has $d_0 = 1$.

More explicitly, for b in a small complex neighborhood of I, the Gaussian representation reads

$$k_b(z) = \int_{(0,\infty)} \int_0^\infty te^{-tu}(4\pi u)^{-3}(\det b)^{-1/2}e^{-z^Tb^{-1}z/(4u)}\,\mathrm{d}u\,\mu(\mathrm{d}t).$$

Choose the neighborhood so that $\mathrm{Re}(z^Tb^{-1}z) \ge c|z|^2$ for real z. Its absolute Gaussian majorant has mass at most C and second moment at most Cu at time u. Integration uses only $\int\mu = M$ and $\int t^{-1}\mu = A_s$. Cauchy's inequality therefore gives Taylor kernels k_ν with

$$k_{I+B} = k_0 + \sum_{|\nu|\ge1} B^\nu k_\nu, \qquad \int(1+|z|^2)|k_\nu(z)|\,\mathrm{d}z \le CR^{-|\nu|}$$

for a fixed $R > 0$. The series converges in weighted L^1 when B is sufficiently small. The same Cauchy argument for the annular symbol estimates gives the constants C_ν used above, with at most exponential growth in $|\nu|$ for each required derivative order. For the variable metric define $b_{j,\nu}(y) = (b_j(y/\kappa_j) - I)^\nu$. Thus, by taking η_0 sufficiently small,

$$\sum_{|\nu|\ge1} C_\nu\|b_{j,\nu}\|_\infty \le C\eta_0, \qquad \sum_{|\nu|\ge1} C_\nu\|D_y b_{j,\nu}\|_\infty \le C.$$

The derivative bound follows by differentiating each monomial and summing $|\nu|\eta_0^{|\nu|-1}$ with its exponential constants. The number of monomials at each degree has only polynomial growth. In the following bilinear estimates b denotes one such scalar coefficient; the constant metric term is $\mathcal{A}_\kappa a$.

Define the midpoint operator by its form

$$\langle T_{\kappa,\nu}(b,a),v\rangle=\frac{\kappa^{-2}}{2}\iint b(y+z/2)[a(y)-a(y+z)][v(y)-v(y+z)]\kappa^{-6}k_\nu(z/\kappa)\,\mathrm{d}z\,\mathrm{d}y. \tag{6.18}$$

With input frequency η and output frequency ξ, its exact bilinear symbol is

$$q_{\kappa,\nu}(\xi,\eta)=\kappa^{-2}\left[\psi_\nu\left(\frac{\kappa(\eta+\xi)}{2}\right)-\psi_\nu\left(\frac{\kappa(\eta-\xi)}{2}\right)\right]. \tag{6.19}$$

Indeed the two differences in the form give the product of two sines; the cosine product formula gives (6.19). It vanishes at both $\xi=0$ and $\eta=0$. If $|\xi|=\rho\le|\eta|/8=\ell/8$, the fundamental theorem of calculus gives

$$|q_{\kappa,\nu}(\xi,\eta)|\le CC_\nu\rho\ell\min\{1,(\kappa\ell)^{-2s-2}\}. \tag{6.20}$$

Differentiating the same formula on the two rescaled annuli, and integrating by parts in their Fourier variables, gives bilinear kernel L^1 bounds with the same right side.

Fix an inhomogeneous Littlewood–Paley decomposition, and write

$$L_l=2^l,\qquad h_l=\kappa L_l,\qquad w_{\kappa,l}=\max(L_l^{-1},\kappa^2L_l),\qquad \mathcal{A}_\kappa=\kappa^{-2}\Sigma(-\kappa^2\Delta_y).$$

We use the row-preserving gauge $d_0=1$. Besides Y_κ, use the weaker Banach space of distributions with norm

$$\|F\|_{Z_\kappa}=\|\Delta_{-1}F\|_\infty+\sup_{l\ge0}w_{\kappa,l}\|\Delta_lF\|_\infty.$$

Both norms include the lowest block; Y_κ embeds continuously in Z_κ.

Decompose $T_{\kappa,\nu}(b,a)$ according to the coefficient and input frequencies. We first estimate the low-coefficient/high-input part. Write $\beta=\xi-\eta$, $L=2^l$, and impose $|\beta|\le L/16$, $|\eta|\asymp L$. After division by the exact input multiplier of $\mathcal{A}_\kappa$, its rescaled symbol is

$$m_h(v,u)=\frac{\psi_\nu(h(u+v/2))-\psi_\nu(hv/2)}{\Sigma(h^2|u|^2)},\qquad u=\eta/L,\quad v=\beta/L,\quad h=\kappa L.$$

Include smooth compact cutoffs to these frequency regions. For the first numerator term, $|u+v/2|$ stays away from zero. If $h\le1$, all fixed-order derivatives in (u,v) are $O(C_\nu h^2)$ and the inverse denominator has derivatives $O(h^{-2})$. If $h\ge1$, the numerator is its finite mass plus a term with bounded rescaled derivatives, and the inverse denominator has bounded derivatives. Fourier integration by parts in (u,v) therefore gives a uniform bilinear kernel L^1 bound for this term.

Estimate the second term without summing the separate blocks of the low coefficient. If $\chi(v)$ is its cutoff and $\phi=\mathcal{F}^{-1}\chi$, the coefficient kernel is

$$\int k_\nu(z)\left[\phi(w)-\frac{\phi(w+hz/2)+\phi(w-hz/2)}{2}\right]\,\mathrm{d}z.$$

The bracket has L^1_w norm at most $C\min(h^2|z|^2,1)$. The absolute second moment and mass of k_ν bound this kernel by $CC_\nu\min(h^2,1)$. The input inverse multiplier has L^1 kernel norm $O(h^{-2})$ for $h\le1$ and $O(1)$ for $h\ge1$. Thus this term also has a uniform bilinear kernel bound. Returning to scale L gives

$$\|\mathcal{P}_{\kappa,\nu,b,l}F\|_\infty\le CC_\nu\|b\|_\infty\|\widetilde{\Delta}_lF\|_\infty.$$

The output is comparable to L. Finite overlap and comparability of the weights on neighboring annuli prove, for $\mathcal{W}=Y,Z$,

$$\|\mathcal{P}_{\kappa,\nu,b}F\|_{\mathcal{W}_\kappa}\le CC_\nu\|b\|_\infty\|F\|_{\mathcal{W}_\kappa}.$$

On Z_κ the map is defined directly by this dyadic series and its finite-overlap bounds; no density of finite truncations in the supremum norm is used. Only input annuli above a fixed lowest scale are assigned to this principal map. The finitely many bounded-frequency interactions are included in the remainder.

Here are the other three frequency regions. In each case $\|\Delta_lb\|_\infty\le CL^{-1}\|Db\|_\infty$ and $\|\Delta_la\|_\infty\le CL^{-1}\|Da\|_\infty$. For high coefficient and low input, factor the low input derivative

by the fundamental theorem of calculus in (6.19). The remaining multiplier is smooth on the high annulus and has bilinear kernel norm

$$CC_\nu L \min\{1, (\kappa L)^{-p-2}\}.$$

Apply it to the whole low-frequency derivative of a; its norm is bounded by $C\|Da\|_\infty$, without a sum of low block norms. The output has size at most $CC_\nu \|Db\|_\infty \|Da\|_\infty \min\{1, (\kappa L)^{-p-2}\}$. Multiplication by $w_{\kappa,l} = L^{-1}\max(1, \kappa^2 L^2)$ bounds it by $CC_\nu \|Db\|_\infty \|Da\|_\infty L^{-1}$.

For comparable coefficient and input frequencies and comparable output, the bilinear kernel norm is $CC_\nu \kappa^{-2} \min((\kappa L)^2, 1)$. Indeed the compact-cutoff inverse kernel is translated in a fixed linear combination of its two variables by the jump term; its second difference has L^1 norm $C\min((\kappa L)^2|z|^2, 1)$. Integration against $|k_\nu|$ gives the claim. The two Lipschitz blocks then have weighted size $CC_\nu \|Db\|_\infty \|Da\|_\infty L^{-1}$.

For comparable input frequencies L and output frequency $\rho = 2^k \le L/8$, the separated-annulus derivative bounds in (6.20) give bilinear kernel norm

$$CC_\nu \rho L \min\{1, (\kappa L)^{-p-2}\}.$$

Use the independent Fourier variables ξ/ρ and η/L; the change back to coefficient/input variables preserves the kernel L^1 norm. After the two Lipschitz blocks the bound is

$$CC_\nu \|Db\|_\infty \|Da\|_\infty \frac{\rho}{L} \min\{1, (\kappa L)^{-p-2}\}.$$

Sum over dyadic $L \ge 8\rho$. If $\kappa\rho \le 1$, the sum is bounded by a constant; if $\kappa\rho \ge 1$, it is bounded by $C(\kappa\rho)^{-p-2}$. Multiplication by $w_{\kappa,k}$ gives at most $CC_\nu \|Db\|_\infty \|Da\|_\infty 2^{-k}$ in both cases.

These estimates give the exact decomposition

$$T_{\kappa,\nu}(b,a) = \mathcal{P}_{\kappa,\nu,b}(\mathcal{A}_\kappa a) + G_{\kappa,\nu}, \qquad \sum_{k\ge K} w_{\kappa,k}\|\Delta_k G_{\kappa,\nu}\|_\infty \le CC_\nu \|Db\|_\infty \|a\|_{W^{1,\infty}} 2^{-K} \quad (K \ge K_*). \tag{6.21}$$

Here K_* is a fixed index depending only on the decomposition. The bounded output blocks are controlled by $CC_\nu \|b\|_{W^{1,\infty}} \|a\|_{W^{1,\infty}}$. For the low–low part, evenness pairs the two opposite jumps: one term contains a second difference of the low-frequency a, and the other a coefficient difference times an input difference. The second moment bounds both. For high–high contributions to a bounded output block, the preceding separated-output estimate with ρ bounded gives a convergent dyadic sum. Thus $G_{\kappa,\nu}$ is bounded in Y_κ and has the displayed uniformly tight tail. The restriction $K \ge K_*$ leaves the amplitude-controlled lowest blocks outside the derivative-only bound.

Apply this to $\mathfrak{a}_j(y) = a_j(y/\kappa_j)$ and the metric expansion coefficients. Preparation gives a uniform bounded Lipschitz norm for $\mathfrak{a}_j$ and $\sum_\nu C_\nu \|D_y b_{j,\nu}\|_\infty \le C$. The small-amplitude bound $\sum_\nu C_\nu \|b_{j,\nu}\|_\infty \le C\eta_0$ makes the sum $\mathcal{P}_j$ a contraction of norm at most $1/2$ on both Y_{κ_j} and Z_{κ_j}. The summed remainder G_j is bounded and tight in Y_{κ_j}. The scalar equation is

$$(I + \mathcal{P}_j)\mathcal{A}_{\kappa_j}\mathfrak{a}_j = F_j - G_j,$$

where F_j is bounded and tight by the preceding source estimates.

To identify its solution without presupposing a summable gradient, first place the actual $\mathcal{A}_\kappa\mathfrak{a}$ in Z_κ. For $l \ge 0$, annular kernel estimates give

$$w_{\kappa,l}\|\Delta_l \mathcal{A}_\kappa \mathfrak{a}\|_\infty \le C 2^l \|\widetilde{\Delta}_l \mathfrak{a}\|_\infty \le C\|D\mathfrak{a}\|_\infty.$$

The lowest block also has a uniform bound. Let ϕ_0 be the Schwartz kernel of Δ_{-1}. Evenness of k_0 shows that the convolution kernel of $\Delta_{-1}\mathcal{A}_\kappa$ is

$$H_\kappa(w) = \kappa^{-2} \int k_0(z) \left[\phi_0(w) - \frac{\phi_0(w+\kappa z) + \phi_0(w-\kappa z)}{2}\right] \mathrm{d}z.$$

Consequently

$$\|H_\kappa\|_1 \le C\kappa^{-2} \int \min(\kappa^2|z|^2, 1) k_0(z)\, \mathrm{d}z \le C \int |z|^2 k_0(z)\, \mathrm{d}z \le C, \qquad \|\Delta_{-1}\mathcal{A}_\kappa\mathfrak{a}\|_\infty \le C\|\mathfrak{a}\|_\infty.$$

This uses cancellation after the low-frequency cutoff; the raw κ^{-2} mass bound would not suffice.

The Neumann series applied to $F_j - G_j$ converges in Y_{κ_j}, hence also in Z_{κ_j}. Uniqueness in Z_{κ_j} identifies that solution with the actual $\mathcal{A}_{\kappa_j}\mathfrak{a}_j$. The same series proves tightness. Indeed a fixed principal map uses only finitely neighboring input/output annuli, so each fixed power preserves tight families. The remainder after m terms has norm at most $2^{1-m}\|F_j - G_j\|_{Y_{\kappa_j}}$. First choose m, then take the dyadic tail of the finite sum, and finally let $m \to \infty$.

The annular derivative bounds for Σ and its positive lower bound give the gradient inverse estimate

$$\|D\Delta_l \mathfrak{a}_j\|_\infty \le C \max(2^{-l}, \kappa_j^2 2^l)\|\widetilde{\Delta}_l \mathcal{A}_{\kappa_j}\mathfrak{a}_j\|_\infty, \qquad l \ge 0. \tag{6.22}$$

For physical frequency $\kappa_j 2^l \le 1$, the inverse derivative has scale 2^{-l}; for $\kappa_j 2^l \ge 1$, it has scale $\kappa_j^2 2^l$. Rescaled annular derivatives give the same bounds for its convolution-kernel L^1 norm. Tightness in Y_{κ_j} therefore gives uniformly vanishing gradient tails. Each finite collection of gradient blocks is equicontinuous by Bernstein's inequality. The lowest solution block is bounded directly by $\|\mathfrak{a}_j\|_\infty$, without inverting the zero frequency. Thus $D\mathfrak{a}_j$ is locally equicontinuous and the phases are locally precompact in C^1.

6.6. **Forced point-normal reconstruction and curvature retention.** Fix a smaller curvature chart whose closure lies in the evaluation plateau. On its physical preimage the localized field agrees with $j_j = f_j N_j$, where N_j is the actual unit normal. Outside this plateau we use only the bounded jets and Lipschitz estimates of the controlled extensions supplied by the preparation.

Set

$$q_{0,j} = \Gamma[b_j]a_j, \qquad e_j = j_j - f_j a_j - q_{0,j}, \qquad r_{0,j} = C[b_j]q_{0,j} + D[b_j]a_j.$$

The zero-mass composition formula (6.10), with the point-normal first-moment envelopes, gives $\|r_{0,j}\|_{L^\infty_x\mathcal{H}} + \|r_{0,j}\|_{L^\infty_x\mathcal{X}} = O(\kappa_j^2)$. Since e_j is complementary, its exact equation is

$$C[b_j]e_j = F_j^\perp, \qquad F_j^\perp = \mathsf{Q}_j(R_j + E_j) - r_{0,j}.$$

The fixed-fiber projection is bounded on both normal spaces, and (6.2) gives $\|F_j^\perp\|_{L^\infty_x\mathcal{X}} = o(\kappa_j^{3/2})$. The genuine complementary inverse first gives

$$\|e_j\|_{L^\infty_x\mathcal{H}} = o(\kappa_j^{3/2}).$$

In the adopted gauge its identity part is I, so

$$e_j = \mathsf{Q}_j K[b_j]\mathsf{Q}_j e_j + F_j^\perp.$$

The integrable $\mathcal{H} \to \mathcal{X}$ kernel envelope then implies

$$\|e_j\|_{L^\infty_x\mathcal{X}} \le C\|e_j\|_{L^\infty_x\mathcal{H}} + \|F_j^\perp\|_{L^\infty_x\mathcal{X}} = o(\kappa_j^{3/2}).$$

The point-normal conclusion uses the $\mathcal{X}$ regularity of the forcing, including the identity term of the inverse.

The formula $e_j = j_j - f_j a_j - \Gamma[b_j]a_j$ also gives bounded physical $C^N_x\mathcal{X}$ jets: differentiate the smooth coefficient and phase, using the integrable point-normal parameter-derivative kernels. No small differentiated forcing is required. Banach-valued tangential interpolation on a smaller buffer yields

$$\|D_x e_j\|_{L^\infty_x\mathcal{X}} \le C\|e_j\|^{1-1/N}_{L^\infty_x\mathcal{X}}\|e_j\|^{1/N}_{C^N_x\mathcal{X}} + C\|e_j\|_{L^\infty_x\mathcal{X}} = o(\kappa_j). \tag{6.23}$$

Indeed $\frac{3}{2}(1 - 1/N) > 1$ for $N > 3$.

The zero mass of Γ_b and its parameter derivatives gives

$$\begin{aligned} D_x q_{0,j}(x) = &\int \Gamma_{b_j(x)}(z)[D_x a_j(x-z) - D_x a_j(x)]\,\mathrm{d}z \\ &+ D_x b_j(x)\int \partial_b\Gamma_{b_j(x)}(z)[a_j(x-z) - a_j(x)]\,\mathrm{d}z. \end{aligned}$$

The second term is $O(\kappa_j^2)$ by two physical Lipschitz bounds and the parameter kernel's first moment. In the first term, divide by κ_j and write the bracket as $D\mathfrak{a}_j(y - \kappa_j z) - D\mathfrak{a}_j(y)$. For bounded z this tends uniformly to zero on each smaller curvature chart by the established

gradient modulus. The global Lipschitz bound and the uniform kernel tails control the remaining z. Hence

$$D_x q_{0,j} = o(\kappa_j) \quad \text{in } L^\infty_x \mathcal{X} \quad \text{on smaller curvature charts,} \qquad \|q_{0,j}\|_{L^\infty_x \mathcal{X}} = O(\kappa_j).$$

At the actual zero, where the tangential localization equals one,

$$n_j(y) := N_j(y/\kappa_j, 0) = \mathfrak{a}_j(y) + \frac{q_{0,j}(y/\kappa_j, 0) + e_j(y/\kappa_j, 0)}{f_j(0)}.$$

Since $f_j(0) \geq c_P$, the value estimate and the derivative estimates divided by κ_j give $n_j - \mathfrak{a}_j \to 0$ in local C^1.

Theorem 6.2 (Curvature retention in stable ambient dimension seven)**.** *Under the preceding preparation, direct source budgets, and the model calculus proved above, the curvature-scaled zero normals are precompact in C^1 on every smaller moving curvature chart. If $|A_{\Sigma_j}(P_j)| \geq c_0 \kappa_j$ at marked points converging in the rescaled charts to P, the limit retains $|A(P)| \geq c_0$ when expressed intrinsically.*

Proof. The phase compactness and point-normal reconstruction give the stated normal compactness by Arzelà–Ascoli. For the rescaled height $H_j(y) = \kappa_j h_j(y/\kappa_j)$, choose the upward normal so its last component stays uniformly positive. Then

$$DH_j = -\frac{n_j^{\mathrm{hor}}}{n_j^7}.$$

Thus C^1 convergence of normals gives C^1 convergence of graph gradients, and local C^2 graph convergence after fixing a vertical constant. With $G_j = I + DH_j \otimes DH_j$, the exact identity is

$$\kappa_j^{-2} |A_{\Sigma_j}|^2 = (G_j^{-1})^{ab} \partial_a n_j \cdot \partial_b n_j.$$

Both its metric and normal-derivative factors converge uniformly. At the marked point it therefore retains the lower curvature bound. All estimates are uniform on the moving charts allowed by the preparation. □

Evenness improves the value of the complementary corrector. Its first moment is zero, and hence

$$q_{0,j}(x) = \int \Gamma_{b_j(x)}(z)[a_j(x - z) - a_j(x) + D_x a_j(x) z] \, \mathrm{d}z.$$

For $|z| \leq R$, the bracket divided by κ_j is bounded by $|z|\omega(\kappa_j R)$, where ω is a common local modulus of $D\mathfrak{a}_j$ on a slightly larger curvature chart. For $|z| > R$ it is bounded by $C|z|$ using the controlled extension. The uniform first-moment tails, first letting $j \to \infty$ and then $R \to \infty$, show $q_{0,j} = o(\kappa_j)$ in $\mathcal{X}$ locally on smaller curvature charts. Since $e_j = o(\kappa_j^{3/2})$, division by $f_j(r)$ is now legitimate on each finite union I of compact intervals contained in E_P: profile convergence gives $\inf_{j \geq j_I, r \in I} f_j(r) > 0$. On the evaluation plateau,

$$N_j(x, r) - a_j(x) = \frac{q_{0,j}(x, r) + e_j(x, r)}{f_j(r)} = o(\kappa_j).$$

Using the same relation at $r = 0$ proves

$$q_{0,j} = o(\kappa_j) \text{ in } \mathcal{X}, \qquad \sup_{r \in I} |N_j(x, r) - N_j(x, 0)| = o(\kappa_j), \tag{6.24}$$

uniformly on each smaller curvature chart. For subsequent weighted integrals, exhaust E_P by such finite unions. The endpoint source weights contain ψ_P^2 and vanish identically outside E_P. On the remaining taper sets the independent full-core physical Lipschitz bound gives $\kappa_j^{-1} |N_j(x, r) - N_j(x, 0)| \leq C_P |r|$. The common weighted kernel majorants and their absolute continuity therefore make these omitted contributions tend to zero as the compact sets exhaust E_P, uniformly in j. No division by f_j at a cutoff edge or convergence of normal derivatives in r is used.

7. Complete stable minimal limits

7.1. **The complete intrinsic carrier.** Let $\Sigma_j \subset \mathbb{R}^n$ be closed embedded C^2 hypersurfaces without boundary. Let $P_j \in \Sigma_j$, $\kappa_j > 0$, $\kappa_j \to 0$, and suppose

$$|A_{\Sigma_j}| \le \kappa_j, \qquad |A_{\Sigma_j}(P_j)| \ge c\kappa_j > 0.$$

Let M_j be the connected component containing P_j, with rescaled inclusion

$$F_j(x) = \kappa_j(x - P_j), \qquad g_j = F_j^* g_{\mathbb{R}^n}.$$

Write $\widehat{A}_j$ for its rescaled second fundamental form. Thus $|\widehat{A}_j| \le 1$ and $|\widehat{A}_j(P_j)| \ge c$. Choose global unit normals ν_j. In the regular-zero application these are $\nabla u_j/|\nabla u_j|$. For a general closed embedded hypersurface in Euclidean space, two-sidedness follows from mod-two intersection parity. A normal-reversing loop on the hypersurface would give, by a small push-off and a short closing normal segment, an ambient loop meeting it transversely exactly once. A transverse disk spanning that ambient loop has a compact one-dimensional preimage of the hypersurface, whose boundary has even cardinality, a contradiction. One may therefore choose either orientation on each component.

Normal compactness (NC).. On every sequence of marked complete small graph charts of the rescaled hypersurfaces, the normal fields in bounded-slope graph coordinates are precompact in C^1 on smaller disks. The statement must hold for moving marked points, not only at P_j, and for every chart encountered in any fixed intrinsic ball. Normal signs may be fixed at the chart center. In the present $n = 7$ application, Theorem 6.2 supplies (NC) with the preparation already established. It is not a consequence of $|\widehat{A}_j| \le 1$ alone.

Theorem 7.1. *Under (NC), after passage to a subsequence there are a connected manifold M, a point $p \in M$, a C^2 immersion $F : M \to \mathbb{R}^n$, and a global C^1 unit normal ν along F such that:*

(1) $F(p) = 0$, the induced metric $g = F^* g_{\mathbb{R}^n}$ is complete, $|A_F| \le 1$, and $|A_F(p)| \ge c$.

(2) There are relatively compact open sets $U_a \Subset U_{a+1}$ exhausting M, and, for all sufficiently large j depending on a, embeddings $\Phi_{j,a} : U_a \to M_j$ taking p to P_j. They can be taken as restrictions of the pointed convergence maps. On every compact subset of U_a,

$$F_j \circ \Phi_{j,a} \longrightarrow F \quad \text{in } C^1, \qquad \nu_j \circ \Phi_{j,a} \longrightarrow \nu \quad \text{in } C^1,$$

$$\Phi_{j,a}^* g_j \longrightarrow g \quad \text{in } C^0, \qquad |\widehat{A}_j|^2 \circ \Phi_{j,a} \longrightarrow |A_F|^2 \quad \text{in } C^0.$$

(3) Every compactly supported scalar test on M has an injectively lifted test on M_j, with convergence of its entire Dirichlet, mass, and curvature-potential integrals as stated below.

Distinct points of M are not identified when their Euclidean images coincide. In particular, the conclusion is an intrinsic carrier, not an assertion about multiplicity-one convergence of ambient subsets.

7.1.1. *Completeness, graph charts, and noncollapse.* Each M_j is intrinsically complete. An intrinsic Cauchy sequence is Euclidean Cauchy. Closedness of Σ_j supplies its ambient limit, and the embedded local graph at that limit converts ambient convergence within the eventual local branch to intrinsic convergence. Indeed, a tail of the intrinsic Cauchy sequence stays within an arbitrarily small intrinsic neighborhood of one of its points; its short connecting paths remain in the ambient graph neighborhood, so the tail lies on that branch. The component is preserved.

Here is an explicit uniform local mechanism, which also works at C^2 regularity. On an intrinsic ball of radius $r_0 = 1/10$ about q, continuation of a chosen normal along a path of length less than r_0 gives

$$\angle(\nu(x), \nu(q)) \le r_0.$$

Consequently tangent-plane projection $\pi_q \circ F$ is a local diffeomorphism there, and its inverse differential has norm at most $(\cos r_0)^{-1} < 2$. Starting at q, lift straight rays in the tangent disk of radius $r_0/4$. Their lifts have length less than $r_0/2$. They cannot stop: a finite-length terminal path is Cauchy, completeness supplies its endpoint, and the inverse function theorem continues it. The local inverses patch uniquely to a graph over that disk. Its slope is at most $\tan r_0$, and its second derivatives are bounded by a dimensional constant via the graph formula for A.

This graph contains $B_g(q, r_0/8)$. To check this without an injectivity assumption, join q to a point of that ball by a path of length less than $r_0/8$. Project the path and homotope it, keeping endpoints fixed, to its straight chord by linear interpolation. Every intermediate path has length less than $r_0/8$; all its lifts remain inside $B_g(q, r_0/4)$. Local inverse uniqueness and continuation give homotopy lifting, and their endpoints coincide. The endpoint therefore lies on the radial graph. These are graph charts of the abstract hypersurface; other ambient sheets are irrelevant.

These charts give uniform lower volume on small intrinsic balls and a uniform packing bound at a fixed small scale. More explicitly, choose $r_* > 0$ so that each intrinsic ball of radius $10r_*$ lies in one of the preceding uniformly bilipschitz graphs. A maximal r_*-separated net has a uniformly bounded number N_* of neighbors within $4r_*$ of each center. A path of length at most R from the marked point can be subdivided into at most $1+R/r_*$ pieces, and nearest net centers along it form a chain in this bounded-degree graph. Thus a fixed intrinsic ball has a covering number bounded in terms of R alone. This argument uses intrinsic paths and estimates no total area in an ambient ball.

Lemma 7.2 (Smooth injectivity check)**.** *For a complete smooth Euclidean hypersurface with* $|A| \le 1$, $\operatorname{inj}(q) \ge \pi/2$ *at every point.*

Proof. The Gauss equation gives sectional curvature at most 1, hence conjugate radius at least π by Rauch comparison. If the injectivity radius at q is less than $\pi/2$, the pointwise cut-point alternative produces a geodesic loop based at q of length $L = 2\operatorname{inj}(q) < \pi$, smooth except possibly at its base point. Its Euclidean curvature along the smooth part is $|A(\dot\gamma, \dot\gamma)| \le 1$. The single corner contributes at most π to total curvature. Fenchel's theorem for closed piecewise smooth curves gives $2\pi \le L + \pi$, a contradiction. Self-intersections of the Euclidean curve do not invalidate Fenchel's theorem. □

This lemma is only asserted in the smooth category here. A bare C^2 immersion induces only a C^1 metric, so one should not silently invoke the smooth exponential-map theory at that regularity. The preceding graph proof and the compactness theorem used next cover the stated C^2 setting directly. The actual prepared regular zero sets are smooth and do satisfy Lemma 7.2.

7.1.2. *Pointed compactness and curvature retention.*

Proof of Theorem 7.1. Apply the immersed-submanifold compactness theorem of Smith [5, Theorem 1.2 and Sections 2.2–2.3, $k = 2$], to (M_j, P_j, F_j) in the fixed pointed ambient space $(\mathbb{R}^n, 0)$. Its hypotheses require bounds for the second fundamental form on intrinsic balls, which here hold globally. It yields a complete connected pointed $C^{1,1}$ immersed limit and convergence maps that are embeddings on each fixed compact set, with local $C^{1,\alpha}$ convergence for $0 < \alpha < 1$. No volume or properness hypothesis is part of this application.

The graph mechanism above explains why this theorem is the intrinsic one appropriate here: chart overlaps are tracked on M_j, and completeness permits continued charts along paths of arbitrary finite intrinsic length. It does not glue charts merely because their ambient images approach one another. Thus a second nearby sheet cannot create an unwanted quotient. A countable exhaustion and a diagonal subsequence handle all marked charts simultaneously.

In a limiting graph chart write $F(x) = (x, h(x))$ and its corresponding prelimit graphs as $(x, h_j(x))$. The slopes are uniformly small; convergence already gives $h_j \to h$ in C^1. Their upward normals obey

$$Dh_j = -\frac{(\nu_j)_{\tan}}{(\nu_j)_n}.$$

By (NC), after a diagonal extraction the right side converges in C^1. Thus $h_j \to h$ in C^2 on smaller disks. These graph charts upgrade the limit to a C^2 immersion. Their transition maps are C^2, being tangent projections of C^2 graphs with invertible differentials. Importantly, the initial pointed convergence maps need not themselves be asserted to have convergent second derivatives.

Choose the global prelimit normals and pass to their values at the marked point. On any finite chain of overlapping carrier charts their signs are fixed by continuity from that choice. They

converge to the same normal on overlaps, since the prelimit fields agree there and the maps are injective on the entire compact chain. Loops introduce no sign ambiguity. Exhaustion therefore yields a global unit normal ν along the immersion, even if F has coincident image sheets.

For precision about the convergence maps, in a chart let $y_j = \pi \circ F_j \circ \Phi_{j,a}$ be its graph coordinate. The established pointed convergence gives $y_j \to y$ in C^1. The graph normals converge in C^1 by (NC); composing with y_j gives $\nu_j \circ \Phi_{j,a} \to \nu$ in C^1. The identity

$$(\Phi_{j,a}^* \widehat{A}_j)(V,W) = -\langle d(\nu_j \circ \Phi_{j,a})(V), d(F_j \circ \Phi_{j,a})(W)\rangle$$

proves uniform convergence of the pulled-back second fundamental forms and their norms. This proves both curvature bounds, including the nonzero mark. Completeness was already obtained for the same induced metric by pointed compactness; upgrading its regularity does not change it. □

If separate analytic arguments prove that every limiting graph has zero mean curvature, they prove minimality of this entire carrier. Interior elliptic regularity for the minimal graph equation then makes its C^2 immersion smooth locally. No new completeness or orientation argument is needed after that upgrade.

7.1.3. *Lifting whole compact tests.* Fix $\psi \in C_c^1(M)$ and take a so large that its support is contained in U_a with positive distance from the boundary. Define

$$\psi_j = \psi \circ \Phi_{j,a}^{-1} \quad \text{on } \Phi_{j,a}(U_a), \qquad \psi_j = 0 \quad \text{elsewhere on } M_j.$$

Because ψ vanishes on a neighborhood of the boundary of a smaller relatively compact carrier domain, this is a compactly supported C^1 test. It is a single function, even when its support requires many coordinates or its ambient image has nearby overlapping sheets. Change of variables, uniform ellipticity on its compact support, and the established convergence give

$$\int_{M_j} \psi_j^2 \, d\mu_j \longrightarrow \int_M \psi^2 \, d\mu,$$
$$\int_{M_j} |\nabla \psi_j|_{g_j}^2 \, d\mu_j \longrightarrow \int_M |\nabla \psi|_g^2 \, d\mu,$$
$$\int_{M_j} |\widehat{A}_j|^2 \psi_j^2 \, d\mu_j \longrightarrow \int_M |A_F|^2 \psi^2 \, d\mu.$$

Smooth approximations may be used where a prelimit analytic inequality is formulated only for smooth tests.

Proposition 7.3 (Sufficient whole-test interface)**.** *Suppose the analytic stability argument, for every such lifted test, proves*

$$\int_{M_j} \left(|\nabla \psi_j|^2 - |\widehat{A}_j|^2 \psi_j^2\right) d\mu_j \geq -r_j(\psi), \qquad r_j(\psi) \longrightarrow 0.$$

Then the complete two-sided limiting immersion is stable:

$$\int_M |A_F|^2 \psi^2 \, d\mu \leq \int_M |\nabla \psi|^2 \, d\mu \quad \textit{for every } \psi \in C_c^1(M).$$

The same conclusion follows from an arbitrary-tolerance version with error at most $o_j(1) + \epsilon \int_M (|\nabla \psi|^2 + \psi^2)$, followed by $\epsilon \downarrow 0$.

Proof. Pass to the limit in the three whole integrals. □

There is an important distinction between this interface and chartwise stability. For a square partition $\sum_\alpha \chi_\alpha^2 = 1$, summing stable inequalities for $\chi_\alpha \psi$ yields only

$$Q(\psi) \geq -\int_M \psi^2 \sum_\alpha |\nabla \chi_\alpha|^2 \, d\mu, \qquad Q(\psi) = \int_M \left(|\nabla \psi|^2 - |A_F|^2 \psi^2\right) d\mu.$$

The displayed error need not vanish on fixed small charts. Thus this argument does not establish the proposition's hypothesis.

7.2. **Minimality of the retained carrier.** Continue in stable ambient dimension 7, with tangent dimension 6. Retain $p_j = 2s_j$, $\lambda_j = 1 - s_j$ and write

$$\mathcal{F}_j = e^{-\sqrt{2}\delta_j} + \lambda_j \delta_j^{-p_j} = o(\kappa_j).$$

This force quantity is distinct from the rescaled inclusions F_j used in the carrier construction.

Use the complete intrinsic carrier and coherent compact exhaustion maps constructed in Theorem 7.1. The convergence supplied by Theorem 6.2 is C^2 for local graphs and C^1 for their zero normals. Coinciding ambient images remain distinct carrier points. We first identify the limiting mean curvature, then pass the stability form on whole compact carrier tests.

7.2.1. *A full-normal residual calculation at physical scale.* Let ρ be a genuine signed distance around a marked zero, oriented so the corresponding layer is increasing. In coordinates $(z,t) \in \mathbb{R}^6 \times \mathbb{R}$ obtained by rotating its marked normal vertically, assume on a cylinder of radius $3R$ that

$$\rho(0,t) = t, \quad \nabla\rho(0,t) = e_7, \quad \|D^2\rho\|_\infty + \|D^3\rho\|_\infty \le C\kappa, \qquad C\kappa R \le c_0. \tag{7.1}$$

Choose $\chi_1 = 1$ on B_1^6, supported in B_2^6, and $\chi_2 = 1$ on $[-1,1]$, supported in $(-2,2)$, both smooth with fixed derivative bounds. For $R \ge 2$, set

$$v(z,t) = \chi_1(z/R)\chi_2(t/R)[q_s(\rho(z,t)) - q_s(t)], \qquad Q^\sharp = q_s(t) + v(z,t).$$

The preparation and profile bounds used here include

$$\int q_s' = 2, \qquad \sum_{a=1}^{4} \|q_s^{(a)}\|_{H^1(\mathbb{R})} + \|tq_s'\|_2 \le C.$$

Taylor expansion from the normal axis gives $|\rho(z,t) - t| \le C\kappa|z|^2$ and $|\nabla\rho(z,t) - e_7| \le C\kappa|z|$. Monotonicity of q_s consequently yields

$$\|v(z,\cdot)\|_1 \le C\kappa|z|^2, \quad \sup_{|z|\le 1} \|D^\alpha v(z,\cdot)\|_2 \le C\kappa \ (|\alpha| \le 3), \quad \|v(z,\cdot)\|_2 \le C\kappa|z|^2 \ (|z| \le 1). \tag{7.2}$$

For example, the first estimate follows by integrating $q_s(t+a) - q_s(t-a)$ with $a = C\kappa|z|^2$; its integral is $4a$. The chain rule and the displayed profile bounds prove the remaining estimates. On the entire normal axis $v = \nabla v = 0$, and on its uncut part $\Delta v(0,t) = q_s'(t)\Delta\rho(0,t)$.

In the Fourier convention $\widehat{(-\Delta)^s f} = |\xi|^{2s}\widehat{f}$, write $c_{7,s} \asymp \lambda$ for the singular-integral constant. The contribution from $|Y| < 1$ to $(-\Delta)^s v(0,t)$ is

$$-\frac{c_{7,s}|\mathbb{S}^6|}{14(2-p)}\Delta v(0,t) + E_{\mathrm{near}}(t), \qquad \|E_{\mathrm{near}}\|_2 \le C\lambda\kappa.$$

Here the coefficient is $1 + O(\lambda)$, and the cubic Taylor remainder is bounded by $Cc_{7,s}\kappa\int_0^1 r^{2-p}\,\mathrm{d}r$. For the far part with $|z| \ge 1/2$, set $K_r(t) = (r^2+t^2)^{-(7+p)/2}$; then $\|K_r\|_2 = Cr^{-6-p-1/2}$. Young's inequality and (7.2) give

$$\|E_{\mathrm{far},1}\|_2 \le C\lambda\kappa \int_{1/2}^{2R} r^{1/2-p}\,\mathrm{d}r \le C_{p_*}\lambda\kappa.$$

This is where $p_* > 3/2$ is used. On $|z| < 1/2$, $|Y| \ge 1$, the normal kernel is separated from zero and has bounded L^1 norm; the final estimate in (7.2) bounds that contribution by $C\lambda\kappa$. Thus

$$\|(-\Delta)^s v(0,\cdot) + \Delta v(0,\cdot)\|_2 \le C\lambda\kappa. \tag{7.3}$$

It was essential to remove the exactly planar part before this estimate.

If $H_0 = \Delta\rho(0,0)$, the parallel-surface formula gives

$$\Delta\rho(0,t) = \sum_{a=1}^{6} \frac{\alpha_a}{1 + t\alpha_a}, \qquad |\Delta\rho(0,t) - H_0| \le C\kappa^2|t|.$$

The norm $\|tq_s'\|_2$ therefore pays the difference by $C\kappa^2$. For any bounded extension Q that equals $q_s \circ \rho$ on the radius-$2R$ cylinder, and $|t| \le T \le R/4$, the exterior discrepancy is pointwise at most $C\lambda R^{-p}$. We have proved

$$(-\Delta)^s Q(0,t) - (Q - Q^3)(0,t) = -H_0 q_s'(t) + r_s(t), \quad \|r_s\|_{L^2(-T,T)} \le C\{\lambda\kappa + \kappa^2 + \lambda\sqrt{T}R^{-p}\}. \tag{7.4}$$

The first two terms are global normal L^2 bounds. For an interval of fixed length centered at a remote normal depth, the same bounds hold provided the interval stays inside $[-R/4, R/4]$. The pointwise exterior estimate then has a uniform constant and no growing $\sqrt{\delta}$ factor. A uniform bound integrated over an interval of length δ would not justify that assertion without this pointwise exterior estimate.

The full-buffer geometry used by the residual. Theorem 5.4, specifically (4.24) and (4.29), supplies the required two inputs at every moving marked zero. On a smaller chamber cylinder of radius $L_j = c\min(\delta_j, \kappa_j^{-1})$, the same-reference stack obeys $|u_j - Q_j| \le C\kappa_j$ with one constant on the entire cylinder. Each retained signed distance also has the bounds (7.1) on its larger genuine buffered tube, including the remote normal depths used below. The latter statement concerns true feet and distance derivatives; it does not enlarge the region where the quantitative chamber estimate is used.

To form the reference for the residual, first trim by actual height. When $\kappa_j\delta_j \to 0$, retain the consecutive profiles whose marked common-frame heights have absolute value at most $L\delta_j$, for a fixed sufficiently large L, and replace the rest by saturated constants. Separation makes this a fixed finite block. The common slope bound gives distance at least $(L/\sqrt{1+\theta^2} - c)\delta_j \ge \delta_j$ to every omitted graph throughout the smaller receiving cylinder. The omitted profile tail is thus $O(e^{-\sqrt{2}\delta_j} + \lambda_j\delta_j^{-p_j}) = o(\kappa_j)$. The retained depths are now genuinely $O_L(\delta_j)$, unlike a mere rank cutoff. They fit inside the larger prepared natural window $L_* \asymp \sqrt{\delta_j/\kappa_j}$ after fixed buffer enlargement. If instead $\kappa_j\delta_j \ge a_0 > 0$, use a smaller fixed curvature-distance cylinder and a single-sheet reference. Omitted exponentials divided by κ_j vanish since $\delta_j \gtrsim \kappa_j^{-1}$, and omitted algebraic tails are covered by the force hypothesis.

Choose the uncorrected distance stack. A value-controlled pin correction of size $O(\mathcal{F}_j) = o(\kappa_j)$ can be removed before defining its residual; no derivative of that correction is needed. Linearity of $(-\Delta)^s$ permits summing the individual residuals first. On a fixed compact set, remote profiles differ from saturated constants by $O(\mathcal{F}_j)$, so the nonlinear reaction mismatch is $O(\mathcal{F}_j)$ as well. Their leading terms $H_{j,k}q_{s_j}'(t_k)$ are $o(\kappa_j)$ since $|t_k| \to \infty$. The first two error terms of (7.4) are $o(\kappa_j)$; its exterior error on each fixed evaluation interval is $O(\lambda_j L_*^{-p_j}) = o(\kappa_j)$. Normal-to-ambient changes of variables have bounded Jacobians because $\kappa_j|t_k| = O_L(\kappa_j\delta_j) \to 0$. The two expanding scales here have different purposes. The bound on $w_j = (u_j - Q_j)/\kappa_j$ uses only the chamber cylinder of radius $L_j = c\min(\delta_j, \kappa_j^{-1})$. The residual of an individual distance profile is computed on a larger genuine-distance cylinder: in the regime $\kappa_j\delta_j \to 0$ take $R_j \asymp \sqrt{\delta_j/\kappa_j}$. All retained offsets are $O_L(\delta_j) = o(R_j)$, so the fixed evaluation intervals remain a distance comparable to R_j from its cutoff. No $O(\kappa_j)$ chamber approximation on that larger cylinder is required. The normalized exterior term is

$$\frac{\lambda_j R_j^{-p_j}}{\kappa_j} \le C\frac{\lambda_j\delta_j^{-p_j}}{\kappa_j}(\kappa_j\delta_j)^{p_j/2} = o(1).$$

In the isolated regime use $R_j \asymp \kappa_j^{-1}$, giving $O(\lambda_j\kappa_j^{p_j-1}) = o(1)$. Passing to a subsequence always places the argument in one of these two regimes.

The finite sum therefore has, whenever $H_j(0)/\kappa_j \to h$,

$$Q_j \to q(t) = \tanh(t/\sqrt{2}), \qquad \kappa_j^{-1}[(-\Delta)^{s_j}Q_j - (Q_j - Q_j^3)] \longrightarrow -hq'(t) \tag{7.5}$$

in distributions on each fixed physical compact set. Here normal C^1 compactness makes $H_j/\kappa_j - h \to 0$ uniformly on fixed physical tangential patches. The new reference is chosen before its residual is formed; no estimate of the fractional Laplacian of the old-minus-new reference is being asserted.

7.2.2. *A bounded resonant equation forces zero mean curvature.*

Lemma 7.4 (Resonant layer forcing)**.** *Let $q(t) = \tanh(t/\sqrt{2})$ and $h \in \mathbb{R}$. If $w \in L^\infty(\mathbb{R}^6 \times \mathbb{R})$ satisfies in distributions*

$$(-\Delta_x - \partial_t^2 + 3q(t)^2 - 1)w = hq'(t),$$

then $h = 0$.

Proof. Set $a(x) = \int w(x,t)q'(t)\,\mathrm{d}t$, a bounded function. Test with $\varphi(x)q'(t)\chi(t/T)$ and let $T \to \infty$. All normal cutoff errors vanish because q' and its derivatives decay exponentially and w is bounded. The identity $(-\partial_t^2 + 3q^2 - 1)q' = 0$ gives $-\Delta_x a = h\int (q')^2$. Testing this with $R^{-6}\varphi(x/R)$, $\int \varphi = 1$, bounds its right side by $C\|a\|_\infty R^{-2}$; let $R \to \infty$. □

Proposition 7.5 (Minimality of the retained limit)**.** *Under the full-buffer preparation specified above and the normal convergence of Theorem 6.2, $H_j/\kappa_j \to 0$ uniformly on compact curvature charts. The complete carrier immersion is minimal.*

Proof. Otherwise recenter at moving points with $H_j(0)/\kappa_j \to h \neq 0$. Put $w_j = (u_j - Q_j)/\kappa_j$. It is uniformly bounded on the expanding cylinder and bounded globally by C/κ_j. The finite stack has a uniformly bounded global extension, so the last global bound has a constant independent of j. Since $L_j \to \infty$, a diagonal weak-star extraction on fixed balls gives a limit $w \in L^\infty(\mathbb{R}^7)$ with the same local bound on every ball. For a fixed compact test, the contribution from outside preparation is

$$C\frac{\lambda_j}{\kappa_j L_j^{p_j}} \le C\left(\frac{\lambda_j \delta_j^{-p_j}}{\kappa_j} + \lambda_j \kappa_j^{p_j - 1}\right) = o(1). \tag{7.6}$$

On a fixed large ball $(-\Delta)^{s_j}\varphi \to -\Delta\varphi$ uniformly. Outside that ball, $|(-\Delta)^{s_j}\varphi(X)| \le C_\varphi \lambda_j (1 + |X|)^{-7-p_j}$. Between the ball and the preparation boundary use boundedness of w_j; outside use (7.6). This proves distributional convergence of the operator without normalized second-derivative bounds. The exact equation is

$$(-\Delta)^{s_j} w_j - \frac{f(Q_j + \kappa_j w_j) - f(Q_j)}{\kappa_j} = -\kappa_j^{-1} R_{Q_j}, \qquad f(v) = v - v^3.$$

The nonlinear remainder on fixed compacts is $O(\kappa_j)$. Equation (7.5) gives the equation in Lemma 7.4, contradicting $h \neq 0$. The moving-point argument proves uniform local smallness, and the C^2 graph convergence passes it to the carrier. Local minimal-graph regularity then upgrades the immersion from C^2 to smooth. □

7.3. From ambient stability to intrinsic stability.

7.3.1. *A canonical phase elimination and its exact identity.* The reach estimate (2.6), including its conclusion about short normal segments, gives

$$\operatorname{reach}(\Sigma_j) \ge c \min(\kappa_j^{-1}, \delta_j) \longrightarrow \infty.$$

Thus, for each fixed P, the depth-$2P$ normal tube of the entire zero set is injective for all sufficiently large j. This includes returning branches of a single connected component. We use this global tube to define one phase operator for the whole compact test.

Choose an even smooth ψ_P, with $0 \le \psi_P \le 1$, with plateau $[-P, P]$ and support in $(-2P, 2P)$, and define globally

$$\begin{aligned} \chi_P(X) &= \psi_P(\operatorname{dist}(X, \Sigma_j))^2, & W &= 3(1 - u_j^2), \\ W_C &= \chi_P W, & V_P &= (1 - \chi_P)W, \\ G_{V_P} &= ((-\Delta)^{s_j} + 2 - V_P)^{-1}. \end{aligned}$$

The cutoff is smooth because of the tube, evenness, and its constant plateau. Well clearing gives $0 \le V_P \le \epsilon_P < 2$, with $\epsilon_P \to 0$ as fixed $P \to \infty$ after the sequence limit. This is one globally defined resolvent, not a collection of separately eliminated chart operators. Put $a = |\nabla u_j|$ on the

cores and $\nu = \nabla u_j/a$. The positive massive inverse and uniqueness for its bounded homogeneous equation give exactly

$$\nabla u_j = G_{V_P}(W_C \nabla u_j), \qquad S = W_C a, \qquad J(X,Y) = S(X) G_{V_P}(X,Y) S(Y) \geq 0. \tag{7.7}$$

Here the inverse is also the bounded-solution inverse. To see this, let P_t denote the fractional heat semigroup. For bounded h with $((-\Delta)^s + 2)h = f$, the semigroup identity gives

$$h = e^{-2T} P_T h + \int_0^T e^{-2t} P_t f \, \mathrm{d}t.$$

The first term tends to zero in L^∞ and the integral tends to $G_s^2 f$. The identity is valid distributionally, or follows first after smoothing and then by bounded convergence. If $((-\Delta)^s + 2 - V_P)h = 0$, it follows that

$$\|h\|_\infty \leq \tfrac{1}{2} \|V_P\|_\infty \|h\|_\infty,$$

so $h = 0$. The positive Neumann series for G_{V_P} converges on L^∞, since $\|G_s^2 V_P\| \leq \|V_P\|_\infty/2 < 1$. Apply this inverse to the differentiated equation to obtain (7.7), with no L^2 assumption on ∇u_j.

The kernel is symmetric and all physical volume densities are retained. The stability inequality for $(-\Delta)^{s_j} + 2 - W$ implies

$$I - \sqrt{W_C}\, G_{V_P} \sqrt{W_C} \geq 0.$$

Write $A_P = (-\Delta)^{s_j} + 2 - V_P$ and $U = \sqrt{W_C}$. The phase operator satisfies $A_P \geq 2 - \epsilon_P > 0$. Ambient stability is the form inequality $A_P - U^2 \geq 0$. Testing it with $A_P^{-1/2} f$, by form-domain density, gives $\|U A_P^{-1/2}\| \leq 1$. Its adjoint has the same norm, so

$$U G_{V_P} U = (A_P^{-1/2} U)^* (A_P^{-1/2} U) \leq I.$$

This proves the displayed inequality on ambient L^2.

Given a nonnegative scalar amplitude η on a coherently lifted compact carrier, extended by zero on all other zero sheets, use the physical test

$$v = \sqrt{W_C}\, a\eta. \tag{7.8}$$

There is no orientation sign in this field. In unitary Fermi coordinates it also has the factor $\sqrt{J}$ of the coordinate Jacobian. Dotting (7.7) with ν gives the exact stable form

$$0 \leq \int \eta(X)^2 \int J(X,Y) \nu(X) \cdot \nu(Y) \, \mathrm{d}Y \, \mathrm{d}X - \iint J(X,Y) \eta(X) \eta(Y) \, \mathrm{d}X \, \mathrm{d}Y. \tag{7.9}$$

The amplitude is constant along each full depth-$2P$ normal fiber. Its zero-foot support is compact in the chosen component, so its ambient tubular support is compact. Consequently the field (7.8) belongs to L^2 even when the entire zero set has infinite area. The row mass is bounded by the positive massive comparison and the bounded source S; all terms below are absolutely integrable. Multiplying the scalar product of (7.7) with $\nu(X)$ by $S(X)\eta(X)^2$ identifies the diagonal $\int W_C a^2 \eta^2$ with the first term of (7.9). Subtracting the kernel pairing of the unsigned test proves that identity directly.

There is no exterior load in this global identity. Any subsequent endpoint truncation introduces a load that must be retained until its absolute bound has been applied.

For any symmetric splitting $J = J_{\mathrm{loc}} + J_{\mathrm{off}}$ into nonnegative kernels, symmetry and $\nu \cdot \nu' = 1 - |\nu - \nu'|^2/2$ show that (7.9) is at most

$$\begin{aligned} &\frac{1}{2} \iint J_{\mathrm{loc}} (\eta(X) - \eta(Y))^2 - \frac{1}{2} \int \eta(X)^2 \int J_{\mathrm{loc}} |\nu(X) - \nu(Y)|^2 \\ &\qquad + \int \eta(X)^2 \int J_{\mathrm{off}} \nu(X) \cdot \nu(Y). \end{aligned} \tag{7.10}$$

The removed off-diagonal product is $-\iint J_{\mathrm{off}} \eta(X) \eta(Y) \leq 0$. In particular simultaneous activity on several ambient branches is harmless for this product. A test with amplitudes $\sigma_i \eta_i$ would have a different sign and is not used here.

7.3.2. *Kernel estimates for the stability limit.* Passage to the limit in the stability form requires second-moment estimates for arbitrary tests, in addition to the on-solution equation (6.1). The following proposition provides them.

Proposition 7.6 (Actual global-resolvent estimates for stability)**.** *The full prepared windows and preceding adjacent-capacity bounds give the following estimates for the one resolvent (7.7), uniformly on moving prepared charts and every fixed finite atlas of a compact intrinsic carrier.*

(1) In the physical flux convention, the self kernel and its transported frozen Fermi kernel J^{fr} obey

$$\sup_{x,r}\int(|x-y|+|r-t|)|J_{ii}^{\mathrm{actual}}-J_{ii}^{\mathrm{fr}}|\,\mathrm{d}y\,\mathrm{d}t\le C_P\kappa_j. \tag{7.11}$$

Both have common local integrable bounds and second-moment tails

$$\sup_{x,r}\int_{|x-y|>R}|x-y|^2J_{ii}\,\mathrm{d}y\,\mathrm{d}t\le C_P(e^{-cR}+\lambda_jR^{1-p_j}),\qquad R\ge 1. \tag{7.12}$$

Endpoint edge weights have uniform absolute continuity in these norms. A quadratic error only on the actual normal does not imply (7.11).

(2) Positive actual cross rows have the sharp massive lower/upper algebraic comparisons with prepared normal masses, and upper exponential error $C_P(1+D)^Le^{-cD}$ for some $c>\sqrt{2}/2$. The errors in the near receiving frame are relative $C\sqrt{d_0}+C\epsilon_P+o_j(P)$, as calculated below. All phase visits, endpoint source factors and both coordinate densities must be included. The scalar adjacent capacity is fixed at one depth P_0 and transfers to the fixed receiving-ray band with a constant independent of later P and d_0.

(3) Unsigned endpoints outside the prepared curvature-scale receiving disks, distant labels, and loads from those curvature-scale truncations have total row

$$C_{P,d_0}\big(\lambda_j\kappa_j^{p_j+1}+\lambda_j\kappa_j^{p_j}/\delta_j\big)+o(\kappa_j^2)=o(\kappa_j^2). \tag{7.13}$$

The summed full-normal angular cross row is

$$\sum_{k\ne i}\int J_{ik}|N_i-N_k|^2\le C_P\kappa_j\mathcal{F}_j+o(\kappa_j^2)=o(\kappa_j^2). \tag{7.14}$$

Every true foot, complete plateau fiber, and distance-level piece used in these estimates lies in its supplied buffer.

These are estimates for the actual all-visit kernels, in addition to the frozen Wiener calculus.

Proof. We first relate the single global phase resolvent to the prepared window estimates. Fix a smaller receiving plateau and a larger curvature-scale window, with a smooth spatial cutoff $\eta_0=1$ on the entire intervening plateau. Include every actual fixed-depth core meeting $\operatorname{supp}\eta_0$. Injectivity of the global tube and completeness of these fibers give the exact identity

$$\sum_i\psi_P(\rho_i)^2=\chi_P\quad\text{on }\operatorname{supp}\eta_0.$$

There is at most one nonzero summand. Thus the local preparation (4.30) uses precisely

$$V_{\mathrm{loc}}=\eta_0V_P,\qquad S_{i,\mathrm{loc}}=\eta_0S_i,\qquad S_i=\mathbf{1}_{\mathrm{core}\,i}\,\chi_PW|\nabla u_j|.$$

In particular its endpoint factors agree with the global ones on every retained full plateau fiber. Choose the two windows with a fixed positive curvature-scale margin, so $\operatorname{supp}(V_P-V_{\mathrm{loc}})$ is at distance $H\ge c\rho/\kappa_j$ from the smaller receiving plateau. Both phase potentials are between zero and the same small ϵ_P, and hence

$$G_{V_P}-G_{V_{\mathrm{loc}}}=G_{V_{\mathrm{loc}}}(V_P-V_{\mathrm{loc}})G_{V_P},\qquad 0\le G_{V_{\mathrm{loc}}},G_{V_P}\le G_{s_j}^{2-\epsilon_P}.$$

This comparison retains all intermediate visits. It does not identify resolvents belonging to different windows.

For the self row, the donor measure is supported on one supplied bounded-slope graph tube and has translation-uniform ball mass $C_P R^6$ for $R \geq 1$. The returning exterior estimate (5.29), applied to the displayed resolvent identity, bounds its first-weighted difference by

$$C_P\{e^{-cH} + \lambda_j H^{-p_j}\} = o(\kappa_j).$$

Endpoint restrictions at this distance have zeroth row mass $C_P\{e^{-cH} + \lambda_j H^{-p_j-1}\} = o(\kappa_j^2)$; their first-weighted row is again $o(\kappa_j)$. These bounds also permit the actual graph row to be extended by zero outside its supplied chart when compared with a global model extension. All donor restrictions and extensions are charged before any action on a test is taken.

We may therefore apply the first-weighted kernel comparison proved in Section 5.3. More precisely, (5.26) and (5.28) compare the actual squared phase with the true-distance self phase, including remote core cutoffs and nonadjacent chambers. The natural window omissions are paid by (5.29) and (5.30). The genuine normal extension (5.31) and the full-path estimate (5.32) then freeze every leg and every intermediate density in the positive phase series. Their conclusion, before acting on the solution normal, is an $O_P(\kappa_j)$ error in the weighted row norm with weight $1 + |Z - Z'|$. This is the required kernel estimate for arbitrary tests, stronger than the ensuing action estimate (5.34). Only bounded sectional second moments enter the full-path argument; it never assumes a finite ambient second moment. Finally the relative preparation of the endpoint factors and the exact physical flux convention (5.33) retain both coordinate densities and the leading tangent density. They have the same first-weighted error. On fixed-depth fibers the weights $|Z - Z'|$ and $|x - y| + |r - t|$ are uniformly comparable. This proves (7.11) for the global actual row and its transported frozen midpoint row.

We record separately the moments needed for arbitrary tests. The positive massive upper comparison, the bounded endpoint coefficients, and the small-slope Fermi distance comparison give a common local integrable majorant for the actual and frozen self kernels. One may use the s-independent radial majorant H_* from the proof of (5.32). After normal integration, a tangential majorant is

$$h_P(z) = C_P \int_{-4P}^{4P} H_*(c(z,t))\, \mathrm{d}t, \qquad \int_{\mathbb{R}^6} (1 + |z|^2) h_P(z)\, \mathrm{d}z < \infty,$$

where the fixed $c > 0$ accounts for the chart distortion. Retaining the order-dependent massive bound gives the sharper tail

$$\sup_{x,r} \int_{|x-y|>R} |x - y|^2 J_{ii}(x, r; y, t)\, \mathrm{d}y\, \mathrm{d}t \leq C_P e^{-cR} + C_P \lambda_j \int_R^\infty u^{-p_j}\, \mathrm{d}u \leq C_P\{e^{-cR} + \lambda_j R^{1-p_j}\}.$$

This proves (7.12). The constants are uniform because $p_j - 1 \geq p_* - 1 > 0$. The same statements hold for the frozen row. These tails tend to zero uniformly as $R \to \infty$; no second-moment error of order $O(\kappa_j)$ is claimed or required.

The flux at its endpoints factors as $\psi_P(r)^2 \psi_P(t)^2$ times uniformly bounded coefficients and the massive majorant. Its zeroth and second weighted normal sections are bounded uniformly in r, t by the sectional estimates used in (5.32). Therefore restriction of either integrated normal endpoint to a set of length tending to zero has vanishing weighted mass, uniformly in the sequence and in the receiving point; at a receiving cutoff edge there is also the factor $\psi_P(r)^2$. This proves the asserted edge absolute continuity, including the bounded weight $|r - t|^2$. It does not divide an unweighted estimate by a vanishing profile factor.

For the remaining global rows let $\mu = S\, \mathrm{d}X$ be the unsigned measure of all fixed-depth cores. By (4.34) and the bounded source factors,

$$\mu(B_R(X)) \leq C_P(R^6 + R^7/\ell_j), \qquad \ell_j \asymp \min(\delta_j, \kappa_j^{-1}), \quad R \geq 1.$$

The same ball estimate holds for $F = G_{s_j}^{2-\epsilon_P} * \mu$, by Tonelli and the fixed bounded mass of the massive kernel. Dyadic integration consequently gives

$$\int_{|Z-X|>H} G_{s_j}^{2-\epsilon_P}(X - Z) F(Z)\, \mathrm{d}Z \leq C_P\{e^{-cH} + \lambda_j(H^{-p_j-1} + \ell_j^{-1} H^{-p_j})\}.$$

The exponential term also follows directly from the uniform bound on F. This is an insertion estimate: use it in the exact resolvent identity above and retain the bounded insertion factor. The identical algebraic bound applies to omitted endpoints, using μ directly. At $H \asymp d_0/\kappa_j$ these are precisely (7.13). To check its normalization, when $\kappa_j\delta_j \le 1$ write

$$\lambda_j\kappa_j^{p_j-2}/\delta_j = \left(\lambda_j\delta_j^{-p_j}/\kappa_j\right)(\kappa_j\delta_j)^{p_j-1} = o(1).$$

When $\kappa_j\delta_j > 1$ it is at most $\lambda_j\kappa_j^{p_j-1} = o(1)$, which also controls the other term. This global estimate uses no count of ambient components. In particular the total difference between the global cross rows and the plateau window rows has mass $o(\kappa_j^2)$, including all exterior phase visits and endpoint tapering. Multiplication by the bounded square of a unit-normal difference preserves that bound.

The cross-row algebraic coefficients and their receiving-frame errors follow from the direct global inequalities $G_{s_j}^2 \le G_{V_P} \le G_{s_j}^{2-\epsilon_P}$, with the full prepared source masses and both endpoint densities. Their calculation is given in (7.16) below, together with transfer of the depth-P_0 capacity to the fixed receiving-ray band. It does not use a new phase elimination or a capacity constant depending on the later testing depth.

For the angular estimate the window rows now satisfy exactly the hypotheses already checked for (4.45): complete plateau fibers, the same phase potential (4.30), buffered gap-excess geometry, and the actual full normal fields. Thus (4.43), (4.44), and (4.45) apply unchanged. In particular the logarithmic comparison retains an $O_P(\lambda_j D^{-p_j-1})$ algebraic row remainder from exterior phase visits; only its exponential error can be made smaller than any prescribed power by enlarging the fixed logarithmic buffer. The algebraic remainder is paid in the same-foot angular estimate by $C_P\kappa_j\lambda_j d^{-p_j}$ and its ordered sum. The finite-label second-moment sum of (4.44) is used only up to curvature distance; it is uniform as $p_j \uparrow 2$ and is not extended to infinitely many labels. Endpoints beyond that distance, and the change from the window resolvent to the global one, have the unsigned budget just proved. We obtain

$$\sum_{k\neq i}\int J_{ik}|N_i - N_k|^2 \le C_P\kappa_j\mathcal{F}_j + o(\kappa_j^2) = o(\kappa_j^2),$$

which is (7.14). All constants are uniform on the fixed finite atlas, since the preceding estimates hold for moving prepared charts with the supplied margins. □

7.3.3. *Receiving-frame comparison and the signed diagonal.* Fix capacity at one sufficiently large depth P_0:

$$\int \kappa_j^{-2}(e^{-\sqrt{2}d_a} + \lambda_j d_a^{-p_j-1})\zeta^2 \le C_0\int(|D\zeta|^2 + \zeta^2). \tag{7.15}$$

These integrals are in curvature coordinates. This is the scalar adjacent-potential capacity already established in Theorem 5.2; it concerns the actual gap geometry and does not depend on the later testing depth P.

Choose a fixed ray band $d_1 > 0$ inside the supplied curvature buffers. Fix the finite receiving atlas and its test partitions on this scale, before choosing $d_0 \ll d_1$. At each receiving zero rotate its tangent plane to horizontal and let

$$D_0 = 0 < D_1 < \cdots < D_K, \qquad d_a = D_a - D_{a-1} \quad (a \ge 1).$$

Thus $D_k = \sum_{a=1}^k d_a$, and every consecutive ray gap is at least δ_j. The comparison first retains $D_k \le d_0/\kappa_j$. Its donor integration disk has radius $R = \sqrt{d_0}/\kappa_j$. These pointwise integration disks are not the supports of a differentiated test partition.

The buffered positive-gap estimate and curvature bound give

$$|Dh_k(0)| \le C\sqrt{\kappa_j D_k}, \qquad |Dh_k| \le C\sqrt{d_0}, \qquad |h_k(z) - D_k| \le C\sqrt{d_0}|z| \quad (|z| \le R).$$

For the first estimate, Taylor's inequality for the nonnegative inter-sheet gap is used in the direction of its negative gradient; the surrounding complete buffer permits the required step. For fixed endpoint depths, the physical endpoint vector differs from (z, D_k) by at most $C\sqrt{d_0}|z|+C_P$.

Its relative length error is therefore $O(\sqrt{d_0}) + O(P/\delta_j)$. The donor surface density differs from one by $O(d_0)$, and the normal Jacobian by $O_P(\kappa_j)$.

All comparisons below are per unit receiving surface measure. If $B_{ik}(x)$ is stored as a coordinate flux density, write

$$\mathcal{B}_{ik}(x) = B_{ik}(x)/J_i(x,0), \qquad d\mu_i = J_i(x,0)\,\mathrm{d}x.$$

Then $B_{ik}(x)\,\mathrm{d}x = \mathcal{B}_{ik}(x)d\mu_i$ exactly. This retains the receiving Jacobian when the pointwise rotated frame is used to estimate a global integral; no varying frame is differentiated. The prepared source mass is

$$\mu_{s_j,P} = \int \chi_P(t) 3(1-q_{s_j}(t)^2) q'_{s_j}(t)\,\mathrm{d}t,$$

up to relative $o_j(P)$ on each fixed-depth fiber after cancelling its common cutoff. The mass has upper and positive lower bounds independent of $P \ge P_0$: it is at most 4, and its depth-one part is bounded below.

Use the sharp free massive coefficients of Lemma 3.9, in particular (3.51), together with $G^2_{7,s_j} \le G_{V_P} \le G^{2-\epsilon_P}_{7,s_j}$. The source masses, distance comparison, and tangential integration give the following one-sided bounds for the actual rows:

$$b_k = \frac{c_{1,s_j}\mu^2_{s_j,P}}{4} D_k^{-p_j-1}, \qquad \omega_j = C\sqrt{d_0} + C\epsilon_P + o_j(P),$$
$$\mathcal{B}_{ik} \ge (1-\omega_j)b_k - r^-_{ik}, \qquad \mathcal{B}_{ik} \le (1+\omega_j)b_k + \mathcal{E}_{ik} + r^+_{ik}, \qquad 0 \le \mathcal{E}_{ik} \le C_P(1+D_k)^L e^{-cD_k}, \quad c > \sqrt{2}/2. \tag{7.16}$$

There is no separately assumed algebraic part of an actual row. The truncation errors r^-_{ik}, r^+_{ik} are nonnegative. For a negative odd-rank row only the lower bound is used; for a positive even-rank row only the upper bound is used. The coefficient is common to all ranks, and $c_{1,s_j}\mu^2_{s_j,P}/(4\lambda_j)$ is bounded independently of $P \ge P_0$. The lower comparison has no negative exponential error.

Extending a finite donor comparison disk to all tangent space loses at most $C_P\lambda_j R^{-p_j-1}$ per label in a lower bound. There are at most $C(1 + d_0/(\kappa_j\delta_j))$ retained labels. This loss, the $r^\pm_{ik}$, and all other actual endpoint omissions have the unsigned budget (7.13). After division by κ_j^2, its second algebraic term is

$$\frac{\lambda_j\kappa_j^{p_j-2}}{\delta_j} = \frac{\lambda_j\delta_j^{-p_j}}{\kappa_j}(\kappa_j\delta_j)^{p_j-1} = o(1) \quad \text{if } \kappa_j\delta_j \le 1.$$

If $\kappa_j\delta_j > 1$, the same term is bounded by $\lambda_j\kappa_j^{p_j-1} = o(1)$, as is the first algebraic term. All fixed P, d_0 constants precede the sequence limit. Remote graphs are not independently extended to justify a lower bound.

We now transfer the capacity to the ray increments with constants fixed at P_0, d_1. For $D_a \le 2d_1/\kappa_j$, put

$$T_a(x) = X_i(x) + D_a(x)n_i(x), \qquad T_0(x) = X_i(x).$$

Implicit differentiation of the intersection equation gives

$$|\nabla_{\Sigma_i} D_a| \le C\sqrt{d_1}, \qquad DT_a = I - D_a A_i + n_i \otimes \nabla D_a.$$

The normal projection onto the supplied receiving graph is the inverse of T_a on its image. The genuine-foot and angle bounds therefore make these maps uniformly bi-Lipschitz, with Jacobians and inverse Jacobians bounded independently of a, j, P, d_0.

The two actual endpoints for increment d_a are $T_{a-1}(x)$ and $T_a(x)$; their Euclidean separation is exactly d_a. Apply the depth-one positive lower construction used in (5.5) and (5.7) to this pair. It gives, on full plateau fibers,

$$e^{-\sqrt{2}d_a} + \lambda_j d_a^{-p_j-1} \le C I^{P_0}_{a-1,a}(T_{a-1}(x)) + C\kappa_j^2.$$

Here $I^{P_0}_{a-1,a}$ denotes the actual averaged adjacent row for these two sheets at the fixed baseline depth, in a fresh chart based on the lower endpoint. All constants use depth-one source masses. For the algebraic part integrate a source disk of radius comparable to d_a. For $d_a \le K\log(1/\kappa_j)$,

the favorable depth-one source half-ball has radius $\sqrt{d_a}$ and does not increase the distance of the exact endpoints by more than $O(1) + O(\kappa_j d_a) = O(1)$. Thus it retains $e^{-\sqrt{2}d_a}$ up to a fixed factor, rather than comparing exponentials of angle-distorted gaps. For larger gaps choose $\sqrt{2}K > 2$ and charge the exponential to $O(\kappa_j^2)$. Adjacency is preserved by the complete ordered family.

Choose $\chi = 1$ on $[0,1]$ and $\chi = 0$ on $[2,\infty)$ and multiply a test by $\chi(\kappa_j D_a/d_1)$. Since $0 \le D_{a-1} \le D_a$, this controls both endpoints and both genuine maps. It is identically one on every increment entering a retained $D_k \le d_0/\kappa_j$. Its physical derivative is $O_{d_1}(\kappa_j)$. Transport the cut-off test using T_{a-1}^{-1} and the fixed d_1-scale square partitions in fresh baseline charts. Bounded map distortion and the square-partition identity give a sum of transported H^1 test norms at most $C_{P_0,d_1,\mathrm{atlas}} \int (|D\zeta|^2 + \zeta^2)$ in curvature coordinates. This constant is independent of a, P, d_0. The baseline actual-row capacity therefore proves (7.15) for every cut-off ray increment, with one fixed constant. No derivative of a shrinking comparison disk or of the pointwise rotated receiving frame is taken.

The signs on either side are $(-1)^k$, starting negative. As D_k increases, the exact common model satisfies

$$\sum_{k=1}^{K} (-1)^k b_k \le 0, \qquad -b_{2a-1} + b_{2a} \le 0,$$

and an unpaired final term is negative. To estimate its relative error, put $q_j = p_j + 1$. Convexity gives

$$D_k^{-q_j} \le k^{-q_j-1} \sum_{a=1}^{k} d_a^{-q_j}.$$

Interchanging nonnegative sums yields

$$\sum_{k\ge 1} \lambda_j D_k^{-q_j} \le \sum_{a\ge 1} \lambda_j d_a^{-q_j} \sum_{k\ge a} k^{-q_j-1} \le C \sum_{a\ge 1} a^{-q_j} \lambda_j d_a^{-q_j}.$$

Apply the uniformly transferred increment capacity to each term and use $\sum_{a\ge 1} a^{-q_j} \le C_{p_*}$. For a finite retained family the same calculation extends its sums by zero. It gives the normalized relative-error form bound

$$C_0'[\sqrt{d_0} + \epsilon_P + o_j(P)] \int (|D\eta|^2 + \eta^2), \tag{7.17}$$

where C_0' is fixed at P_0, d_1 and the fixed receiving atlas; the number of branches does not enter it.

Only positive even-rank rows require the exponential upper error. Absorb its fixed polynomial into a rate still satisfying $c > \sqrt{2}/2$. Then, for even $k \ge 2$,

$$e^{-cD_k} \le e^{-(ck-\sqrt{2})\delta_j} \frac{1}{k} \sum_{a=1}^{k} e^{-\sqrt{2}d_a}.$$

This follows from Jensen and $D_k = \sum_{a=1}^{k} d_a \ge k\delta_j$. The transferred increment capacity and the geometric series show that its normalized form tends to zero for fixed P, d_0. Apply the odd lower and even upper bounds in (7.16); the signed scalar off-row diagonal has nonpositive upper limit up to (7.17). The same argument applies separately on both sides of the receiver. All omitted ranks retain their far budget.

Finally $\nu_i = \sigma_i N_i$, so replacing $\nu_i \cdot \nu_k = \sigma_i\sigma_k N_i \cdot N_k$ by $\sigma_i\sigma_k$ costs half the unsigned angular row (7.14). That normalized error vanishes for fixed P. The same upper bound therefore holds for the full-normal diagonal in (7.10).

7.3.4. *Matching self energy and self angular energy.* The first-moment discrepancy (7.11) gives, after a fixed displacement cutoff R, a second-moment discrepancy at most $C_P R\kappa_j$. Then (7.12) removes the cutoff, first taking $j \to \infty$ and then $R \to \infty$. The same truncation handles bounded normal-depth differences and smooth cutoff edges. Thus actual self second moments converge to the frozen ones without needing an $O(\kappa_j)$ second-moment rate.

In an orthonormal tangent frame the frozen kernel is rotationally invariant. Define

$$\beta_{s,P} = \frac{1}{12}\int\int\int_{\mathbb{R}^6} |z|^2 S_{s,P}(r) G_{s,P}((0,r),(z,t)) S_{s,P}(t)\, \mathrm{d}z\, \mathrm{d}r\, \mathrm{d}t, \qquad S_{s,P} = \chi_P W_{q_s} q_s'. \tag{7.18}$$

The energy tensor, including its factor $1/2$, is $\beta_{s,P} I_6$; arbitrary tangent coordinates give the inverse metric and its exact surface density. For fixed P, the phase Neumann series contracts uniformly and each finite path converges as $s \to 1$. Local integrable massive bounds remove the diagonal singularity; (7.12) removes the large displacement cutoff. Therefore $\beta_{s_j,P} \to \beta_P$. There are uniform constants $0 < c < C$ such that

$$c \le \beta_P \le C \qquad (P \ge P_0). \tag{7.19}$$

For the lower bound restrict both normal variables to depth one and z to a fixed annulus, then use $G_{s,P} \ge G_s^2$ and positive source masses. For the upper bound use $G_{s,P} \le G_s^{1.9}$ and its uniformly bounded sectional second moment, together with $\int S_{s,P} \le C$. No P-dependent coefficient is lost at this step.

For a compact intrinsic test, use curvature coordinates $y = \kappa_j x$ and an orthonormal tangent frame at the receiving foot. On bounded physical displacements, the C^1 convergence of the coherent test lifts gives

$$\kappa_j^{-1}[\eta_j(x) - \eta_j(x - z)] \longrightarrow d\eta_y(z).$$

Both amplitudes are constant along their full normal fibers. On each fixed compact carrier neighborhood their differences are bounded by $C\kappa_j|z|$ for pairs in the controlled graph. Consequently the common second moments justify passage to the limit in the gradient term: first restrict the displacement, pass to the limit, and then remove the restriction using (7.12).
The normal-depth and cutoff-edge passage. Fix P and let $E_P = \{t : \psi_P(t) > 0\}$. Exhaust E_P by finite unions of closed intervals compactly contained in E_P. On each such union, the profile convergence gives $\inf_{j,t} f_j(t) > 0$ for all sufficiently large j, so (6.24) applies uniformly. The C^1 convergence of the zero-depth normals then yields, on bounded displacements and these inner normal sets,

$$\kappa_j^{-1}[\nu_j(x,r) - \nu_j(x - z, t)] \longrightarrow d\nu_y(z).$$

The orientation sign is constant on this local graph, so replacing N_j by the physical normal ν_j leaves its squared difference unchanged. The independent full-core estimate $\mathrm{Lip}_{x,r} N_j \le C_P \kappa_j$ bounds that difference divided by κ_j, squared, by $C_P(|z|^2 + |r - t|^2)$. The endpoint source contains ψ_P^2 and vanishes off E_P. Its uniform convergence and the weighted absolute continuity from Proposition 7.6 make the omitted normal sets uniformly negligible. Thus one may pass to the sequence limit on the inner sets, exhaust E_P, and remove the displacement cutoff. This argument does not assume that ψ_P is positive on the entire interval $(-2P, 2P)$ and never divides by its value at a cutoff edge. Outside the controlled graph use the bounded normal difference and the $o(\kappa_j^2)$ row mass, rather than a global Lipschitz extension.

The receiving surface measure contributes κ_j^{-6}, each squared difference contributes κ_j^2, and the remaining normal and donor integrals give the same tensor (7.18). The exact physical flux densities therefore give

$$\begin{aligned} \kappa_j^4 \frac{1}{2} \iint J_{\mathrm{loc}} (\eta_j(X) - \eta_j(Y))^2 &\longrightarrow \beta_P \int_M |\nabla \eta|^2 \, \mathrm{d}\mu, \\ \kappa_j^4 \frac{1}{2} \int \eta_j(X)^2 \int J_{\mathrm{loc}} |\nu(X) - \nu(Y)|^2 &\longrightarrow \beta_P \int_M |A|^2 \eta^2 \, \mathrm{d}\mu. \end{aligned} \tag{7.20}$$

Here $|d\nu|^2 = |A|^2$. In arbitrary coordinates both contractions use the same inverse metric and the same limiting surface density; no chart-dependent coefficient remains.

7.3.5. *One compact carrier test, with no fixed IMS error.* Let $\eta \ge 0$ be smooth and compactly supported on M. Choose relatively compact carrier domains containing its support and a larger intrinsic neighborhood, and use one exhaustion embedding defined beyond their closures, as in 7.1.3. Its image contains a fixed intrinsic neighborhood of the lifted support for all large j: on the larger compact domain the pulled-back metrics are uniformly equivalent to the limit

metric, so every path from the support to that domain's boundary has a fixed positive length. A shorter prelimit path cannot leave its image. Lift η using this one embedding and extend by zero elsewhere on the actual zero set. The lift is C^1, compactly supported, and vanishes near the image boundary; smooth approximation is permitted in the stability form.

Choose a fixed small curvature-coordinate radius r with $2r$ below both the intrinsic graph radius and the neighborhood width just obtained, and use the symmetric selector

$$J_{\mathrm{loc}}(X,Y) = J(X,Y)\vartheta\left(\frac{\kappa_j d_{\Sigma_j}(\pi X,\pi Y)}{r}\right), \quad \vartheta = 1 \text{ on } [0,1], \quad \vartheta = 0 \text{ on } [2,\infty), \quad 0 \le \vartheta \le 1.$$

Here π is the true zero foot and intrinsic distance between different components is infinite. No derivative of this selector is taken. Every local pair lies in the same controlled graph at the receiving foot, and all sufficiently small same-graph displacements lie in its plateau. Its transition and omitted same-graph pairs have physical separation at least cr/κ_j and row mass $o(\kappa_j^2)$. Thus the off diagonal consists of the other ordered local branches, up to an already controlled remainder.

Apply (7.10) once to this whole lifted test. Its active–active off-diagonal products are nonpositive regardless of how many nearby ambient branches meet the compact carrier. The self integrals are single global symmetric integrals. A finite atlas and a partition of the *receiving measure* can be used to evaluate their limits, but no partition is differentiated in those self terms. For the small relative cross error only, one may use $\sqrt{\theta_a}\eta$ in a fixed square partition; its derivative cost is multiplied by (7.17), and hence tends to zero with that error. Fibers over the support and its intrinsic neighborhood are full fibers; a core is counted once even if many atlas charts meet it. Disjointness of different physical cores follows from $\delta_j \gg P$. These facts are precisely why a coherent carrier test cannot be replaced by separate single-chart stability tests.

Theorem 7.7 (Whole-carrier stable minimal limit)**.** *With the stated preparation and the kernel estimates established in this subsection, the retained complete, connected, two-sided immersion $F : M^6 \to \mathbb{R}^7$ is smooth and minimal and satisfies*

$$\int_M |A|^2\phi^2 \, \mathrm{d}\mu \le \int_M |\nabla\phi|^2 \, \mathrm{d}\mu \qquad (\phi \in C_c^\infty(M)).$$

It retains its nonzero curvature mark. Neither properness nor ambient multiplicity-one convergence is assumed or inferred.

Proof. Minimality is Proposition 7.5. Multiply the upper bound (7.10) by κ_j^4. The self terms converge to $\beta_P \int_M (|\nabla\eta|^2 - |A|^2\eta^2)$ by (7.20). The signed diagonal is nonpositive in the upper limit up to (7.17); all far and angular errors vanish. Freeze P_0, d_1 and the compact atlas first, then choose d_0 small, P large, and finally j large. The form error is arbitrarily small and $\beta_P \ge c > 0$. If the limiting Jacobi integral were negative, this order of choices would give a negative upper bound for the nonnegative actual stable form, a contradiction. Hence the inequality holds for nonnegative η. For an arbitrary $\phi \in C_c^\infty(M)$, the function $|\phi|$ belongs to H^1 and $|\nabla|\phi|| = |\nabla\phi|$ almost everywhere. Nonnegative mollification in a finite atlas, followed by a fixed cutoff in a slightly larger compact set, gives nonnegative smooth compact tests converging to $|\phi|$ in H^1. The curvature potential is bounded there, so its integral also converges. Passing to the limit gives the claimed inequality for ϕ.

The mark is supplied by Theorems 6.2 and 7.1; it has not been consumed in identifying minimality or stability. □

8. Stable rigidity and the dimension-eight conclusion

8.1. **Stable classification near the local limit.** We now combine the separation, preparation, and capacity estimates with curvature retention. The resulting complete intrinsic carrier is both minimal and stable, so the stable Bernstein theorem gives the required contradiction.

Proposition 8.1 (Prepared-sequence exclusion)**.** *Under Theorem 1.3 and the preceding analytic estimates, there is no sequence of nonclassified bounded stable solutions u_j on $\mathbb{R}^7$ with $s_j \to 1$,*

the moving-center preparation of Proposition 2.8, and

$$0 < \kappa_j := \sup_{\Sigma_j} |A_j| \longrightarrow 0.$$

Proof. Choose $z_j \in \Sigma_j$ with $|A_j(z_j)| \geq \kappa_j/2$; no attainment of the supremum is required. Lemma 2.10 and Theorem 3.14 give $\delta_j \to \infty$ and (1.4). Theorem 5.4 therefore supplies the prepared normal equation and source budgets, and Theorem 6.2 supplies (NC) on every smaller moving chart needed in a fixed intrinsic ball.

Rescale the component through z_j by $x \mapsto \kappa_j(x - z_j)$. Theorems 7.1 and 7.7 give, on the same pointed carrier (M, p), a smooth complete connected boundaryless two-sided stable minimal immersion $F : M^6 \to \mathbb{R}^7$ with

$$|A_F| \leq 1, \qquad |A_F(p)| \geq \tfrac{1}{2}.$$

In particular its stability holds for every compact intrinsic test, as required by (1.3). Theorem 1.3 applies without an additional area-growth or properness assumption and gives $A_F \equiv 0$, contradicting the retained mark. □

Theorem 8.2 (Unrestricted stable classification near one)**.** *Under Theorems 1.2 and 1.3, there is $s_0 < 1$, which may be increased to exceed $3/4$, such that every bounded stable entire solution of (1.1) in $\mathbb{R}^7$ is classified whenever $s \in (s_0, 1)$.*

Proof. If there were no such threshold, for each sufficiently large integer j we could choose a nonclassified bounded stable solution at an order $s_j \in (1 - 1/j, 1)$. Proposition 2.8 gives the global regular zero geometry, moving-center preparation, and finite curvature suprema $\kappa_j \to 0$ after discarding finitely many terms. Proposition 2.11 excludes $\kappa_j = 0$, so the remaining sequence satisfies all hypotheses of Proposition 8.1, a contradiction. Increasing the resulting threshold to a number above $3/4$ preserves the conclusion. □

The proof is a uniform near-one statement. Classification at each fixed order under a threshold $\eta(s)$ would not imply it, since $\eta(s)$ could vanish faster than the curvature of the selected counterexamples. The sequence argument above retains the uniform analytic estimates at every scale, rather than choosing an order-dependent curvature threshold without control.

8.2. **Directional limits and global minimality.** Fix $s \in (s_0, 1)$ and a solution u as in Theorem 1.1. No estimate in the remaining argument needs to be uniform as $s \to 1$.

Lemma 8.3 (End equations and dimensionally correct stability)**.** *Let u be bounded, solve (1.1) in $\mathbb{R}^8$, and satisfy $\partial_8 u > 0$. The end profiles*

$$u^{\pm}(x') = \lim_{t \to \pm\infty} u(x', t), \qquad x' \in \mathbb{R}^7,$$

are bounded entire solutions in $\mathbb{R}^7$ and are stable there. Moreover $u^-(x') < u(x', t) < u^+(x')$ for every finite t.

Proof. Monotonicity and boundedness give the limits. Comparing a finite t with $t - 1$ and $t + 1$ gives both strict inequalities. Proposition 2.2 and moving-center compactness give local smooth convergence of vertical translates to the uniquely determined invariant functions $U^{\pm}(x', t) = u^{\pm}(x')$. The positive bounded derivative $w = \partial_8 u$ solves the Jacobi equation and satisfies the weighted integrability hypothesis of Lemma 2.5. Thus u is stable in $\mathbb{R}^8$, and Corollary 2.4 passes this stability to $U^{\pm}$.

The operator on a function independent of the last coordinate is exactly the seven-dimensional Fourier-normalized operator. To see this in the kernel form, first symmetrize the singular integral; its quadratic singularity is absolutely integrable. Then use

$$\int_{\mathbb{R}} \frac{\mathrm{d}t}{(r^2 + t^2)^{(8+2s)/2}} = \frac{\sqrt{\pi}\,\Gamma((7+2s)/2)}{\Gamma((8+2s)/2)}\, r^{-7-2s}.$$

Multiplication by $c_{8,s}$ gives exactly $c_{7,s}$, so the end equations and their potentials are unchanged.

For stability, fix $\phi \in C_c^\infty(\mathbb{R}^7)$ and nonzero $\chi \in C_c^\infty(\mathbb{R})$, test with $\Phi_T(x',t) = \phi(x')\chi(t/T)$, and divide by T. With the unitary Fourier transform the kinetic term becomes

$$\int_{\mathbb{R}^7}\int_{\mathbb{R}} (|\xi|^2 + |\rho|^2/T^2)^s |\widehat{\phi}(\xi)|^2 |\widehat{\chi}(\rho)|^2 \,\mathrm{d}\rho\,\mathrm{d}\xi.$$

For $T \geq 1$, $(a+b)^s \leq a^s + b^s$ supplies an integrable majorant, and the limit is $\|\chi\|_2^2 \int |\xi|^{2s}|\widehat{\phi}|^2$. The divided potential term is exactly $\|\chi\|_2^2 \int (1 - 3(u^{\pm})^2)\phi^2$. Dividing by $\|\chi\|_2^2$ proves seven-dimensional stability. □

For a bounded open domain $\Omega \subset \mathbb{R}^d$, put

$$K_s(v;\Omega) = \frac{c_{d,s}}{4} \iint_{\mathbb{R}^{2d}\setminus(\Omega^c\times\Omega^c)} \frac{|v(x)-v(y)|^2}{|x-y|^{d+2s}}\,\mathrm{d}x\,\mathrm{d}y, \tag{8.1}$$

$$J_s(v;\Omega) = K_s(v;\Omega) + \int_\Omega W(v)\,\mathrm{d}x, \qquad W(v) = \tfrac{1}{4}(1-v^2)^2. \tag{8.2}$$

A global minimizer minimizes this energy against smooth compact perturbations in every such domain. Equivalently, it minimizes against every finite-energy competitor agreeing with it outside a compact subset of the domain, by the truncation and density argument below. No finiteness of the energy on all of $\mathbb{R}^d$ is asserted.

Lemma 8.4 (Wells and layers minimize)**.** *The wells and every rotated translate of q_s minimize* (8.2) *against compact perturbations in every dimension.*

Proof. For the wells the energy is zero and its terms are nonnegative. For a layer, let $a = 1-2s$ and write its harmonic extension as Q. Let $d_s > 0$ be the dimension-independent extension constant defined by

$$L_s v = d_s \left[-\lim_{z\downarrow 0} z^a \partial_z (Ev) \right].$$

In the convention of [3, formula (1.4)], the extension of our layer solves their boundary equation with reaction $\widetilde{f} = (1+a)(v-v^3)/d_s$ and potential $\widetilde{W} = (1+a)W/d_s$. This reaction is smooth on the relevant value interval, so the required Hölder exponent can be chosen with $\max\{0, 1-2s\} < \gamma < 1$. Theorem 2.11(a) of that reference applies in every dimension. On each bounded comparison domain, multiplying its extension energy by d_s gives precisely

$$\frac{d_s}{2}\int z^a |\nabla Q|^2 + \int W(Q(\cdot,0)).$$

Their restriction to competitors in $[-1,1]$ is harmless here: projection onto this interval decreases the Dirichlet integrand and the quartic potential. To justify smooth admissible comparisons, fix a bounded comparison region containing the compact perturbation with a positive margin. On its compact closure the continuous extension Q stays a positive distance from both wells. Choose smooth contractions approximating the projection, taking values in $[-1,1]$ and equal to the identity on this compact range of Q. Apply them to the competitor inside the comparison region and keep Q outside. These definitions agree on the margin, where the competitor already equals Q. The chain rule and dominated convergence give convergence of the weighted Dirichlet and boundary-potential energies. Passing to the weighted energy limit gives unrestricted compact extension comparisons. Proposition 2.5 of [6] therefore transfers the minimizing property to (8.2).

The normalization agrees exactly. In [6, formula (1.2)], the symmetric-second-difference constant is $c_{d,s}/2$, and its trace energy coefficient is half that constant, hence $c_{d,s}/4$. Restoring the constants suppressed after formula (2.6) in that reference gives the compatible extension coefficient $d_s/2$. No change of equation or potential is hidden in the equivalence.

For a finite-energy compact competitor, first project its values onto $[-1,1]$. This fixes the background outside the perturbation set and does not increase either energy term. Mollify the bounded projected competitor and blend it with the smooth background using a cutoff equal to one near that set and supported inside Ω. The resulting smooth compact competitors remain in $[-1,1]$ and converge in the compact-perturbation H^s norm and in potential energy. Thus the

same comparison holds for every admissible finite-energy competitor. Finally, if the literature comparison is given on a ball containing the support, its energy difference equals the difference on Ω, since the competitor agrees with the background outside that support. □

Proposition 8.5 (Minimizing-end bridge, [6, Lemma 6.1])**.** *Let $u \in C^2(\mathbb{R}^d, [-1,1])$ solve $L_s u = u - u^3$ and satisfy $\partial_d u > 0$. If its two pointwise directional end profiles are minimizers in dimension $d-1$, then u is a minimizer in dimension d.*

This is the stated application of [6, Lemma 6.1]. That lemma applies for all $s \in (0,1)$ and in arbitrary dimension; its hypotheses include the displayed regularity, range, and strict monotonicity. The pointwise ends need not be constant, and no uniform convergence to them is required. Its energy convention is the one matched above. The cubic satisfies the bistable hypotheses used there, and Proposition 2.2 supplies both the regularity and the bound $|u| \le 1$ here.

Theorem 8.2 and Lemma 8.3 classify both ends of the given solution in $\mathbb{R}^8$ as wells or layers in $\mathbb{R}^7$. Lemma 8.4 and Proposition 8.5 therefore make u a minimizer in $\mathbb{R}^8$.

8.3. **Monotone blowdowns and the proof of the main theorem.** There is a zero of u at which to center the blowdown. Indeed, strict order rules out $u^- \equiv 1$ and $u^+ \equiv -1$. Every remaining classified lower end takes a negative value, and every remaining classified upper end takes a positive value. Pointwise convergence to those ends gives both signs of u, possibly at different horizontal points. A continuous path joining these points supplies a zero. Translate it to the origin.

For $u_\varepsilon(x) = u(x/\varepsilon)$ use the fixed-order energy

$$F_\varepsilon(v;\Omega) = \varepsilon^{2s-1} K_s(v;\Omega) + \varepsilon^{-1} \int_\Omega W(v)\,\mathrm{d}x. \tag{8.3}$$

The unnormalized-kernel conventions are matched separately. Savin uses coefficient $1/4$ on the full interaction-pair integral, so the matching energy is $J_s/c_{8,s}$ with potential $W/c_{8,s}$. Savin–Valdinoci use coefficient $1/2$ on that integral, so the matching energy is $2J_s/c_{8,s}$ with potential $2W/c_{8,s}$. These positive multipliers preserve minimizers at fixed s; both rescaled potentials are admissible double wells, and only the positive surface-tension constant changes. The numbered results of [8] used below refer to the version arXiv:1007.2114v3. A change of variables gives

$$F_\varepsilon(u_\varepsilon; B_R) = \varepsilon^7 J_s(u; B_{R/\varepsilon}).$$

By [8, Theorem 1.3], applied on B_{L+2}, $J_s(u; B_L) \le C_s L^7$ for $L \ge 1$. Hence the displayed scaling identity bounds $F_\varepsilon(u_\varepsilon; B_R)$ for each fixed R. Corollary 1.7(ii) of [8] gives a sequence $\varepsilon_j \downarrow 0$ for which

$$u_{\varepsilon_j} \longrightarrow 2\chi_E - 1 \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^8). \tag{8.4}$$

Since $s > 1/2$, its limiting functional is a positive multiple of classical perimeter, and the same corollary makes E minimizing against every compact perturbation.

For nontriviality, use $u(0) = 0$ and the evenness of W. Applying [8, Theorem 1.4] to u and $-u$, with $\theta_1 = -1/2$ and $\theta_2 = 1/2$, and then rescaling gives

$$|\{u_{\varepsilon_j} > 1/2\} \cap B_r| \ge c_s r^8, \qquad |\{u_{\varepsilon_j} < -1/2\} \cap B_r| \ge c_s r^8$$

for every fixed $r > 0$ and all sufficiently large j. The L^1 convergence in (8.4) implies convergence in measure of these two threshold sets to E and E^c, respectively, inside B_r. Thus both E and E^c have positive density in every ball centered at zero. In the measure-theoretic representative, $0 \in \partial E$ and E is neither empty nor full.

Lemma 8.6 (The blowdown retains monotonicity)**.** *The perimeter minimizer obtained in (8.4) satisfies $D_8 \chi_E \ge 0$ in distributions.*

Proof. For every nonnegative $\varphi \in C_c^\infty(\mathbb{R}^8)$,

$$-\int u_{\varepsilon_j} \partial_8 \varphi \ge 0.$$

Pass to the L^1_{loc} limit and use $\int \partial_8 \varphi = 0$. The resulting inequality is $-2\int \chi_E \partial_8 \varphi \ge 0$. □

Proposition 8.7 (Generalized-graph Bernstein consequence)**.** *Let $E \subset \mathbb{R}^N$, $1 \le N \le 8$, be a nontrivial set of locally finite perimeter which minimizes classical perimeter against every compactly supported change. If, after a rotation, $D_N\chi_E \ge 0$ in distributions, then E is a halfspace modulo a Lebesgue-null set.*

Proof. All sets and containments are understood modulo null sets. For $N = 1$, monotonicity and nontriviality give an upper half-line. For $N \ge 2$, slicing gives

$$E = \{(x', t) : t > \psi(x')\}, \qquad \psi : \mathbb{R}^{N-1} \longrightarrow [-\infty, +\infty].$$

Set $P = \{\psi = +\infty\}$ and $M = \{\psi = -\infty\}$. We use the generalized-solution results recorded in [9, proof of Theorem 7.1, Step 3]: P and M minimize perimeter in $\mathbb{R}^{N-1}$; if both are null, ψ agrees almost everywhere with an entire classical minimal graph. The same step recalls that every nontrivial entire perimeter minimizer in ambient dimension at most seven is a halfspace, and that a nontrivial entire perimeter minimizer containing a halfspace is itself a halfspace.

Neither P nor M is conull, since these alternatives would make E empty or full. If M has positive measure, it is therefore a halfspace, and $M \times \mathbb{R} \subset E$ gives the conclusion by halfspace rigidity. If P has positive measure, use $P \times \mathbb{R} \subset E^c$ and the same rigidity for the perimeter minimizer E^c. In the remaining case the classical Bernstein theorem in base dimension $N-1 \le 7$ makes ψ affine. These are precisely the geometric hypotheses used in the cited step; no conical homogeneity of E is required. □

Apply Proposition 8.7 to E and write $\partial E = \{y : y \cdot e = 0\}$ for a unit vector e. Corollary 1.7(iii) of [8], applied on $\overline{B_1} \subset B_2$ with any fixed transition threshold in $(0, 1)$, implies

$$b_j := \sup\{|y \cdot e| : y \in B_1,\ u_{\varepsilon_j}(y) = 0\} \longrightarrow 0.$$

The set in this supremum is nonempty since it contains zero. Set $R_j = \varepsilon_j^{-1}$ and $\alpha_j = b_j + j^{-1}$. Rescaling this inclusion yields

$$\{u = 0\} \cap B_{R_j} \subset \{x : |x \cdot e| < \alpha_j R_j\}, \qquad R_j \longrightarrow \infty, \quad \alpha_j \longrightarrow 0. \tag{8.5}$$

The minimizer of J_s is also a minimizer of Savin's functional with potential $W/c_{8,s}$. This potential is C^2, positive on $(-1, 1)$, vanishes with its first derivative at both wells, and has positive second derivative there. Theorem 1.1 of [7] therefore applies with slab widths $\alpha_j R_j$ and gives that u is one-dimensional. This theorem holds in every ambient dimension for each fixed $s \in (1/2, 1)$; it requires precisely the sequential flatness in (8.5). Its arbitrary-minimizer corollary, Theorem 1.2, is restricted to dimensions at most seven.

Proof of Theorem 1.1. The preceding fixed-order argument proves that u is one-dimensional. Write $u(x) = h(e \cdot x - a)$ with $|e| = 1$. Since $\partial_8 u = e_8 h' > 0$, one has $e_8 \ne 0$; reversing the direction if necessary gives $e_8 > 0$ and $h' > 0$. The two directional limits are then the constants $h(\pm\infty)$. Their stable classification in Theorem 8.2, together with their strict order, makes them -1 and 1, respectively. The increasing-layer uniqueness recalled in Section 1 identifies h with a translate of q_s, which is absorbed into a. This is the asserted form, with $e_8 > 0$. □

8.4. **Scope and uniformity of the argument.** CL7 supplies the moving-center phase-field preparation, while SB7 excludes the nonflat stable minimal immersion produced by curvature rescaling. Their roles are distinct. The intermediate estimates are uniform over moving zero centers and all charts in each fixed compact intrinsic region; every depth, buffer, and atlas is fixed before the corresponding sequence limit. The final minimizing-end and blowdown arguments instead use one fixed $s \in (s_0, 1)$ and require no uniform constants as $s \uparrow 1$.

The quantitative results from the preceding sections remain essential: the force estimate alone does not give the source capacities or normal compactness, and bounded curvature alone does not retain a nonzero curvature mark. Likewise, chartwise stability does not replace the whole-carrier stability of Theorem 7.7. The conclusion is an existential interval near $s = 1$; no assertion is made here at $s = 1/2$, for all fractional orders, or in higher ambient dimensions.

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